

Conventional and Special Core Analysis

Client: Mellitah Oil & Gas

Well #: CE03-NC41

Draft Final Report

Report No.: 01/PET1-1&2/2024

LPI Work Order No: 282/PET1-1&2/2024

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RE: Conventional & Special Core Analysis for Well #: CE03-NC41

Dear Mr. Nafati Tarhuni \ Geology Superintendent

With reference to your letter dated on January 02, 2024 and direct work order No. DWO/RES/26-23 dated on April 30, 2024, concerned with your request to carry out conventional and Special core analysis for well **CE03-NC41**, we thus hereby enclose a digital copy of the draft final report of the conventional and special core analysis which contains all the results performed on plugs from the subject well except for the relative permeability tests.

We hope that the report meets your requirements. Your remarks and suggestions shall be highly appreciated.

Muhieddin Ibrahim Hamza
Manager, Petroleum Engineering Research Dept.

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Summary

Summary

This report presents the results of the conventional core analysis for the well **CE03-NC41**. This well is in the concession 41 in Bahr Essalam Field - "C" Structure Central East Area and the cores were recovered from El-Garia formation.

A Total of (333 Horizontal and 112 Vertical) one and half inch diameter core plugs were selected from four full diameter cores, having a total core length 333 ft, from the well #: CE03-NC41, these plugs were drilled by our staff.

Conventional core analysis measurements have been done on core samples taken from the mentioned well, as per scope of work, these measurements were conducted by Petroleum Research Centre (PRC) laboratories and finally the results were processed and interpreted.

- **The first part of this report includes the conventional core analysis as follows:**

- The measurement of porosity, grain density, horizontal permeability and vertical permeability under ambient conditions for the core samples.
- The measurement of porosity, grain density and air permeability for three hundred and thirty-three horizontal core plug samples of 1.5 inches, in addition of that, one hundred and twelve vertical core plug samples were selected to perform the measurement of air permeability. Then fluid Saturation was measured by Dean-Stark Method. After that the statistical analysis for the measurement's results were calculated and sample selection was carried out to perform the special core analysis on the representative samples.

- **The second part of the report includes the special core analysis as follows: -**

- Firstly, the porosity and permeability at various overburden pressures (5 pressures) were measured and the pore volume compressibility was determined from the data derived for each overburden pressure. After that, the electrical properties of the selected plug samples were measured, which include formation resistivity factor, cementation factor at ambient conditions and at different overburden pressures. Air-Brine capillary pressure by porous plate was measured in conjunction with resistivity index and saturation exponent at ambient conditions and at overburden pressure. The air-water capillary pressure by ultra-speed centrifuge, (at ambient conditions) was carried out and the calculations were made using Hassler technique.

- Ten samples were selected for the measurements of high pressure (up to 60,000 psi) mercury injection capillary pressure, drainage & imbibition, and pore size distribution.
- Five samples were selected to measure gas-water and gas-oil relative permeability at reservoir conditions by using SMAX system, the test was carried out by using crude and mineral oil at reservoir temperature and overburden pressure. Relative permeability relationships at reservoir conditions were calculated from collected data, sample and fluid parameters by using *Sendra* software.

NOTE: -

- Most of the vuggy and friable samples are excluded from the selection, in addition to that, fractured and the samples with low permeability (less than 2 mD) were excluded.

Part One

Part One

Conventional Core Analysis

Methods and Experiments Tests According to API RP 40, Recommended Practice for Core Analysis

ANALYTICAL PROCEDURES AND EQUIPMENT SPECIFICATION

Sample Preparation: -

Four full diameter cores from well No. **CE03-NC41** with total length of (333) feet were received in our laboratory, as per scope of work.

Core pieces were chipped from the full diameter core to determine residual fluid saturation, then one and half inch diameter core plugs were drilled in horizontal and vertical directions as requested by Mellitah Oil Company using diamond core bit with water as the bit coolant and lubricant.

The samples were extracted of hydrocarbons using toluene, leached of salt using methanol, and oven dried at 80 °C. Silver Nitrate solution was used to ensure the removal of salts existing in the pore spaces.

Dean Stark Analysis: -

The equipment consisted of a flask, a sample chamber with side arm for collection of water and a condenser with a silica gel filter fitted to the top as an atmospheric water trap. The solvent used for extraction was toluene; this has been previously de-watered by distillation and refluxed in each extraction unit for at least twelve hours to remove any water in the system. Water if present, accumulated in the trap, was removed from the system and preliminary recycling continued. The core chips were taken from the core, weighed and transferred immediately to clean, pre-weighed tares. The tares with samples were sealed without delay in distillation apparatus and the system was refluxed for a minimum of 24 hours. Water accumulation in the graduated side arm was recorded, and when the samples stabilised were removed. Upon removal from extraction apparatus, the samples were tested for fluorescence under ultraviolet light to check for hydrocarbon presence.

The samples were subsequently transferred to a standard oven at 60 C°, and dried at constant weight. Once dried and weighed, the samples with tares were placed in individual soxhlets and cleaned with methanol to remove salts.

The methanol in the soxhlet chamber was tested with silver nitrate solution to ensure that the salt derived from formation water had been removed from the sample.

Finally, the samples were returned to the oven and again dried to constant weight and subjected to porosity measurements in the manner discussed below.

The weight of the chips before and after the extraction, the chip pore volume and the water collected were used to calculate residual fluid saturations using oil gravity of **29.4 API**.

POROSITY, PERMEABILITY AND GRAIN DENSITY MEASUREMENTS: -

The clean, dry samples were subjected to various analyses to determine porosity, permeability and grain density values where possible. The experimental procedures and equipment specifications are discussed below: -

1. Helium Gas Expansion Porosimeter: -

Core analysis laboratory of Petroleum Research Center use a twin cell helium expansion gas porosimeter for the plug samples grain volume measurement. The porosimeter operates using the principle of **Boyle's Law**.

A sealed reference chamber in the instrument is filled with helium gas at ambient temperature to a pressure of 100 psig. A sample is placed in another sealed chamber, connected to the reference chamber by a two-way valve. This valve when opened allows the gas in the reference chamber to expand into the combined volume of the two chambers. From Boyle's Law, the volume of the sample chamber can be calculated when the volume of the reference chamber, the initial pressure and the final pressure are known.

The instrument must be calibrated before using the porosimeter for analysis. This is done by running a series of stainless-steel blanks of known volumes, to build up a graph of blank volume versus the inverse of the final pressure.

The resulted calibration graph has to be entered into a computer program which performs a linear regression, producing an equation relating the grain volume of any sample run to the final pressure reading. The porosity and the grain density were then calculated by determining the bulk volume, and the weight of the sample.

As a quality check, a suite of standard samples of known porosities and grain densities were measured every 20 samples.

The instrument used by our laboratory has the following specifications: -

Pressure indicator with remote pressure transducer

- | | | |
|---------------|---|-------------------------------|
| 1- Range | : | 0-100 psi |
| 2- Resolution | : | 0.01 psi |
| 3- Accuracy | : | $\pm 0.1\%$ of the full scale |

Precision Pressure Controller

- 1- Range : 0-100 psi
- 2- Resolution : Better than 0.01 psi
- 3- Repeatability : Better than 0.02 psi

2. Positive Displacement Pump: -

The positive displacement mercury pump is used to determine the bulk volume of each sample. The unit consists of a stainless-steel high-pressure pycnometer (sample container); it is attached to the volumetric pump cylinder. As the hand wheel connected to the pump is wound in or out, mercury is charged or withdrawn from the sample chamber. The volume of mercury displaced is measured linearly by a digital transducer connected to the pump. In practice, mercury is allowed into the pycnometer to a set reference mark, and the transducer set to zero. This process should be repeated several times to ensure accuracy. The mercury is subsequently released back into the pump, and the sample is placed and sealed in the pycnometer. After that mercury is allowed back into the pycnometer up to the previously determined reference mark, and the display shown on the digital transducer noted. This value is later converted to the actual volume by using a simple correction factor derived from calibration volumes. The mercury is cleaned, and the pump reset to zero at least once every four samples, depending on the sample type.

3. Calculation of Porosity: -

The porosity of a plug sample is calculated by using the following equations: -

$$\text{Pore volume (cc)} = \text{Bulk volume} - \text{Grain volume}$$

$$\text{Porosity (\%)} = \frac{\text{Pore volume}}{\text{Bulk volume}} \times 100$$

$$\text{Grain density (gm/cc)} = \frac{\text{Sample weight}}{\text{Grain volume}}$$

4. Nitrogen Gas Permeability: -

The clean, dry sample is placed in the Hassler-type core holder, and an overburden pressure of 200 psig is applied to the cell, compressing the rubber sleeve around the sample. Nitrogen gas pressure is applied to one end face of the sample, whilst the other end face is open to atmospheric pressure, causing the gas to flow through the sample. The flow of gas is measured at the low pressure (downstream) end face of the sample. The flow rate is measured by passing the gas through one of three laminar flow orifices and measuring the differential pressure developed across the orifice.

The transducers and flow orifices are calibrated using a dead weight tester and soap film meter at least once every three months, or when any component is altered or replaced.

The equipment specifications are as follow: -

High pressure precision regulator

- 1- Range : 0-30 psi
- 2- Resolution : Better than 0.01 psi

High pressure transducer and readout (upstream)

- 1- Range : 0-100 psi
- 2- Resolution : 0.01 psi
- 3- Accuracy : ± 0.1 % of the full range

Low pressure transducer and readout (upstream)

- 1- Range : 0-15 psi
- 2- Resolution : 0.001 psi
- 3- Accuracy : ± 0.1 % of full scale

The length of the sample, upstream and downstream pressures, flow rate, viscosity of nitrogen, barometric pressure and temperature are entered into Darcy's equation for gas permeability, and the permeability of the sample calculated.

$$Kg = \frac{2000 \times P_b \times Q_g \times \mu_g \times L}{(P_1^2 - P_2^2) \times A}$$

Where:

- K_g** = Gas permeability (mD)
P_b = Barometric pressure (atm)
P₁ = Upstream pressure (atm)
P₂ = Downstream pressure (atm)
Q_g = Gas flow rate (cc/sec.)
μ_g = Gas viscosity (cp)
L = Sample length (cm)
A = Cross-sectional area (sq.cm)

5. Klinkenberg Permeability:-

The Klinkenberg permeability correction is calculated using the mathematical relationship between **K_g** and **K_I** as follows:

$$K_I = 0.68 (K_g)^{1.06} \dots\dots\dots \text{(Empirical Correction)}$$

This empirical correction is valid for permeability ranges from 0.01 up to 174 mD and from 175 to 7500 mD using the following table:

(K _g) Permeability Range, mD	(K _I) Klinkenberg Permeability, mD
175-299	K _g – 20
300-699	K _g – 30
700-999	K _g – 40
1000-1699	K _g – 50
1700-2299	K _g – 60
2300-5799	K _g – 80
5800-7500	K _g – 100

Internal Quality Control: -

As previously stated, all pressure gauges and transducers are regularly checked and calibrated using dead weight tester and wet soap flow meters. In order to maintain a rigid quality assurance programme, a series of four standard check plugs whose dimensions, porosity and grain density are known, are run every data set (maximum of 20 samples) to check repeatability.

The following criteria are used to check repeatability of all data by using the same equipment with different operator.

Parameter	Acceptable Error
Grain volume	$\pm 0.2 \%$
Grain density	$\pm 0.02 \text{ gm/cc}$
Porosity	$\pm 0.2 \%$
Permeability	$\pm 4 \%$

Conventional Core Analysis

Conventional Core Analysis Results And Profiles

Conventional Core Analysis Results

Client: Mellitah Oil & Gas
Well : CE03-NC41

Concession: 41
Formation : El Garia

Report No: 01/PET1-1&2 /2024
Field: Bahr Essalam

Sample No.	Depth (ft.in)	Air Permeability, mD No Klinkenberg correction @200 psi		Air Permeability, mD With Klinkenberg correction @200 psi		Porosity (%)	Density, (gm/cc)		Remarks
		Horizontal	Vertical	Horizontal	Vertical		Bulk	Grain	
Core # 1 (8356'-8446') El Garia Formation									
1	8358.06	1.04	0.995	0.709	0.676	8.66	2.55	2.70	Short
2	8359.03	22.1		18.1		8.26	2.56	2.70	Short+ Vuggy
3	8360.06	91.0		81.1		8.50	2.56	2.70	Vuggy
4	8361.08	1.80	1.59	1.30	1.11	6.33	2.59	2.70	Vuggy
5	8362.02	5.29		3.97		6.49	2.59	2.70	Vuggy
6	8363.07	0.510		0.335		4.15	2.63	2.70	Vuggy
7	8364.08	3.26		2.38		5.65	2.61	2.70	Vuggy
8	8365.03	5.60		4.22		7.82	2.57	2.70	Vuggy
9	8366.03	1.08		0.740		9.41	2.54	2.70	Vuggy
10	8367.06	19.0	2.93	15.5	2.13	9.49	2.54	2.70	Short+ Vuggy
11	8368.06	4.10		3.04		7.59	2.57	2.70	Vuggy
12	8369.09	16.1		13.0		9.73	2.53	2.70	Short+broken
13	8370.05	1.10	10.3	0.75	8.09	5.46	2.61	2.70	Short+Vuggy
14	8371.04	2.81		2.03		7.18	2.58	2.70	Vuggy
15	8372.05	1.10		0.751		6.67	2.59	2.70	Vuggy
16	8373.07	5.22	5.36	3.92	4.03	8.86	2.55	2.70	Vuggy
17	8374.05	7.77		5.97		8.69	2.55	2.70	Vuggy
18	8375.05	4.74		3.54		7.79	2.57	2.70	Vuggy

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		Horizontal	Vertical	Horizontal	Vertical		Bulk	Grain	
19	8376.07	10.2	0.306	8.01	0.194	4.22	2.63	2.70	Vuggy
20	8377.04	0.352		0.225		6.74	2.59	2.70	Short
21	8378.11	5.47		4.12		7.46	2.57	2.70	Vuggy
22	8379.08	4.17	4.28	3.09	3.18	8.14	2.56	2.70	Vuggy
23	8380.04	6.52		4.96		9.08	2.54	2.70	Vuggy
24	8081.05	6.12		4.64		9.25	2.54	2.70	Vuggy
25	8382.06	21.7	5.70	17.7	4.30	10.6	2.52	2.70	Vuggy
26	8383.06	49.7		42.7		12.7	2.48	2.70	Vuggy
27	8384.07	4.58	1.85	3.41	1.30	9.84	2.53	2.70	Vuggy
28	8385.03	5.57		4.20		10.13	2.53	2.70	Vuggy
29	8386.03	4.02		2.98		9.98	2.53	2.70	Vuggy
30	8387.09	8.85	16.79	6.86	13.5	11.3	2.51	2.70	Vuggy
31	8388.03	73.3		64.5		13.6	2.47	2.70	Vuggy
32	8389.08	136		124		14.8	2.45	2.70	Vuggy
33	8390.07	93.8	27.30	83.7	22.6	13.5	2.47	2.70	Vuggy
34	8391.04	129		117		14.1	2.46	2.70	Vuggy
35	8392.03	25.0		20.6		13.5	2.47	2.70	Vuggy

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		Horizontal	Vertical	Horizontal	Vertical		Bulk	Grain	
36	8393.08	15.6	31.9	12.5	26.7	13.5	2.47	2.70	Vuggy
37	8394.03	24.6		20.3		12.0	2.49	2.70	Vuggy
38	8395.04	8.65		6.70		11.0	2.51	2.70	Vuggy
39	8396.08	45.1	41.1	38.6	34.9	12.3	2.49	2.70	Vuggy
40	8397.04	41.0		34.9		12.5	2.49	2.70	Vuggy
41	8398.04	36.5		30.8		11.6	2.50	2.69	Vuggy
42	8399.06	21.4	3.20	17.5	2.33	12.4	2.48	2.69	Vuggy
43	8400.03	8.28		6.39		10.9	2.51	2.70	Vuggy
44	8401.04	11.7		9.21		10.9	2.51	2.70	Vuggy
45	8402.07	8.05	19.7	6.20	16.0	10.4	2.52	2.70	Vuggy
46	8403.02	2.43		1.74		7.8	2.56	2.70	Vuggy
47	8404.02	11.5		9.03		10.4	2.52	2.69	Vuggy
48	8405.07	42.5	4.72	36.2	3.53	14.3	2.45	2.69	Vuggy
49	8406.05	28.1		23.3		12.3	2.49	2.69	Vuggy
50	8407.05	39.6		33.6		12.2	2.49	2.69	Vuggy
51	8408.08	13.8	8.60	11.0	6.65	11.9	2.49	2.69	Vuggy
52	8409.03	5.60		4.23		10.4	2.51	2.69	Vuggy

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		Horizontal	Vertical	Horizontal	Vertical		Bulk	Grain	
53	8410.04	8.87		6.88		11.9	2.49	2.69	Vuggy
54	8411.06	13.8	15.8	11.0	12.7	12.9	2.47	2.69	Vuggy
55	8412.02	9.68		7.55		13.9	2.46	2.69	Vuggy
56	8413.07	17.0		13.7		14.0	2.45	2.69	Vuggy
57	8414.09	31.2	12.5	26.1	9.93	14.6	2.44	2.69	Vuggy
58	8415.03	1.29		0.890		13.5	2.46	2.69	Vuggy
59	8416.03	24.1		19.8		14.7	2.44	2.69	Vuggy
60	8417.08	28.1	32.9	23.3	27.6	16.2	2.42	2.69	Vuggy
61	8418.05	7.22		5.53		12.9	2.47	2.68	Vuggy
62	8419.06	13.2		10.5		12.7	2.47	2.69	Vuggy
63	8420.08	21.8	20.0	17.8	16.3	15.0	2.43	2.68	Vuggy
64	8421.03	7.65		5.88		12.5	2.47	2.68	Vuggy
65	8422.04	8.07		6.22		14.2	2.45	2.68	Vuggy
66	8423.07	9.45	11.8	7.35	9.29	12.8	2.47	2.68	Vuggy
67	8424.03	22.9		18.8		15.6	2.42	2.69	Vuggy
68	8425.04	56.4		48.9		15.9	2.42	2.69	Vuggy
69	8426.04	10.3	7.69	8.04	5.91	13.1	2.47	2.69	Vuggy

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		Horizontal	Vertical	Horizontal	Vertical		Bulk	Grain	
70	8427.03	7.25		5.56		11.9	2.48	2.68	Vuggy
71	8428.03	7.76		5.96		11.4	2.50	2.69	Vuggy
72	8429.06	26.9	10.2	22.3	8.00	13.9	2.46	2.69	Vuggy
73	8430.02	16.6		13.4		12.2	2.49	2.69	Vuggy
74	8431.02	29.2		24.3		15.3	2.44	2.69	Vuggy
75	8432.08	98.5	123.0	88.2	112	17.7	2.39	2.69	Vuggy
76	8433.03	69.3		60.8		22.2	2.31	2.69	Vuggy
77	8434.05	55.2		47.7		19.6	2.36	2.69	Vuggy
78	8435.08	32.3	88.2	27.1	78.5	15.5	2.42	2.68	Vuggy
79	8436.04	25.4		21.0		20.4	2.35	2.69	
80	8437.06	51.7		44.5		21.0	2.34	2.70	
81	8438.08	30.3	32.8	25.3	27.5	20.3	2.35	2.70	
82	8439.03	34.6		29.1		19.1	2.37	2.69	
83	8440.06	56.3		48.8		20.0	2.35	2.69	Vuggy
84	8441.08	99.6	244	89.3	224	18.9	2.38	2.70	Vuggy
85	8442.02	142		130		21.2	2.34	2.69	Vuggy
86	8443.04	270		250		19.8	2.36	2.70	Vuggy
87	8444.06	165	84.0	152	74.5	21.1	2.33	2.69	Vuggy
88	8445.02	93.2		83.1		22.4	2.31	2.69	Vuggy

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		Horizontal	Vertical	Horizontal	Vertical		Bulk	Grain	
Core # 2 (8446'-8536') El Garia Formation									
89	8446.04	54.6		47.2		23.0	2.30	2.69	Vuggy
90	8447.07	97.5		87.3		22.8	2.30	2.69	Short
91	8448.07	46.4	23.4	39.8	19.3	20.9	2.34	2.69	Vuggy
92	8449.03	60.1		52.3		22.4	2.31	2.69	
93	8450.06	18.9		15.3		23.3	2.30	2.69	
94	8451.08	97.9	53.1	87.6	45.8	23.3	2.30	2.69	Vuggy
95	8452.05	90.8		81.0		23.5	2.29	2.69	Vuggy
96	8453.09	531		501		23.3	2.30	2.69	Vuggy
97	8454.09	161	121.2	148	110	24.4	2.28	2.70	Vuggy
98	8455.04	99.0		88.7		23.9	2.29	2.69	Vuggy
99	8456.07	35.0		29.5		19.4	2.37	2.70	Vuggy
100	8457.07	55.5	118.5	48.0	107	23.1	2.30	2.69	
101	8458.02	112		101		28.6	2.21	2.70	
102	8459.03	44.4		37.9		24.5	2.28	2.69	Vuggy
103	8460.08	---	51.2	----	44.1	30.8	2.17	2.69	(H Broken)+Vuggy
104	8461.05	81.0		71.7		22.9	2.30	2.69	Vuggy
105	8462.03	17.4		14.1		21.2	2.32	2.68	
106	8463.07	274	202	254	182	29.2	2.19	2.69	Short+Vuggy

Conventional Core Analysis Results

Client: Mellitah Oil & Gas
Well : CE03-NC41

Concession: 41
Formation : El Garia

Report No: 01/PET1-1&2 /2024
Field: Bahr Essalam

Sample No.	Depth (ft.in)	Air Permeability, mD No Klinkenberg correction @ 200 psi		Air Permeability, mD With Klinkenberg correction @ 200 psi		Porosity (%)	Density, (gm/cc)		Remarks
		Horizontal	Vertical	Horizontal	Vertical		Bulk	Grain	
107	8464.05	314		284		28.7	2.21	2.69	Vuggy
108	8465.05	62.8		54.7		25.6	2.26	2.69	
109	8466.08	264	205	244	185	29.3	2.19	2.69	Vuggy
110	8467.05	381		351		22.4	2.32	2.70	Fracured+Vuggy
111	8468.08	363		333		23.2	2.30	2.69	Vuggy
112	8469.10	37.3	265	31.5	245	26.8	2.24	2.69	Vuggy+Fracured
113	8470.03	23.5		19.3		21.8	2.31	2.68	Fractured
114	8471.07	32.4		27.1		22.4	2.31	2.69	
115	8472.08	58.4	57.0	50.7	49.4	23.5	2.29	2.69	Broken edge
116	8473.03	48.1		41.3		23.0	2.31	2.70	
117	8474.05	135		123		26.9	2.24	2.69	Vuggy
118	8475.10	82.3	182	72.9	162	21.4	2.33	2.69	Vuggy
119	8476.04	94.9		84.8		25.4	2.26	2.69	Vuggy
120	8477.06	216		196		26.1	2.27	2.72	Vuggy
121	8478.04	155	140	143	128	23.7	2.29	2.69	Vuggy
122	8479.06	61.0		53.1		24.7	2.27	2.69	Vuggy
123	8480.06	34.3		28.8		25.0	2.27	2.69	Short+Vuggy
124	8481.10	38.4	101	32.5	90.7	21.7	2.33	2.69	Vuggy

Conventional Core Analysis Results

Client: Mellitah Oil & Gas
Well : CE03-NC41

Concession: 41
Formation : El Garia

Report No: 01/PET1-1&2 /2024
Field: Bahr Essalam

Sample No.	Depth (ft.in)	Air Permeability, mD No Klinkenberg correction @ 200 psi		Air Permeability, mD With Klinkenberg correction @ 200 psi		Porosity (%)	Density, (gm/cc)		Remarks
		Horizontal	Vertical	Horizontal	Vertical		Bulk	Grain	
125	8482.00	21.8		17.9		24.6	2.27	2.69	Short
126	8483.10	39.4		33.4		24.4	2.28	2.69	
127	8484.07	37.2	29.1	31.4	24.2	22.0	2.32	2.70	Vuggy
128	8485.03	108		97.22		25.3	2.27	2.70	
129	8486.05	18.0		14.5		19.2	2.37	2.70	Vuggy
130	8487.07	136	187	124	167	24.6	2.28	2.70	Vuggy
131	8488.05	46.57		39.87		25.8	2.25	2.69	
132	8489.08	33.4		28.1		23.1	2.30	2.69	
133	8490.07	29.6	46.7	24.7	39.97	19.8	2.36	2.69	
134	8491.04	42.0		35.8		24.4	2.28	2.69	Vuggy
135	8492.03	45.0		38.4		22.0	2.32	2.70	
136	8493.07	83.3	110	73.8	99.0	24.6	2.28	2.70	Vuggy
137	8494.03	42.4		36.1		22.8	2.30	2.69	Broken Edge
138	8495.06	20.2		16.5		18.1	2.39	2.70	
139	8496.07	140	69.1	128	60.6	25.2	2.27	2.69	Vuggy
140	8497.04	55.2		47.7		20.3	2.35	2.70	Vuggy
141	8498.03	41.0		34.8		18.2	2.39	2.70	
142	8499.10	25.1	21.5	20.7	17.6	18.6	2.38	2.69	

Conventional Core Analysis Results

Client: Mellitah Oil & Gas
Well : CE03-NC41

Concession: 41
Formation : El Garia

Report No: 01/PET1-1&2 /2024
Field: Bahr Essalam

Sample No.	Depth (ft.in)	Air Permeability, mD No Klinkenberg correction @ 200 psi		Air Permeability, mD With Klinkenberg correction @ 200 psi		Porosity (%)	Density, (gm/cc)		Remarks
		Horizontal	Vertical	Horizontal	Vertical		Bulk	Grain	
143	8500.07	30.3		25.3		21.8	2.32	2.69	
144	8501.04	49.5		42.6		23.2	2.29	2.69	
145	8502.05	46.1	64.1	39.5	56.0	23.9	2.28	2.69	
146	8503.03	26.0		21.5		23.6	2.29	2.69	
147	8504.04	52.2		45.0		24.4	2.28	2.69	
148	8505.10	73.4	54.4	64.5	47.0	25.3	2.27	2.69	
149	8506.06	90.7		80.9		29.5	2.19	2.69	
150	8507.09	84.6		75.1		25.5	2.26	2.69	
151	8508.10	50.3	34.7	43.3	29.1	23.1	2.30	2.69	
152	8509.03	14.5		11.6		18.2	2.38	2.69	
153	8510.06	23.9		19.7		23.2	2.30	2.69	
154	8511.05	47.4	34.2	40.6	28.7	24.7	2.28	2.70	
155	8512.02	20.2		16.4		23.0	2.30	2.69	
156	8513.05	51.0		43.9		27.4	2.23	2.70	
157	8514.08	54.4	31.6	47.0	26.4	25.2	2.27	2.70	
158	8515.04	69.7		61.1		25.1	2.27	2.70	
159	8516.04	32.6		27.3		22.2	2.32	2.69	
160	8517.08	75.4	26.2	66.5	21.7	26.3	2.25	2.69	

Conventional Core Analysis Results

Client: Mellitah Oil & Gas
Well : CE03-NC41

Concession: 41
Formation : El Garia

Report No: 01/PET1-1&2 /2024
Field: Bahr Essalam

Sample No.	Depth (ft.in)	Air Permeability, mD No Klinkenberg correction @ 200 psi		Air Permeability, mD With Klinkenberg correction @ 200 psi		Porosity (%)	Density, (gm/cc)		Remarks
		Horizontal	Vertical	Horizontal	Vertical		Bulk	Grain	
161	8518.03	29.9		24.9		25.3	2.26	2.68	
162	8519.05	23.3		19.2		19.6	2.35	2.68	
163	8520.07	10.9	8.65	8.58	6.69	19.4	2.36	2.69	
164	8521.04	4.89		3.66		16.3	2.41	2.68	
165	8522.05	8.06		6.21		17.8	2.38	2.69	
166	8523.08	7.77	10.8	5.98	8.49	16.1	2.40	2.67	Fractured
167	8524.05	46.3		39.6		22.6	2.31	2.69	
168	8525.08	30.0		25.0		22.8	2.30	2.69	
169	8526.09	53.0	104	45.7	93.6	25.1	2.27	2.69	
170	8527.04	35.3		29.7		23.1	2.30	2.69	
171	8528.05	36.2		30.5		25.6	2.25	2.69	
172	8529.07	77.1	34.9	68.0	29.4	30.8	2.17	2.69	Vuggy
173	8530.04	36.8		31.0		24.7	2.27	2.69	Vuggy
174	8531.04	56.5		49.0		26.9	2.24	2.69	
175	8532.09	56.6	67.2	49.1	58.9	26.6	2.24	2.69	Vuggy
176	8533.07	28.0		23.2		25.0	2.27	2.70	Vuggy
177	8534.06	58.0		50.3		22.6	2.31	2.70	
178	8535.05	39.7	55.6	33.7	48.1	24.7	2.28	2.69	

Conventional Core Analysis Results

Client: Mellitah Oil & Gas
Well : CE03-NC41

Concession: 41
Formation : El Garia

Report No: 01/PET1-1&2 /2024
Field: Bahr Essalam

Sample No.	Depth (ft.in)	Air Permeability, mD No Klinkenberg correction @200 psi		Air Permeability, mD With Klinkenberg correction @200 psi		Porosity (%)	Density, (gm/cc)		Remarks
		Horizontal	Vertical	Horizontal	Vertical		Bulk	Grain	
Core # 3 (8536'-8604') El Garia Formation									
179	8537.06	56.1		48.6		25.2	2.27	2.70	Vuggy
180	8538.08	72.0		63.3		24.6	2.28	2.70	Vuggy
181	8539.04	95.5	84.1	85.4	74.6	26.4	2.25	2.70	Vuggy
182	8540.02	115		104		24.9	2.27	2.69	Vuggy
183	8541.06	42.8		36.5		23.8	2.31	2.71	Short
184	8542.04	44.8	41.2	38.3	35.0	24.5	2.28	2.70	Vuggy
185	8543.07	85.6		76.0		25.6	2.26	2.69	Vuggy
186	8544.05	61.0		53.1		24.3	2.28	2.69	
187	8545.06	143	121	131	110	26.7	2.24	2.69	
188	8546.02	393		363		25.0	2.27	2.69	Vuggy
189	8547.02	201		181		24.9	2.27	2.69	Vuggy
190	8548.08	120	208	108	188	24.9	2.27	2.69	Vuggy
191	8549.02	228		208		24.4	2.28	2.69	Vuggy
192	8550.05	146		134		21.9	2.32	2.69	Vuggy
193	8551.07	2462	3791	2382	3711	22.3	2.32	2.70	Vuggy
194	8552.03	1952		1892		23.3	2.30	2.70	Vuggy
195	8553.05	704		664		24.9	2.27	2.69	Vuggy
196	8554.07	286	2795	266	2715	23.7	2.29	2.69	Vuggy

Conventional Core Analysis Results

Client: Mellitah Oil & Gas
Well : CE03-NC41

Concession: 41
Formation : El Garia

Report No: 01/PET1-1&2 /2024
Field: Bahr Essalam

Sample No.	Depth (ft.in)	Air Permeability, mD No Klinkenberg correction @ 200 psi		Air Permeability, mD With Klinkenberg correction @ 200 psi		Porosity (%)	Density, (gm/cc)		Remarks
		Horizontal	Vertical	Horizontal	Vertical		Bulk	Grain	
197	8555.02	268		248		23.6	2.29	2.69	Vuggy
198	8556.03	402		372		23.6	2.29	2.68	Vuggy
199	8557.07	243	91.1	223	81.2	22.4	2.31	2.69	Vuggy
200	8558.05	620		590		25.4	2.28	2.71	Vuggy
201	8559.05	216		196		23.5	2.30	2.70	Vuggy
202	8560.08	739	288	699	268	25.3	2.27	2.70	Vuggy
203	8561.03	484		454		24.2	2.29	2.70	Vuggy
204	8562.06	56.9		49.3		25.6	2.26	2.70	Vuggy
205	8563.08	35.7	42.5	30.1	36.2	26.1	2.25	2.70	
206	8564.04	113		102		26.6	2.25	2.70	Vuggy
207	8565.04	107		96.5		26.3	2.25	2.70	Vuggy
208	8566.09	76.7	65.3	67.7	57.0	25.2	2.27	2.69	Vuggy
209	8567.06	36.2		30.6		25.3	2.27	2.69	
210	8568.05	17.4		14.1		21.9	2.32	2.69	
211	8569.09	47.4	25.4	40.6	21.0	23.8	2.29	2.69	Fractured+Vuggy
212	8570.04	4.43		3.29		14.3	2.45	2.69	
213	8571.06	19.3		15.7		19.4	2.37	2.70	
214	8572.04	20.7	79.5	16.9	70.3	20.5	2.35	2.69	

Conventional Core Analysis Results

Client: Mellitah Oil & Gas
Well : CE03-NC41

Concession: 41
Formation : El Garia

Report No: 01/PET1-1&2 /2024
Field: Bahr Essalam

Sample No.	Depth (ft.in)	Air Permeability, mD No Klinkenberg correction @ 200 psi		Air Permeability, mD With Klinkenberg correction @ 200 psi		Porosity (%)	Density, (gm/cc)		Remarks
		Horizontal	Vertical	Horizontal	Vertical		Bulk	Grain	
215	8573.03	73.5		64.7		23.2	2.31	2.70	Vuggy
216	8574.06	19.5		15.9		22.6	2.31	2.70	
217	8575.06	12.6	26.7	9.98	22.1	22.6	2.31	2.70	
218	8576.04	23.2		19.1		24.0	2.29	2.70	
219	8577.04	33.2		27.9		26.4	2.25	2.70	Broken Edge
220	8578.08	27.8	32.8	23.1	27.5	25.6	2.26	2.70	
221	8579.05	28.9		24.1		25.9	2.26	2.70	
222	8580.05	35.0		29.5		24.5	2.28	2.70	
223	8581.07	27.1	66.7	22.5	58.4	23.8	2.30	2.70	
224	8582.03	40.3		34.3		25.1	2.27	2.70	
225	8583.04	35.6		30.0		25.2	2.27	2.70	
226	8584.07	106	50.2	95.0	43.1	26.8	2.25	2.70	Vuggy
227	8585.02	126		114		25.9	2.26	2.70	
228	8586.03	108		97.1		26.1	2.26	2.70	Broken Edge
229	8587.10	78.7	71.8	69.6	63.1	25.5	2.27	2.70	Vuggy
230	8588.04	13.7		10.9		19.3	2.37	2.70	
231	8589.05	2.06		1.46		13.0	2.44	2.66	
232	8590.10	18.7	15.5	15.1	12.4	19.7	2.36	2.70	

Conventional Core Analysis Results

Client: Mellitah Oil & Gas
Well : CE03-NC41

Concession: 41
Formation : El Garia

Report No: 01/PET1-1&2 /2024
Field: Bahr Essalam

Sample No.	Depth (ft.in)	Air Permeability, mD No Klinkenberg correction @ 200 psi		Air Permeability, mD With Klinkenberg correction @ 200 psi		Porosity (%)	Density, (gm/cc)		Remarks
		Horizontal	Vertical	Horizontal	Vertical		Bulk	Grain	
233	8591.04	51.7		44.5		22.6	2.31	2.69	
234	8592.04	18.0		14.6		20.6	2.35	2.70	
235	8593.07	181	54.6	161	47.2	20.5	2.35	2.70	Vuggy
236	8594.03	144		132		21.9	2.33	2.70	Vuggy
237	8595.02	152		140		22.6	2.32	2.70	Vuggy
238	8596.06	34.9	40.9	29.3	34.7	20.3	2.36	2.70	Vuggy
239	8597.04	189		169		22.9	2.31	2.70	Vuggy
240	8598.04	42.2		35.9		20.8	2.35	2.70	Vuggy
241	8599.08	27.0	13.6	22.4	10.8	20.6	2.35	2.70	Vuggy
242	8600.03	39.4		33.4		19.8	2.37	2.70	Vuggy
243	8601.04	68.8		60.3		22.0	2.33	2.71	Vuggy
244	8602.08	32.5	19.8	27.2	16.1	21.7	2.34	2.71	Vuggy

Conventional Core Analysis Results

Client: Mellitah Oil & Gas
Well : CE03-NC41

Concession: 41
Formation : El Garia

Report No: 01/PET1-1&2 /2024
Field: Bahr Essalam

Sample No.	Depth (ft.in)	Air Permeability, mD No Klinkenberg correction @200 psi		Air Permeability, mD With Klinkenberg correction @200 psi		Porosity (%)	Density, (gm/cc)		Remarks
		Horizontal	Vertical	Horizontal	Vertical		Bulk	Grain	
Core # 4 (8604'-8696') El Garia Formation									
245	8606.04	77.6		68.5		23.7	2.30	2.71	Vuggy
246	8607.08	44.2	11.7	37.8	9.22	21.5	2.34	2.70	Vuggy
247	8608.00	75.8		66.9		19.6	2.37	2.70	Vuggy
248	8609.03	99.1		88.8		21.5	2.34	2.71	Vuggy
249	8610.05	119	5.56	108	4.19	23.6	2.30	2.70	Vuggy
250	8611.03	84.3		74.8		20.5	2.35	2.70	Vuggy
251	8612.04	143		131		20.4	2.36	2.70	Vuggy
252	8613.07	95.5	60.1	85.4	52.2	21.2	2.34	2.70	Vuggy
253	8614.02	240		220		22.4	2.32	2.70	Vuggy
254	8615.06	102		92		22.0	2.33	2.71	Vuggy
255	8616.05	390	81.9	360	72.5	24.1	2.29	2.70	Vuggy
256	8617.02	146		134		22.1	2.33	2.71	Vuggy
257	8618.02	84.5		75.0		21.5	2.33	2.70	Vuggy
258	8619.05	113	54.0	102	46.7	22.3	2.32	2.70	Vuggy
259	8620.02	55.2		47.8		20.1	2.36	2.70	Vuggy
260	8621.08	74.4		65.5		20.1	2.35	2.69	Vuggy
261	8622.08	182	39.9	162	33.8	20.9	2.34	2.69	Vuggy
262	8623.02	290		270		19.1	2.37	2.70	Vuggy

Conventional Core Analysis Results

Client: Mellitah Oil & Gas
Well : CE03-NC41

Concession: 41
Formation : El Garia

Report No: 01/PET1-1&2 /2024
Field: Bahr Essalam

Sample No.	Depth (ft.in)	Air Permeability, mD No Klinkenberg correction @ 200 psi		Air Permeability, mD With Klinkenberg correction @ 200 psi		Porosity (%)	Density, (gm/cc)		Remarks
		Horizontal	Vertical	Horizontal	Vertical		Bulk	Grain	
263	8624.05	42.6		36.2		19.6	2.36	2.69	Vuggy
264	8625.06	94.6	66.6	84.5	58.2	20.2	2.36	2.71	Vuggy
265	8626.02	325		295		21.5	2.33	2.69	Vuggy
266	8627.05	95.8		85.6		20.6	2.35	2.70	Vuggy
267	8628.07	62.9	47.3	54.9	40.5	20.8	2.34	2.69	Vuggy
268	8629.03	127		115		21.1	2.34	2.70	Vuggy
269	8630.06	141		129		20.2	2.35	2.69	Vuggy
270	8631.07	223	30.5	203	25.5	21.9	2.32	2.69	Vuggy
271	8632.02	136		124		21.6	2.33	2.69	Vuggy
272	8633.05	173		160		20.9	2.34	2.70	Vuggy
273	8634.06	186	35.5	166	29.9	21.2	2.34	2.70	Vuggy
274	8635.06	109		98		21.1	2.34	2.70	Vuggy
275	8636.06	225		205		24.2	2.29	2.70	
276	8637.06	230	13.0	210	10.4	24.1	2.28	2.69	
277	8638.02	98.4		88.1		25.2	2.27	2.70	
278	8639.01	38.1		32.3		24.9	2.27	2.70	
279	8640.08	36.8	28.3	31.0	23.6	22.4	2.32	2.69	
280	8641.03	68.7		60.2		23.5	2.29	2.69	Fractured

Conventional Core Analysis Results

Client: Mellitah Oil & Gas
Well : CE03-NC41

Concession: 41
Formation : El Garia

Report No: 01/PET1-1&2 /2024
Field: Bahr Essalam

Sample No.	Depth (ft.in)	Air Permeability, mD No Klinkenberg correction @ 200 psi		Air Permeability, mD With Klinkenberg correction @ 200 psi		Porosity (%)	Density, (gm/cc)		Remarks
		Horizontal	Vertical	Horizontal	Vertical		Bulk	Grain	
281	8642.05	50.6		43.5		23.1	2.31	2.70	
282	8643.07	53.2	33.1	45.9	27.8	22.1	2.32	2.70	
283	8644.01	70.0		61.4		21.9	2.33	2.70	
284	8645.06	29.7		24.8		21.6	2.33	2.70	
285	8646.06	30.8	16.4	25.7	13.2	21.4	2.34	2.70	
286	8647.02	46.5		39.8		23.1	2.31	2.70	
287	8648.06	28.1		23.4		23.3	2.31	2.70	
288	8649.06	30.9	11.3	25.8	8.91	21.9	2.33	2.70	
289	8650.05	30.7		25.6		22.3	2.32	2.70	Vuggy
290	8651.07	31.6		26.4		21.5	2.34	2.70	
291	8652.07	42.4	12.9	36.1	10.2	22.4	2.32	2.70	
292	8653.04	28.4		23.6		21.4	2.34	2.70	Vuggy
293	8654.05	19.5		15.8		20.5	2.35	2.70	Vuggy
294	8655.11	37.3	28.0	31.5	23.3	21.9	2.33	2.70	Vuggy
295	8656.04	29.4		24.4		20.3	2.36	2.71	Vuggy
296	8657.06	32.1		26.9		21.9	2.33	2.70	Vuggy
297	8658.02	47.7	40.6	40.9	34.4	21.7	2.33	2.70	Vuggy
298	8659.01	58.5		50.8		22.8	2.31	2.70	Vuggy

Conventional Core Analysis Results

Client: Mellitah Oil & Gas
Well : CE03-NC41

Concession: 41
Formation : El Garia

Report No: 01/PET1-1&2 /2024
Field: Bahr Essalam

Sample No.	Depth (ft.in)	Air Permeability, mD No Klinkenberg correction @ 200 psi		Air Permeability, mD With Klinkenberg correction @ 200 psi		Porosity (%)	Density, (gm/cc)		Remarks
		Horizontal	Vertical	Horizontal	Vertical		Bulk	Grain	
299	8660.08	51.8		44.7		21.1	2.34	2.70	Vuggy
300	8661.08	33.3	23.7	28.0	19.5	20.1	2.36	2.70	Vuggy
301	8662.03	35.1		29.5		19.0	2.37	2.70	Vuggy
302	8663.07	16.8		13.5		15.0	2.44	2.69	Vuggy
303	8664.08	23.1	47.2	18.9	40.4	17.2	2.40	2.69	Vuggy
304	8665.02	81.5		72.2		16.6	2.42	2.70	Vuggy
305	8666.03	56.6		49.0		20.4	2.35	2.70	Vuggy
306	8667.05	32.3	33.6	27.1	28.2	20.0	2.36	2.70	Vuggy
307	8668.03	68.0		59.6		20.5	2.35	2.70	Vuggy
308	8669.06	83.2		73.7		22.0	2.32	2.70	Vuggy
309	8670.07	84.5	23.5	75.0	19.3	20.7	2.34	2.69	Vuggy
310	8671.02	45.9		39.3		20.5	2.35	2.70	Vuggy
311	8672.08	110.7		99.8		20.6	2.35	2.70	Vuggy
312	8673.09	28.9	5.04	24.0	3.78	19.5	2.36	2.70	Vuggy
313	8674.02	90.9		81.0		16.5	2.42	2.70	
314	8675.04	18.9		15.4		19.8	2.36	2.70	
315	8676.06	56.6	84.6	49.0	75.0	23.4	2.30	2.70	Vuggy
316	8677.02	83.1		73.7		23.6	2.30	2.70	Vuggy

Conventional Core Analysis Results

Client: Mellitah Oil & Gas
Well : CE03-NC41

Concession: 41
Formation : El Garia

Report No: 01/PET1-1&2 /2024
Field: Bahr Essalam

Sample No.	Depth (ft.in)	Air Permeability, mD No Klinkenberg correction @ 200 psi		Air Permeability, mD With Klinkenberg correction @ 200 psi		Porosity (%)	Density, (gm/cc)		Remarks
		Horizontal	Vertical	Horizontal	Vertical		Bulk	Grain	
317	8678.07	124		113		23.5	2.30	2.70	Vuggy
318	8679.06	249	132	229	121	26.8	2.24	2.70	Vuggy
319	8680.02	314		284		25.7	2.26	2.70	Vuggy
320	8681.05	290		270		28.3	2.22	2.70	Vuggy
321	8682.01	338	109	308	97.9	27.6	2.24	2.71	Vuggy+Short
322	8683.04	511		481		24.1	2.30	2.71	Vuggy
323	8684.04	946		906		27.6	2.24	2.71	Vuggy
324	8685.07	3.21	6.09	2.34	4.61	16.3	2.43	2.71	Cemented Fracture
325	8686.04	1.29		0.890		13.2	2.48	2.71	Vuggy
326	8687.08	0.495		0.323		11.3	2.52	2.71	
327	8689.07	0.928	1.73	0.629	1.21	13.8	2.49	2.73	Vuggy
328	8690.05	82.6		73.2		20.8	2.35	2.71	Vuggy
329	8691.08	83.2		73.8		22.2	2.33	2.71	Vuggy
330	8692.08	3.19	8.66	2.32	6.70	11.8	2.50	2.71	Vuggy
331	8693.05	5.07		3.80		17.3	2.41	2.71	Vuggy
332	8694.05	14.8		11.8		19.1	2.38	2.71	Vuggy
333	8695.08	17.2	50.2	13.9	43.1	21.1	2.34	2.70	Vuggy

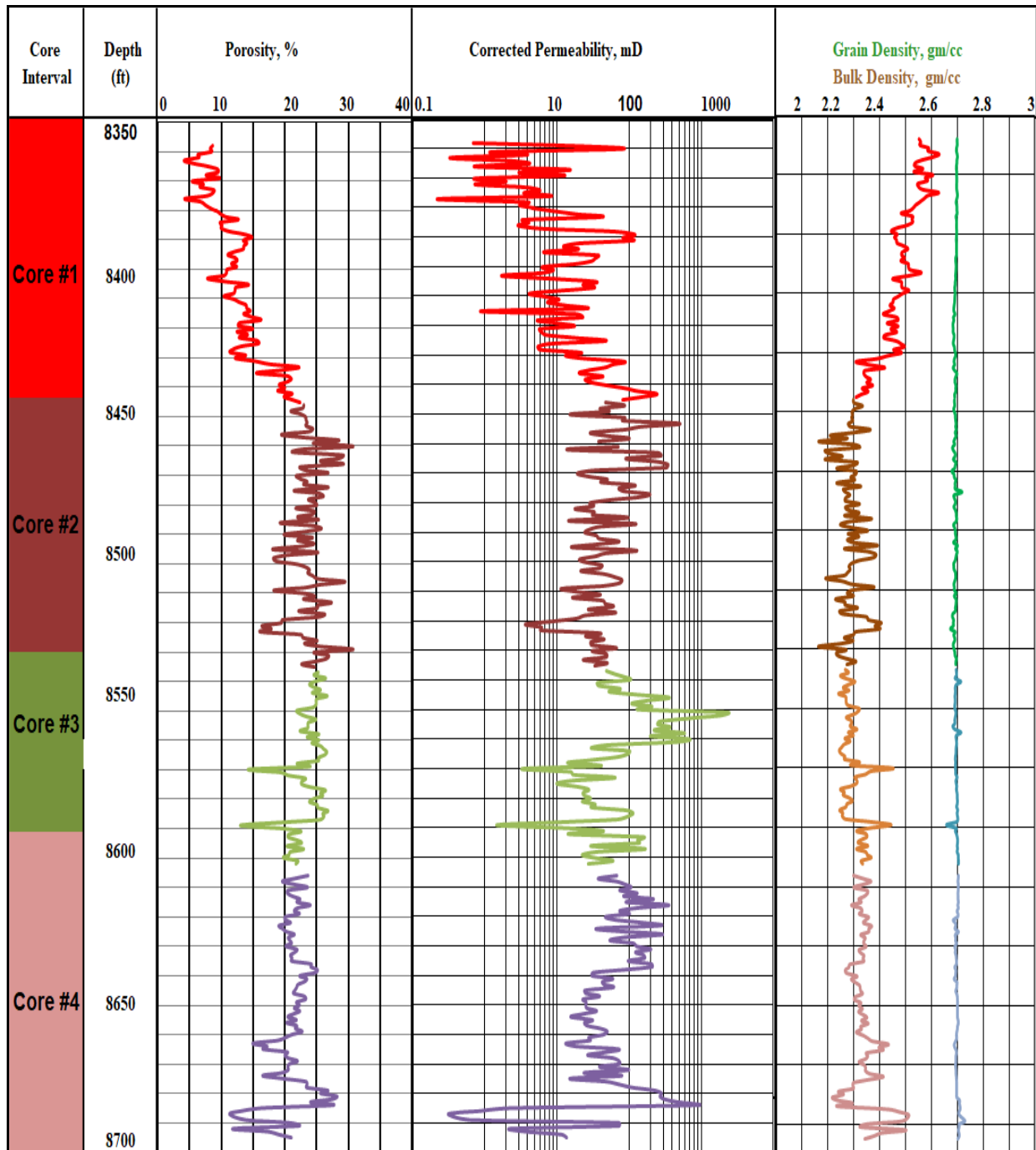
Conventional Core Analysis Profile

Client: Mellitah Oil & Gas

Formation: El Garia

Well: CE03-NC41

Core #: 1, 2, 3 & 4



Conventional Core Analysis Results

Dean-Stark Analysis Results

Client: Mellitah Oil & Gas

Concession: 41

Well: CE03-NC41

Formation: El Garia

Number of Samples: 25

Sample No.	Depth (ft)	Saturation (Pore Volume)			Remarks
		Oil %	Water %	Gas %	
1	8359.9	0.75	40.26	58.99	
3	8439.08	10.09	64.02	25.89	
7	8540.9	5.74	56.54	37.72	
9	8549.08	10.26	24.12	65.62	
10	8555.08	21.28	41.49	37.22	
11	8561.08	18.32	80.08	1.60	
12	8567.08	12.96	9.90	77.15	
13	8573.08	10.44	0.84	88.72	
14	8579.08	18.23	79.12	2.65	
15	8585.08	12.67	10.02	77.31	
16	8594.9	18.75	58.13	23.12	
17	8597.08	6.21	60.92	32.87	
18	8606.08	39.03	0.00	60.97	
19	8611.08	32.47	14.63	52.90	
20	8617.08	43.75	42.05	14.21	
21	8623.08	14.08	33.07	52.85	
22	8629.08	42.45	13.66	43.89	
23	8635.08	13.54	0.00	86.46	
24	8641.08	17.04	33.16	49.80	
25	8647.08	13.57	6.74	79.69	
26	8653.08	2.20	35.93	61.87	
27	8662.08	8.73	13.19	78.08	
28	8668.08	21.37	24.84	53.79	
31	8686.08	7.52	39.34	53.14	
32	8692.08	14.01	33.42	52.56	

Conventional Core Analysis

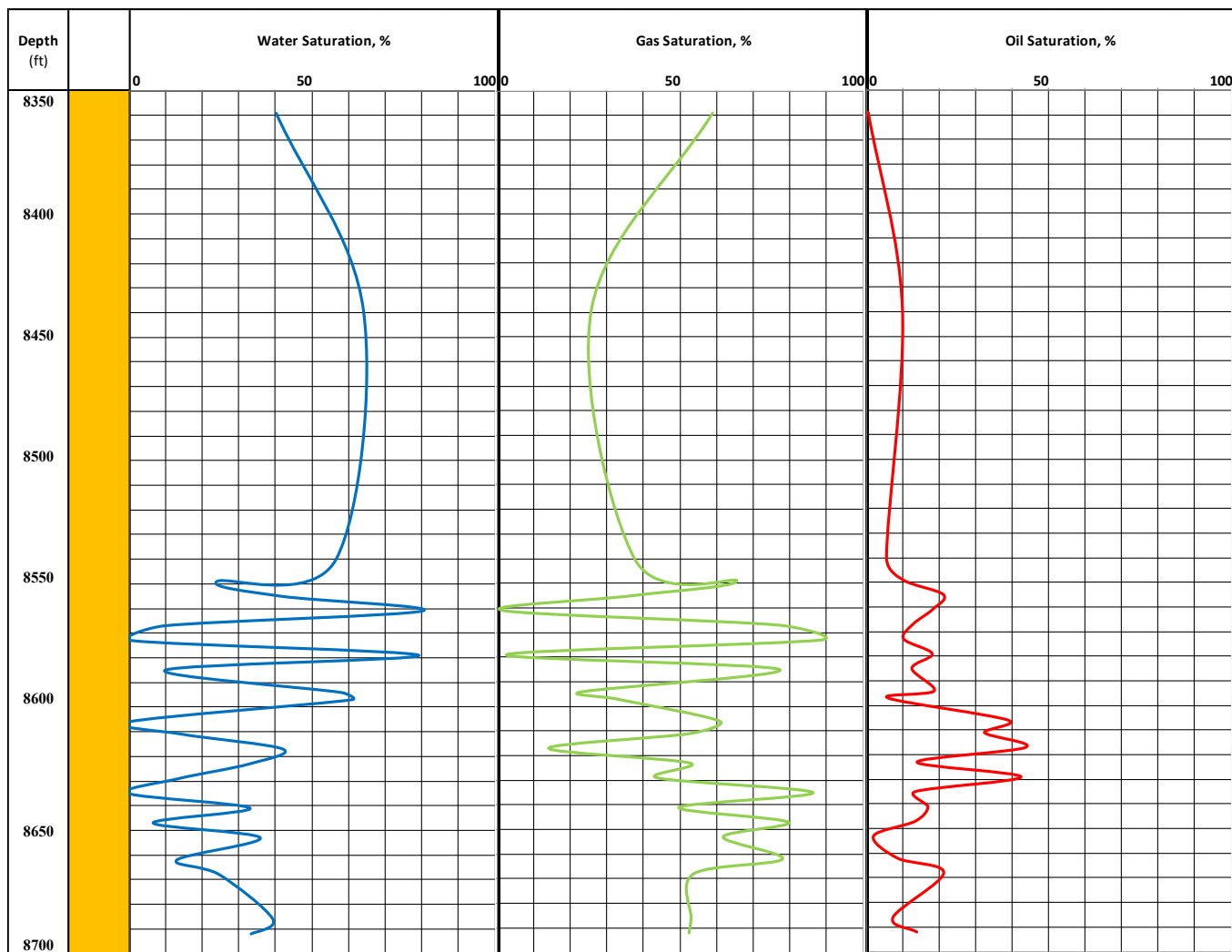
Dean-Stark Analysis Profile

Client : Mellitah Oil & Gas Company

Formation: El Garia

Well : CE03-NC41

Core #: 1, 2, 3 & 4



Statistical Analysis

Statistical Analysis of Conventional Core Analysis Data

Client: Mellitah Oil & Gas

Well #: CE03-NC41

Formation: El Garia

Core # 1 (8357-8446) ft – 100% recovery
Core # 2 (8446-8536) ft – 100% recovery
Core # 3 (8536-8604) ft – 97% recovery
Core # 4 (8604-8695) ft – 100% recovery

The Following Correlations are Presented: -

1. Horizontal Permeability Vs Porosity.
2. Horizontal Permeability Vs Vertical Permeability.
4. Frequency and Cumulative Frequency Distribution of Horizontal Permeability.
5. Frequency and Cumulative Frequency Distribution of Porosity.
6. Frequency and Cumulative Frequency Distribution of Grain Density.

All Porosities are reported in percentage.

All Permeabilities are reported in millidarcy.

Horizontal Permeability vs. Porosity

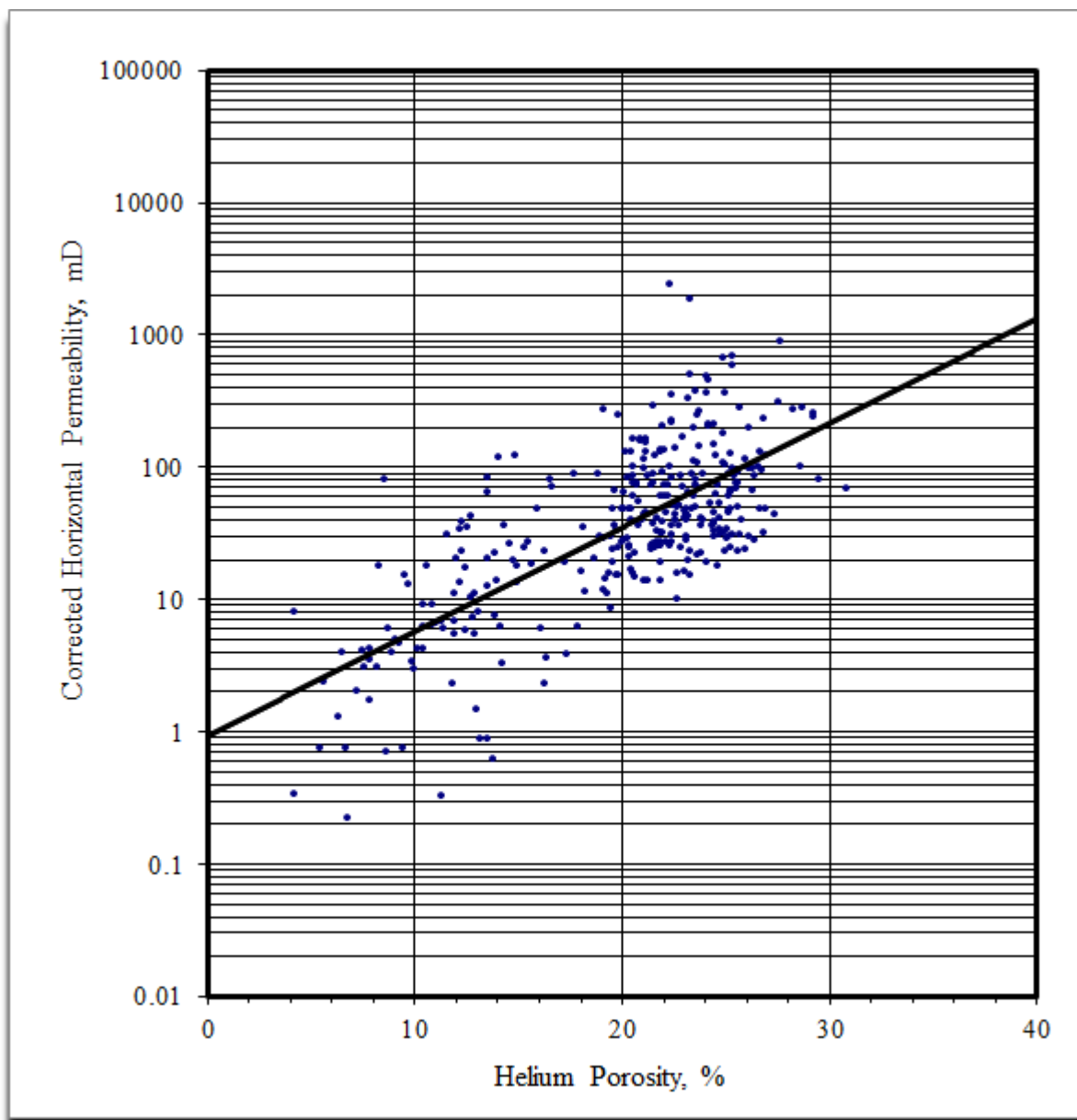
Client : Mellitah Oil & Gas

Formation: El Garia

Well : CE03-NC41

Number of Samples: 332

$$\text{Log}(K_h) = \text{Log}(0.936) + 0.079 \times \Phi$$

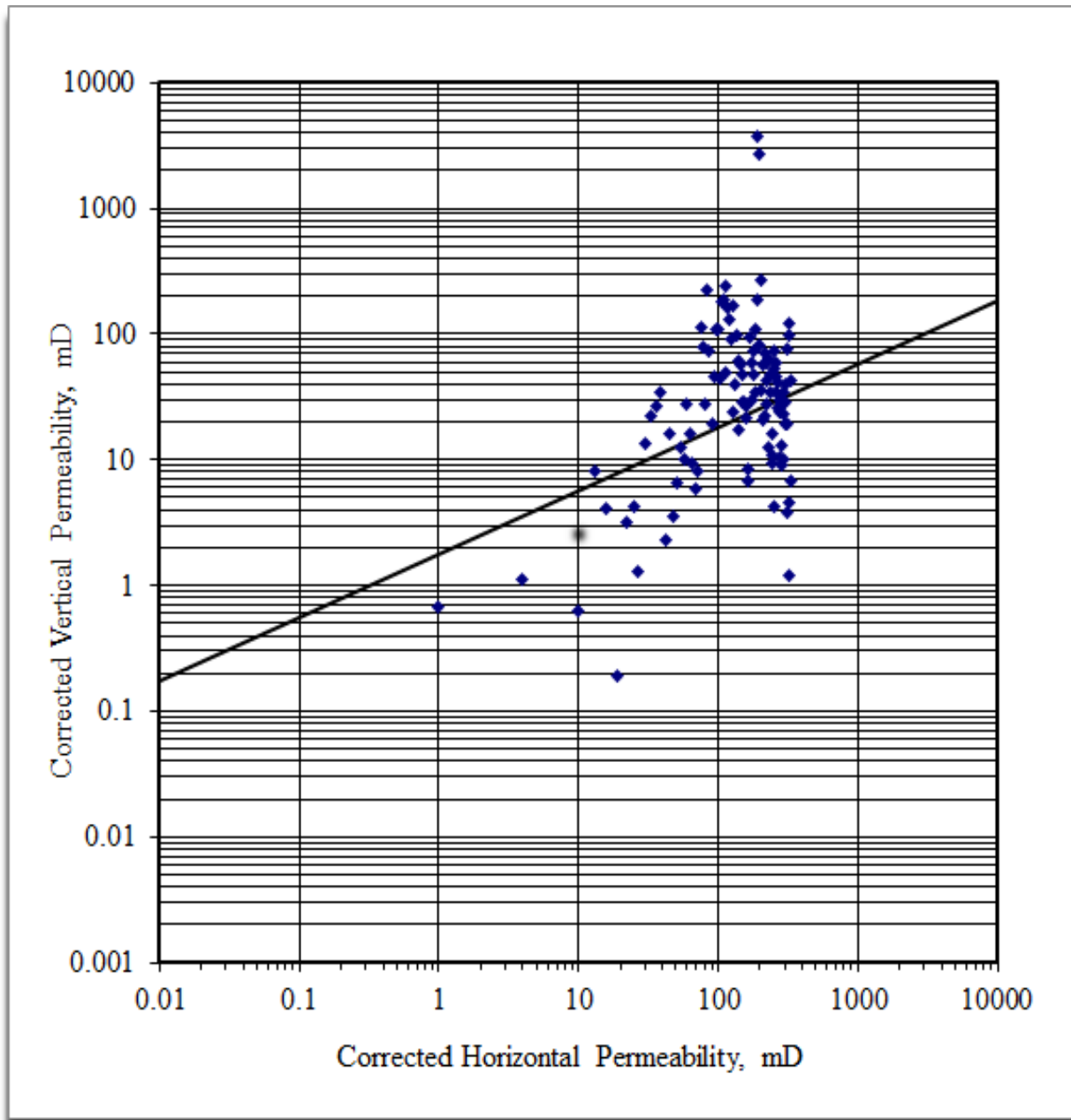


Horizontal Permeability vs. Vertical Permeability

Client : Mellitah Oil & Gas
Well : CE03-NC41

Formation: El Garia
Number of Samples: 111

$$\text{Log}(K_v) = \text{Log}(1.764) + 0.504 \times \text{Log}(K_h)$$



Horizontal Permeability Frequency Distribution

Client: Mellitah Oil & Gas

WELL #: CE03-NC41

Formation: El Garia

HORIZONTAL PERMEABILITY ANALYSIS SUMMARY

Number of samples	332
Minimum	0.225
Maximum	2382
Arithmetic mean	86.5
Geometric mean	34.7
Harmonic mean	8.69

GROUPING BY PERMEABILITY RANGE

Permeability Range (mD)	Samples in range	Permeability Arith. (mD)	Permeability Geom. (mD)	Frequency (%)	Cumulative Frequency (%)
0.16-0.32	1	0.225	0.000	0.301	0.30
0.32-0.64	3	0.429	0.408	0.904	1.20
0.64-1.25	6	0.789	0.786	1.81	3.01
1.25-2.5	7	1.94	1.89	2.11	5.12
2.5-5	16	3.82	3.78	4.82	9.94
5-10	21	7.05	6.95	6.33	16.27
10-20	38	15.4	15.2	11.4	27.71
20-40	82	29.7	29.1	24.7	52.41
40-80	66	56.8	55.6	19.9	72.29
80-160	53	107	105	15.96	88.25
160-320	25	228	224	7.530	95.78
320-640	9	423	415	2.711	98.5
640-1280	3	1135	749	0.904	99.4
1280-2560	2	2137	2123	0.602	100.0

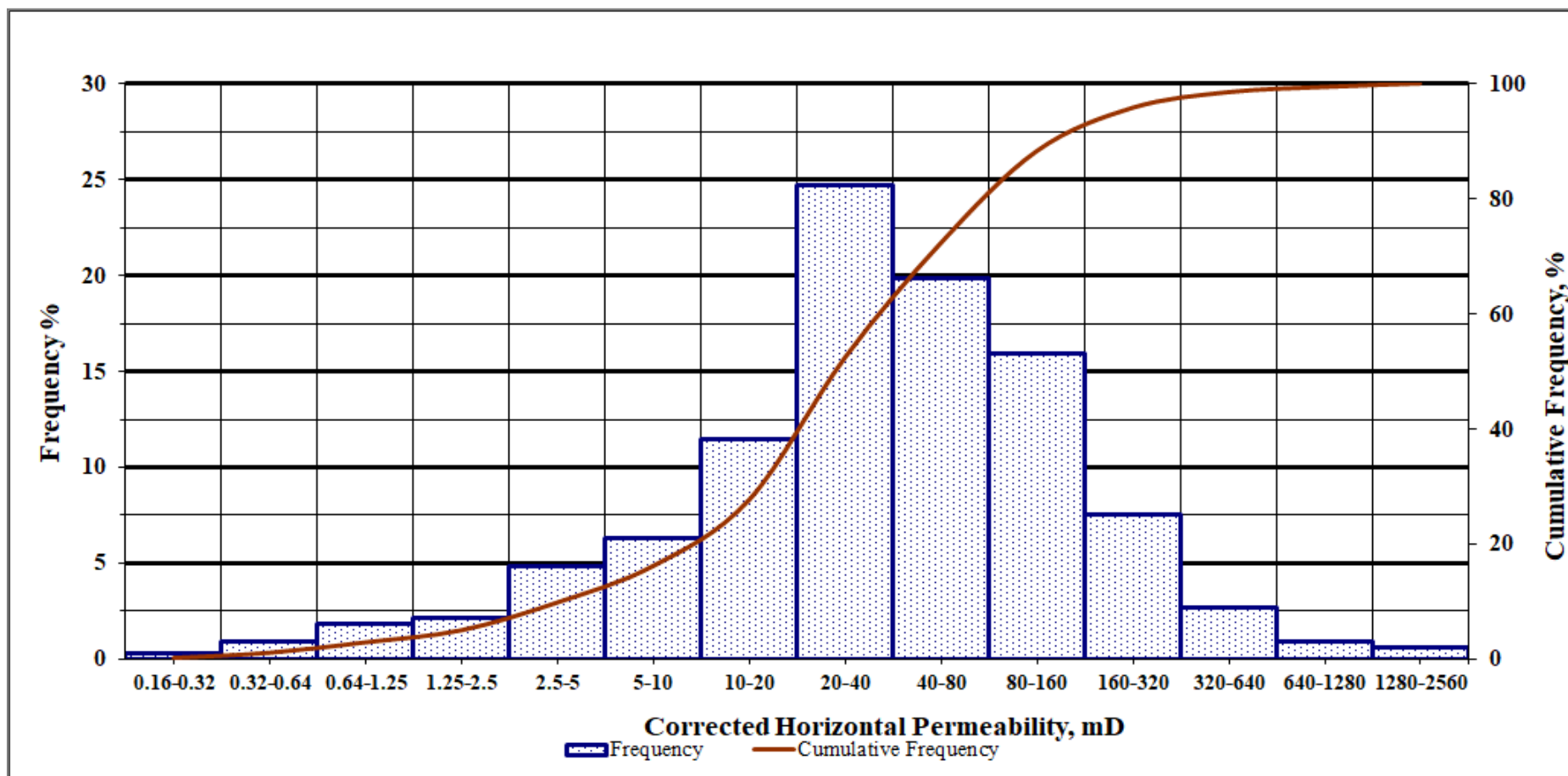
Horizontal Permeability Frequency Distribution

Client : Mellitah Oil & Gas

Well : CE03-NC41

Formation: El Garia

Number of Samples: 332



Porosity Frequency Distribution

Client: Mellitah Oil & Gas

WELL #: CE03-NC41

Formation: El Garia

POROSITY ANALYSIS SUMMARY

Number of samples	333
Minimum	4.15
Maximum	30.82
Arithmetic mean	19.96

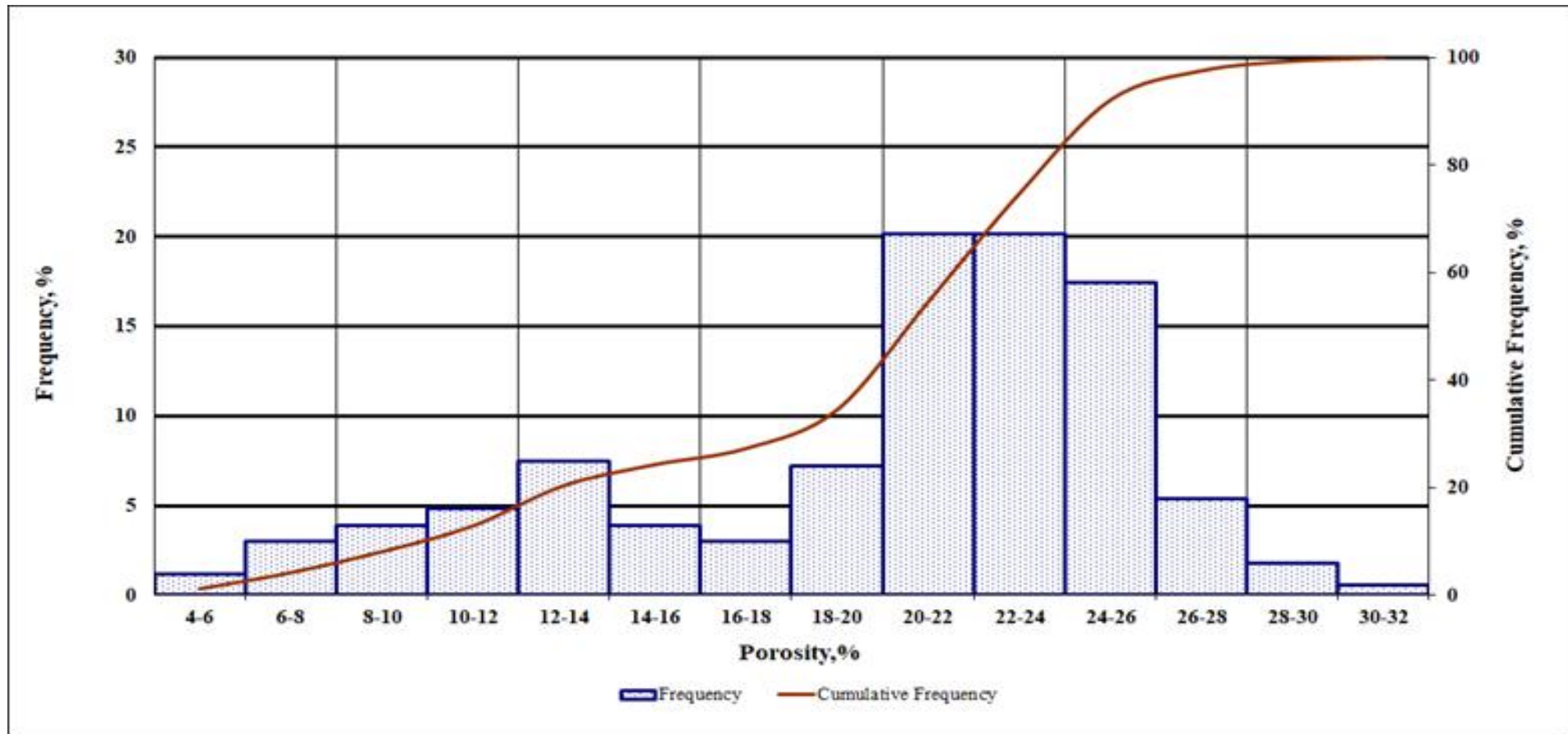
GROUPING BY POROSITY RANGE

Porosity Range (%)	Samples in range	Mean Porosity (%)	Frequency (%)	Cumulative Frequency (%)
4-6	4	4.872	1.201	1.20
6-8	10	7.188	3.00	4.20
8-10	13	9.068	3.90	8.11
10-12	16	11.11	4.805	12.91
12-14	25	12.99	7.508	20.42
14-16	13	14.86	3.90	24.32
16-18	10	16.81	3.003	27.33
18-20	24	19.24	7.207	34.53
20-22	67	21.05	20.120	54.65
22-24	67	22.98	20.120	74.77
24-26	58	24.95	17.417	92.19
26-28	18	26.68	5.405	97.60
28-30	6	28.94	1.802	99.40
30-32	2	30.82	0.601	100.00

Porosity Frequency Distribution

Client : Mellitah Oil & Gas
Well : CE03-NC41

Formation: El Garia
Number of Samples: 333



Grain Density Frequency Distribution

Client: Mellitah Oil & Gas

WELL #: CE03-NC41

Formation: El Garia

GRAIN DENSITY ANALYSIS SUMMARY

Number of samples	333
Minimum	2.66
Maximum	2.73
Arithmetic mean	2.70

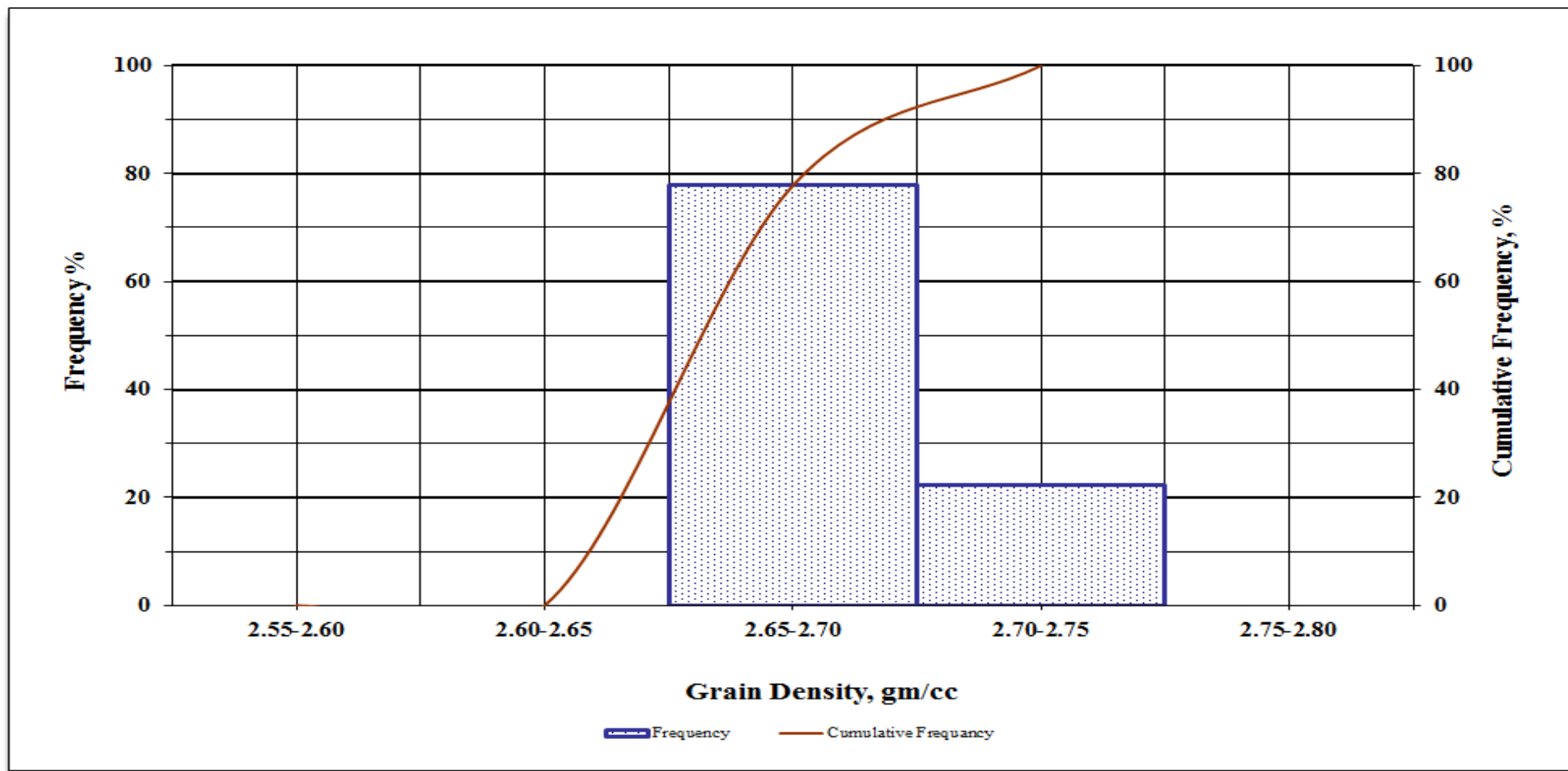
GROUPING BY DENSITY RANGE

Grain Density Range	Samples in range	Mean Grain Density G/C.C	Frequency (%)	Cumulative Frequency (%)
2.65-2.70	259	2.69	77.78	77.78
2.70-2.75	74	2.70	22.22	100.00

Grain Density Frequency Distribution

Client : Mellitah Oil & Gas
Well : CE03-NC41

Formation: El Garia
Number of Samples: 333



Hydraulic Flow Units

Hydraulic (flow) Units

The hydraulic units for this well were found to be three units as shown in figures 1 and 2. The vuggy, fractured and low permeability samples, samples with air permeability less than 1 mD, were excluded from the selection.

Table 1 shows the selected plug samples and their hydraulic units.

The analysis of this data was performed by using the following equations which were obtained by Kozeny and Carman:

$$RQI = 0.0314 (K/\Phi)^{1/2}$$

$$\Phi_z = (\Phi / (1 - \Phi))$$

$$FZI = RQI / \Phi_z$$

Where:

RQI Reservoir Quality Index

Φ_z Normalized porosity index

FZI Flow Zone Indicator

Sample Units Distribution

Client : Mellitah Oil & Gas
Well : CE03-NC41

Formation: El Garia
Concession: 41

Sample No.	Depth, (ft.in)	Klinkenberg Horizontal Permeability	Porosity (%)	Φ_z	RQI	FZI	Unit
231	8589.05	1.46	12.97	0.149	0.106	0.708	1
217	8575.10	9.98	22.64	0.293	0.209	0.713	
164	8521.04	3.66	16.34	0.195	0.149	0.760	
93	8450.06	15.29	23.29	0.304	0.254	0.838	
165	8522.10	6.21	17.83	0.217	0.185	0.854	
163	8520.07	8.58	19.44	0.241	0.209	0.864	
221	8579.05	24.06	25.92	0.350	0.303	0.865	
220	8578.10	23.10	25.58	0.344	0.298	0.868	
218	8576.04	19.08	24.04	0.317	0.280	0.884	
155	8512.02	16.44	22.98	0.298	0.266	0.890	
210	8568.10	14.06	21.89	0.280	0.252	0.898	
216	8574.06	15.88	22.64	0.293	0.263	0.899	
212	8570.04	3.29	14.26	0.166	0.151	0.907	
161	8518.00	24.95	25.26	0.338	0.312	0.924	
105	8462.03	14.06	21.21	0.269	0.256	0.949	
153	8510.06	19.66	23.21	0.302	0.289	0.956	
205	8563.10	30.11	26.08	0.353	0.337	0.956	
146	8503.03	21.47	23.60	0.309	0.299	0.969	
223	8581.07	22.50	23.85	0.313	0.305	0.974	
230	8588.00	10.93	19.26	0.239	0.237	0.991	
171	8528.05	30.52	25.64	0.345	0.343	0.993	
234	8592.04	14.56	20.60	0.260	0.264	1.017	
225	8583.00	30.00	25.18	0.337	0.343	1.018	
209	8567.06	30.58	25.30	0.339	0.345	1.019	
287	8648.06	23.38	23.32	0.304	0.314	1.034	
156	8513.10	43.95	27.35	0.376	0.398	1.057	
222	8580.05	29.46	24.46	0.324	0.345	1.064	
278	8639.01	32.28	24.88	0.331	0.358	1.080	

Table (1)

Sample Units Distribution

Client : Mellitah Oil & Gas
Well : CE03-NC41

Formation: El Garia
Concession: 41

Sample No.	Depth, (ft.in)	Klinkenberg Horizontal Permeability	Porosity (%)	Φ_z	RQI	FZI	Unit
224	8582.03	34.25	25.06	0.334	0.367	1.098	2
214	8572.04	16.87	20.47	0.257	0.285	1.108	
168	8525.08	25.03	22.85	0.296	0.329	1.110	
314	8675.04	15.35	19.82	0.247	0.276	1.118	
178	8535.05	33.66	24.69	0.328	0.367	1.118	
232	8590.10	15.13	19.73	0.246	0.275	1.119	
131	8488.05	39.87	25.79	0.347	0.390	1.124	
152	8509.03	11.61	18.21	0.223	0.251	1.127	
126	8483.05	33.36	24.44	0.324	0.367	1.134	
132	8489.08	28.07	23.12	0.301	0.346	1.150	
174	8531.04	48.97	26.90	0.368	0.424	1.151	
213	8571.06	15.65	19.38	0.240	0.282	1.174	
170	8527.04	29.71	23.08	0.300	0.356	1.188	
114	8471.07	27.14	22.39	0.288	0.346	1.198	
143	8500.07	25.31	21.79	0.279	0.338	1.214	
288	8649.06	25.81	21.86	0.280	0.341	1.219	
284	8645.06	24.78	21.60	0.276	0.336	1.220	
159	8516.04	27.34	22.16	0.285	0.349	1.225	
154	8511.05	40.65	24.71	0.328	0.403	1.227	
149	8506.06	80.85	29.54	0.419	0.520	1.239	
79	8436.04	20.99	20.39	0.256	0.319	1.244	
169	8526.09	45.74	25.13	0.336	0.424	1.263	
285	8646.06	25.73	21.39	0.272	0.344	1.265	
290	8651.07	26.41	21.50	0.274	0.348	1.270	
157	8514.08	46.98	25.16	0.336	0.429	1.276	
162	8519.05	19.15	19.57	0.243	0.311	1.277	
279	8640.08	31.04	22.39	0.288	0.370	1.282	

Table (1-Continue)

Sample Units Distribution

Client : Mellitah Oil & Gas

Formation: El Garia

Well : CE03-NC41

Concession: 41

Table (1-Continue)

Sample No.	Depth, (ft.in)	Klinkenberg Horizontal Permeability	Porosity (%)	Φ_z	RQI	FZI	Unit
145	8502.05	39.47	23.91	0.314	0.403	1.284	3
147	8504.04	45.01	24.42	0.323	0.426	1.320	
138	8495.06	16.48	18.05	0.220	0.300	1.362	
286	8647.02	39.80	23.10	0.300	0.412	1.372	
81	8438.08	25.30	20.31	0.255	0.350	1.375	
291	8652.07	36.14	22.40	0.289	0.399	1.382	
160	8517.08	66.45	26.30	0.357	0.499	1.398	
144	8501.04	42.57	23.23	0.303	0.425	1.404	
116	8473.03	41.31	22.99	0.299	0.421	1.409	
133	8490.07	24.66	19.84	0.248	0.350	1.414	
167	8524.05	39.62	22.55	0.291	0.416	1.429	
281	8642.05	43.50	23.15	0.301	0.430	1.429	
151	8508.10	43.28	23.08	0.300	0.430	1.433	
142	8499.10	20.69	18.63	0.229	0.331	1.445	
186	8544.05	53.10	24.29	0.321	0.464	1.447	
158	8515.04	61.13	25.15	0.336	0.490	1.457	
101	8458.02	100.95	28.60	0.401	0.590	1.472	
135	8492.03	38.44	21.97	0.282	0.415	1.475	
148	8505.10	64.54	25.26	0.338	0.502	1.485	
100	8457.07	47.99	23.13	0.301	0.452	1.503	
233	8591.04	44.51	22.56	0.291	0.441	1.515	
150	8507.09	75.05	25.49	0.342	0.539	1.575	
282	8643.07	45.88	22.09	0.283	0.453	1.597	
177	8534.06	50.33	22.63	0.293	0.468	1.601	
82	8439.03	29.07	19.11	0.236	0.387	1.640	
92	8449.03	52.29	22.44	0.289	0.479	1.657	
80	8437.06	44.50	21.05	0.267	0.457	1.713	

Table (1-Continue)

Sample Units Distribution

Client : Mellitah Oil & Gas
Well : CE03-NC41

Formation: El Garia
Concession: 41

Sample No.	Depth, (ft.in)	Klinkenberg Horizontal Permeability	Porosity (%)	Φ_z	RQI	FZI	Unit
108	8465.05	90.46	25.60	0.344	0.590	1.715	3
277	8638.02	88.10	25.19	0.337	0.587	1.745	
128	8485.03	97.22	25.31	0.339	0.615	1.816	
283	8644.01	61.39	21.90	0.280	0.526	1.874	
227	8585.02	114.4	25.93	0.350	0.659	1.883	
187	8545.06	130.7	26.66	0.363	0.695	1.913	
141	8498.03	34.85	18.17	0.222	0.435	1.958	
275	8636.06	205.3	24.20	0.319	0.915	2.864	
276	8637.06	210.1	24.13	0.318	0.926	2.913	
313	8674.02	80.99	16.51	0.198	0.695	3.517	

Table (1-Continue)

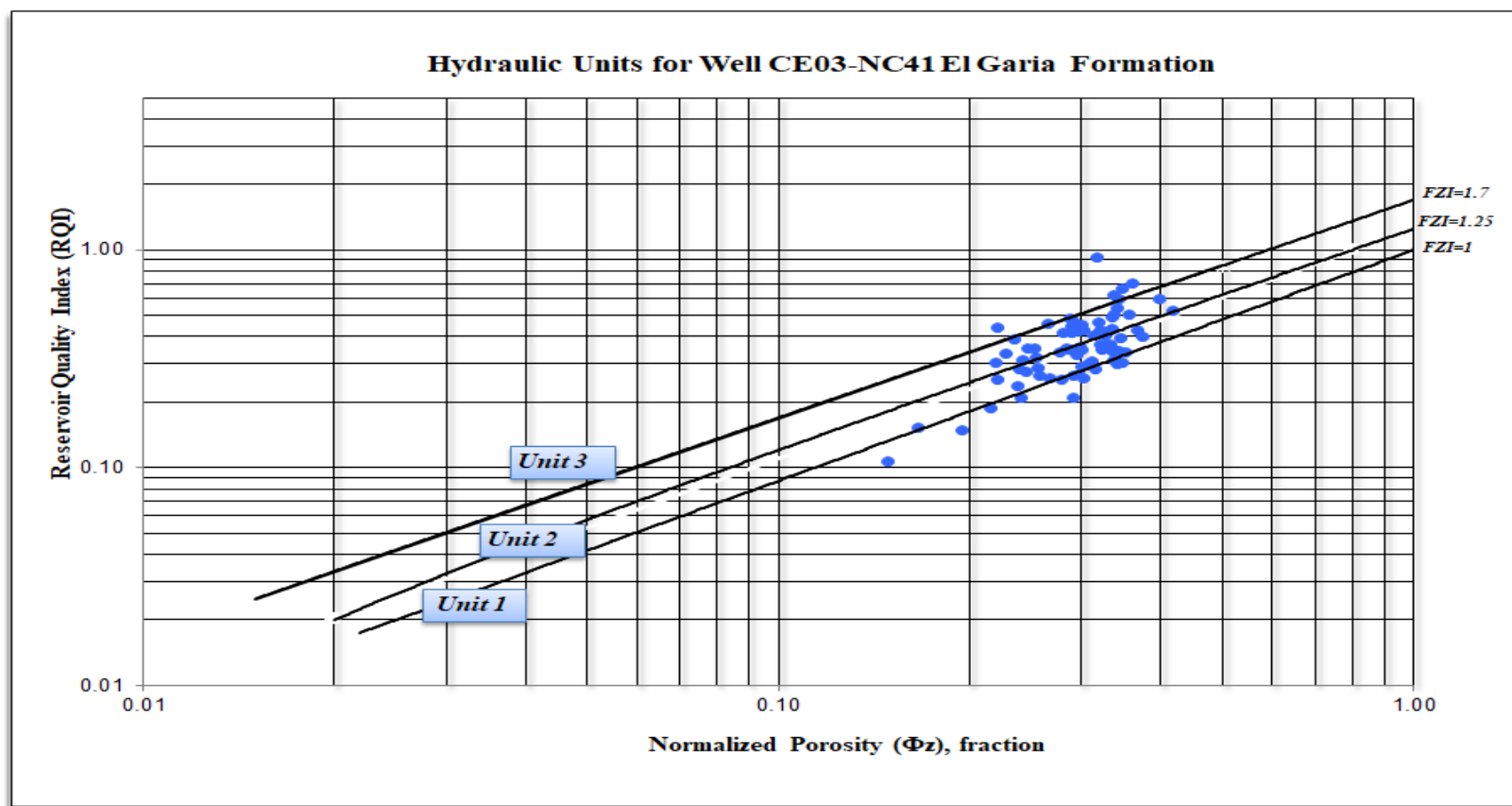
Sample Units Distribution

Client : Mellitah Oil & Gas

Well : CE03-NC41

Formation: El Garia

Concession: 41



Sample Units Distribution

Client : Mellitah Oil & Gas

Well : CE03-NC41

Formation: El Garia

Concession: 41

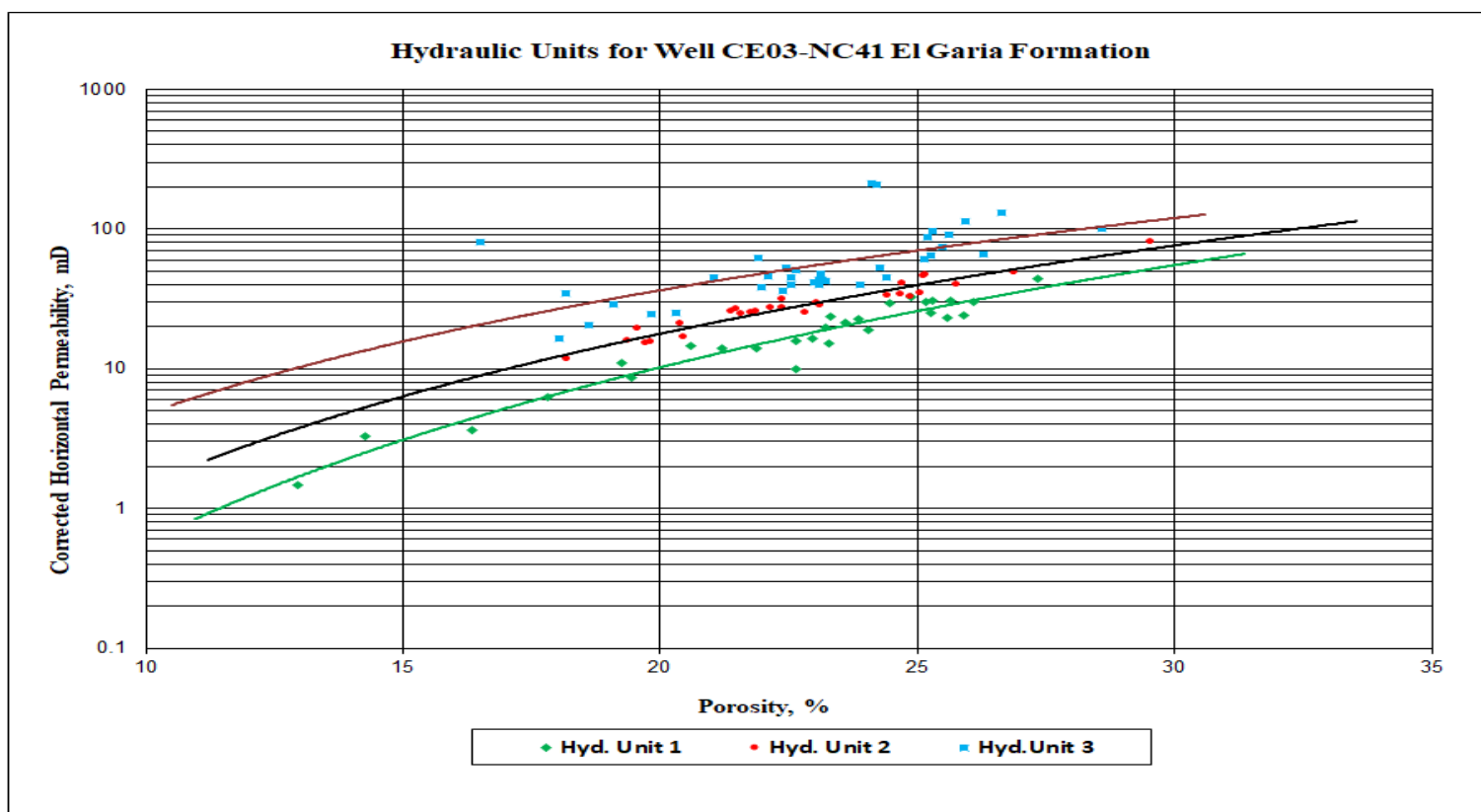


Figure -2

Part Two Special Core Analysis

Section One

Porosity and Permeability as a function of Overburden Pressure

POROSITY AND PERMEABILITY AS A FUNCTION OF OVERBURDEN PRESSURE:-

Ten plug samples, representing El-Garia formations, were selected from the well No. **CE03-NC41**, to investigate the effect of overburden pressure on effective porosity and permeability. The clean dry samples were placed in a **CMS-300** instrument which is fully automated permeameter and porosimeter used to determine porosity and permeability. The effective porosity and permeability were measured at pressure values of 800, 1500, 2500, 3858 and 4500 psi.

The base porosity was measured without overburden pressure and the base permeability was measured at 200 psig overburden pressure, the measurements and the calculations of these petrophysical properties are based on the techniques described by “API, Recommended practice 40 February, 1998 section 5,6” (Reference 1). The results are presented in tables **2.1.1** through **2.1.10** and in figures **2.1.1** through **2.1.10**.

POROSITY AND PERMEABILITY AS A FUNCTION OF OVERBURDEN PRESSURE

Company	: Mellitha Oil & Gas	Formation	: Elgaria
Well #	: CE03-NC41	Concession	: 41
Sample #	: 80	Sample Depth (ft.in)	: 8437.06

Pressure (psi)	Porosity (%)	Permeability mD	Relative to base porosity	Relative to base Permeability
0	21.00*		1.0000	
200	20.94	51.70**	0.9973	1.0000
800	20.79	50.061	0.9899	0.9683
1500	20.64	48.563	0.9827	0.9393
2500	20.46	46.983	0.9745	0.9088
3858	20.28	45.540	0.9659	0.8809
4500	20.22	45.049	0.9626	0.8713

Table (2.1.1)

* Base porosity

** Base permeability

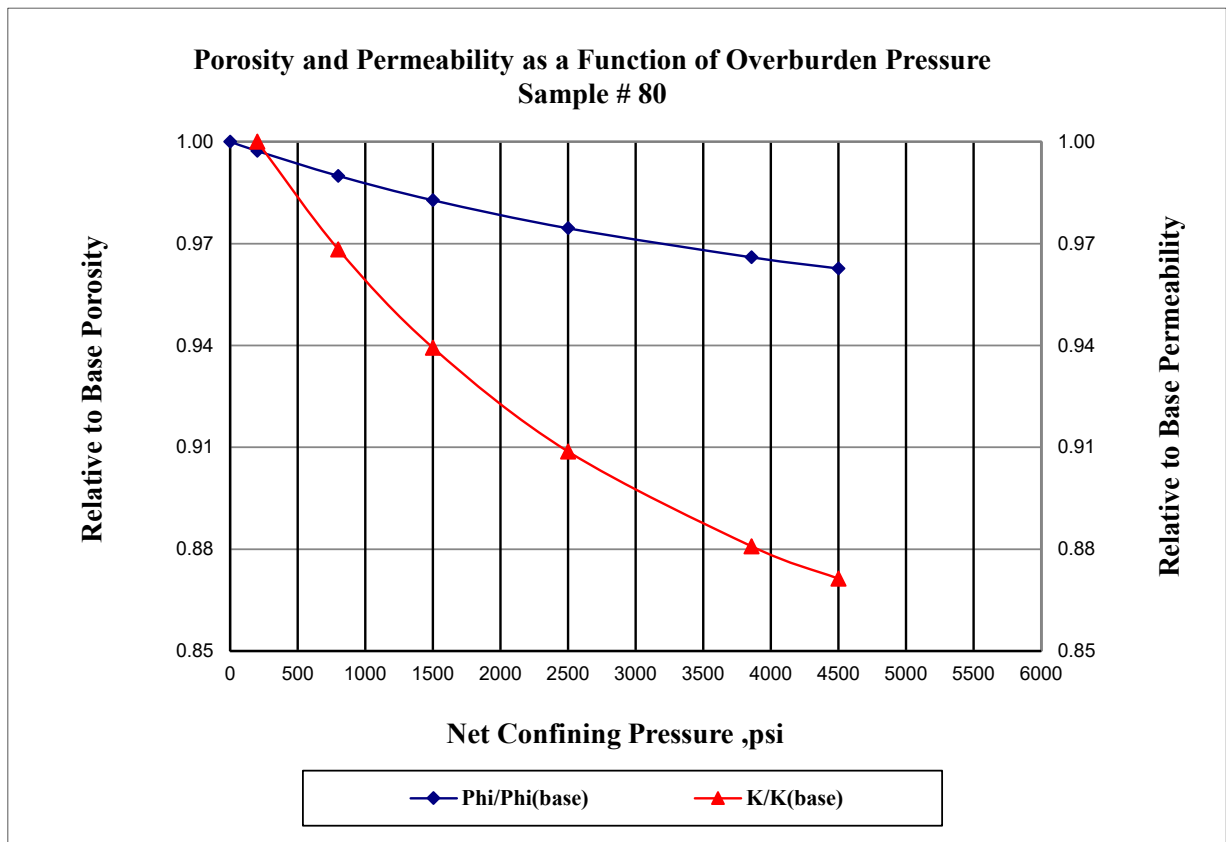


Figure (2.1.1)

POROSITY AND PERMEABILITY AS A FUNCTION OF OVERBURDEN PRESSURE

Company	: Mellitha Oil & Gas	Formation	: Elgaria
Well #	: CE03-NC41	Concession	: 41
Sample #	: 93	Sample Depth (ft.in)	: 8450.06

Pressure (psi)	Porosity (%)	Permeability mD	Relative to base porosity	Relative to base Permeability
0	23.30*		1.0000	
200	23.26	18.90**	0.9983	1.0000
800	23.15	18.570	0.9935	0.9826
1500	23.04	18.261	0.9887	0.9662
2500	22.91	17.926	0.9831	0.9484
3858	22.76	17.606	0.9769	0.9315
4500	22.70	17.493	0.9744	0.9256

Table (2.1.2)

* Base porosity

** Base permeability

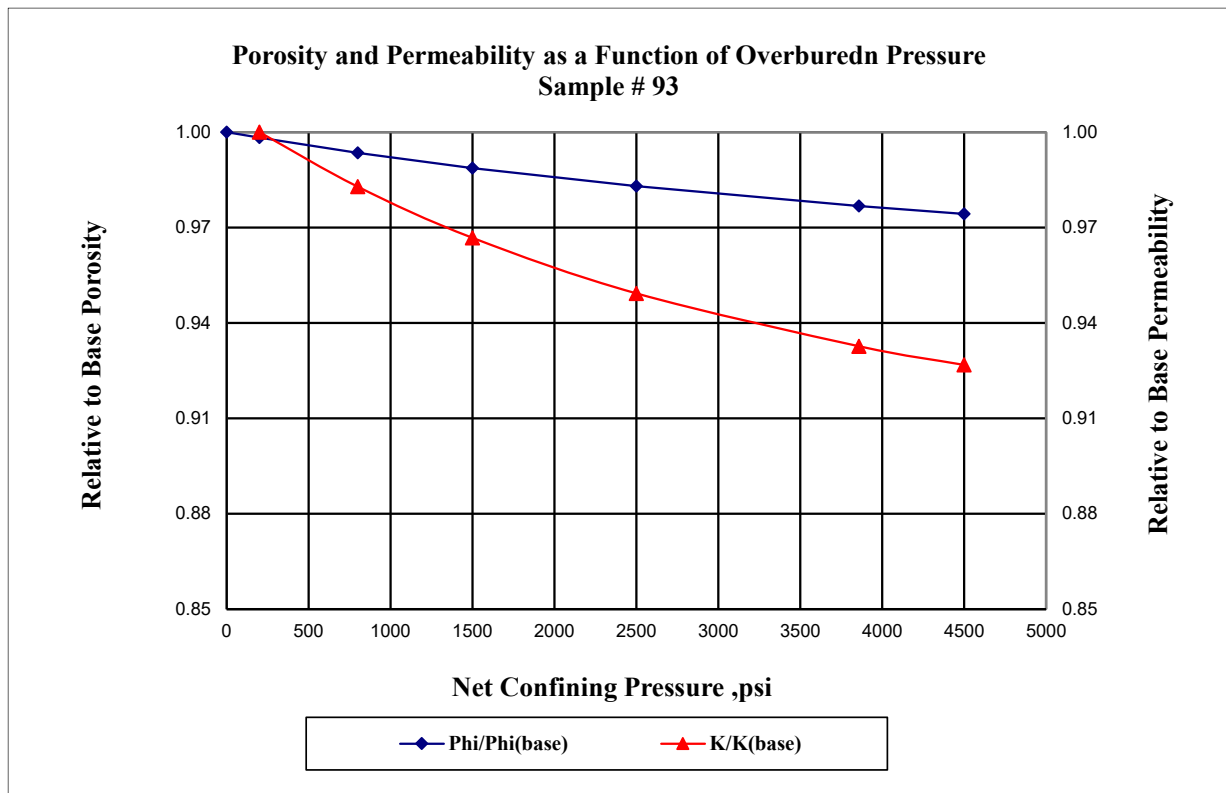


Figure (2.1.2)

POROSITY AND PERMEABILITY AS A FUNCTION OF OVERBURDEN PRESSURE

Company	: Mellitha Oil & Gas	Formation	: Elgaria
Well #	: CE03-NC41	Concession	: 41
Sample #	: 108	Sample Depth (ft.in)	: 8465.05

Pressure (psi)	Porosity (%)	Permeability mD	Relative to base porosity	Relative to base Permeability
0	25.58*		1.000	
200	25.55	62.795**	0.999	1.000
800	25.46	61.236	0.995	0.975
1500	25.37	59.794	0.992	0.952
2500	25.26	58.250	0.988	0.928
3858	25.14	56.812	0.983	0.905
4500	25.08	56.313	0.981	0.897

Table (2.1.3)

* Base porosity

** Base permeability

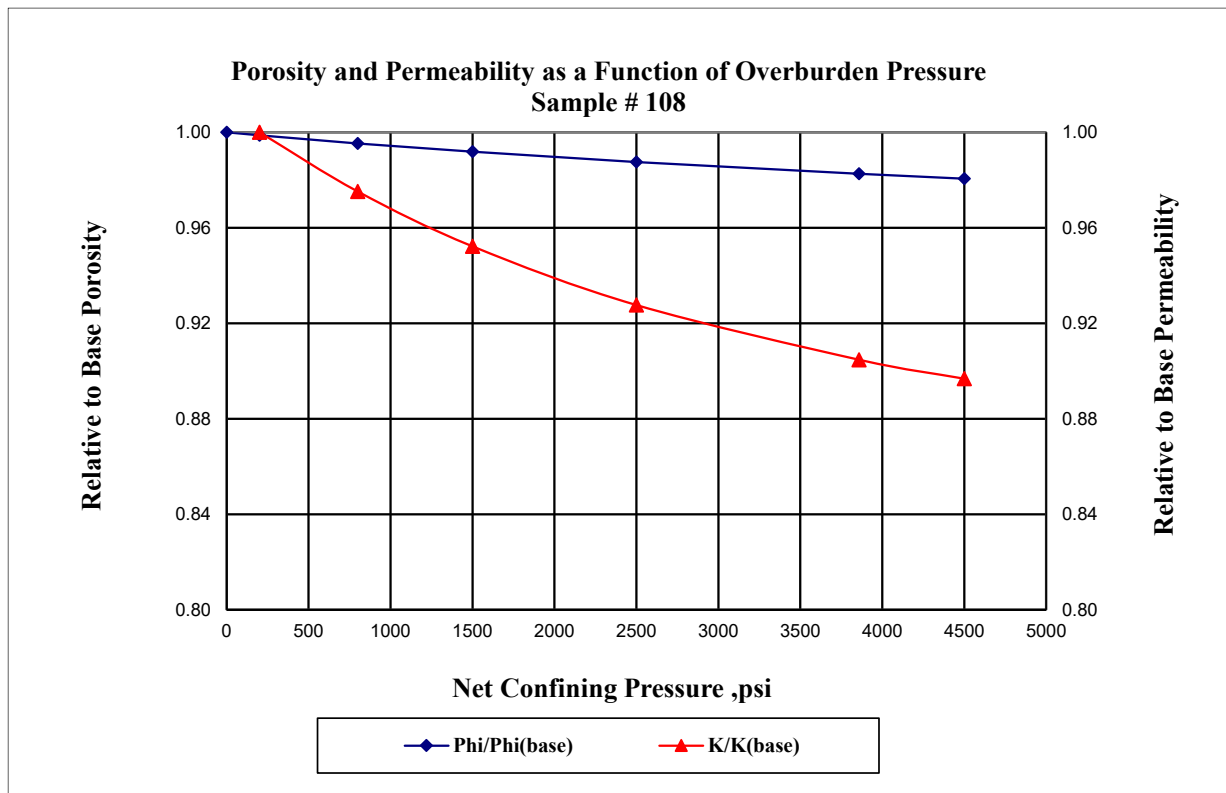


Figure (2.1.3)

POROSITY AND PERMEABILITY AS A FUNCTION OF OVERBURDEN PRESSURE

Company	: Mellitha Oil & Gas	Formation	: Elgaria
Well #	: CE03-NC41	Concession	: 41
Sample #	: 114	Sample Depth (ft.in)	: 8471.07

Pressure (psi)	Porosity (%)	Permeability mD	Relative to base porosity	Relative to base Permeability
0	22.40*		1.000	
200	22.35	32.40**	0.998	1.000
800	22.20	31.208	0.991	0.963
1500	22.05	30.130	0.985	0.930
2500	21.89	29.003	0.977	0.895
3858	21.71	27.988	0.969	0.864
4500	21.64	27.645	0.966	0.853

Table (2.1.4)

* Base porosity

** Base permeability

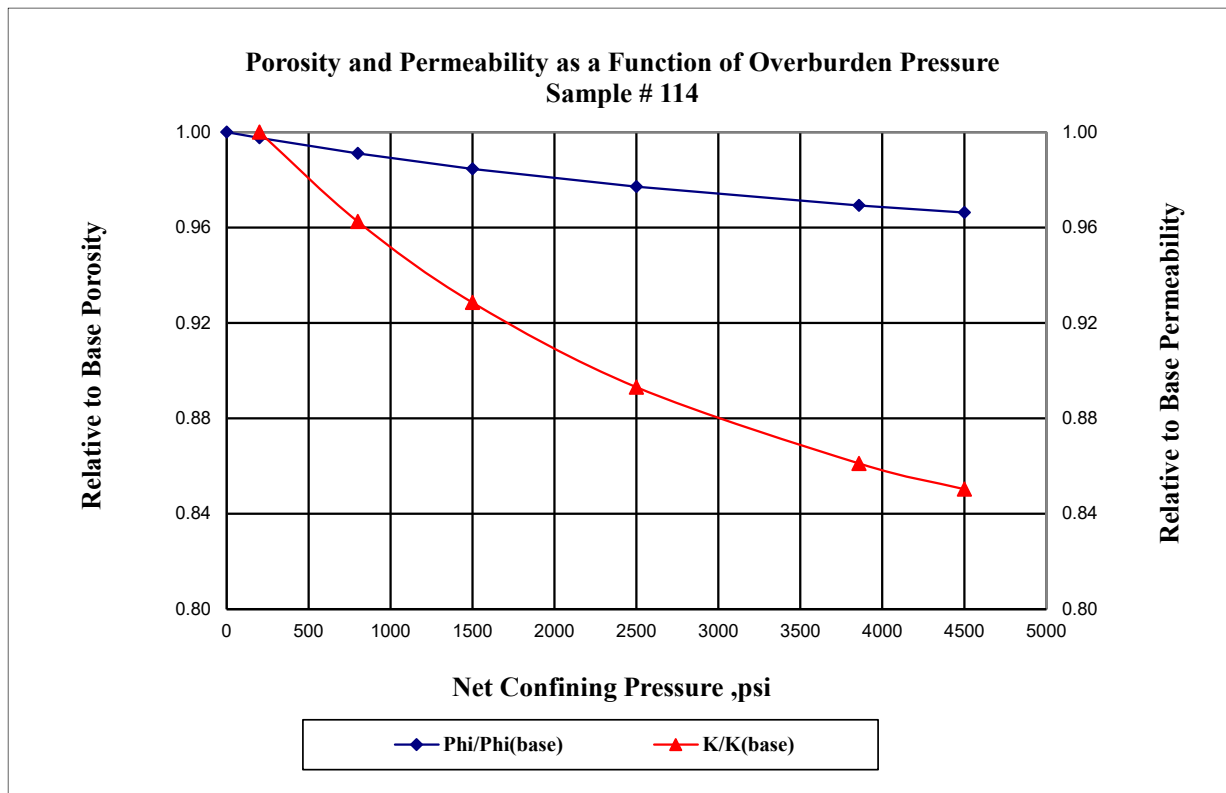


Figure (2.1.4)

POROSITY AND PERMEABILITY AS A FUNCTION OF OVERBURDEN PRESSURE

Company	: Mellitha Oil & Gas	Formation	: Elgaria
Well #	: CE03-NC41	Concession	: 41
Sample #	: 126	Sample Depth (ft.in)	: 8483.10

Pressure (psi)	Porosity (%)	Permeability mD	Relative to base porosity	Relative to base Permeability
0	24.40*		1.000	
200	24.34	39.40**	0.998	1.000
800	24.19	38.235	0.991	0.970
1500	24.04	37.168	0.985	0.943
2500	23.86	36.038	0.978	0.914
3858	23.67	35.002	0.970	0.887
4500	23.60	34.647	0.967	0.878

Table (2.1.5)

* Base porosity

** Base permeability

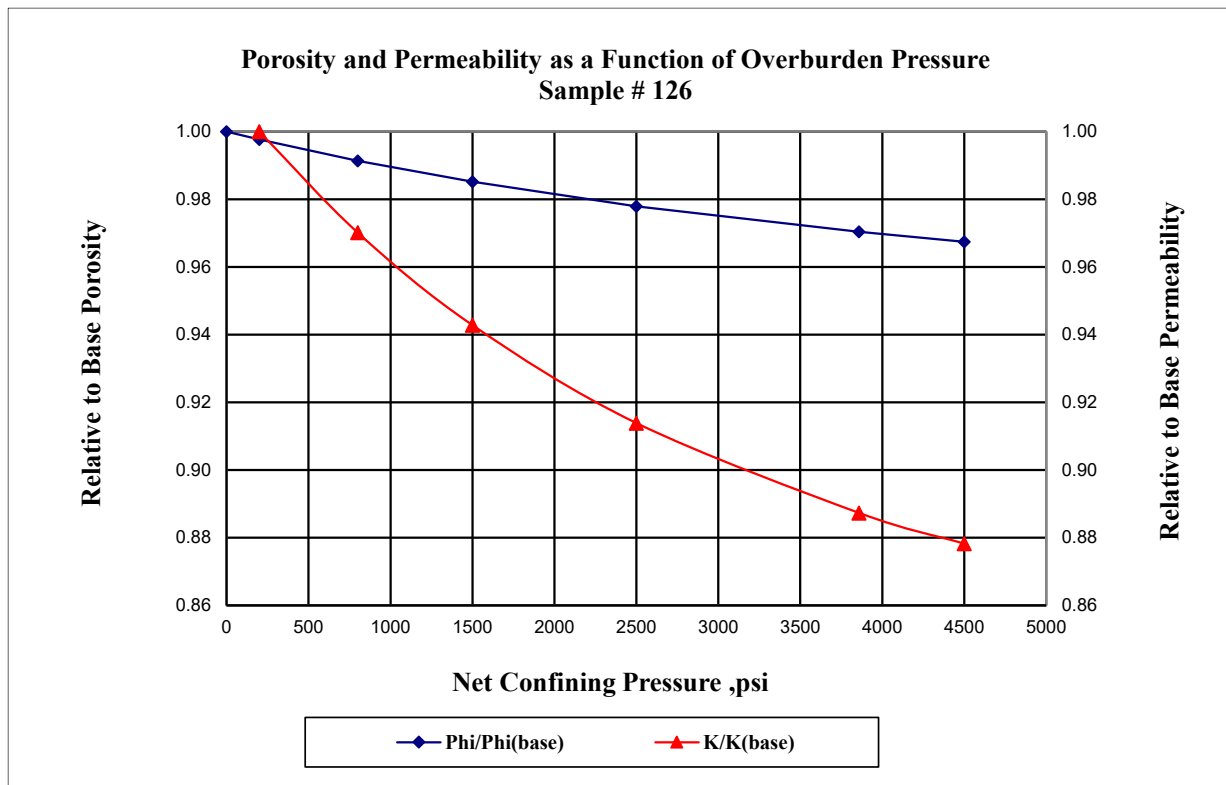


Figure (2.1.5)

POROSITY AND PERMEABILITY AS A FUNCTION OF OVERBURDEN PRESSURE

Company	: Mellitha Oil & Gas	Formation	: Elgaria
Well #	: CE03-NC41	Concession	: 41
Sample #	: 164	Sample Depth (ft.in)	: 8521.04

Pressure (psi)	Porosity (%)	Permeability mD	Relative to base porosity	Relative to base Permeability
0	16.30*		1.000	
200	16.28	4.890**	0.999	1.000
800	16.21	4.780	0.995	0.977
1500	16.15	4.678	0.990	0.957
2500	16.06	4.569	0.985	0.934
3858	15.97	4.467	0.980	0.914
4500	15.93	4.432	0.977	0.906

Table (2.1.6)

* Base porosity

** Base permeability

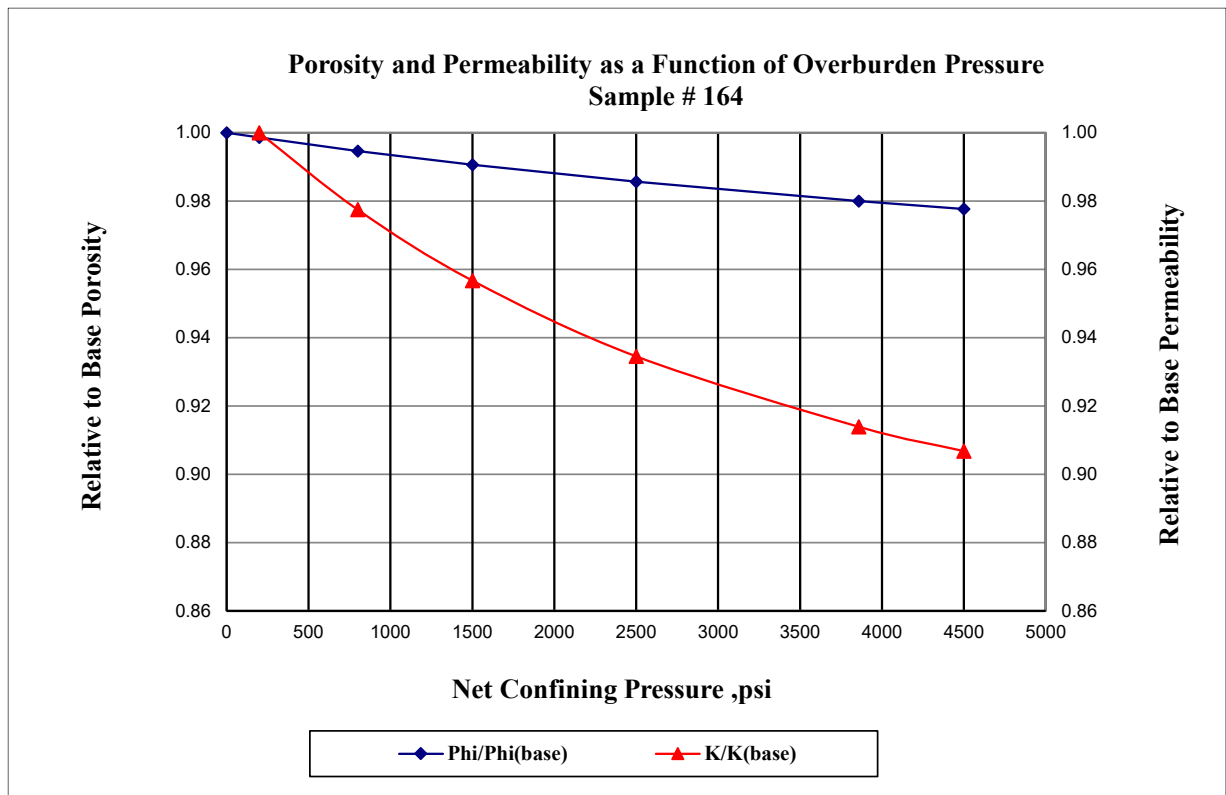


Figure (2.1.6)

POROSITY AND PERMEABILITY AS A FUNCTION OF OVERBURDEN PRESSURE

Company	: Mellitha Oil & Gas	Formation	: Elgaria
Well #	: CE03-NC41	Concession	: 41
Sample #	: 227	Sample Depth (ft.in)	: 8585.02

Pressure (psi)	Porosity (%)	Permeability mD	Relative to base porosity	Relative to base Permeability
0	25.90*		1.000	
200	25.84	126.00**	0.998	1.000
800	25.69	122.401	0.992	0.972
1500	25.54	119.088	0.986	0.946
2500	25.36	115.564	0.979	0.918
3858	25.18	112.311	0.972	0.892
4500	25.10	111.191	0.969	0.884

Table (2.1.7)

* Base porosity

** Base permeability

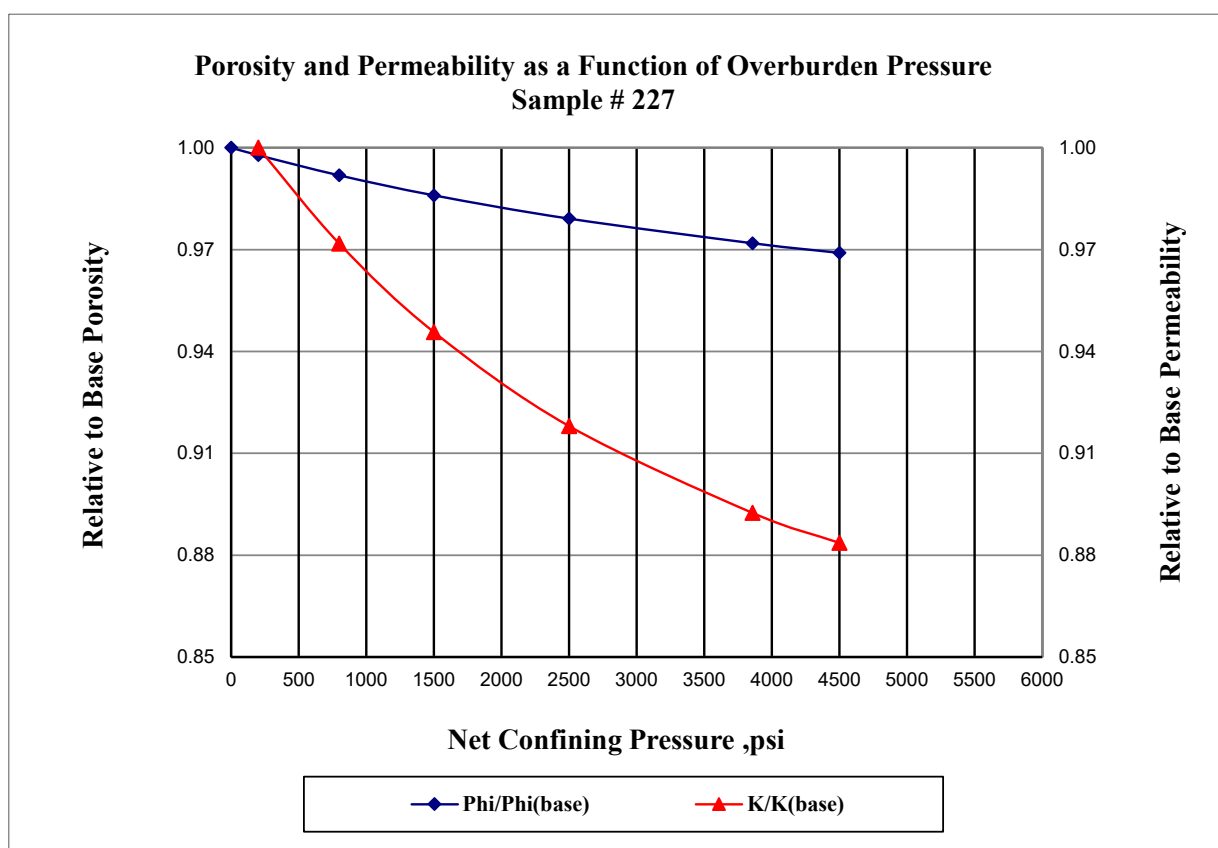


Figure (2.1.7)

POROSITY AND PERMEABILITY AS A FUNCTION OF OVERBURDEN PRESSURE

Company	: Mellitha Oil & Gas	Formation	: Elgaria
Well #	: CE03-NC41	Concession	: 41
Sample #	: 233	Sample Depth (ft.in)	: 8591.04

Pressure (psi)	Porosity (%)	Permeability mD	Relative to base porosity	Relative to base Permeability
0	22.60*		1.000	
200	22.56	51.700**	0.998	1.000
800	22.46	50.924	0.994	0.985
1500	22.35	50.192	0.989	0.971
2500	22.22	49.386	0.983	0.955
3858	22.09	48.609	0.977	0.940
4500	22.03	48.329	0.975	0.935

Table (2.1.8)

* Base porosity

** Base permeability

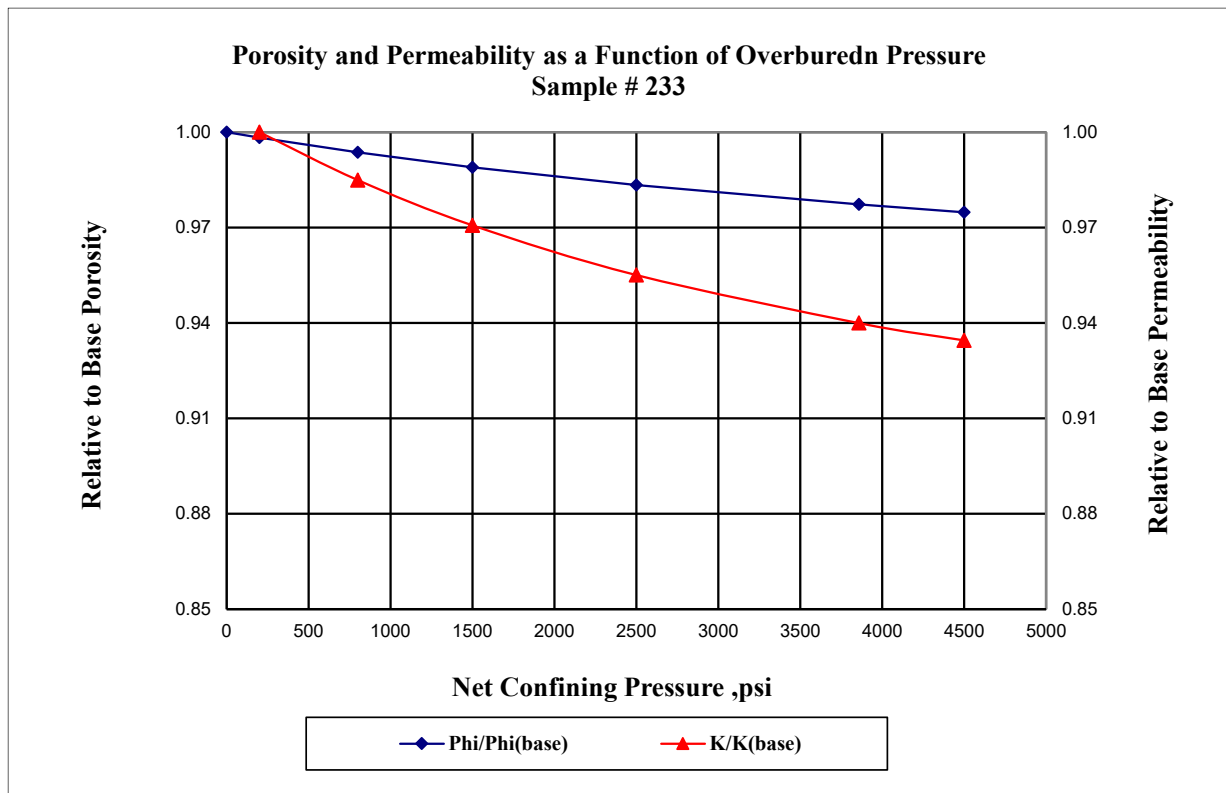


Figure (2.1.8)

POROSITY AND PERMEABILITY AS A FUNCTION OF OVERBURDEN PRESSURE

Company	: Mellitha Oil & Gas	Formation	: Elgaria
Well #	: CE03-NC41	Concession	: 41
Sample #	: 275	Sample Depth (ft.in)	: 8636.06

Pressure (psi)	Porosity (%)	Permeability mD	Relative to base porosity	Relative to base Permeability
0	24.20*		1.000	
200	24.15	225.00**	0.998	1.000
800	24.02	220.387	0.993	0.979
1500	23.89	216.087	0.987	0.960
2500	23.73	211.442	0.981	0.940
3858	23.57	207.064	0.974	0.920
4500	23.50	205.528	0.971	0.913

Table (2.1.9)

* Base porosity

** Base permeability

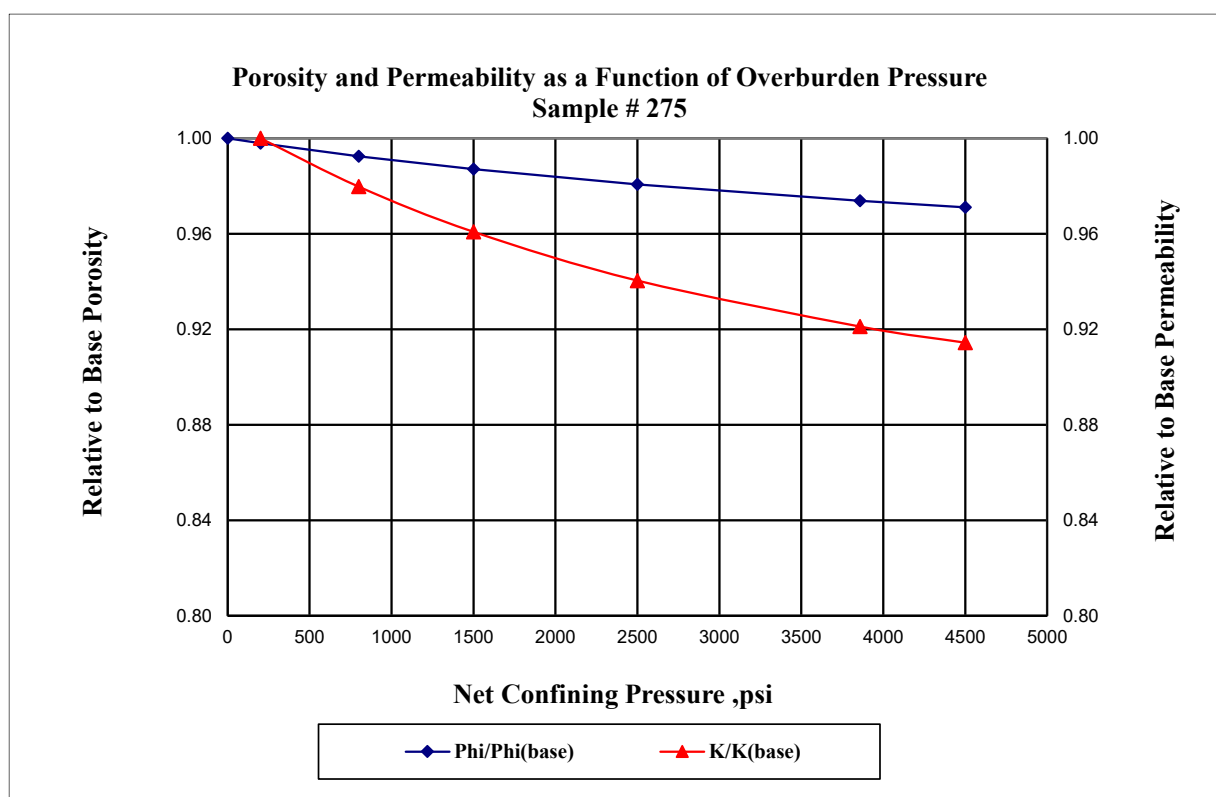


Figure (2.1.9)

POROSITY AND PERMEABILITY AS A FUNCTION OF OVERBURDEN PRESSURE

Company	: Mellitha Oil & Gas	Formation	: Elgaria
Well #	: CE03-NC41	Concession	: 41
Sample #	: 278	Sample Depth (ft.in)	: 8639.01

Pressure (psi)	Porosity (%)	Permeability mD	Relative to base porosity	Relative to base Permeability
0	24.90*		1.000	
200	24.84	38.100**	0.998	1.000
800	24.69	37.271	0.992	0.978
1500	24.54	36.502	0.985	0.958
2500	24.36	35.676	0.978	0.936
3858	24.18	34.904	0.971	0.916
4500	24.10	34.635	0.968	0.909

Table (2.1.10)

* Base porosity

** Base permeability

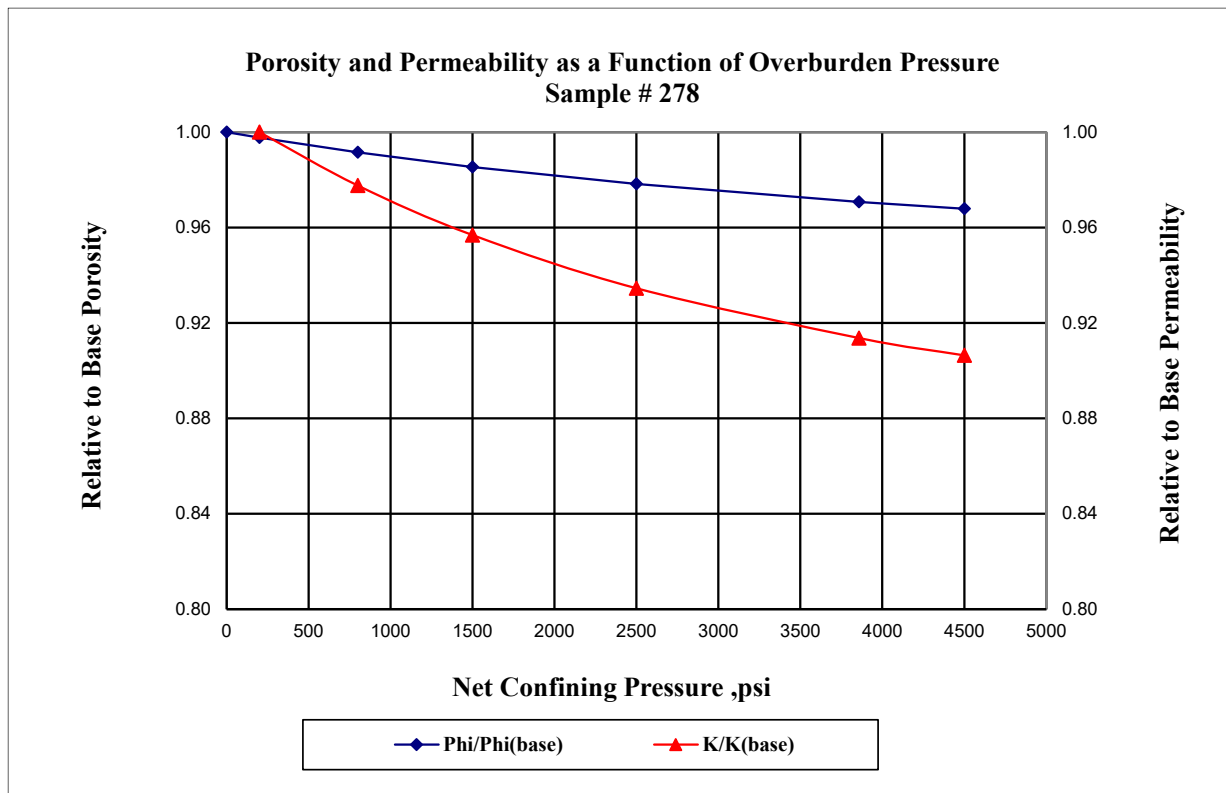


Figure (2.1.10)

Section Two

Pore Volume Compressibility

PORE VOLUME COMPRESSIBILITY

The data derived from porosity at overburden pressure was used to calculate pore volume compressibility. Using the reduction in pore volume related to the hydrostatic net overburden pressure, a value of the hydrostatic pore volume compressibility was calculated for each overburden pressure. This value was converted to uniaxial pore volume compressibility for each overburden pressure using Teew's equation shown below (Reference 2) and assuming a value of 0.35 for the Poisson's ratio. The results are presented in tables 2.2.1 through 2.2.10 and in figures 2.2.1 through 2.2.10.

$$C_{p_{uni}} = 1/3 [(1+v) / (1-v)] C_{p_{hyd}}$$

Where:

$C_{p_{uni}}$ = uniaxial pore volume compressibility, $pv/pv/psi$

$C_{p_{hyd}}$ = hydrostatic pore volume compressibility, $pv/pv/psi$

v = Poisson's ratio (assumed to be 0.35 for Carbonate formations)

PORE VOLUME COMPRESSIBILITY AS A FUNCTION OF OVERBURDEN PRESSURE

Company	: Mellitha Oil & Gas	Formation	: Elgaria
Well #	: CE03-NC41	Concession	: 41
Sample #	: 80	Sample Depth(ft.in)	: 8437.06

Pressure, Psi	Porosity (%)	Pore Volume Compressibility	
		Uniaxial	Hydrostatic
		pv/pv/psi x 10 ⁻⁶	
0	21.00*		
200	20.94	11.645	16.820
800	20.79	9.907	14.309
1500	20.64	8.272	11.948
2500	20.46	6.507	9.399
3858	20.28	4.880	7.049
4500	20.22	4.332	6.257

Table (2.2.1)

* Base porosity

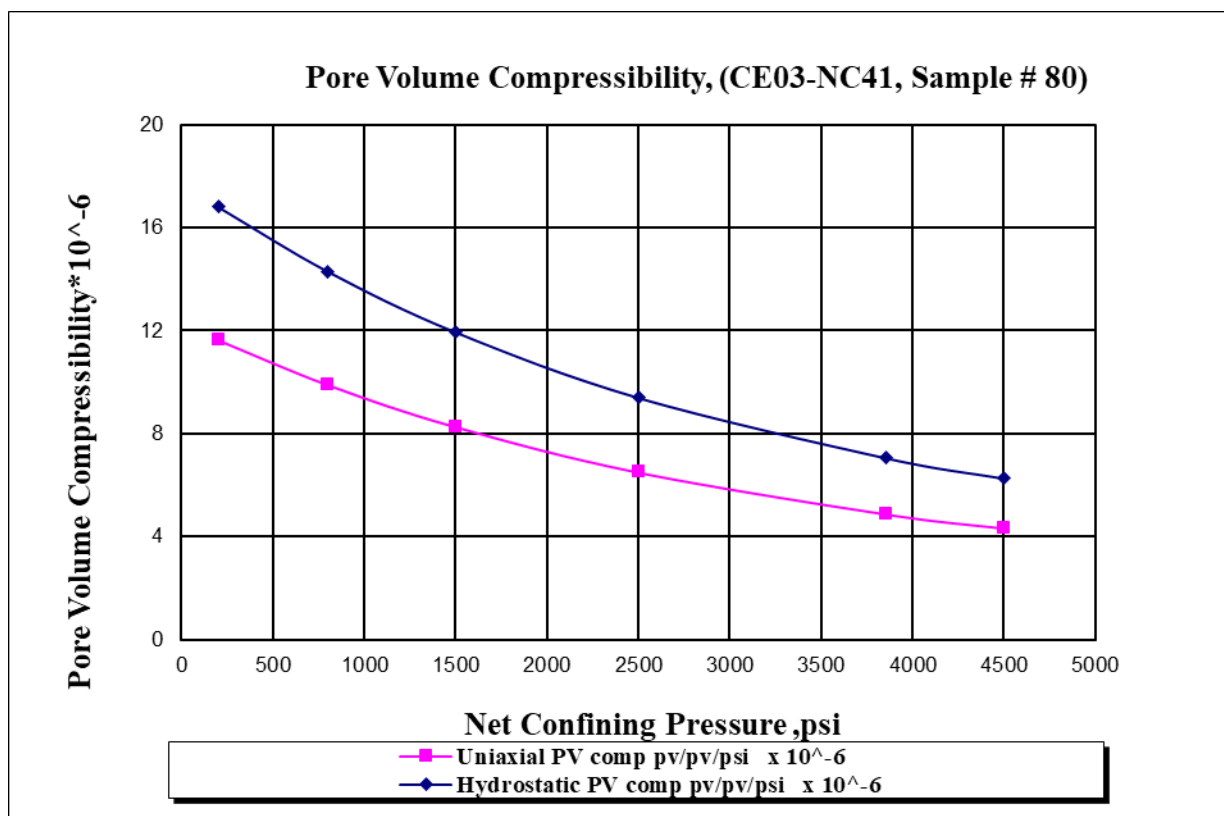


Figure (2.2.1)

PORE VOLUME COMPRESSIBILITY AS A FUNCTION OF OVERBURDEN PRESSURE

Company	: Mellitha Oil & Gas	Formation	: Elgaria
Well #	: CE03-NC41	Concession	: 41
Sample #	: 93	Sample Depth(ft.in)	: 8450.06

Pressure, Psi	Porosity (%)	Pore Volume Compressibility	
		Uniaxial	Hydrostatic
		pv/pv/psi x 10 ⁻⁶	
0	23.30*		
200	23.26	7.550	10.905
800	23.15	6.554	9.467
1500	23.04	5.617	8.113
2500	22.91	4.605	6.651
3858	22.76	3.670	5.302
4500	22.70	3.355	4.846

Table (2.2.2)

* Base porosity

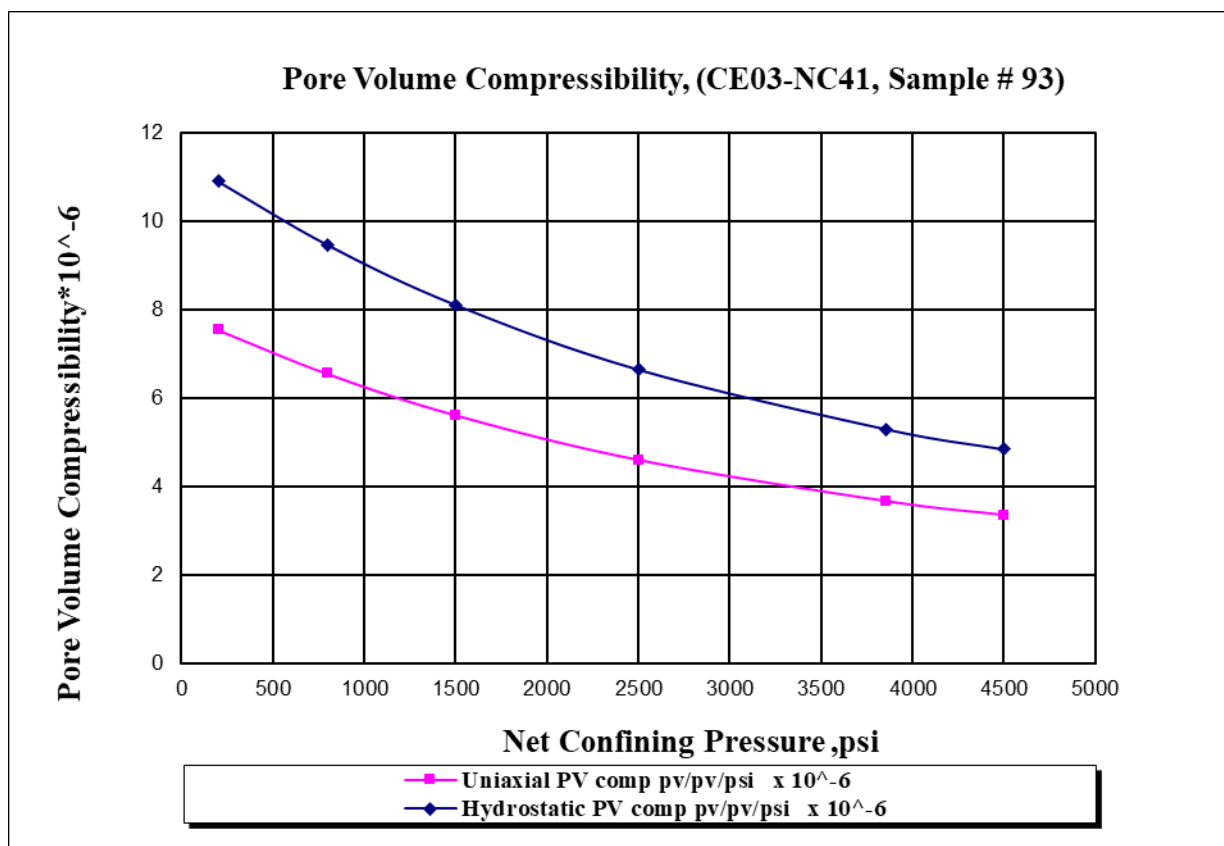


Figure (2.2.2)

PORE VOLUME COMPRESSIBILITY AS A FUNCTION OF OVERBURDEN PRESSURE

Company	: Mellitha Oil & Gas	Formation	: Elgaria
Well #	: CE03-NC41	Concession	: 41
Sample #	: 108	Sample Depth(ft.in)	: 8465.05

Pressure, Psi	Porosity (%)	Pore Volume Compressibility	
		Uniaxial	Hydrostatic
		pv/pv/psi x 10 ⁻⁶	
0	25.58*		
200	25.55	5.611	8.105
800	25.46	4.967	7.174
1500	25.37	4.360	6.298
2500	25.26	3.704	5.350
3858	25.14	3.098	4.474
4500	25.08	2.893	4.178

Table (2.2.3)

* Base porosity

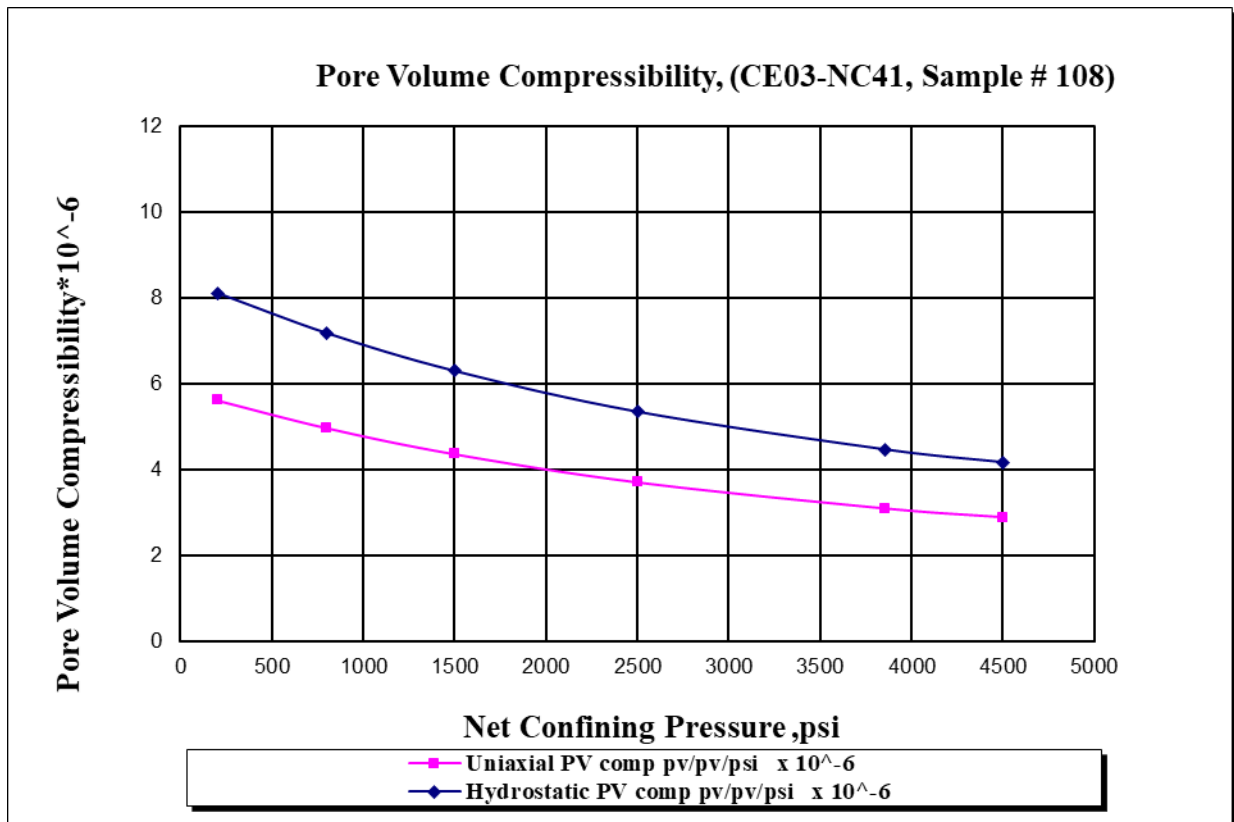


Figure (2.2.3)

PORE VOLUME COMPRESSIBILITY AS A FUNCTION OF OVERBURDEN PRESSURE

Company	: Mellitha Oil & Gas	Formation	: Elgaria
Well #	: CE03-NC41	Concession	: 41
Sample #	: 114	Sample Depth(ft.in)	: 8471.07

Pressure, Psi	Porosity (%)	Pore Volume Compressibility	
		Uniaxial	Hydrostatic
		pv/pv/psi x 10 ⁻⁶	
0	22.40*		
200	22.35	10.440	15.079
800	22.20	8.920	12.884
1500	22.05	7.490	10.819
2500	21.89	5.947	8.590
3858	21.71	4.524	6.535
4500	21.64	4.044	5.842

Table (2.2.4)

* Base porosity

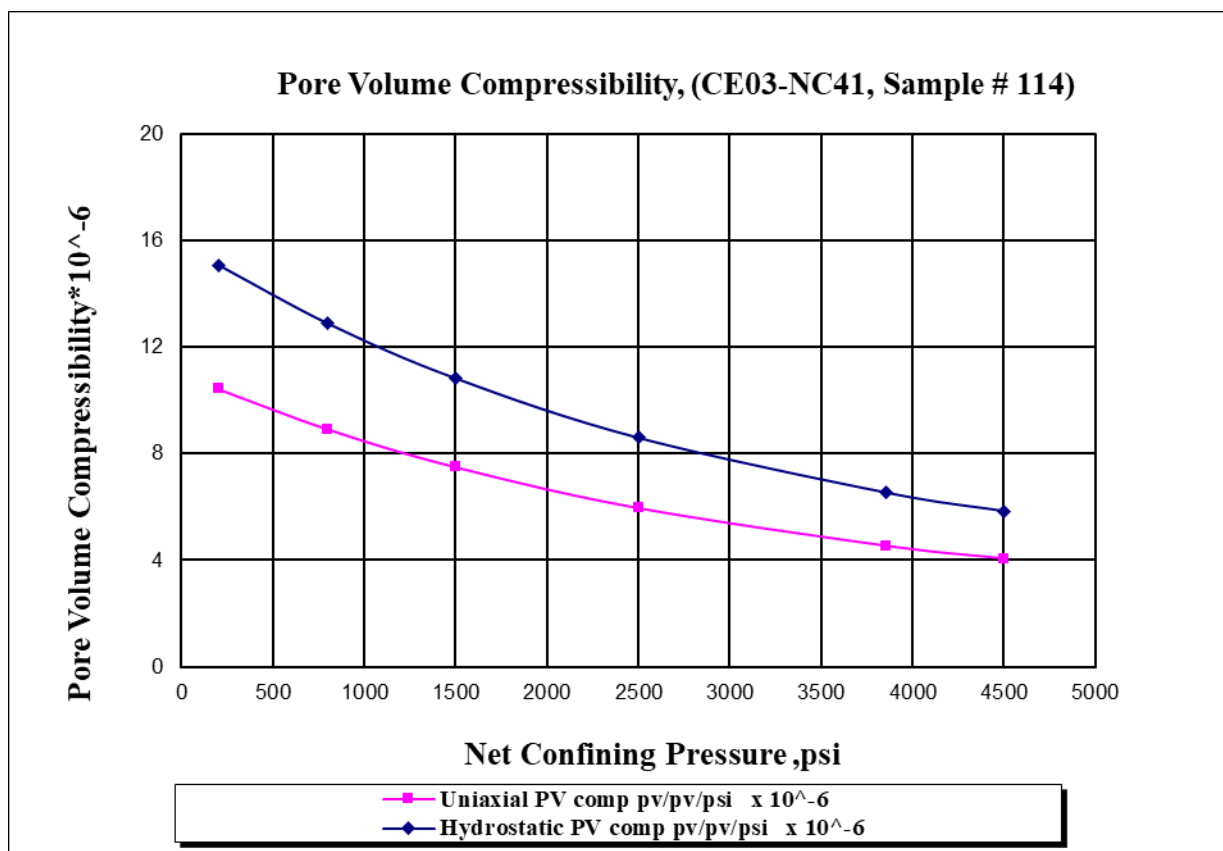


Figure (2.2.4)

PORE VOLUME COMPRESSIBILITY AS A FUNCTION OF OVERBURDEN PRESSURE

Company	: Mellitha Oil & Gas	Formation	: Elgaria
Well #	: CE03-NC41	Concession	: 41
Sample #	: 126	Sample Depth(ft.in)	: 8483.10

Pressure, Psi	Porosity (%)	Pore Volume Compressibility	
		Uniaxial	Hydrostatic
		pv/pv/psi x 10 ⁻⁶	
0	24.40*		
200	24.34	10.378	14.991
800	24.19	8.870	12.812
1500	24.04	7.451	10.762
2500	23.86	5.919	8.549
3858	23.67	4.506	6.509
4500	23.60	4.030	5.820

Table (2.2.5)

* Base porosity

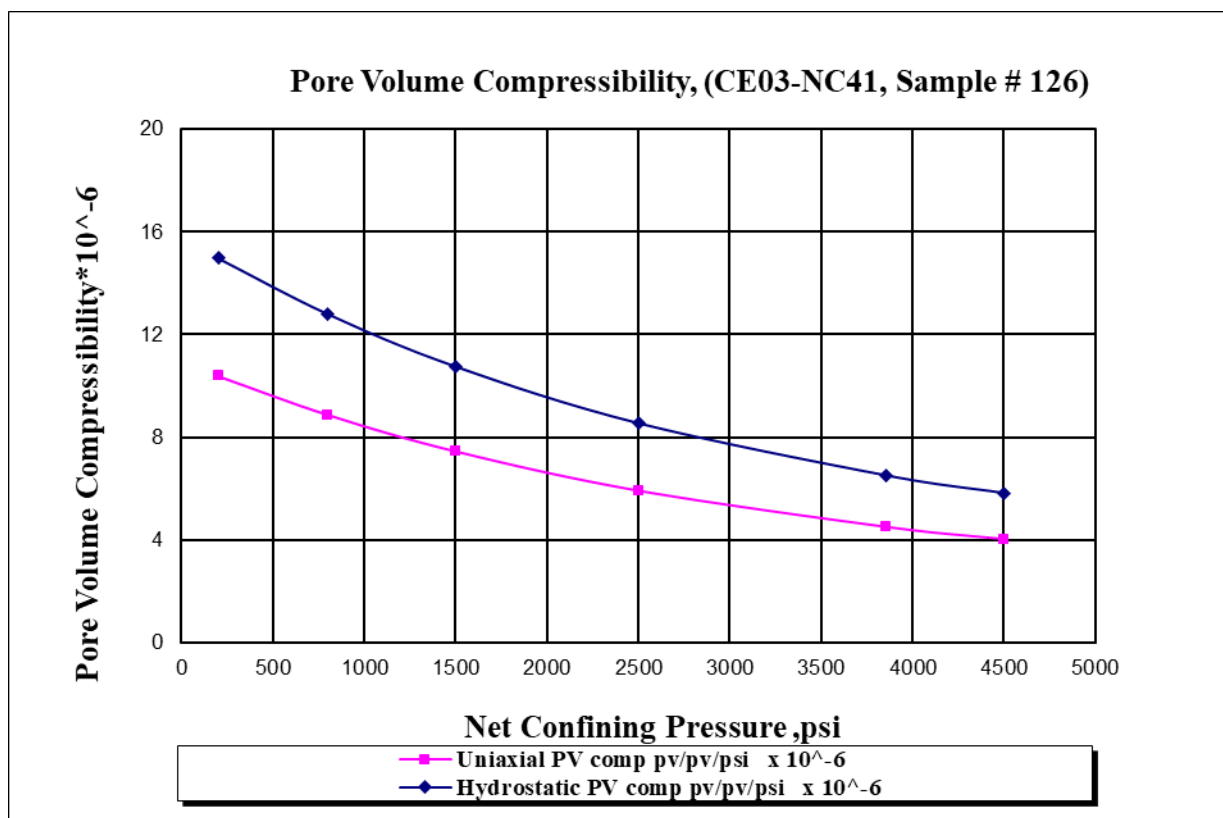


Figure (2.2.5)

PORE VOLUME COMPRESSIBILITY AS A FUNCTION OF OVERBURDEN PRESSURE

Company	: Mellitha Oil & Gas	Formation	: Elgaria
Well #	: CE03-NC41	Concession	: 41
Sample #	: 164	Sample Depth(ft.in)	: 8521.04

Pressure, Psi	Porosity (%)	Pore Volume Compressibility	
		Uniaxial	Hydrostatic
		pv/pv/psi x 10 ⁻⁶	
0	16.30*		
200	16.28	5.600	8.088
800	16.21	4.957	7.160
1500	16.15	4.352	6.287
2500	16.06	3.698	5.342
3858	15.97	3.094	4.469
4500	15.93	2.890	4.174

Table (2.2.6)

* Base porosity

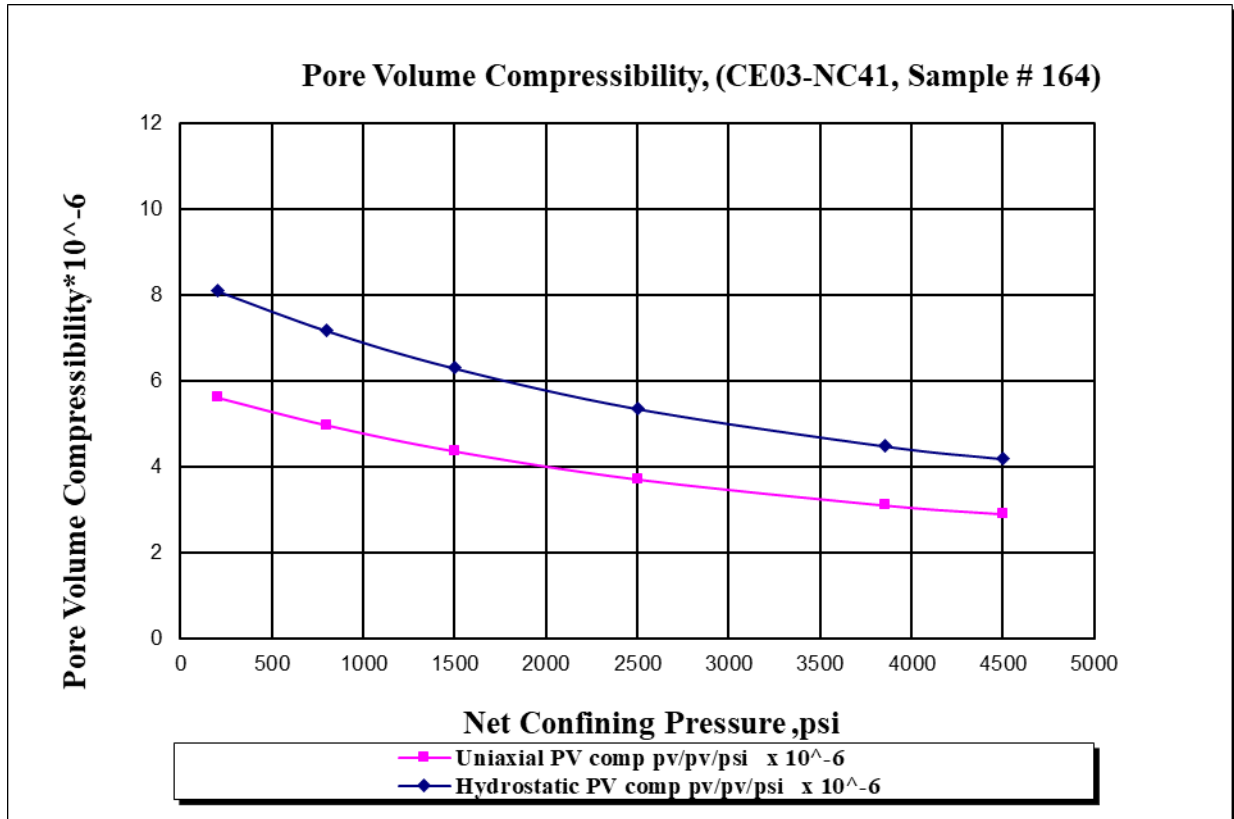


Figure (2.2.6)

PORE VOLUME COMPRESSIBILITY AS A FUNCTION OF OVERBURDEN PRESSURE

Company	: Mellitha Oil & Gas	Formation	: Elgaria
Well #	: CE03-NC41	Concession	: 41
Sample #	: 227	Sample Depth(ft.in)	: 8585.02

Pressure, Psi	Porosity (%)	Pore Volume Compressibility	
		Uniaxial	Hydrostatic
		pv/pv/psi x 10 ⁻⁶	
0	25.90*		
200	25.84	9.778	14.124
800	25.69	8.378	12.101
1500	25.54	7.061	10.200
2500	25.36	5.640	8.146
3858	25.18	4.329	6.252
4500	25.10	3.886	5.614

Table (2.2.7)

* Base porosity

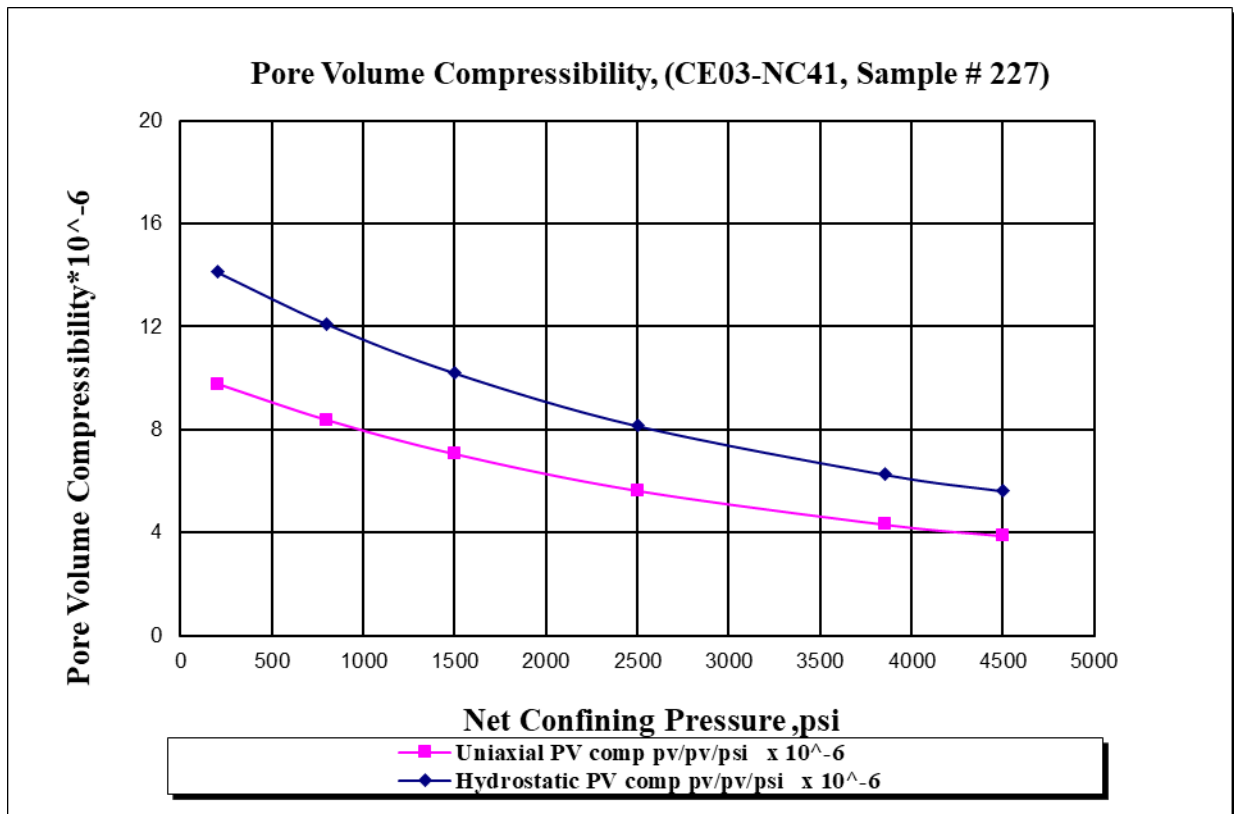


Figure (2.2.7)

PORE VOLUME COMPRESSIBILITY AS A FUNCTION OF OVERBURDEN PRESSURE

Company	: Mellitha Oil & Gas	Formation	: Elgaria
Well #	: CE03-NC41	Concession	: 41
Sample #	: 233	Sample Depth(ft.in)	: 8591.04

Pressure, Psi	Porosity (%)	Pore Volume Compressibility	
		Uniaxial	Hydrostatic
		pv/pv/psi x 10 ⁻⁶	
0	22.60*		
200	22.56	7.395	10.682
800	22.46	6.427	9.284
1500	22.35	5.516	7.968
2500	22.22	4.533	6.547
3858	22.09	3.625	5.236
4500	22.03	3.318	4.793

Table (2.2.8)

* Base porosity

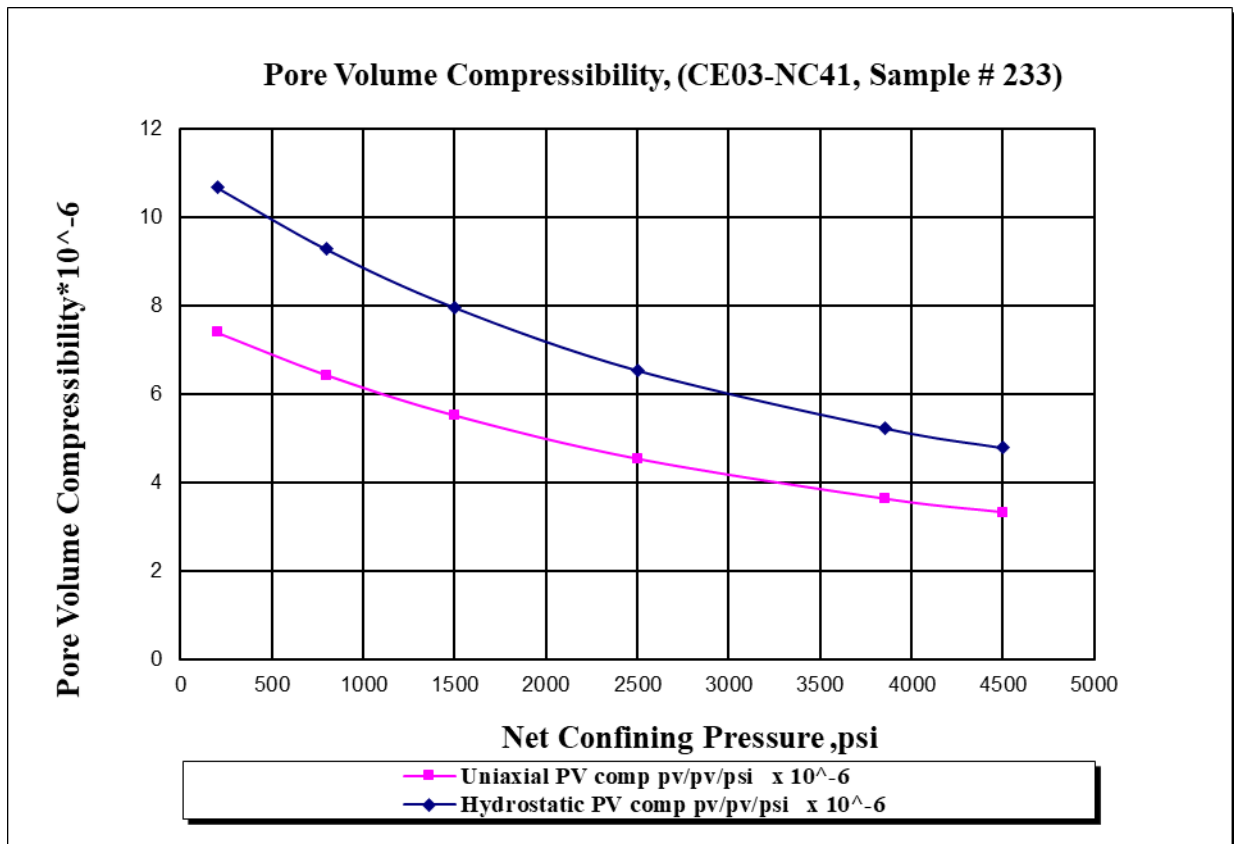


Figure (2.2.8)

PORE VOLUME COMPRESSIBILITY AS A FUNCTION OF OVERBURDEN PRESSURE

Company	: Mellitha Oil & Gas	Formation	: Elgaria
Well #	: CE03-NC41	Concession	: 41
Sample #	: 275	Sample Depth(ft.in)	: 8636.06

Pressure, Psi	Porosity (%)	Pore Volume Compressibility	
		Uniaxial	Hydrostatic
		pv/pv/psi x 10 ⁻⁶	
0	24.20*		
200	24.15	8.897	12.851
800	24.02	7.656	11.059
1500	23.89	6.490	9.374
2500	23.73	5.230	7.555
3858	23.57	4.068	5.876
4500	23.50	3.676	5.310

Table (2.2.9)

* Base porosity

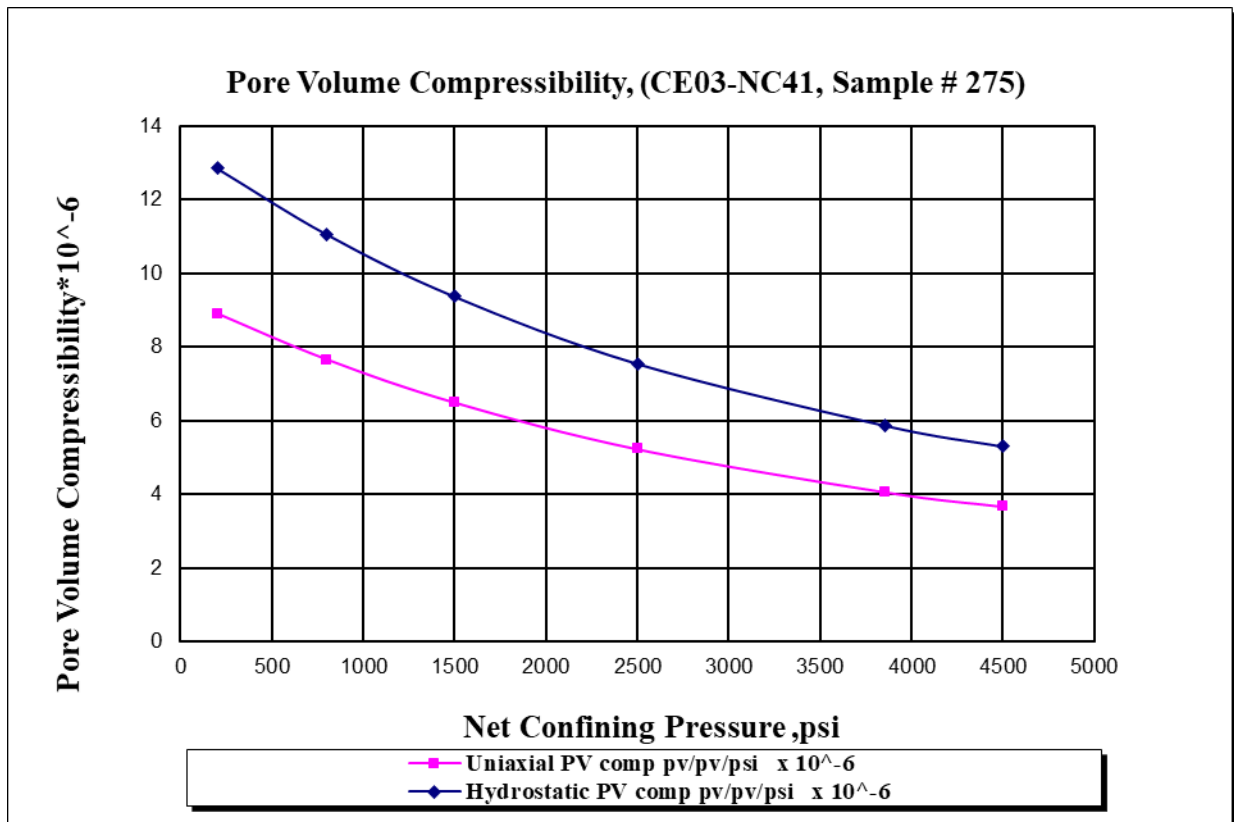


Figure (2.2.9)

PORE VOLUME COMPRESSIBILITY AS A FUNCTION OF OVERBURDEN PRESSURE

Company	: Mellitha Oil & Gas	Formation	: Elgaria
Well #	: CE03-NC41	Concession	: 41
Sample #	: 278	Sample Depth(ft.in)	: 8639.01

Pressure, Psi	Porosity (%)	Pore Volume Compressibility	
		Uniaxial	Hydrostatic
		pv/pv/psi x 10 ⁻⁶	
0	24.90*		
200	24.84	9.984	14.422
800	24.69	8.547	12.345
1500	24.54	7.195	10.393
2500	24.36	5.735	8.284
3858	24.18	4.390	6.340
4500	24.10	3.936	5.685

Table (2.2.10)

* Base porosity

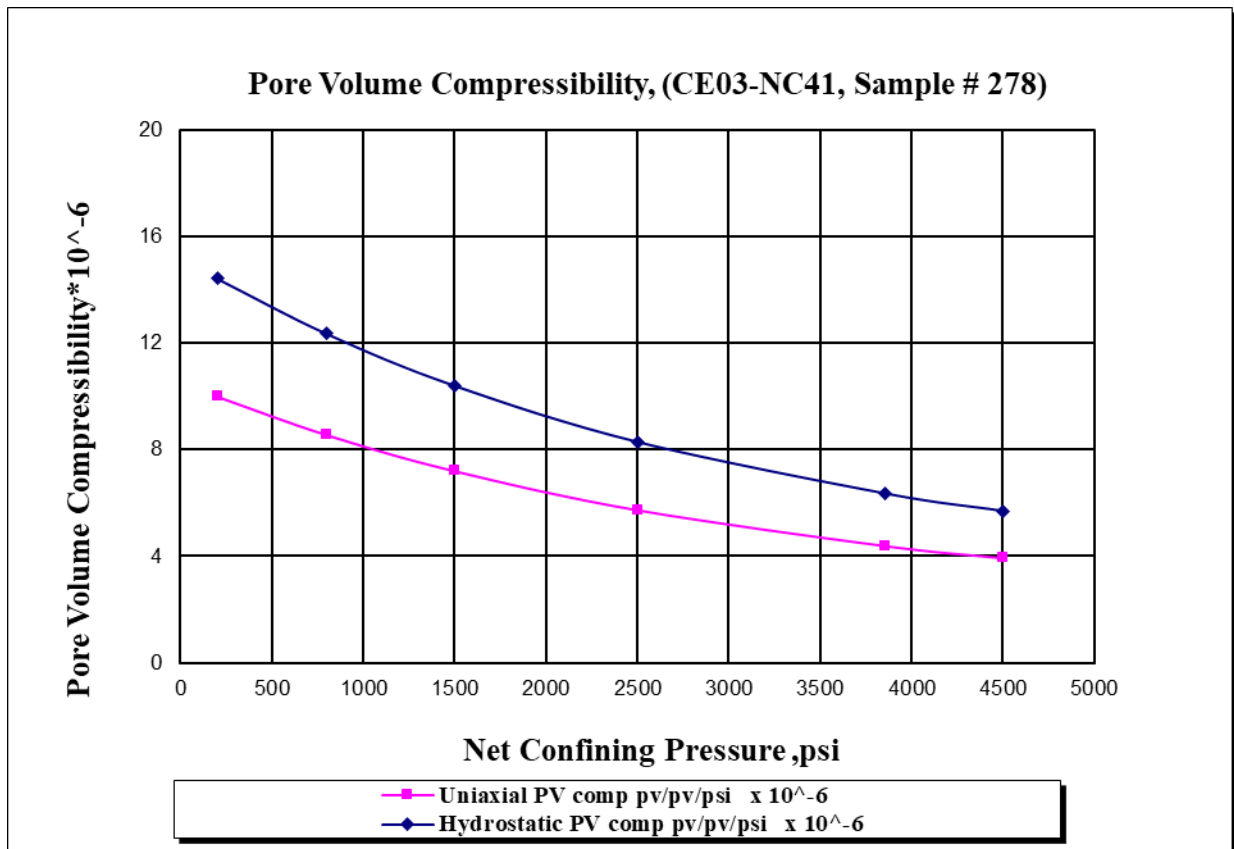


Figure (2.2.10)

Section Three

Formation Resistivity Factor (ambient conditions)

FORMATION RESISTIVITY FACTOR (at ambient conditions):

Ten samples were selected from the well **CE03-NC41**, Elgaria formation, to perform formation resistivity factor measurements at ambient conditions. The clean, dry samples were loaded in a stainless-steel pressurized cell and evacuated for 24 hours. A sodium chloride solution of (70,000 ppm) was introduced at the end of the period, followed by pressurising the system at 2500 psig for 24 hours to assist penetration. Gravimetric checks were made to ensure complete saturation of pore spaces.

The brine saturated plugs were placed in turn between electrodes at 1 Khz frequency and their electrical resistances were measured in consecutive days until ionic equilibrium was achieved between the fluid and the rock samples.

A four-point electrical device was used to measure the resistance, and the observed phase angel was set to zero.

Finally, the resistance values were used to calculate specific resistivity (R_o) of the plug samples, and formation factor was obtained by dividing the specific resistivity (R_o) of the plug sample by water resistivity ($R_w = 0.0965 \text{ ohm.m @ } 77^\circ\text{F}$) for the solution mentioned above. The generated data for the well are presented in table **2.3.1** and in figure **2.3.1**.

Formation Resistivity Factor (ambient conditions)

Company	: Mellitah Oil and Gas	Salinity	: 70,000 ppm
Well	: CE03-NC41	Overburden pressure	: 0 psig
Formation	: Elgaria	Concession	: 41
No of Samples	: 10	Rw @ 77 deg. F	: 0.0965 ohm.m

Sample #	Core Depth, (ft.in)	Porosity (%)	Air Permeability (mD)	Formation Factor	Cementation Exponent "m"
80	8437.06	21.00	51.70	18.03	1.86
93	8450.06	23.30	18.90	15.89	1.88
108	8465.05	25.58	62.80	15.22	2.00
114	8471.07	22.40	32.40	15.63	1.85
126	8483.10	24.40	39.40	15.19	1.93
164	8521.04	16.30	4.890	24.87	1.82
227	8585.02	25.90	126.0	12.70	1.85
233	8591.04	22.60	51.70	17.00	1.89
275	8636.06	24.20	225.0	15.55	1.91
278	8639.01	24.90	38.10	13.14	1.85

Table (2.3.1)

Formation Factor-Porosity Correlation

Company	: Mellitah Oil and Gas	Salinity	: 70,000 ppm
Well	: CE03-NC41	Overburden pressure	: 0 psig
Formation	: Elgaria	Concession	: 41
No. of Samples	: 10	Rw @ 77 deg. F	: 0.0965 ohm.m

	<u>Forced fit</u>	<u>Free fit</u>
a =	1.0	2.34
m =	1.88	1.30

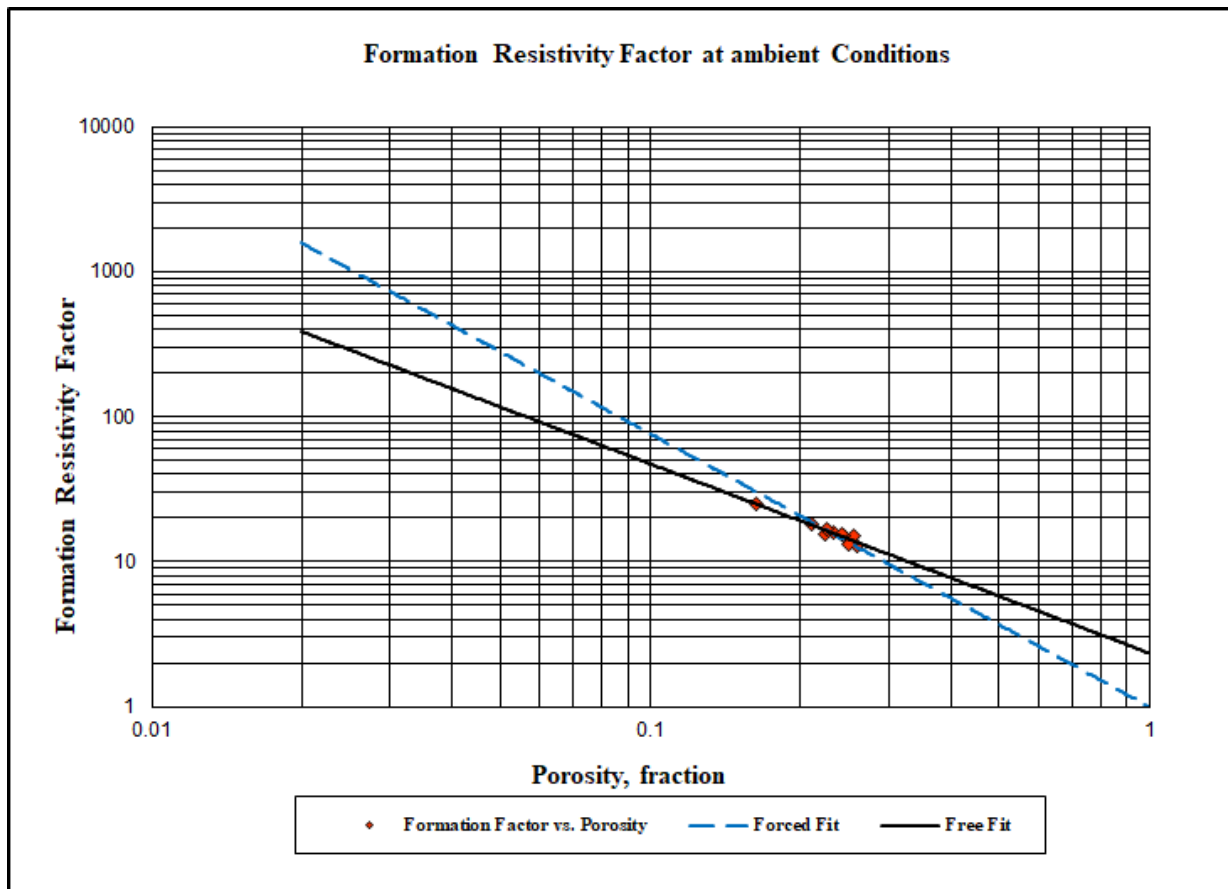


Figure (2.3.1)

Section Four Formation Resistivity Factor at Overburden Pressure

FORMATION RESISTIVITY FACTOR (at overburden pressure)

Samples previously used for the formation factor measurements at ambient conditions were scheduled for formation factor testing at overburden conditions.

The fully saturated samples were individually placed between stainless steel end pieces, in a rubber sleeve and immersed in oil within a hydrostatic core holder. All connecting pipe work and contact surfaces were fully saturated with simulated formation brine to ensure electrical continuity. The overburden pressure was applied on the cores up to 400 psig in the manner described in Nieto paper (Reference 3).

The volume of brine expelled being monitored to determine the reduction in pore volume. The resistance values were measured at each requested pressure (400, 1000, 1500, 2000, 2700 and 3858 psig). The samples were allowed to stabilise before their resistances were measured. On achieving equilibrium, samples the electrical resistances were measured by using a four-point electrical device and the observed phase angel was set to zero. The resistance values were used to calculate specific resistivity (R_o) of the plug samples, and formation factor was obtained by dividing the specific resistivity (R_o) of the plug sample by water resistivity (R_w). The generated data are presented in tables 2.4.1 through 2.4.6 and in figures 2.4.1 through 2.4.8.

Formation Resistivity Factor (overburden Pressure)

Company	: Mellitah Oil and Gas	Salinity	: 70,000 ppm
Well	: CE03-NC41	Overburden pressure	: 400 psig
Aquifers	: Elgaria	Concession	: 41
No of Samples	: 10	Rw @ 77 deg. F	: 0.0965 ohm-m

Sample #	Shifted Depth, (ft.in)	Porosity (%)	Formation Factor	Cementation Exponent "m"
80	8437.06	21.00	17.7	1.84
93	8450.06	23.30	14.5	1.84
108	8465.05	25.58	13.9	1.93
114	8471.07	22.40	15.9	1.85
126	8483.10	24.40	13.8	1.86
164	8521.04	16.30	24.1	1.76
227	8585.02	25.90	10.2	1.72
233	8591.04	22.60	14.2	1.78
275	8636.06	24.20	15.4	1.93
278	8639.01	24.90	11.46	1.75

Table (2.4.1)

Formation Factor-Porosity Correlation

Company	: Mellitah Oil and Gas	Salinity	: 70,000 ppm
Well	: CE03-NC41	Overburden pressure	: 400 psig
Formation	: Elgaria	Concession	: 41
No of Samples	: 10	Rw @ 77 deg. F	: 0.0965 ohm-m

	<u>Forced fit</u>	<u>Free fit</u>
a =	1.00	1.46
m =	1.82	1.57

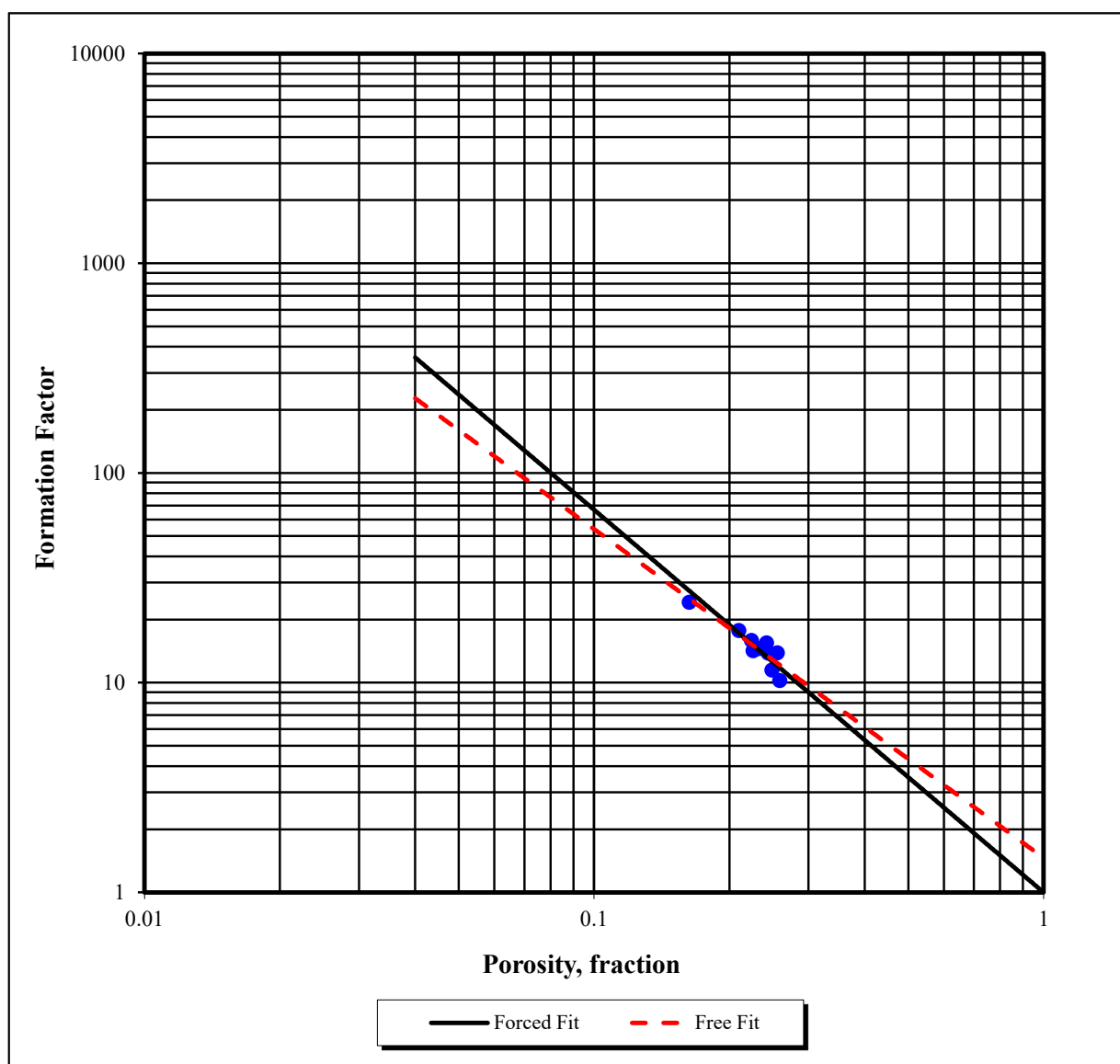


Figure (2.4.1)

Formation Resistivity Factor (overburden Pressure)

Company	: Mellitah Oil and Gas	Salinity	: 70,000 ppm
Well	: CE03-NC41	Overburden pressure	: 1000 psig
Aquifers	: Elgaria	Concession	: 41
No of Samples	: 10	Rw @ 77 deg. F	: 0.0965 ohm-m

Sample #	Shifted Depth, (ft.in)	Porosity (%)	Formation Factor	Cementation Exponent "m"
80	8437.06	20.36	19.2	1.86
93	8450.06	22.93	15.5	1.86
108	8465.05	25.20	14.5	1.94
114	8471.07	22.17	16.1	1.84
126	8483.10	24.13	14.3	1.87
164	8521.04	16.22	24.6	1.76
227	8585.02	25.48	10.7	1.73
233	8591.04	22.37	14.7	1.80
275	8636.06	23.92	16.0	1.94
278	8639.01	24.62	12.1	1.78

Table (2.4.2)

Formation Factor-Porosity Correlation

Company	: Mellitah Oil and Gas	Salinity	: 70,000 ppm
Well	: CE03-NC41	Overburden pressure	: 1000 psig
Formation	: Elgaria	Concession	: 41
No of Samples	: 10	Rw @ 77 deg. F	: 0.0965 ohm-m

	<u>Forced fit</u>	<u>Free fit</u>
a =	1.00	1.51
m =	1.84	1.56

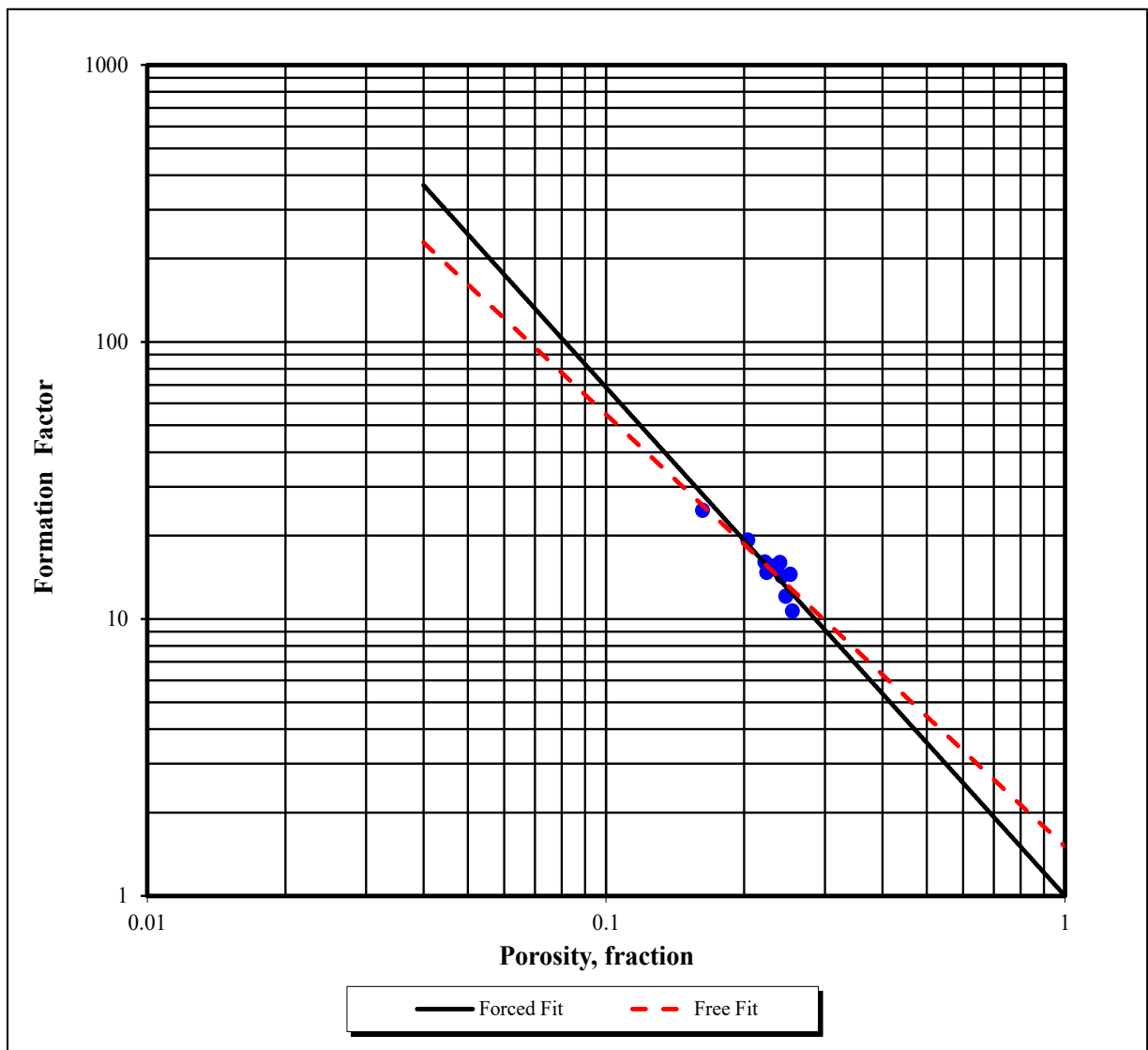


Figure (2.4.2)

Formation Resistivity Factor (overburden Pressure)

Company	: Mellitah Oil and Gas	Salinity	: 70,000 ppm
Well	: CE03-NC41	Overburden pressure	: 1500 psig
Aquifers	: Elgaria	Concession	: 41
No of Samples	: 10	Rw @ 77 deg. F	: 0.0965 ohm-m

Sample #	Shifted Depth, (ft.in)	Porosity (%)	Formation Factor	Cementation Exponent "m"
80	8437.06	20.18	19.6	1.86
93	8450.06	22.79	15.7	1.86
108	8465.05	24.91	14.7	1.93
114	8471.07	22.01	16.5	1.85
126	8483.10	23.97	14.5	1.87
164	8521.04	16.18	25.2	1.77
227	8585.02	25.35	10.9	1.74
233	8591.04	22.25	14.8	1.79
275	8636.06	23.77	16.2	1.94
278	8639.01	24.48	12.3	1.78

Table (2.4.3)

Formation Factor-Porosity Correlation

Company	: Mellitah Oil and Gas	Salinity	: 70,000 ppm
Well	: CE03-NC41	Overburden pressure	: 1500 psig
Formation	: Elgaria	Concession	: 41
No of Samples	: 10	Rw @ 77 deg. F	: 0.0965 ohm-m

	<u>Forced fit</u>	<u>Free fit</u>
a =	1.00	1.46
m =	1.84	1.59

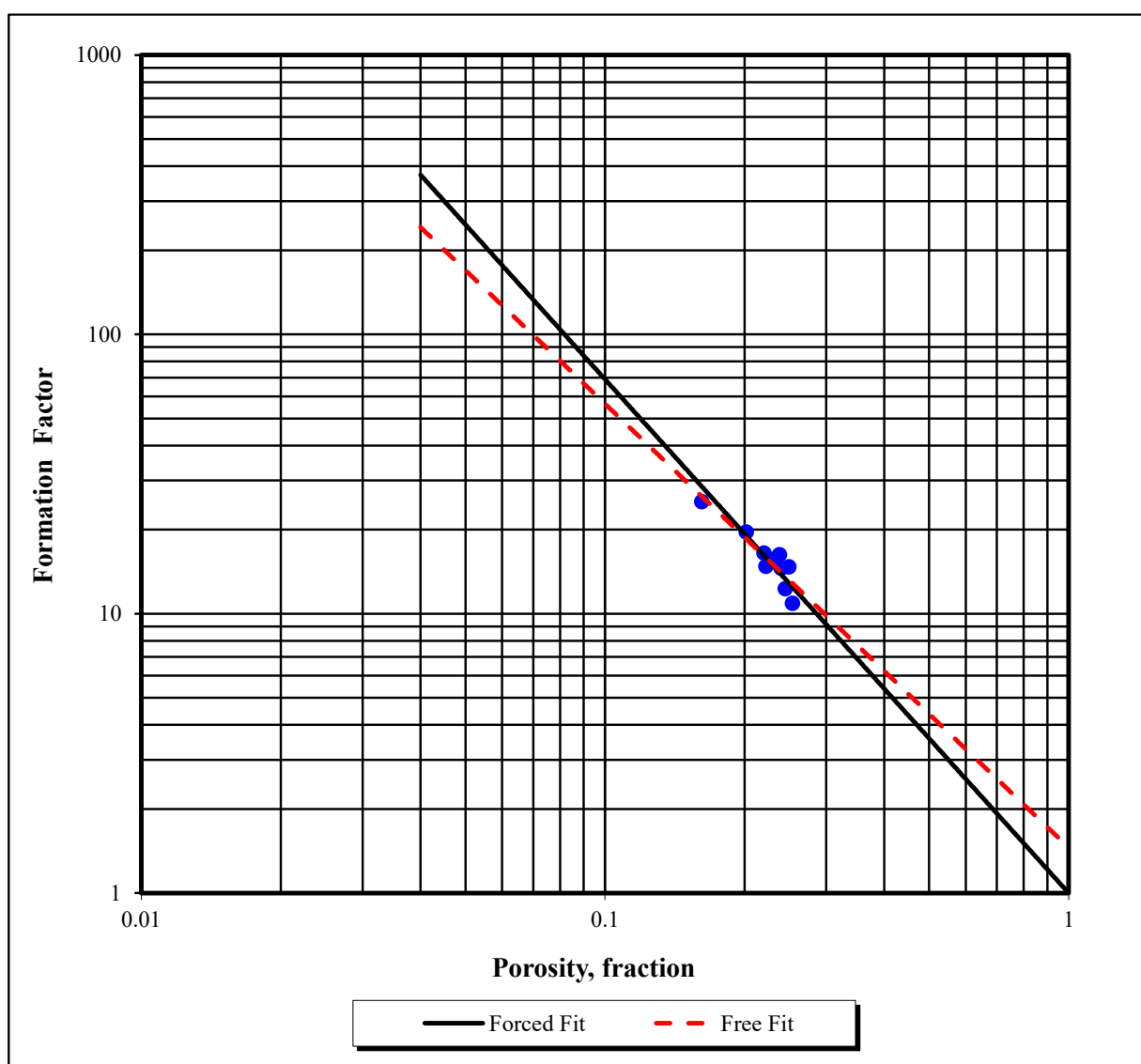


Figure (2.4.3)

Formation Resistivity Factor (overburden Pressure)

Company	: Mellitah Oil and Gas	Salinity	: 70,000 ppm
Well	: CE03-NC41	Overburden pressure	: 2000 psig
Aquifers	: Elgaria	Concession	: 41
No of Samples	: 10	Rw @ 77 deg. F	: 0.0965 ohm-m

Sample #	Shifted Depth, (ft.in)	Porosity (%)	Formation Factor	Cementation Exponent "m"
80	8437.06	20.05	19.9	1.86
93	8450.06	22.68	15.7	1.86
108	8465.05	24.68	15.0	1.93
114	8471.07	21.83	16.7	1.85
126	8483.10	23.83	14.5	1.87
164	8521.04	16.14	25.7	1.78
227	8585.02	25.24	11.1	1.75
233	8591.04	22.15	14.8	1.79
275	8636.06	23.64	16.4	1.94
278	8639.01	24.25	12.87	1.80

Table (2.4.4)

Formation Factor-Porosity Correlation

Company	: Mellitah Oil and Gas	Salinity	: 70,000 ppm
Well	: CE03-NC41	Overburden pressure	: 2000 psig
Formation	: Elgaria	Concession	: 41
No of Samples	: 10	Rw @ 77 deg. F	: 0.0965 ohm-m

	<u>Forced fit</u>	<u>Free fit</u>
a =	1.00	1.42
m =	1.84	1.61

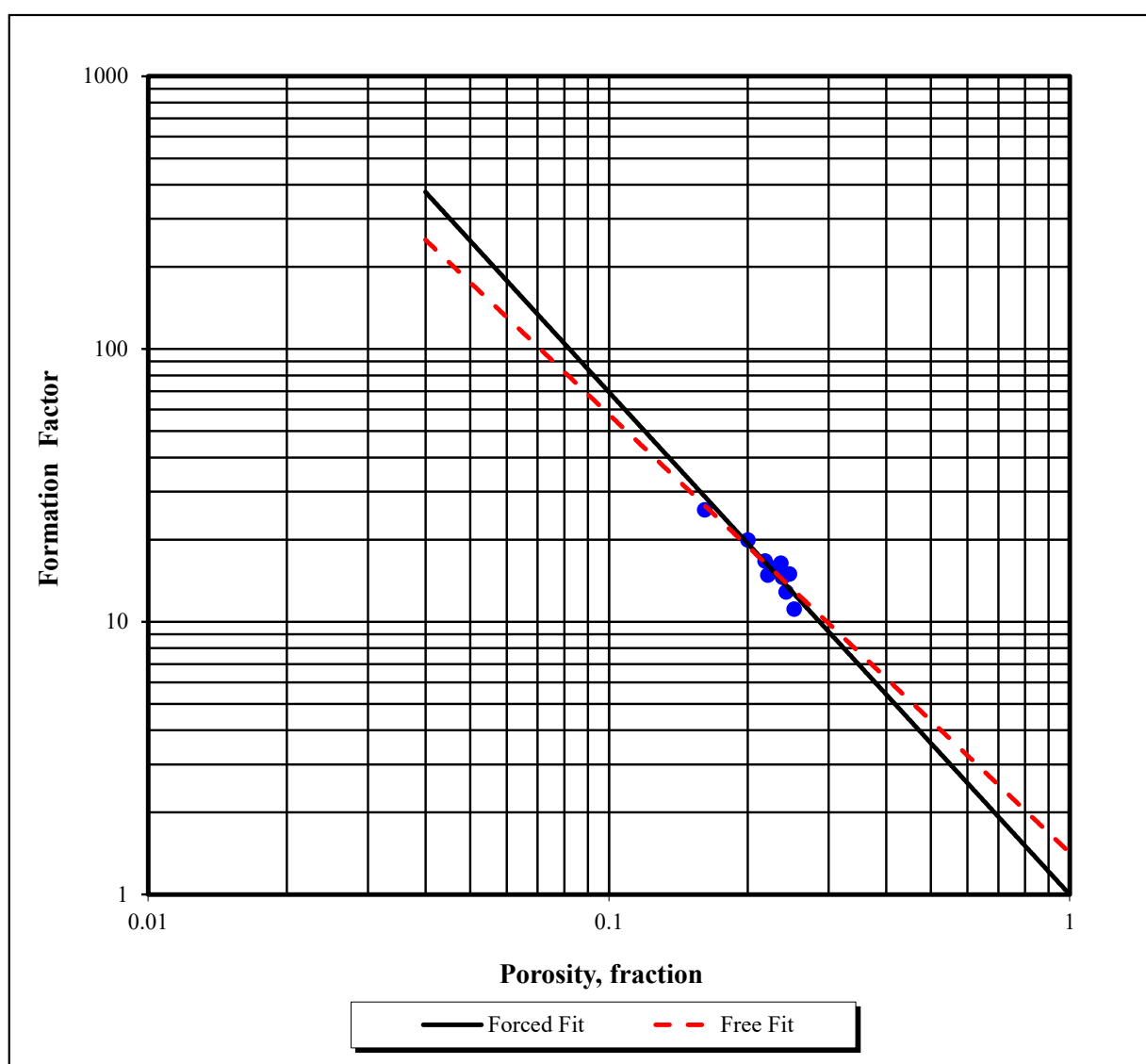


Figure (2.4.4)

Formation Resistivity Factor (overburden Pressure)

Company	: Mellitah Oil and Gas	Salinity	: 70,000 ppm
Well	: CE03-NC41	Overburden pressure	: 2700 psig
Aquifers	: Elgaria	Concession	: 41
No of Samples	: 10	Rw @ 77 deg. F	: 0.0965 ohm-m

Sample #	Shifted Depth, (ft.in)	Porosity (%)	Formation Factor	Cementation Exponent "m"
80	8437.06	19.94	20.20	1.86
93	8450.06	22.55	15.94	1.86
108	8465.05	24.50	14.96	1.92
114	8471.07	21.66	16.91	1.85
126	8483.10	23.68	14.78	1.87
164	8521.04	16.10	26.71	1.80
227	8585.02	25.07	11.24	1.75
233	8591.04	22.04	14.47	1.77
275	8636.06	23.46	16.62	1.94
278	8639.01	24.09	13.56	1.83

Table (2.4.5)

Formation Factor-Porosity Correlation

Company	: Mellitah Oil and Gas	Salinity	: 70,000 ppm
Well	: CE03-NC41	Overburden pressure	: 2700 psig
Formation	: Elgaria	Concession	: 41
No of Samples	: 10	Rw @ 77 deg. F	: 0.0965 ohm-m

	<u>Forced fit</u>	<u>Free fit</u>
a =	1.00	1.31
m =	1.84	1.66

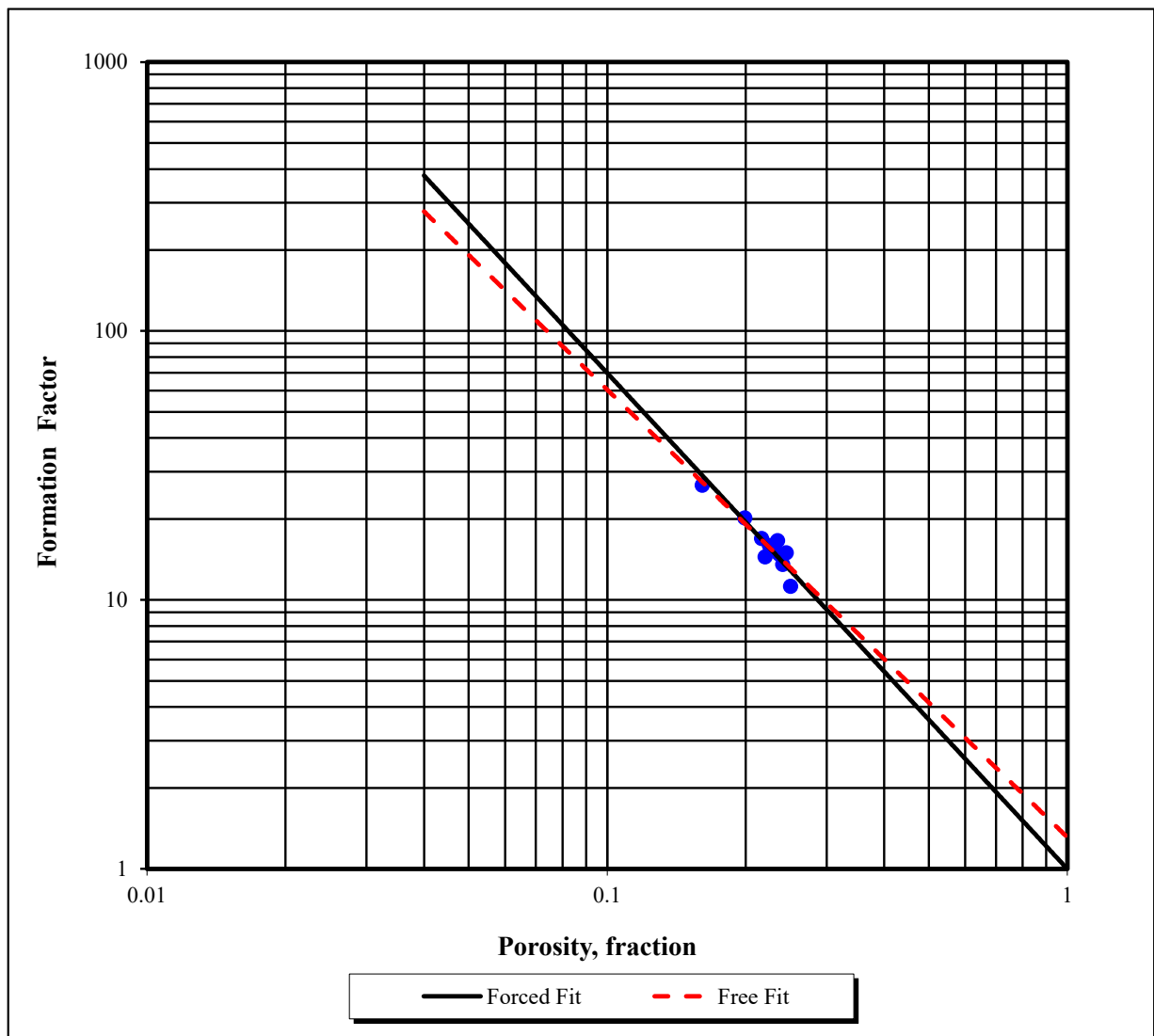


Figure (2.4.5)

Formation Resistivity Factor (overburden Pressure)

Company	: Mellitah Oil and Gas	Salinity	: 70,000 ppm
Well	: CE03-NC41	Overburden pressure	: 3858 psig
Aquifers	: Elgaria	Concession	: 41
No of Samples	: 10	Rw @ 77 deg. F	: 0.0965 ohm-m

Sample #	Shifted Depth, (ft.in)	Porosity (%)	Formation Factor	Cementation Exponent "m"
80	8437.06	19.74	20.6	1.86
93	8450.06	22.38	16.0	1.85
108	8465.05	24.23	15.1	1.92
114	8471.07	21.48	17.0	1.84
126	8483.10	23.40	14.9	1.86
164	8521.04	16.05	28.2	1.83
227	8585.02	24.61	11.6	1.75
233	8591.04	21.88	14.6	1.76
275	8636.06	23.22	16.9	1.94
278	8639.01	23.96	13.93	1.84

Table (2.4.6)

Formation Factor-Porosity Correlation

Company	: Mellitah Oil and Gas	Salinity	: 70,000 ppm
Well	: CE03-NC41	Overburden pressure	: 3858 psig
Formation	: Elgaria	Concession	: 41
No of Samples	: 10	Rw @ 77 deg. F	: 0.0965 ohm-m

	<u>Forced fit</u>	<u>Free fit</u>
a =	1.00	1.13
m =	1.84	1.76

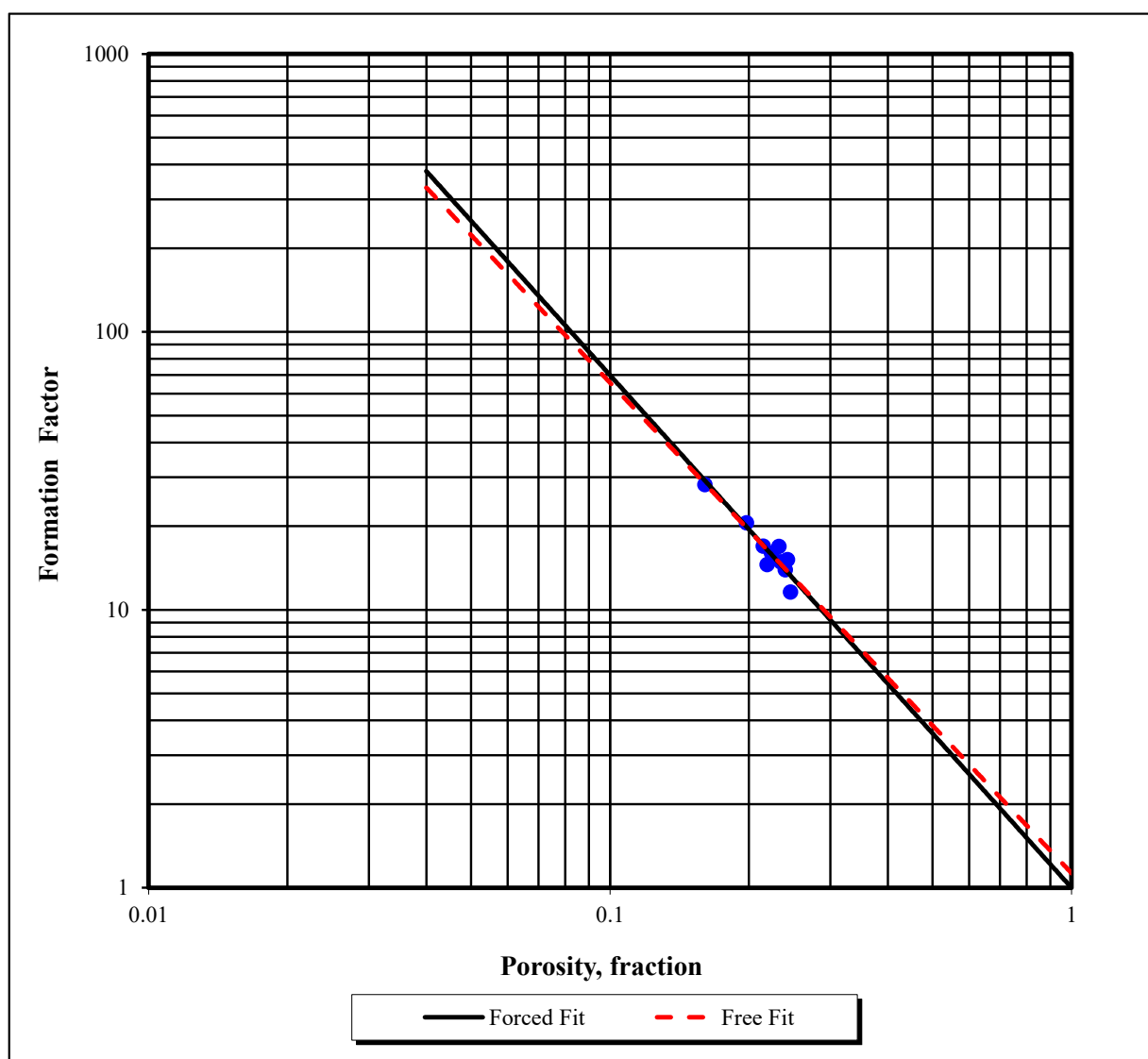


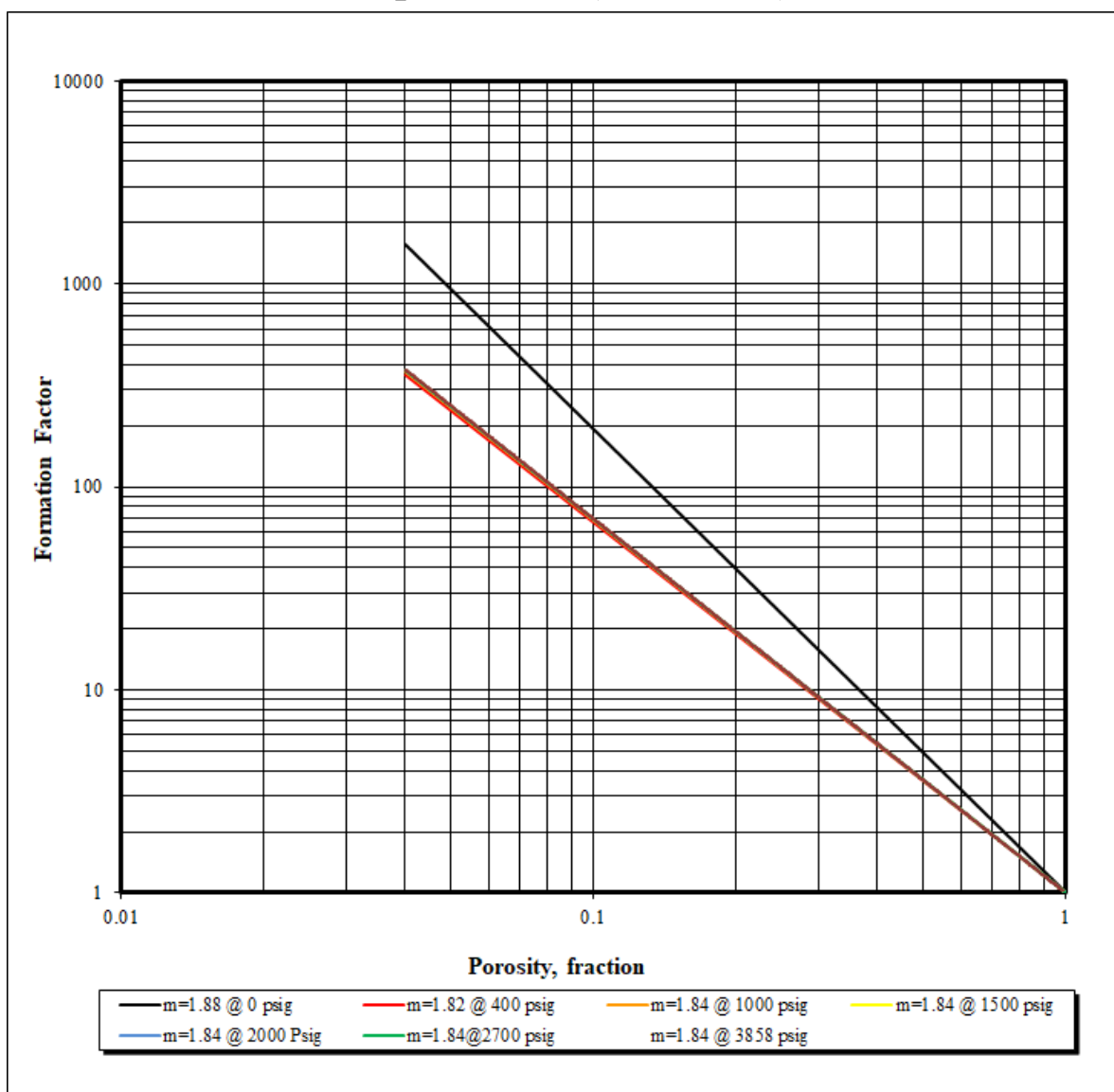
Figure (2.4.6)

Formation Factor-Porosity Correlation

Company	: Mellitah Oil and Gas	Salinity	: 70,000 ppm
Well	: CE03-NC41	Overburden pressure	: 400-3858 psig
Formation	: Elgaria	Concession	: 41
No of Samples	: 10	Rw @ 77 deg. F	: 0.0965 ohm-m

O.B.P.	0 psig	400 psig	1000 psig	1500 psig	2000 psig	2700 psig	3858 psig
m	m = 1.88	m =1.82	m =1.84	m = 1.84	m = 1.84	m =1.84	m=1.84

Composite Plot (Forced Fit)



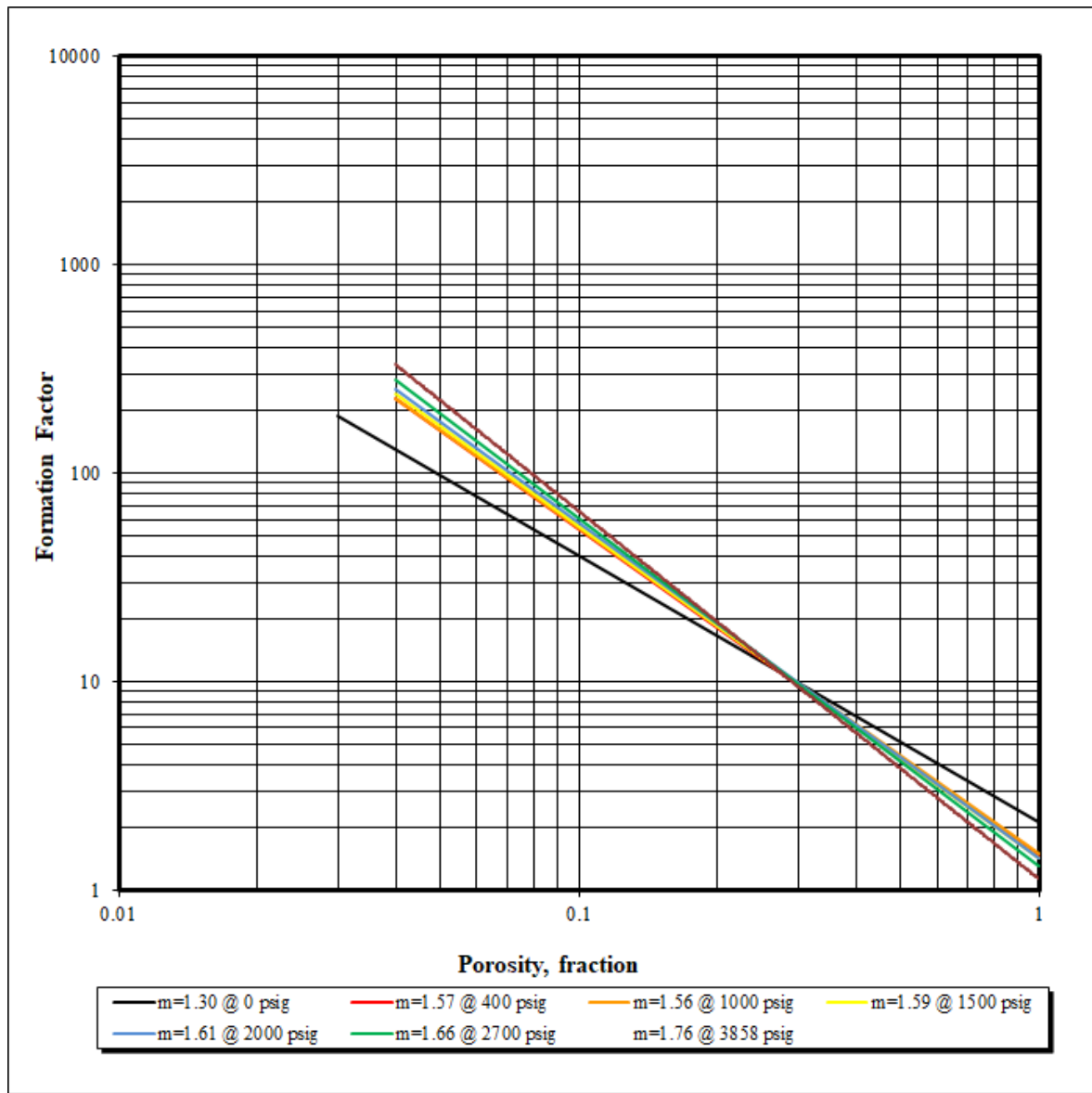
(Figure 2.4.7)

Formation Factor-Porosity Correlation

Company	: Mellitah Oil and Gas	Salinity	: 70,000 ppm
Well	: CE03-NC41	Overburden pressure	: 400-3858 psig
Formation	: Elgaria	Concession	: 41
No of Samples	: 10	Rw @ 77 deg. F	: 0.0965 ohm-m

O.B.P.	0 psig	400 psig	1000 psig	1500 psig	2000 psig	2700 psig	3858 psig
m	m = 1.30	m = 1.57	m = 1.56	m = 1.59	m = 1.61	m = 1.66	m = 1.76

Composite Plot (Free Fit)



(Figure 2.4.8)

Section Five Formation Resistivity Index at (ambient conditions)

FORMATION RESISTIVITY INDEX (at ambient conditions):-

Ten samples were selected from the well **CE03-NC41** to perform formation resistivity index measurements. This test was conducted after formation resistivity factor measurements. The fully saturated samples were placed on a semi-permeable porous diaphragm, in a capillary pressure apparatus, to bring the samples to lower saturation. Air humidified by water was admitted at a controlled pressure to the apparatus and the volume expelled was monitored. When equilibrium saturation had been attained, the samples were removed from the cell, weighed and their electrical resistances were measured by using a four-point electrical device and the observed phase angle was set to zero.

Sample dry weight, the fully saturated weight and the weight after each de-saturation step were used to calculate the average saturation of the sample. The resistance was used to calculate the sample resistivity (R_t) and the resistivity was divided by the sample resistivity at one hundred percent liquid saturation (R_o) which yield resistivity index (RI).

The results of all the samples are presented in tables **2.5.1** through **2.5.10** and figures **2.5.1** through **2.5.11**.

Formation Resistivity Index, (at ambient conditions)

Sample #: 80	Porosity (%): 21.00	Air Permeability(mD): 51.70
Company: Mellitah	Well: CE-03-NC41	Sample Depth(ft.in): 8437.06

Table 2.5.1: Laboratory Results

Resistivity Index (RI)	Brine Saturation (Sw), fraction
1.000	1.0000
1.002	0.9445
1.132	0.9131
1.445	0.7564
3.031	0.4748
7.389	0.2749
18.72	0.1458
55.29	0.0902
62.47	0.0840

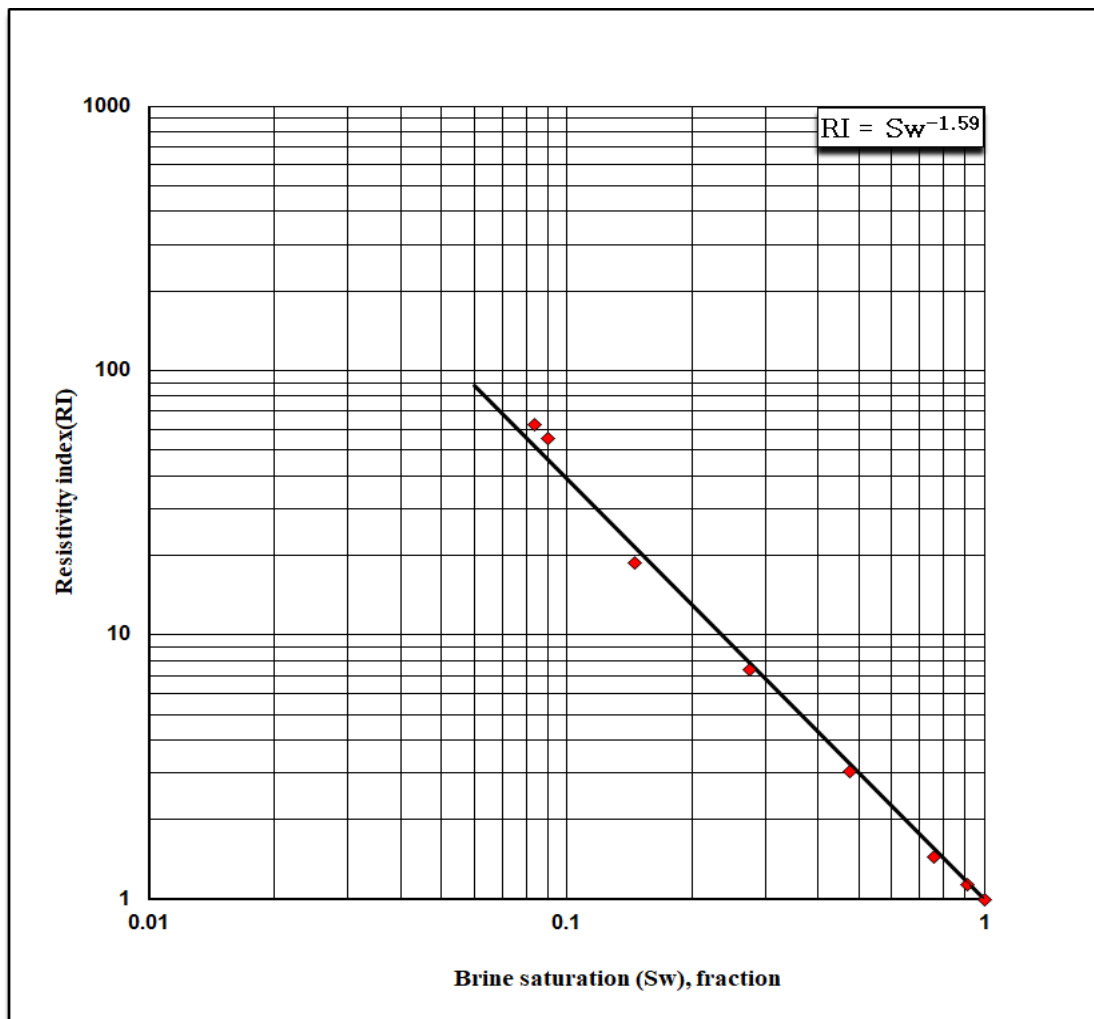


Figure (2.5.1)

Formation Resistivity Index, (at ambient conditions)

Sample #: 93	Porosity (%): 23.3	Air Permeability(mD):18.9
Company: Mellitah	Well: CE-03-NC41	Sample Depth(ft.in): 8450.06

Table 2.5.2: Laboratory Results

Resistivity Index (RI)	Brine Saturation (Sw), fraction
1.000	1.0000
1.001	0.9617
1.032	0.9489
1.099	0.9294
1.239	0.7947
2.842	0.4735
11.544	0.2374
20.284	0.1715
22.934	0.1407

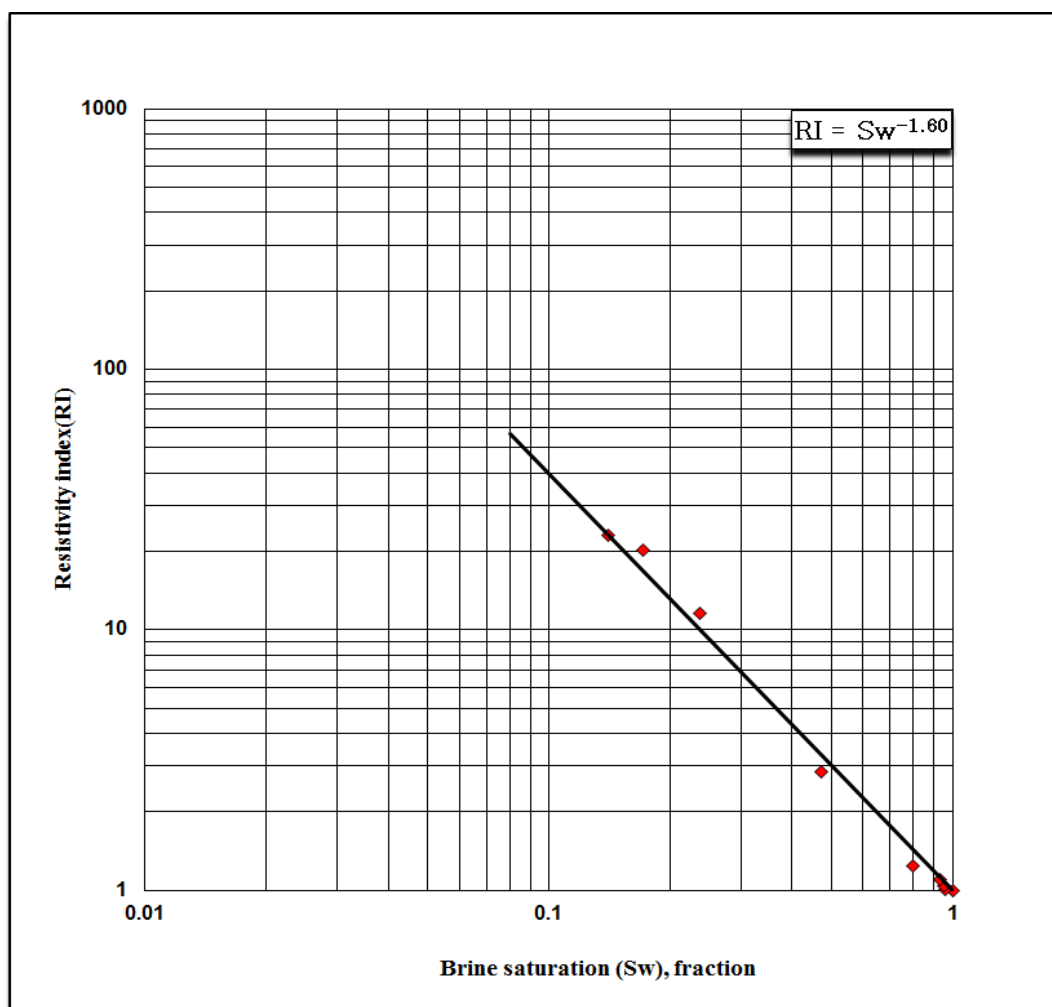


Figure (2.5.2)

Formation Resistivity Index, (at ambient conditions)

Sample #: 108	Porosity (%): 25.6	Air Permeability(mD): 62.8
Company: Mellitah	Well: CE-03-NC41	Sample Depth(ft.in): 8465.05

Table 2.5.3: Laboratory Results

Resistivity Index (RI)	Brine Saturation (Sw), fraction
1.000	1.0000
1.004	0.9583
1.118	0.9207
1.590	0.7359
3.426	0.4591
7.282	0.2881
18.210	0.1864
28.481	0.1431
35.960	0.1134

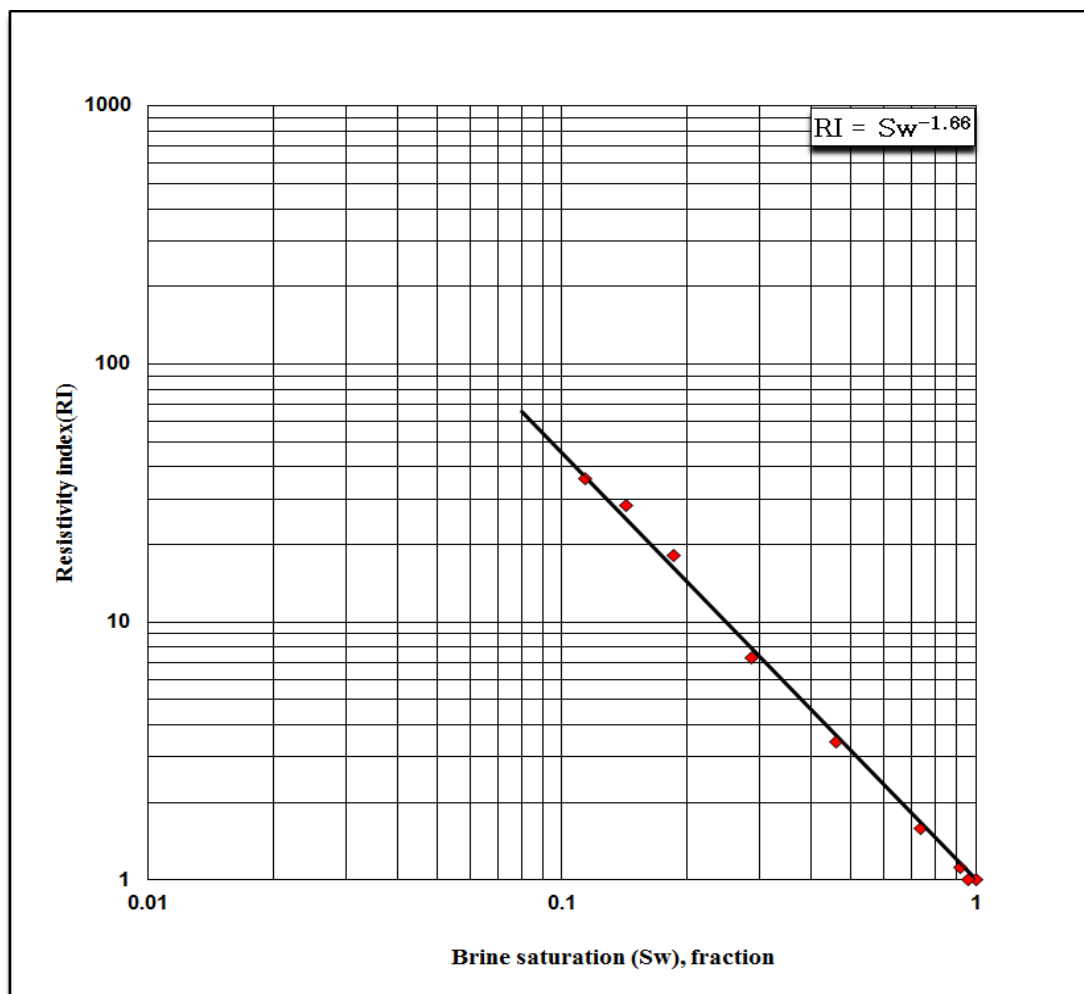


Figure (2.5.3)

Formation Resistivity Index, (at ambient conditions)

Sample #: 114	Porosity (%): 22.4	Air Permeability(mD): 32.4
Company: Mellitah	Well: CE-03-NC41	Sample Depth(ft.in): 8471.07

Table 2.5.4: Laboratory Results

Resistivity Index (RI)	Brine Saturation (Sw), fraction
1.000	1.0000
1.002	0.9592
1.070	0.9489
1.127	0.9061
2.010	0.6176
7.022	0.3068
27.318	0.1533
48.544	0.0991
62.074	0.0778

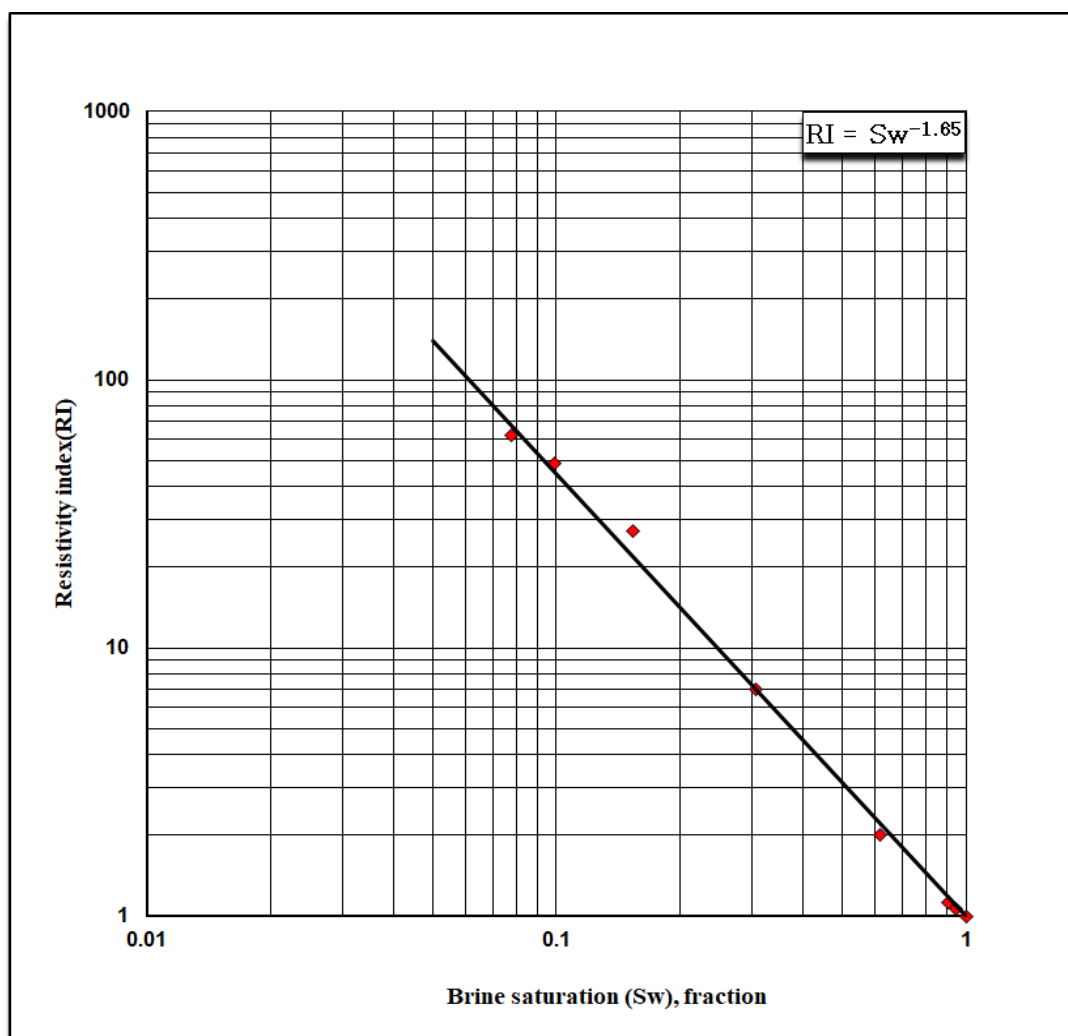


Figure (2.5.4)

Formation Resistivity Index, (at ambient conditions)

Sample #: 126	Porosity (%): 24.4	Air Permeability(mD): 39.4
Company: Mellitah	Well: CE-03-NC41	Sample Depth(ft.in): 8483.10

Table 2.5.5: Laboratory Results

Resistivity Index (RI)	Brine Saturation (Sw), fraction
1.000	1.0000
1.002	0.9539
1.039	0.9457
1.105	0.8883
2.411	0.5641
7.154	0.2986
22.311	0.1559
61.967	0.0959
85.080	0.0672

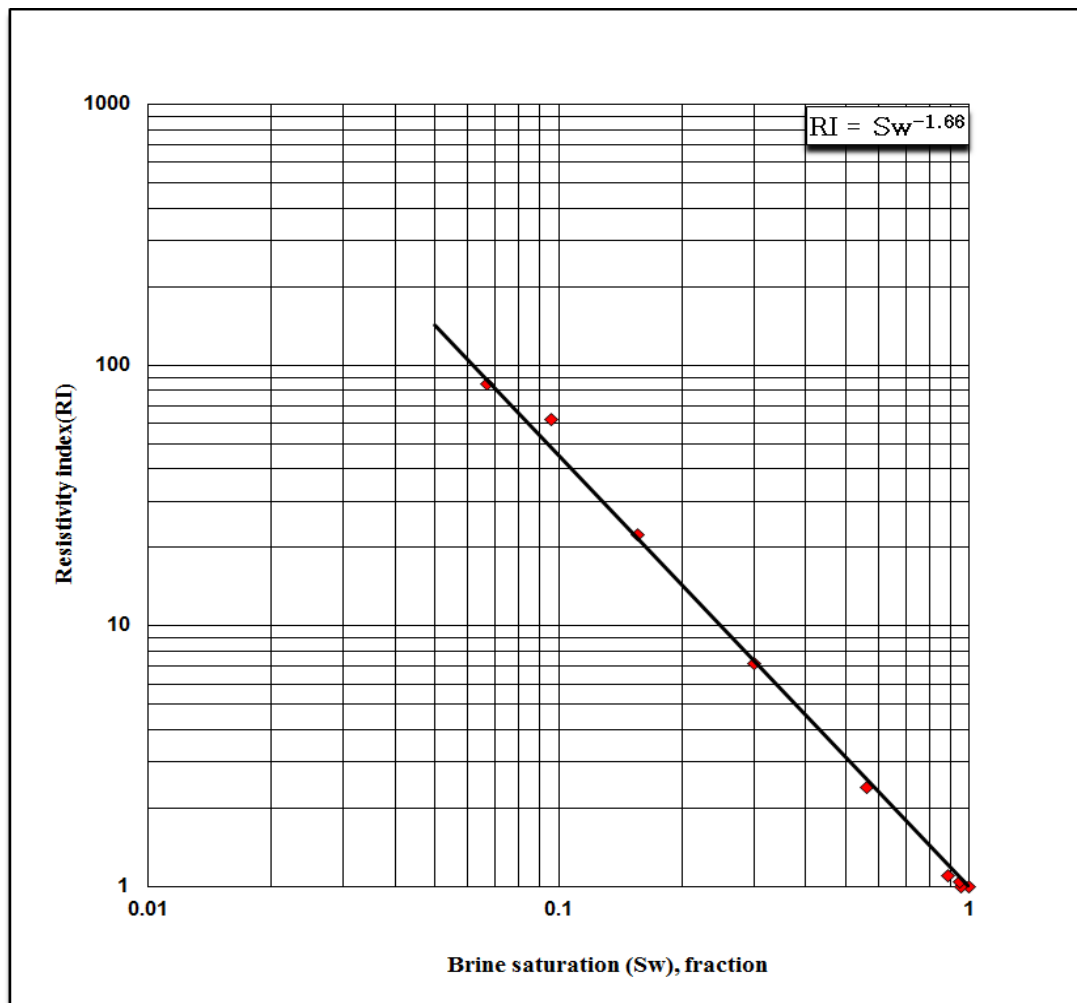


Figure (2.5.5)

Formation Resistivity Index, (at ambient conditions)

Sample #: 164	Porosity (%): 16.3	Air Permeability(mD): 4.89
Company: Mellitah	Well: CE-03-NC41	Sample Depth(ft.in): 8521.04

Table 2.5.6: Laboratory Results

Resistivity Index (RI)	Brine Saturation (Sw), fraction
1.000	1.0000
1.001	0.9909
1.059	0.9773
1.133	0.9587
1.169	0.9205
1.444	0.7936
3.517	0.3701
11.297	0.2263
13.935	0.1805

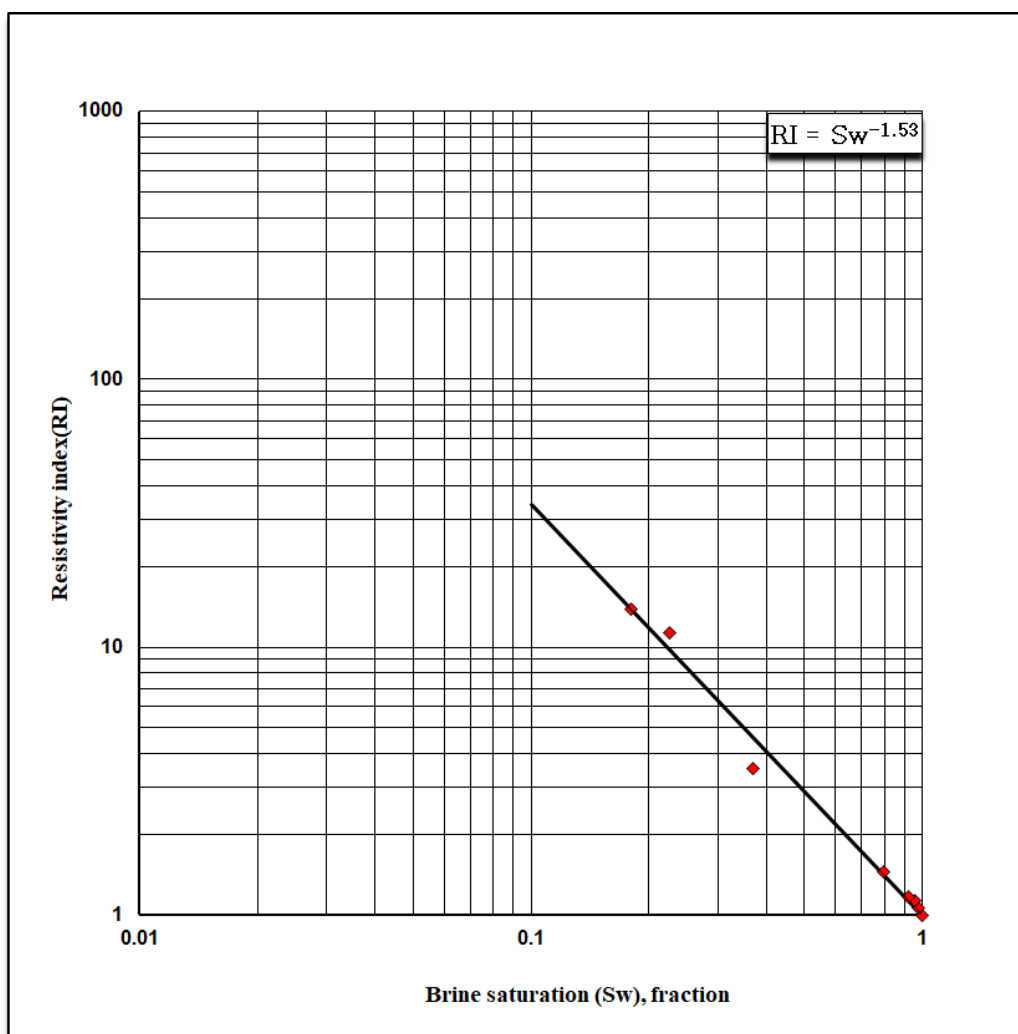


Figure (2.5.6)

Formation Resistivity Index, (at ambient conditions)

Sample #: 227	Porosity (%): 25.9	Air Permeability(mD): 126
Company: Mellitah	Well: CE-03-NC41	Sample Depth(ft.in): 8585.02

Table 2.5.7: Laboratory Results

Resistivity Index (RI)	Brine Saturation (Sw), fraction
1.000	1.0000
1.041	0.9233
1.252	0.8839
1.955	0.6548
3.828	0.4334
7.037	0.2936
22.421	0.1682
38.473	0.1209
41.807	0.0966

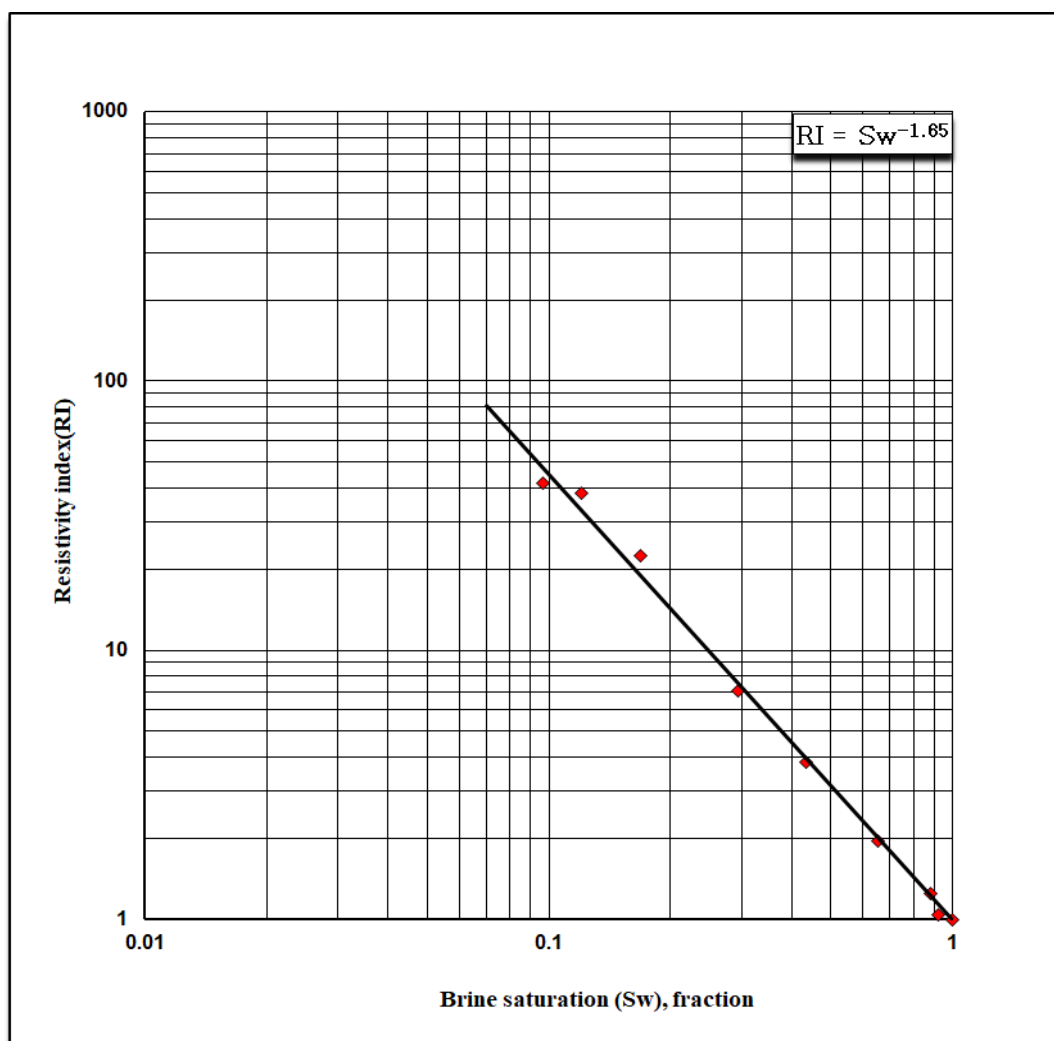


Figure (2.5.7)

Formation Resistivity Index, (at ambient conditions)

Sample #: 233	Porosity (%): 22.6	Air Permeability(mD): 51.7
Company: Mellitah	Well: CE-03-NC41	Sample Depth(ft.in): 8591.04

Table 2.5.8: Laboratory Results

Resistivity Index (RI)	Brine Saturation (Sw), fraction
1.000	1.0000
1.004	0.9672
1.156	0.9449
1.406	0.8158
3.859	0.4502
8.033	0.2829
21.251	0.1804
43.889	0.1233
54.692	0.0959

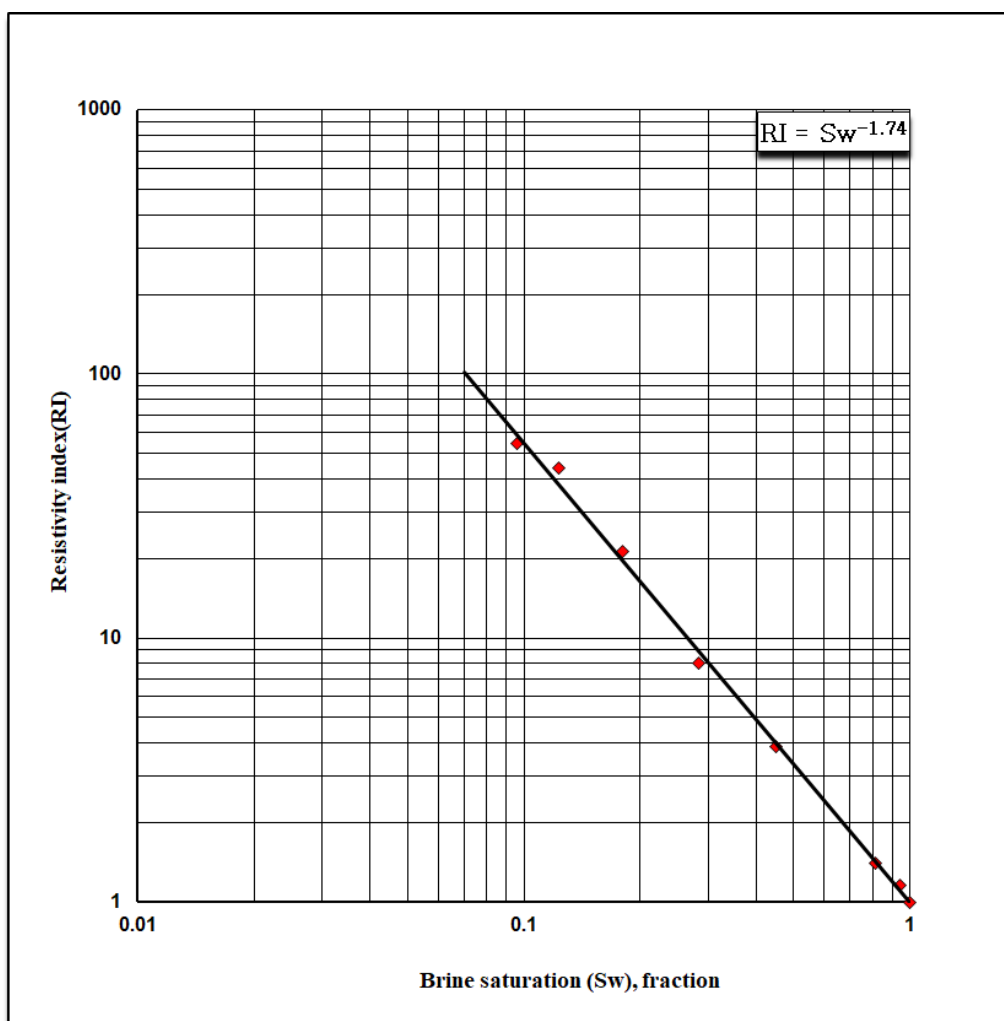


Figure (2.5.8)

Formation Resistivity Index, (at ambient conditions)

Sample #: 275	Porosity (%): 24.2	Air Permeability(mD): 225
Company: Mellitah	Well: CE-03-NC41	Sample Depth(ft.in): 8636.06

Table 2.5.9: Laboratory Results

Resistivity Index (RI)	Brine Saturation (Sw), fraction
1.000	1.0000
1.117	0.8999
1.734	0.7110
3.981	0.4633
6.505	0.3144
12.738	0.2418
25.987	0.1807
42.578	0.1429
44.566	0.1391

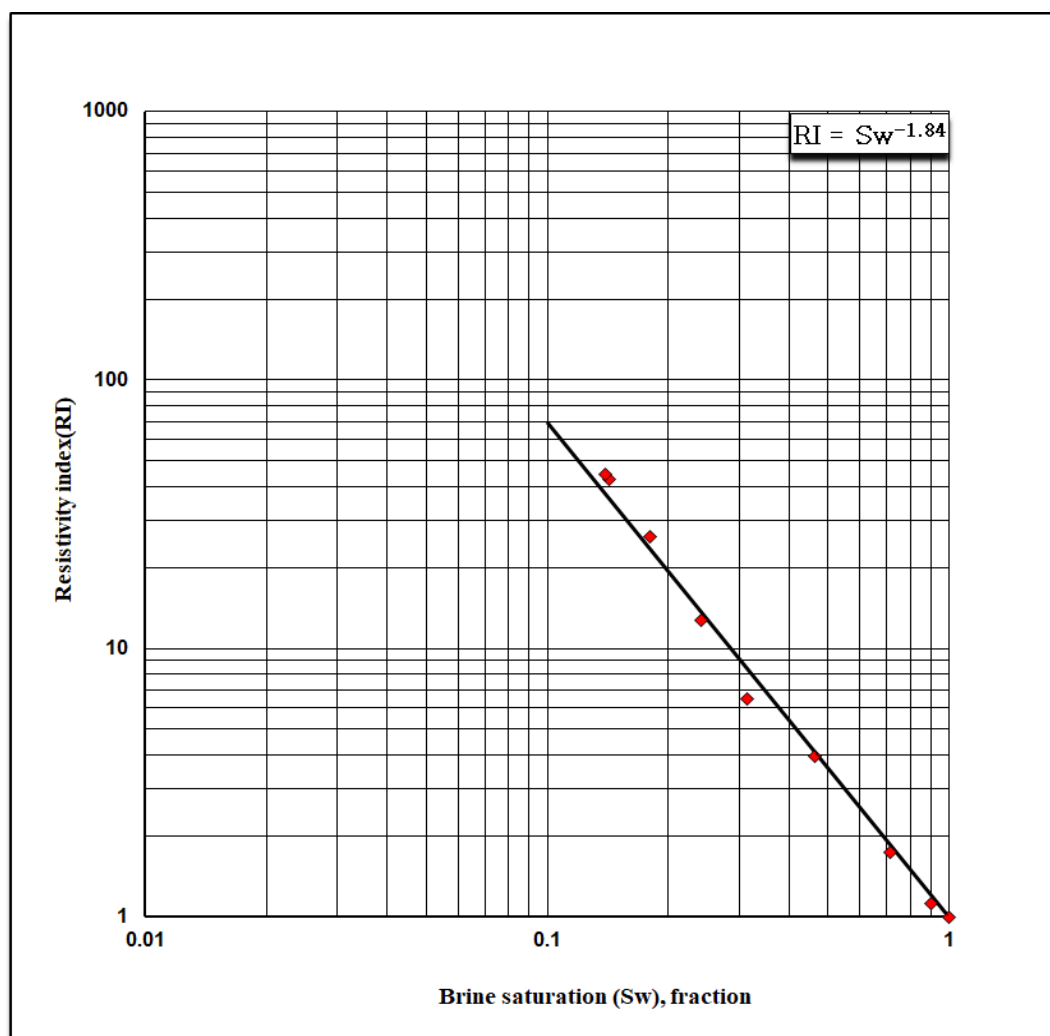


Figure (2.5.9)

Formation Resistivity Index, (at ambient conditions)

Sample #: 278	Porosity (%): 24.9	Air Permeability(mD): 38.1
Company: Mellitah	Well: CE-03-NC41	Sample Depth(ft.in): 8639.01

Table 2.5.10: Laboratory Results

Resistivity Index (RI)	Brine Saturation (Sw), fraction
1.000	1.0000
1.034	0.9480
1.149	0.9257
1.459	0.7846
2.161	0.5995
3.002	0.4718
6.539	0.3237
10.370	0.2380
17.652	0.2004

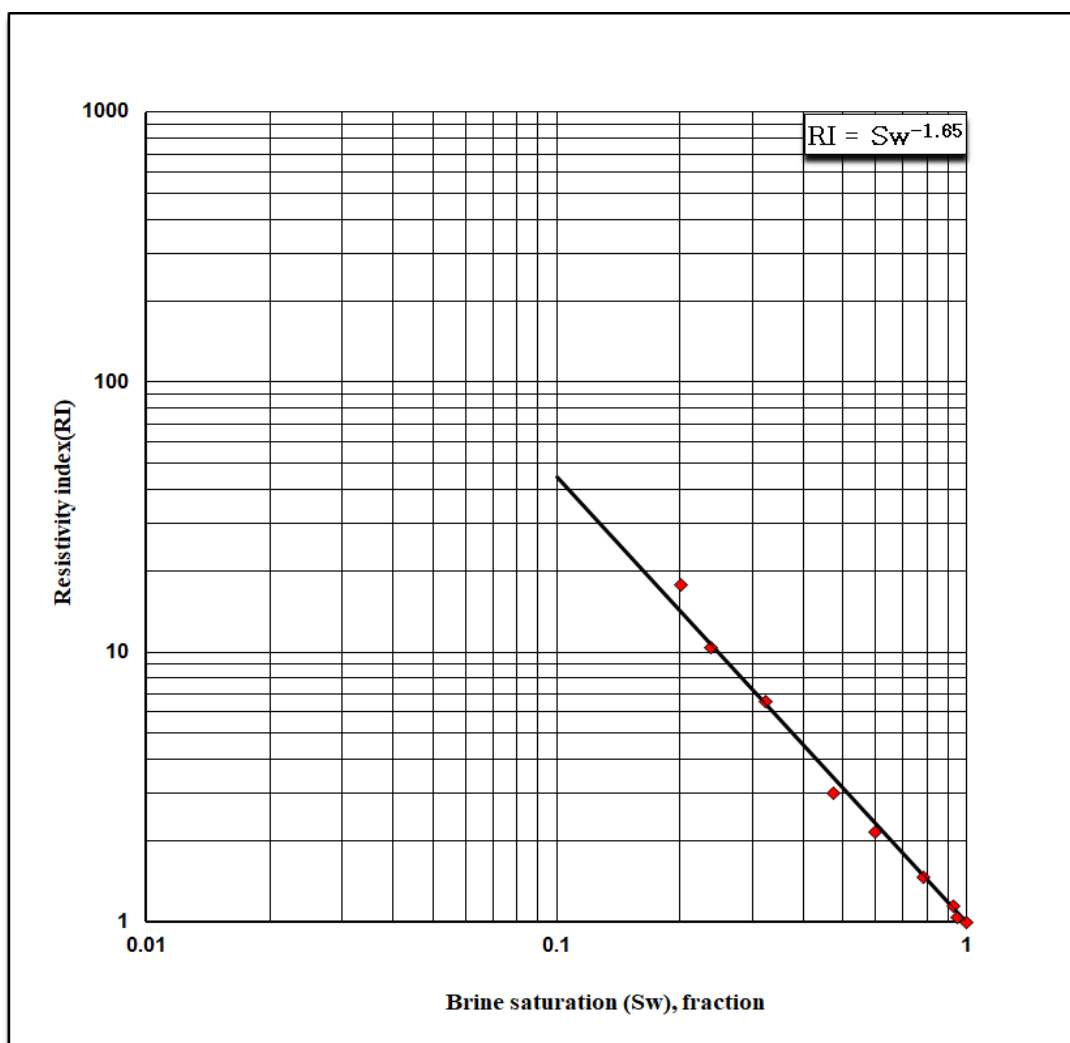


Figure (2.5.10)

Formation Resistivity Index, (at ambient conditions)

Company	: Mellitah Oil and Gas	Salinity	: 70,000 ppm
Well	: CE-03-NC41	Concession	: 41
Formation	: Elgaria	Rw @ 77 deg. F	: 0.0965 ohm-m
No of Samples	: 10	Saturation exponent(n)	: 1.71

Composite Plot

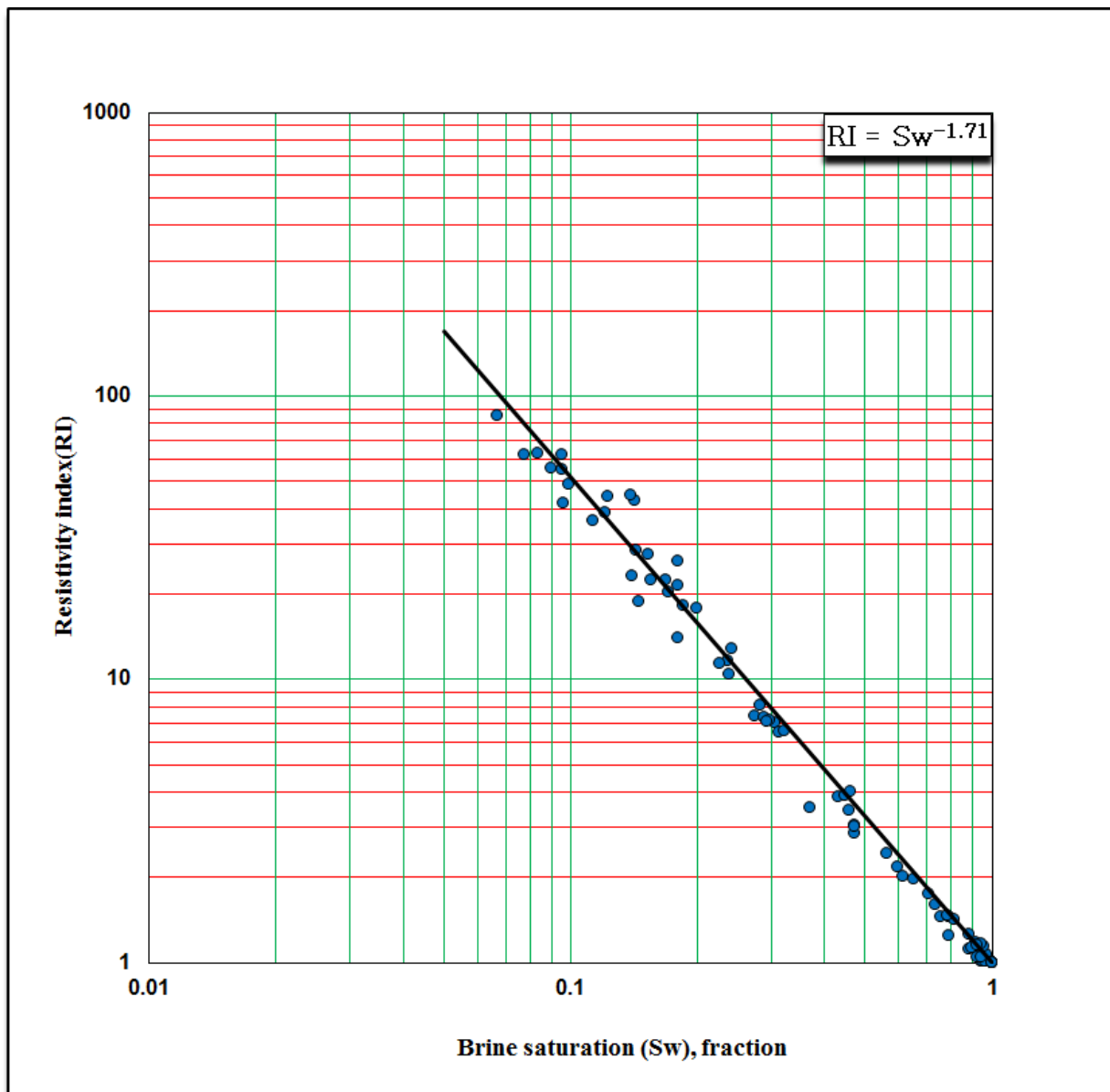


Figure (2.5.11)

Section Six Formation Resistivity Index at Overburden Pressure

FORMATION RESISTIVITY INDEX (at overburden pressure):-

Ten samples were selected from the well **CE03-NC41, Elgaria formation**, to perform formation resistivity index measurements at overburden pressure. This test was conducted after formation resistivity factor measurements. The fully saturated samples were individually placed in a hydrostatic core holder with a semi-permeable porous diaphragm, and an overburden pressure (3858 psig) was applied in a capillary pressure apparatus.

To bring the samples to lower saturation, an air humidified by water was admitted at a controlled pressure to the apparatus and the volume expelled was monitored. When equilibrium saturation had been attained, the volume out was recorded, and the electrical resistances were measured by using a four-point electrical device and the observed phase angle was set to zero. This step was repeated for each capillary pressure value (1, 2, 4, 8, 15, 35, 60 and 120 psi).

Sample dry weight, the fully saturated weight and the volume out after each de-saturation step were used to calculate the average saturation of the sample. The resistance was used to calculate the sample resistivity (R_t) and the resistivity was divided by the sample resistivity at one hundred percent liquid saturation (R_o) which yield resistivity index (RI).

The results of the ten samples are presented in tables **(2.6.1 and 2.6.10)** and figures **(2.6.1-2.6.11)**.

Formation Resistivity Index at Overburden Pressure (3858 psig)

Company: Mellitah Oil and Gas	(O.B.P) Porosity (%): 20.28	(O.B.P.) Permeability (mD): 45.54
Sample #: 80	Well: CE03-NC41	Sample Depth(ft.in): 8437.06

Table 2.6.1: Laboratory Results

Resistivity Index (RI)	Brine Saturation (Sw), fraction
1.000	1.0000
1.205	0.9417
1.213	0.9334
1.479	0.8667
3.203	0.6084
9.245	0.3668
51.682	0.2002
119.96	0.1169
293.17	0.0753

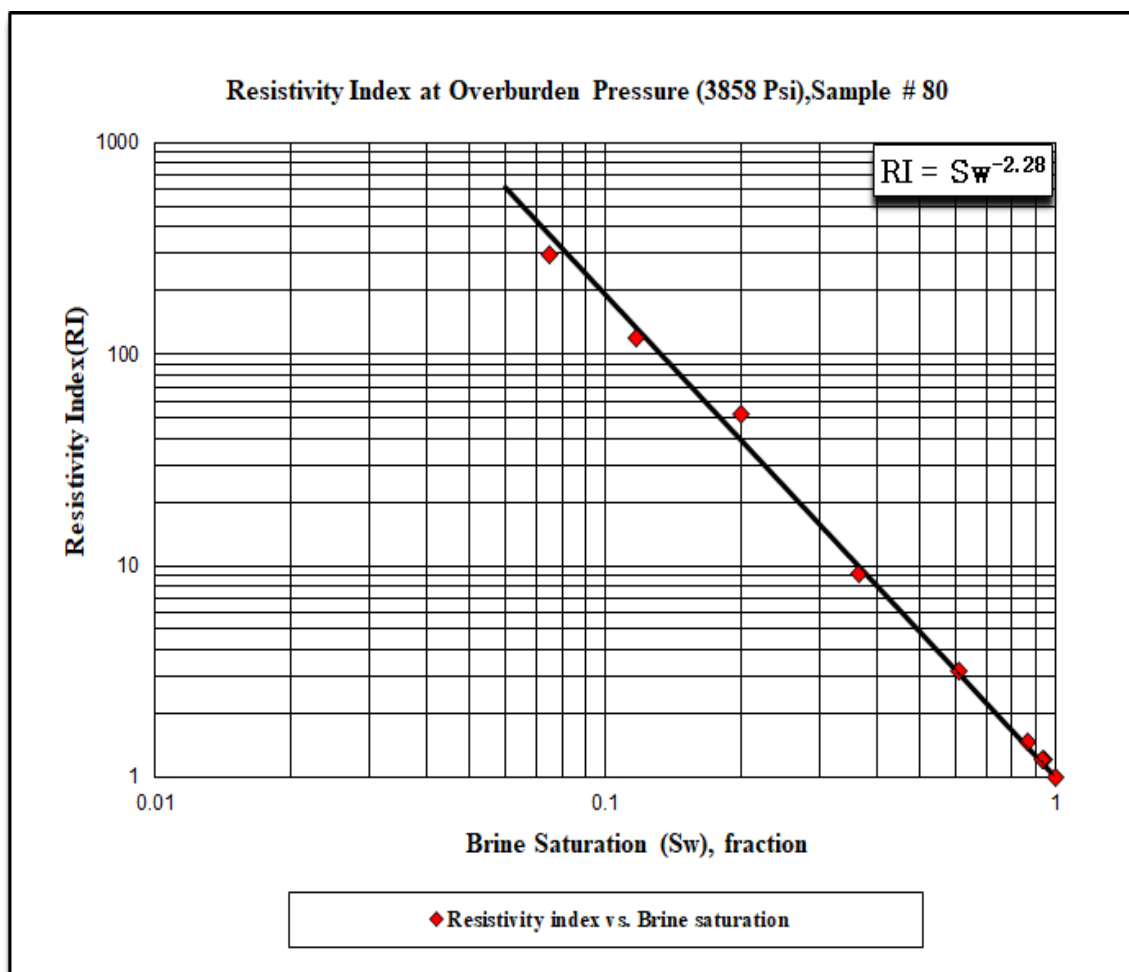


Figure (2.6.1)

Formation Resistivity Index at Overburden Pressure (3858 psig)

Company: Mellitah Oil and Gas	(O.B.P) Porosity (%): 22.76	(O.B.P.) Permeability (mD): 17.61
Sample #: 93	Well: CE03-NC41	Sample Depth(ft.in): 8450.06

Table 2.6.2: Laboratory Results

Resistivity Index (RI)	Brine Saturation (Sw), fraction
1.000	1.0000
1.058	0.9925
1.059	0.9850
1.083	0.9737
1.479	0.8274
4.096	0.4673
30.24	0.1747
73.97	0.1071
194.05	0.0846

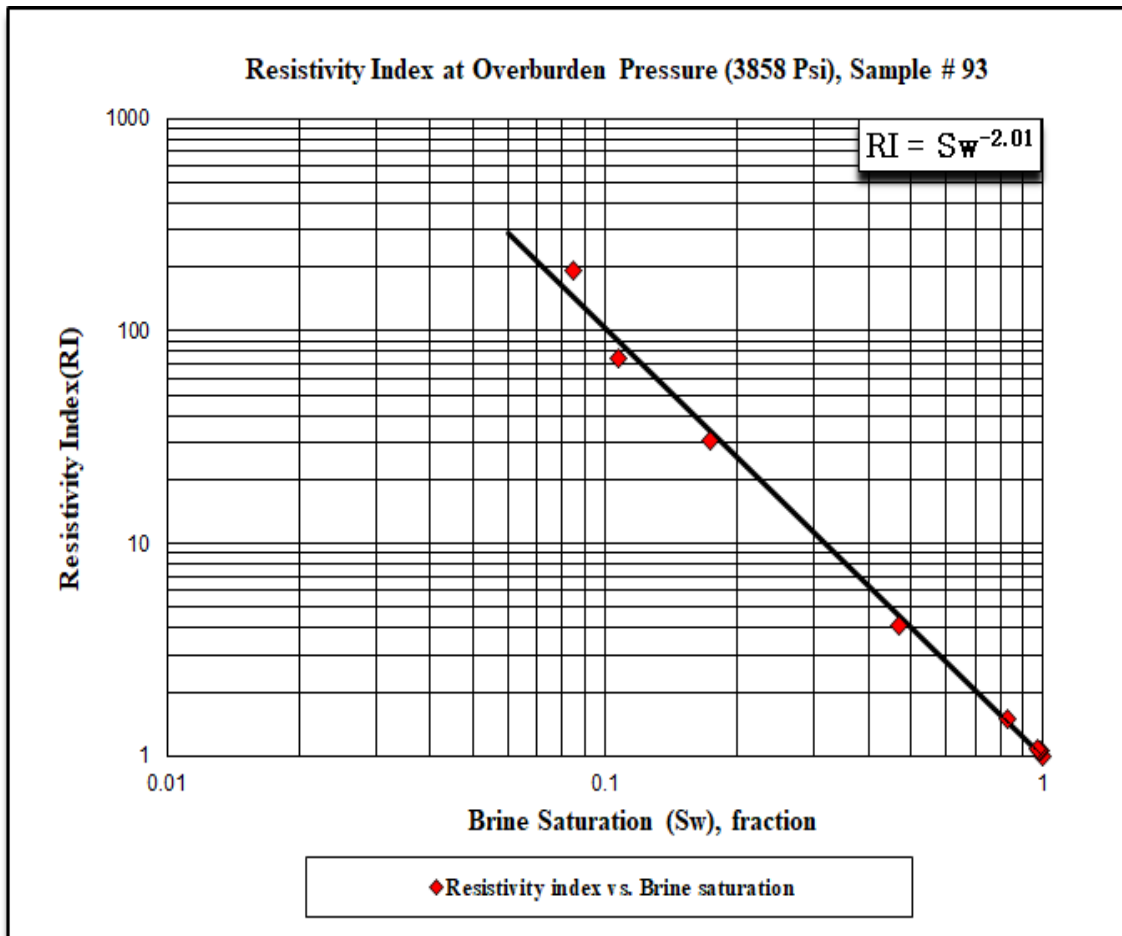


Figure (2.6.2)

Formation Resistivity Index at Overburden Pressure (3858 psig)

Company: Mellitah Oil and Gas	(O.B.P) Porosity (%): 25.14	(O.B.P.) Permeability (mD): 56.81
Sample #: 108	Well: CE03-NC41	Sample Depth(ft.in): 8465.05

Table 2.6.3: Laboratory Results

Resistivity Index (RI)	Brine Saturation (Sw), fraction
1.000	1.0000
1.050	0.9694
1.052	0.9661
1.069	0.9634
1.082	0.9568
1.115	0.9269
33.916	0.2221
75.230	0.1423
173.78	0.1091

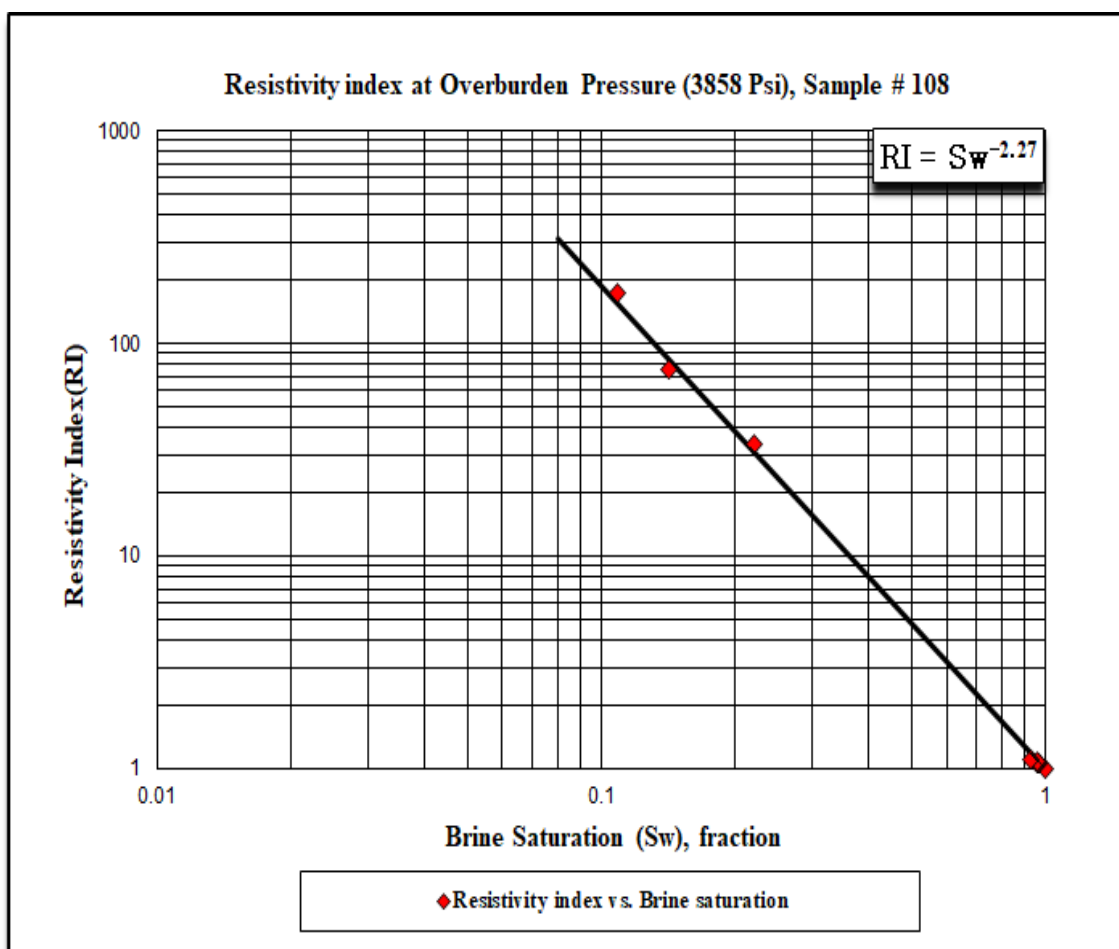


Figure (2.6.3)

Formation Resistivity Index at Overburden Pressure (3858 psig)

Company: Mellitah Oil and Gas	(O.B.P) Porosity (%): 21.71	(O.B.P.) Permeability (mD): 27.98
Sample #: 114	Well: CE03-NC41	Sample Depth(ft.in): 8471.07

Table 2.6.4: Laboratory Results

Resistivity Index (RI)	Brine Saturation (Sw), fraction
1.000	1.0000
1.092	0.9646
1.149	0.9577
1.184	0.9462
1.389	0.9154
1.554	0.8462
47.29	0.2233
73.90	0.1157
103.8	0.0772

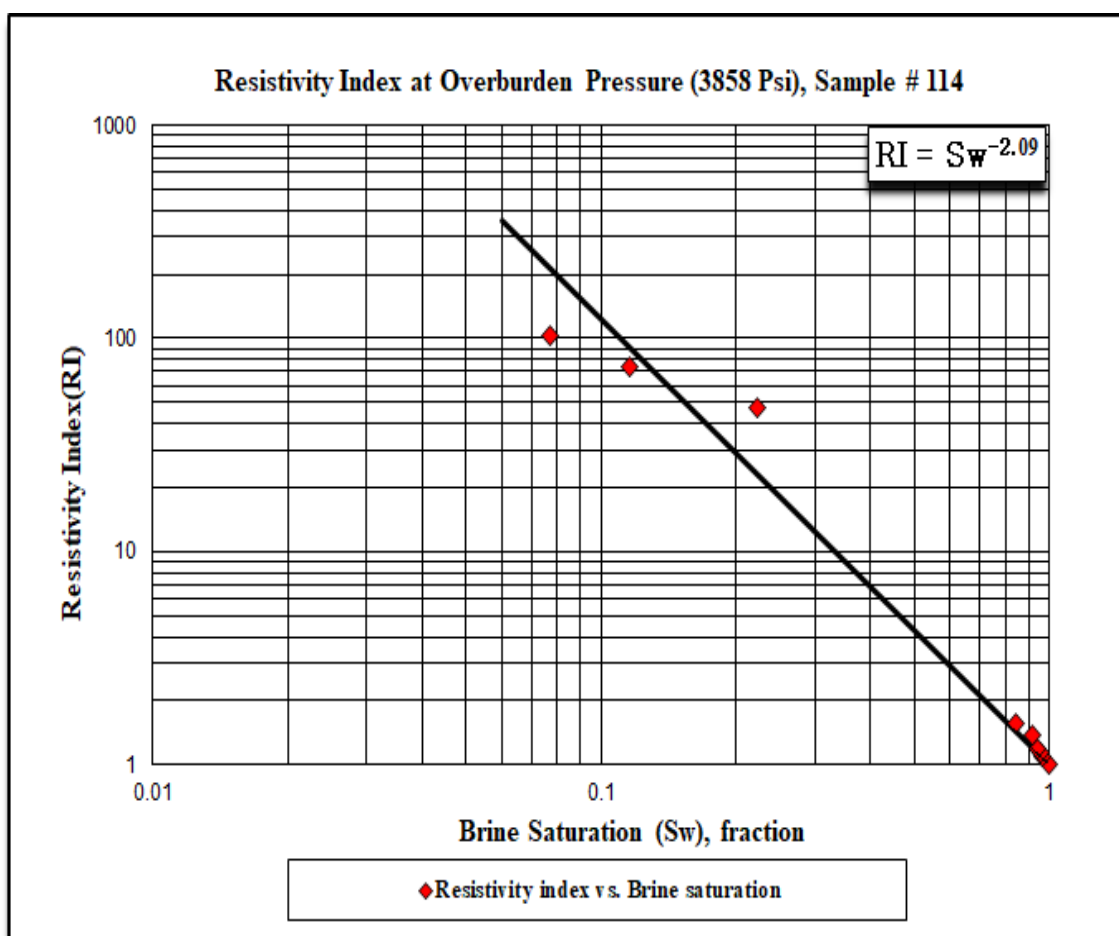


Figure (2.6.4)

Formation Resistivity Index at Overburden Pressure (3858 psig)

Company: Mellitah Oil and Gas	(O.B.P) Porosity (%) = 23.67	(O.B.P.) Permeability (mD): 35.00
Sample #: 126	Well: CE03-NC41	Sample Depth(ft.in): 8483.10

Table 2.6.5: Laboratory Results

Resistivity Index (RI)	Brine Saturation (Sw), fraction
1.000	1.0000
1.021	0.9639
1.025	0.9495
1.052	0.9279
1.314	0.8558
2.271	0.6250
7.776	0.2789
75.67	0.1202
190.02	0.0625

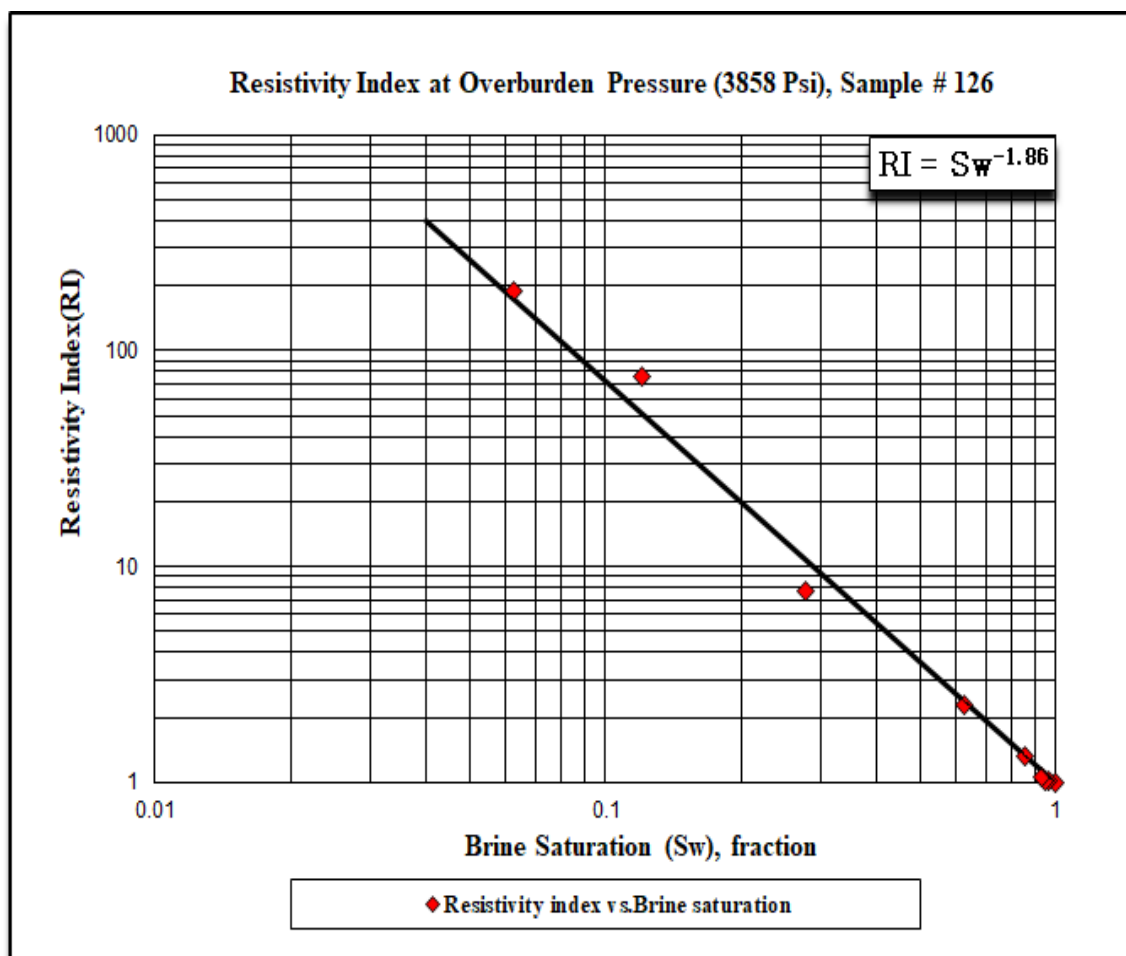


Figure (2.6.5)

Formation Resistivity Index at Overburden Pressure (3858 psig)

Company: Mellitah Oil and Gas	(O.B.P) Porosity (%): 15.97	(O.B.P.) Permeability (mD): 4.46
Sample #: 164	Well: CE03-NC41	Sample Depth(ft.in): 8521.04

Table 2.6.6: Laboratory Results

Resistivity Index (RI)	Brine Saturation (Sw), fraction
1.000	1.0000
1.080	0.9912
1.107	0.9824
1.205	0.9647
1.277	0.9118
2.069	0.7884
12.651	0.3651
43.619	0.2064
66.680	0.1843

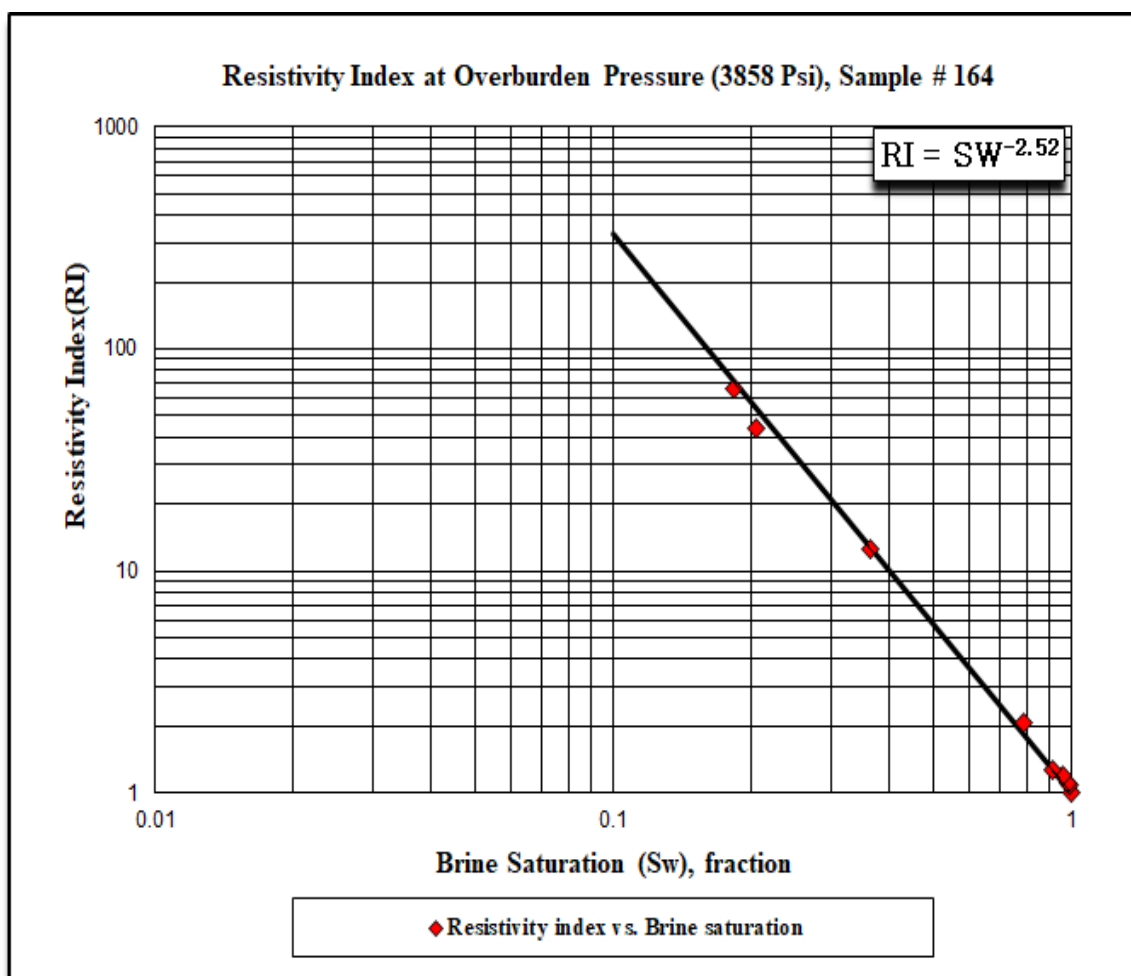


Figure (2.6.6)

Formation Resistivity Index at Overburden Pressure (3858 psig)

Company: Mellitah Oil and Gas	(O.B.P) Porosity (%): 25.18	(O.B.P.) Permeability (mD): 112.31
Sample #:227	Well: CE03-NC41	Sample Depth(ft.in): 8585.02

Table 2.6.7: Laboratory Results

Resistivity Index (RI)	Brine Saturation (Sw), fraction
1.000	1.0000
1.063	0.9361
1.067	0.9233
1.489	0.7763
3.183	0.5271
6.782	0.3801
20.321	0.2267
41.897	0.1373
121.243	0.0925

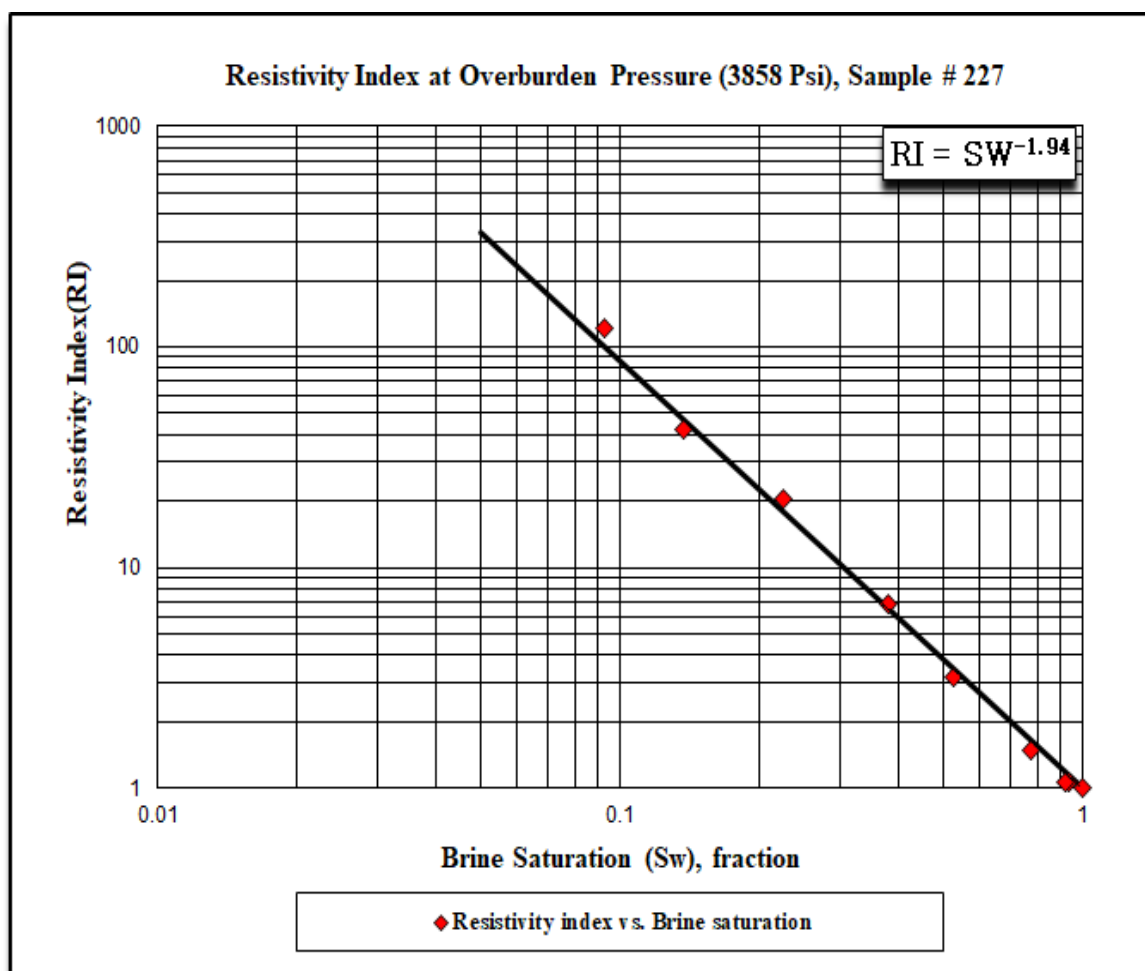


Figure (2.6.7)

Formation Resistivity Index at Overburden Pressure (3858 psig)

Company: Mellitah Oil and Gas	(O.B.P) Porosity (%): 22.09	(O.B.P.) Permeability (mD): 48.61
Sample #: 233	Well: CE03-NC41	Sample Depth(ft.in): 8591.04

Table 2.6.8: Laboratory Results

Resistivity Index (RI)	Brine Saturation (Sw), fraction
1.000	1.0000
1.002	0.9689
1.004	0.9619
1.017	0.9557
1.044	0.9339
7.709	0.3445
27.042	0.1797
52.957	0.1214
108.019	0.0902

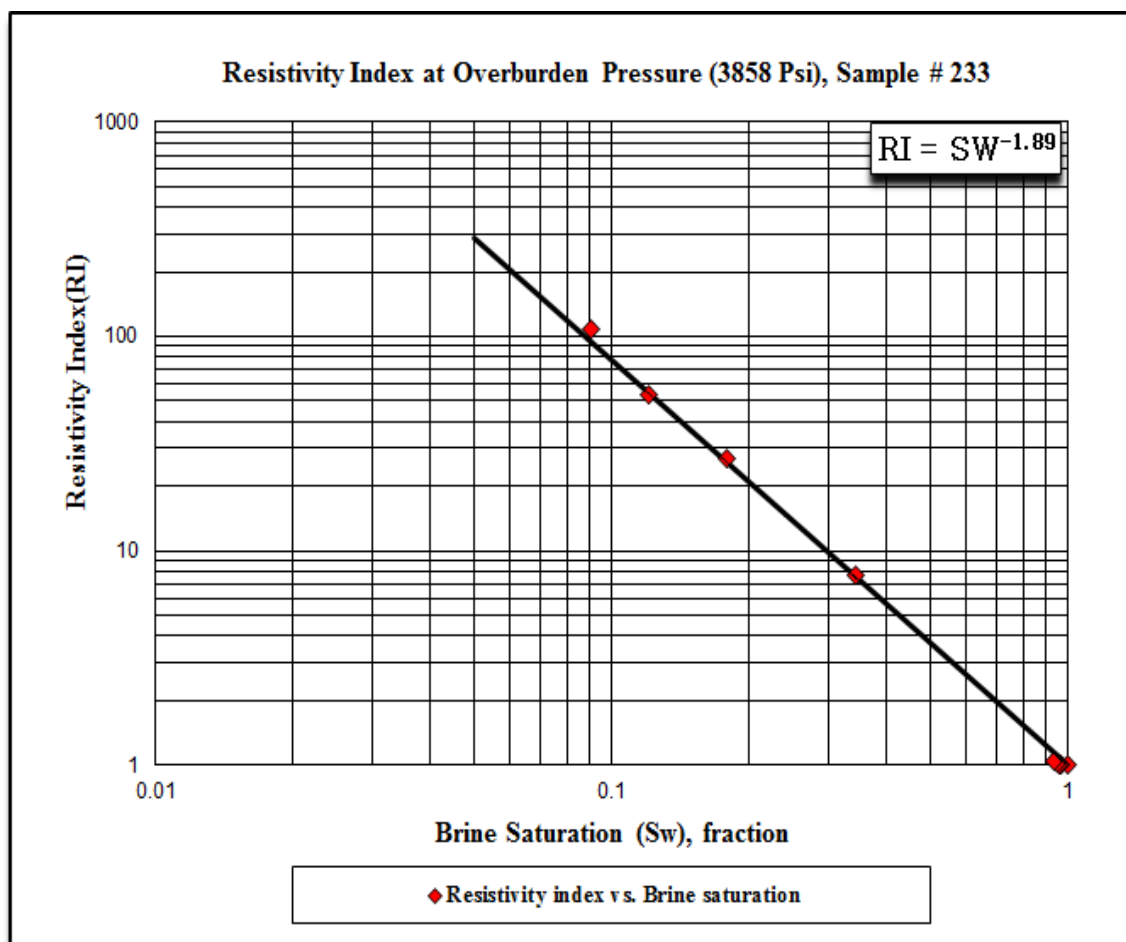


Figure (2.6.8)

Formation Resistivity Index at Overburden Pressure (3858 psig)

Company: Mellitah Oil and Gas	(O.B.P) Porosity (%): 23.57	(O.B.P.) Permeability (mD): 207.06
Sample #: 275	Well: CE03-NC41	Sample Depth(ft.in): 8636.06

Table 2.6.9: Laboratory Results

Resistivity Index (RI)	Brine Saturation (Sw), fraction
1.000	1.0000
1.592	0.9708
1.712	0.9598
1.995	0.7953
3.081	0.6345
9.950	0.3422
25.379	0.2106
40.730	0.1668
113.070	0.1302

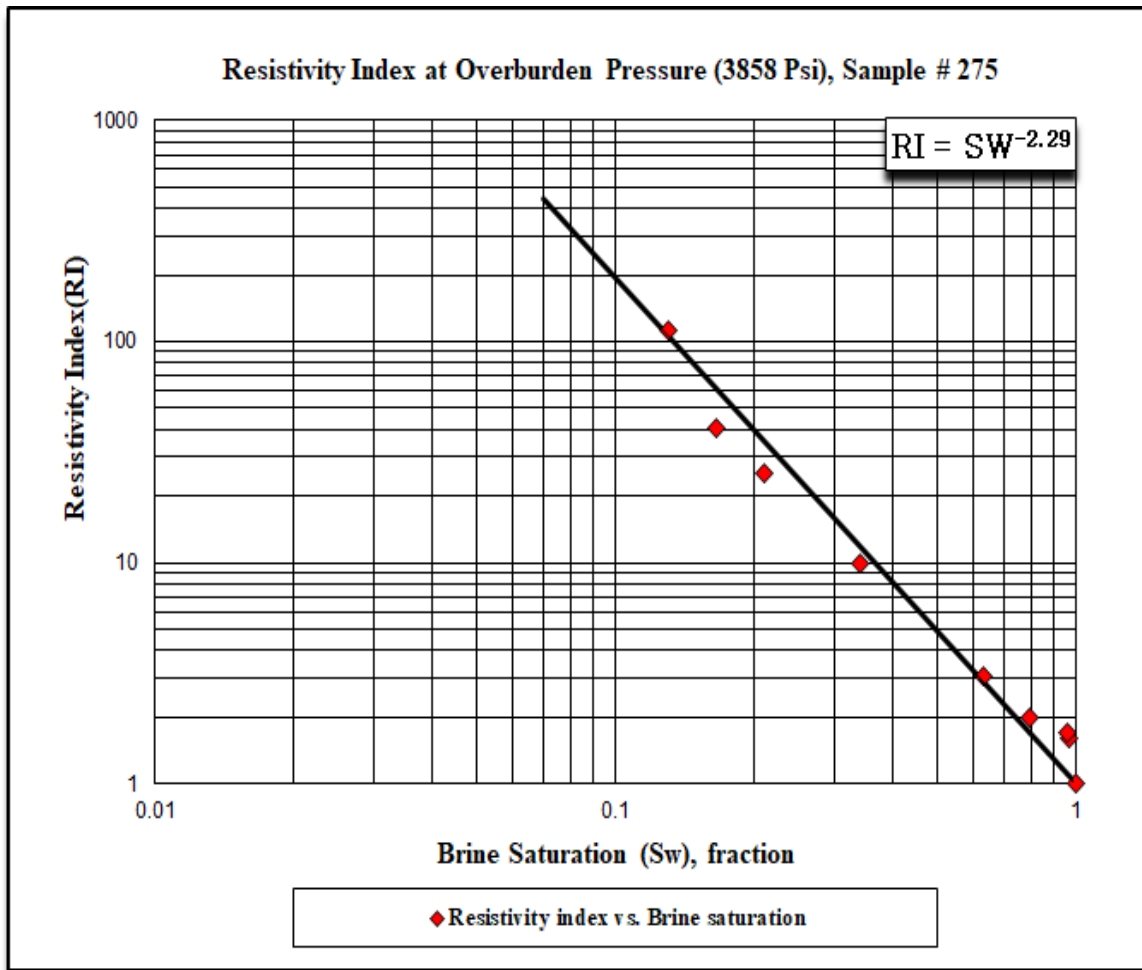


Figure (2.6.9)

Formation Resistivity Index at Overburden Pressure (3858 psig)

Company: Mellitah Oil and Gas	(O.B.P) Porosity (%): 24.18	(O.B.P.) Permeability (mD): 34.90
Sample #: 278	Well: CE03-NC41	Sample Depth(ft.in): 8639.01

Table 2.6.10: Laboratory Results

Resistivity Index (RI)	Brine Saturation (Sw), fraction
1.000	1.0000
1.001	0.9745
1.002	0.9600
1.015	0.9272
1.368	0.7381
2.442	0.5707
6.440	0.3452
18.170	0.2360
35.796	0.1706

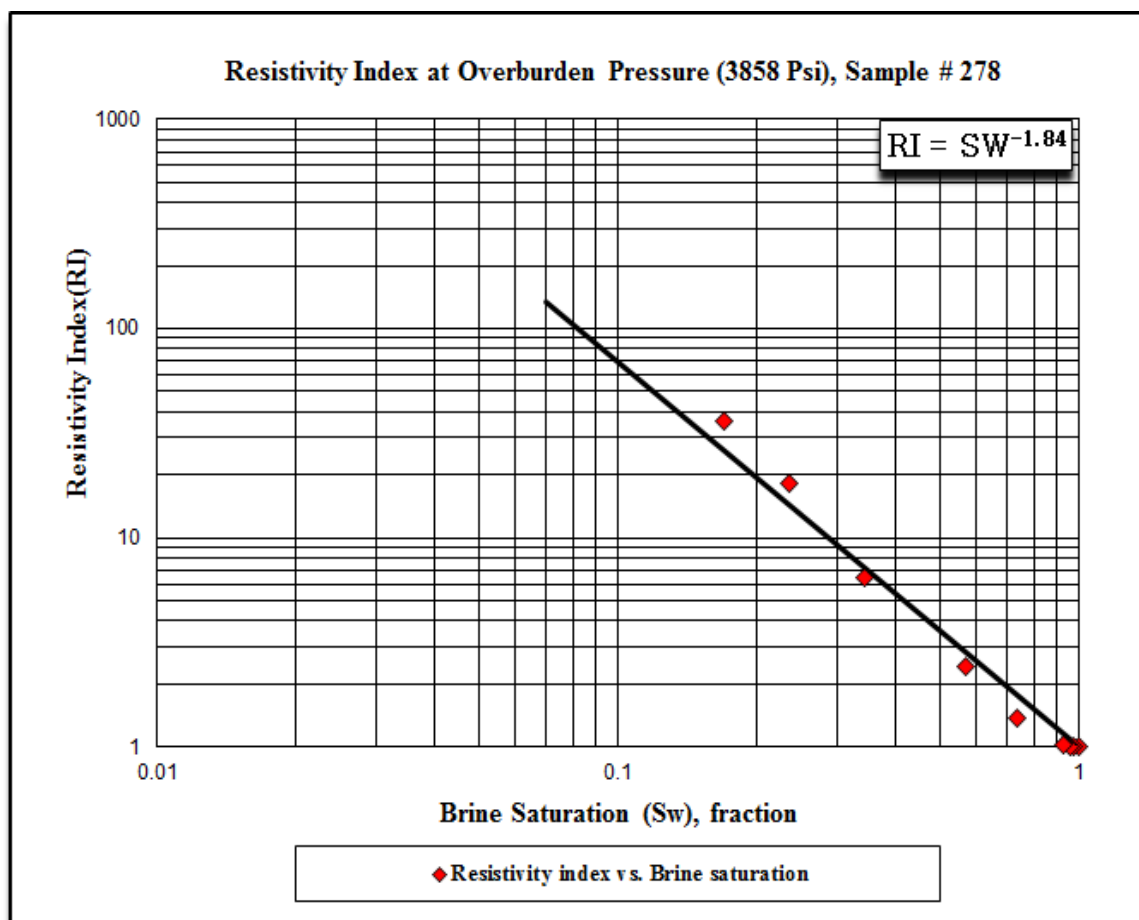


Figure (2.6.10)

Formation Resistivity Index, (at Overburden Pressure)

Company	: Mellitah Oil and Gas	Salinity	: 70,000 ppm
Well	: CE-03-NC41	Concession	: 41
Formation	: Elgaria	Rw @ 77 deg. F	: 0.0965 ohm-m
No of Samples	: 10	Saturation exponent(n)	: 2.06

Composite Plot

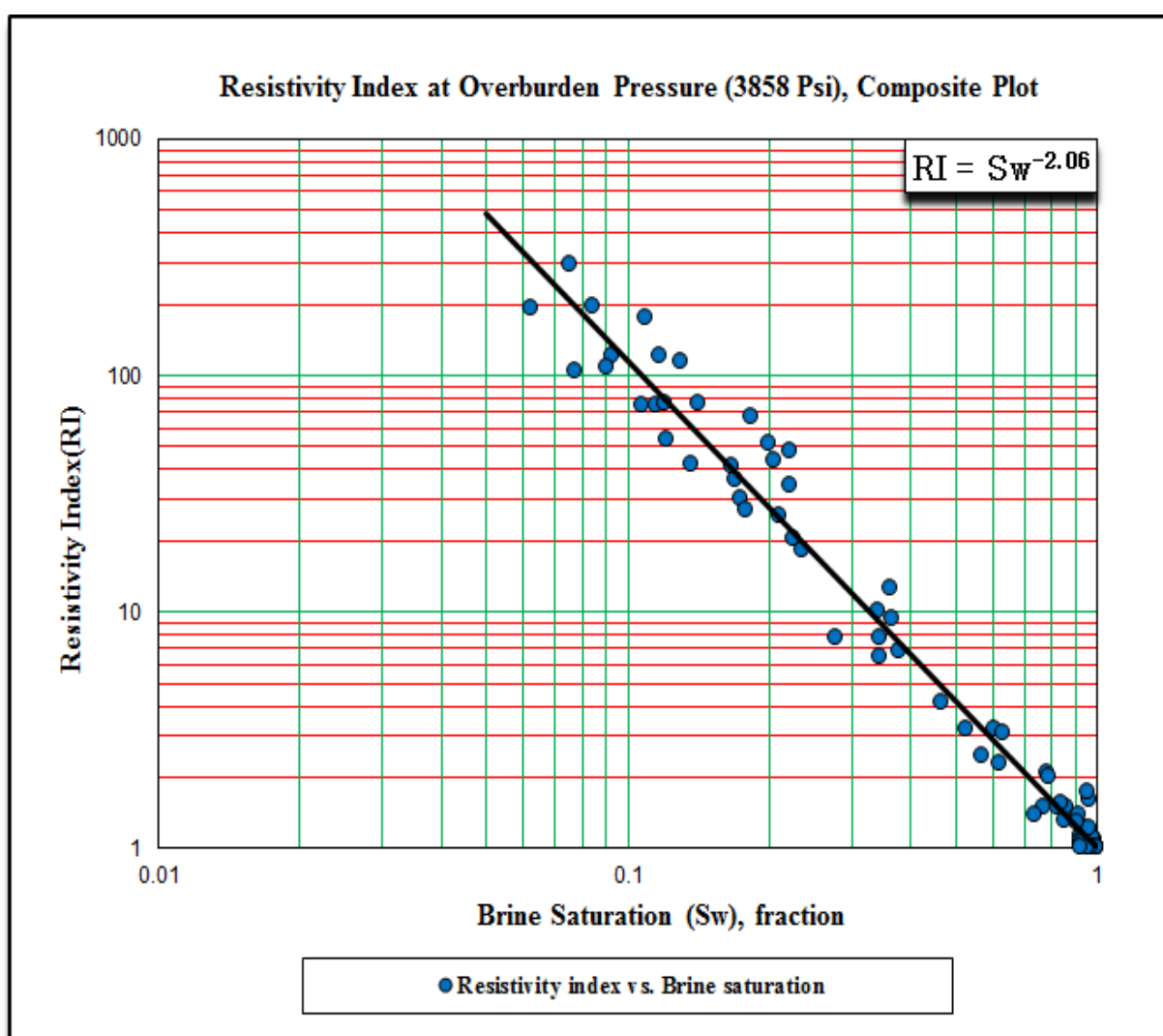


Figure (2.6.11)

Section Seven

Capillary Pressure

by Using Ultra Speed Centrifuge

ULTRA SPEED CENTRIFUGE CAPILLARY PRESSURE:-

Air – Water (drainage cycle):-

Ten plug samples were selected from the well **CE03-NC41** representing Elgaria formation to perform centrifuge capillary pressure. The brine saturated plug samples were placed in Beckman Ultracentrifuge drainage vessels under air. With samples in place, the temperature was set to ambient temperature. The rotor speed was increased until the first designated speed (421 RPM) equivalent to 1 psi was achieved. The centrifuge was maintained at the required speed for a minimum of four hours. When water production was ceased, Samples were off – loaded and the volume of water expelled was recorded. This volume and the pore volume of sample were used to calculate the average liquid saturation. This process was repeated until a speed (4614 RPM) equivalent to 120 psi, was reached.

Speed in RPM, length of sample, density difference between air and water and the radius of rotation were used to calculate the capillary pressure at the sample inner face. The local saturation at the inner face of the sample was calculated using Hassler technique (Reference 4). The generated data are presented in tables (2.7.1) through (2.7.10) and in figures (2.7.1) through (2.7.10).

Air-Water Drainage Cycle

Company	: Mellitah	Core Porosity, %	: 21.00
Well No.	: CE03-NC41	Core Air Permeability, (mD)	: 51.70
Sample No.	: 80	Irreducible Water Saturation, %	: 9.42
Salinity, ppm	: 70,000	Temperature in Chamber, C°	: 25
Brine Density(g/cc)	: 1.0495	Sample Depth (ft.in)	: 8437.06

Rotor Speed RPM	Ambient Conditions Capillary Pressure , psia	Liquid Saturation fraction
0.0	0	1.0000
421	1	0.9420
596	2	0.8910
842	4	0.7900
1191	8	0.5410
1631	15	0.2970
2492	35	0.1400
3262	60	0.1005
4614	120	0.0942

Table (2.7.1)

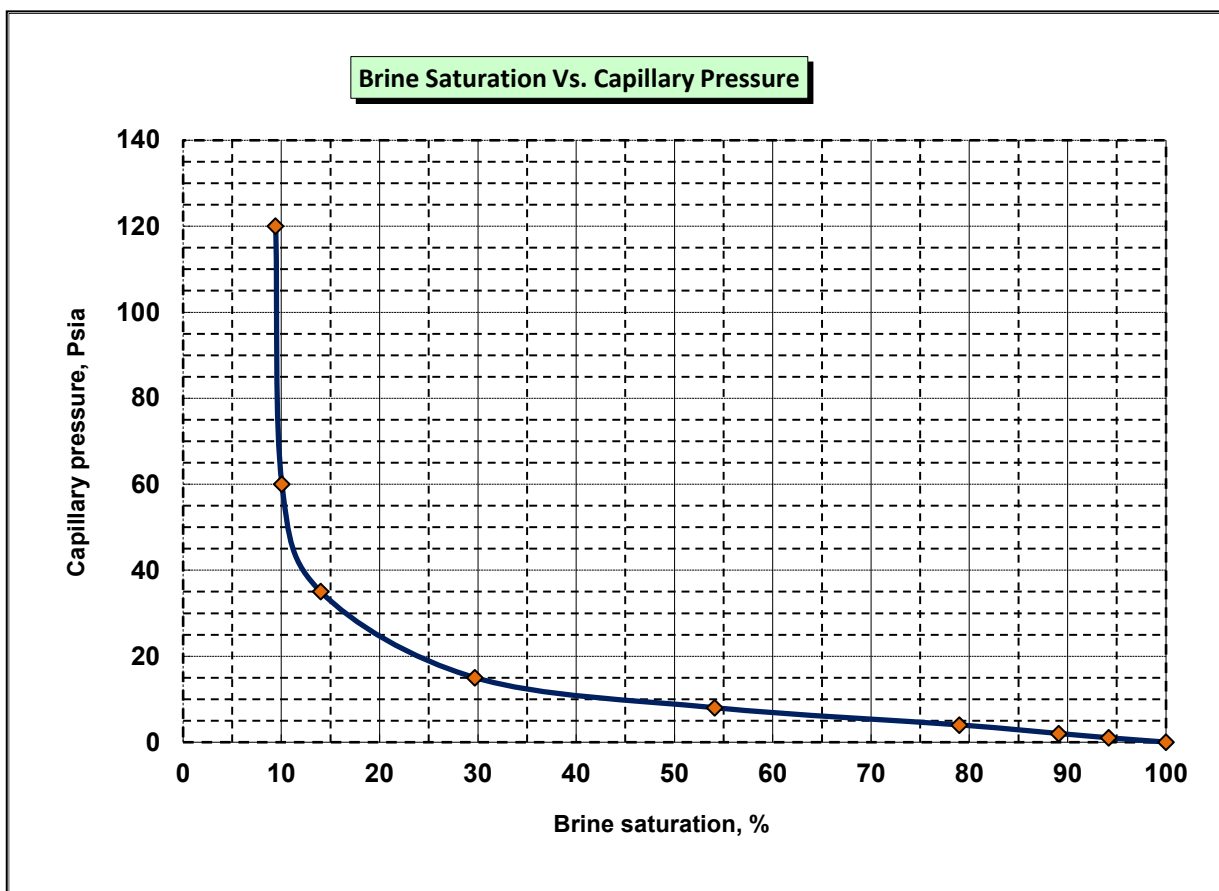


Figure (2.7.1)

Air-Water Drainage Cycle

Company	: Mellitah	Core Porosity, %	: 23.30
Well No.	: CE03-NC41	Core Air Permeability, (mD)	: 18.90
Sample No.	: 93	Irreducible Water Saturation, %	: 14.80
Salinity, ppm	: 70,000	Temperature in Chamber, C°	: 25
Brine Density(g/cc)	: 1.0495	Sample Depth (ft.in)	: 8450.06

Rotor Speed RPM	Ambient Conditions Capillary Pressure , psia	Brine Saturation fraction
0.0	0	1.0000
421	1	0.9825
596	2	0.9711
842	4	0.9540
1191	8	0.8649
1631	15	0.4310
2492	35	0.2550
3262	60	0.1760
4614	120	0.1480

Table (2.7.2)

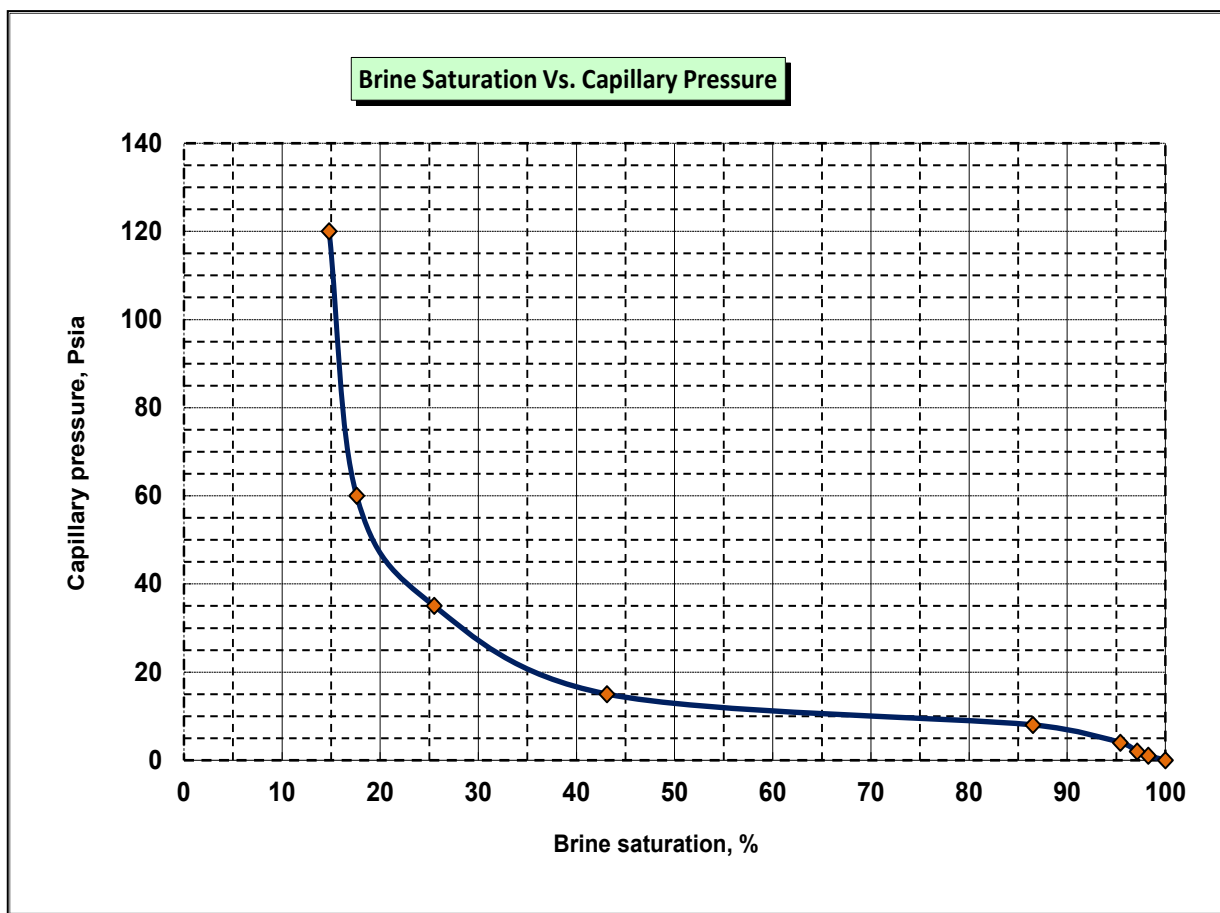


Figure (2.7.2)

Air-Water Drainage Cycle

Company	: Mellitah	Core Porosity, %	: 25.6
Well No.	: CE03-NC41	Core Air Permeability, (mD)	: 62.8
Sample No.	: 108	Irreducible Water Saturation, %	: 12.60
Salinity, ppm	: 70,000	Temperature in Chamber, C°	: 25
Brine Density(g/cc)	: 1.0495	Sample Depth (ft.in)	: 8465.05

Rotor Speed RPM	Ambient Conditions Capillary Pressure , psia	Brine Saturation fraction
0.0	0	1.0000
421	1	0.9510
596	2	0.8340
842	4	0.5170
1191	8	0.2850
1631	15	0.1970
2492	35	0.1530
3262	60	0.1390
4614	120	0.1260

Table (2.7.3)

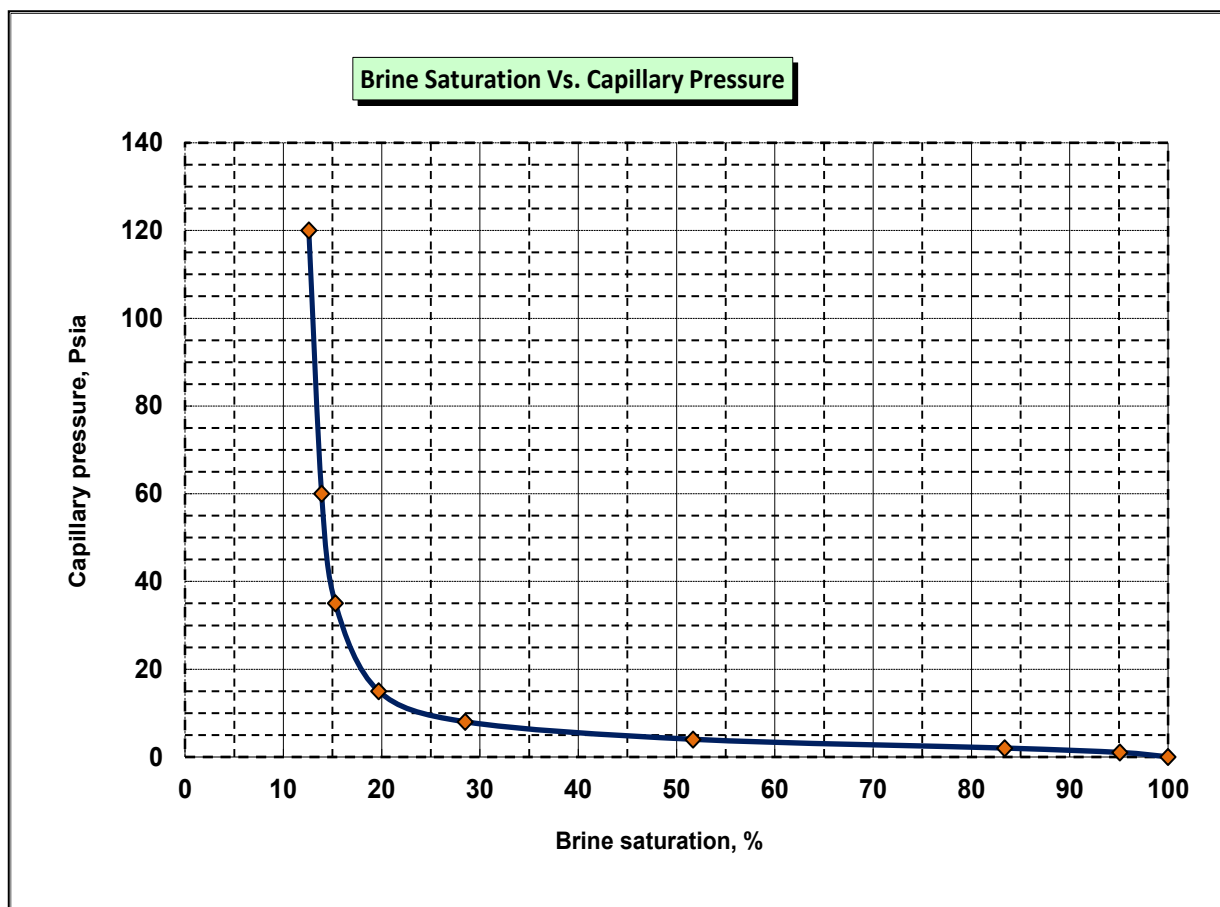


Figure (2.7.3)

Air-Water Drainage Cycle

Company	: Mellitah	Core Porosity, %	: 22.40
Well No.	: CE03-NC41	Core Air Permeability, (mD)	: 32.4
Sample No.	: 114	Irreducible Water Saturation, %	: 8.74
Salinity, ppm	: 70,000	Temperature in Chamber, C°	: 25
Brine Density(g/cc)	: 1.0495	Sample Depth (ft.in)	: 8471.07

Rotor Speed RPM	Ambient Conditions Capillary Pressure, psia	Brine Saturation fraction
0.0	0	1.0000
421	1	0.9740
596	2	0.9578
842	4	0.8142
1191	8	0.5870
1631	15	0.3470
2492	35	0.1631
3262	60	0.1026
4614	120	0.0874

Table (2.7.4)

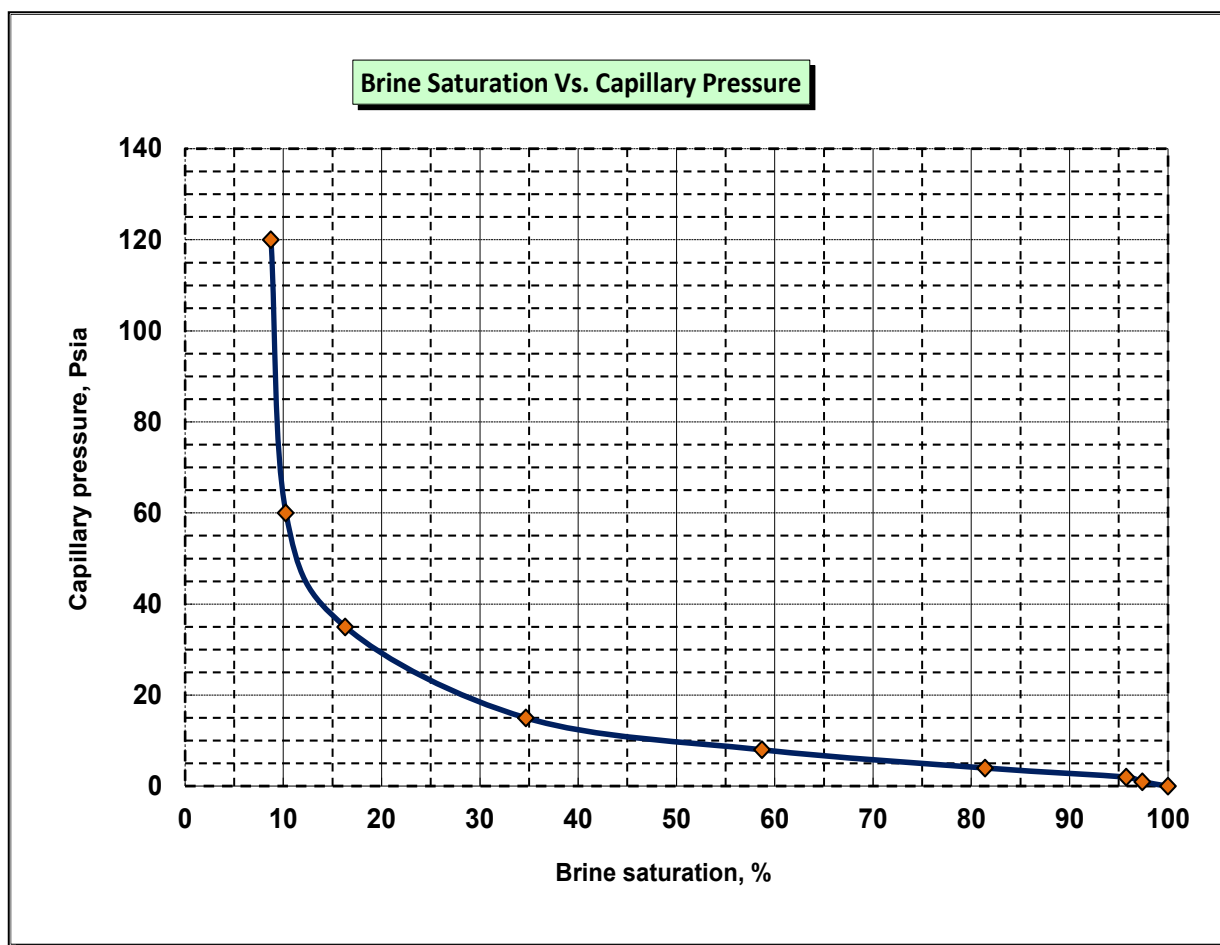


Figure (2.7.4)

Air-Water Drainage Cycle

Company	: Mellitah	Core Porosity, %	: 24.4
Well No.	: CE03-NC41	Core Air Permeability, (mD)	: 39.4
Sample No.	: 126	Irreducible Water Saturation, %	: 7.91
Salinity, ppm	: 70,000	Temperature in Chamber, C°	: 25
Brine Density(g/cc)	: 1.0495	Sample Depth (ft.in)	: 8483.10

Rotor Speed RPM	Ambient Conditions Capillary Pressure, psia	Brine Saturation fraction
0.0	0	1.0000
421	1	0.9470
596	2	0.8520
842	4	0.7440
1191	8	0.5800
1631	15	0.3140
2492	35	0.1657
3262	60	0.1040
4614	120	0.0791

Table (2.7.5)

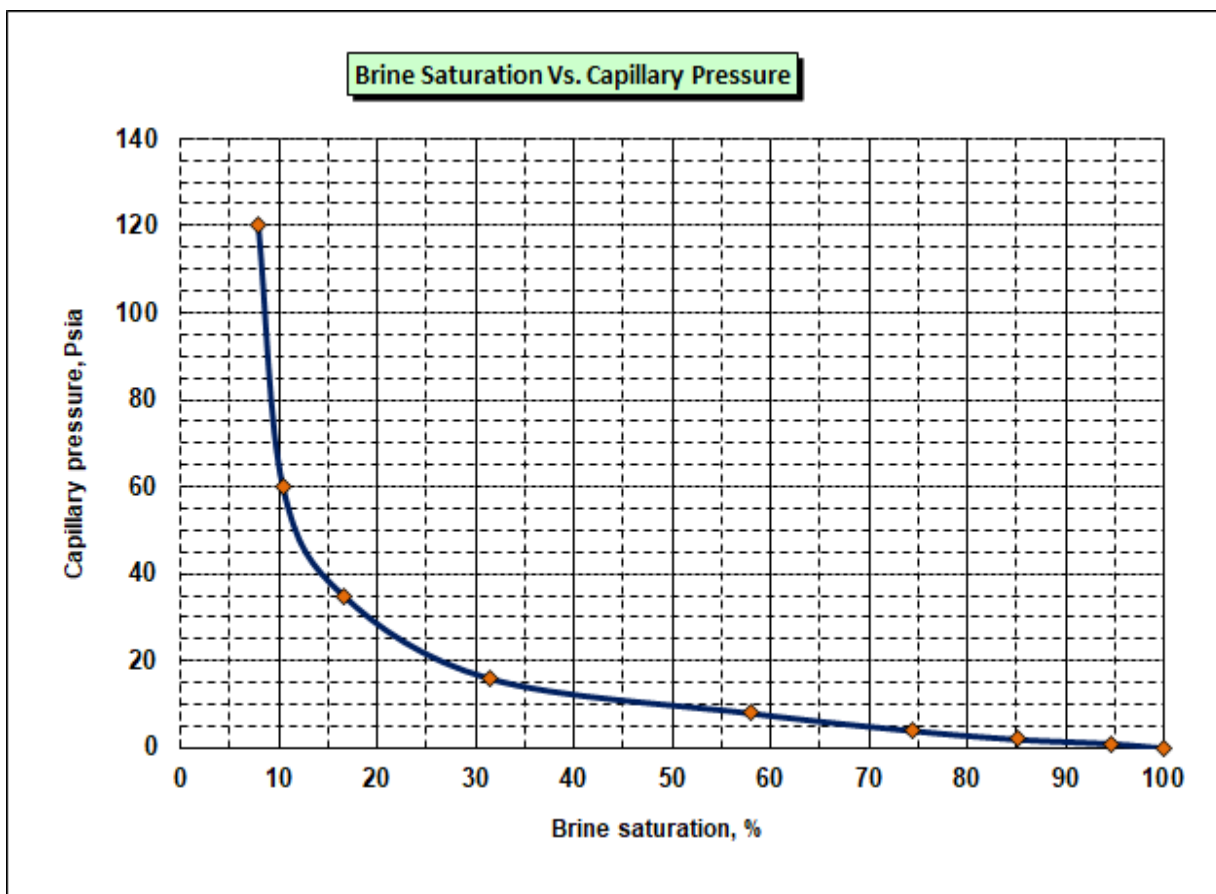


Figure (2.7.5)

Air-Water Drainage Cycle

Company	: Mellitah	Core Porosity, %	: 16.3
Well No.	: CE03-NC41	Core Air Permeability, (mD)	: 4.89
Sample No.	: 164	Irreducible Water Saturation, %	: 18.30
Salinity, ppm	: 70,000	Temperature in Chamber, C°	: 25
Brine Density(g/cc)	: 1.0495	Sample Depth (ft.in)	: 8521.04

Rotor Speed RPM	Ambient Conditions Capillary Pressure, psia	Brine Saturation fraction
0.0	0	1.0000
421	1	0.9760
596	2	0.9664
842	4	0.9435
1191	8	0.8674
1631	15	0.4530
2492	35	0.2970
3262	60	0.2291
4614	120	0.1830

Table (2.7.6)

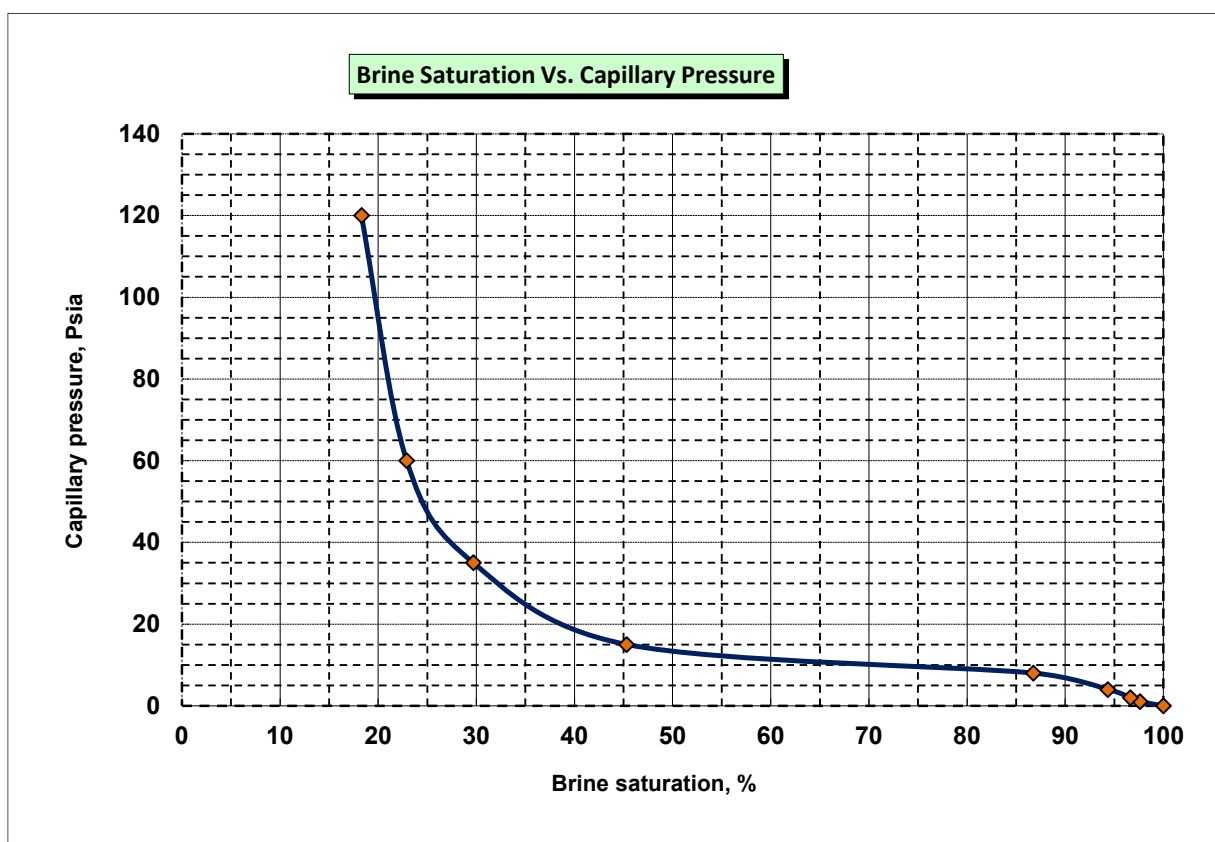


Figure (2.7.6)

Air-Water Drainage Cycle

Company	: Mellitah	Core Porosity, %	: 25.9
Well No.	: CE03-NC41	Core Air Permeability, (mD)	: 126.0
Sample No.	: 227	Irreducible Water Saturation, %	: 10.11
Salinity, ppm	: 70,000	Temperature in Chamber, C°	: 25
Brine Density(g/cc)	: 1.0495	Sample Depth (ft.in)	: 8585.02

Rotor Speed RPM	Ambient Conditions Capillary Pressure, psia	Brine Saturation fraction
0.0	0	1.0000
421	1	0.9670
596	2	0.9340
842	4	0.8317
1191	8	0.5640
1631	15	0.3640
2492	35	0.2140
3262	60	0.1287
4614	120	0.1011

Table (2.7.7)

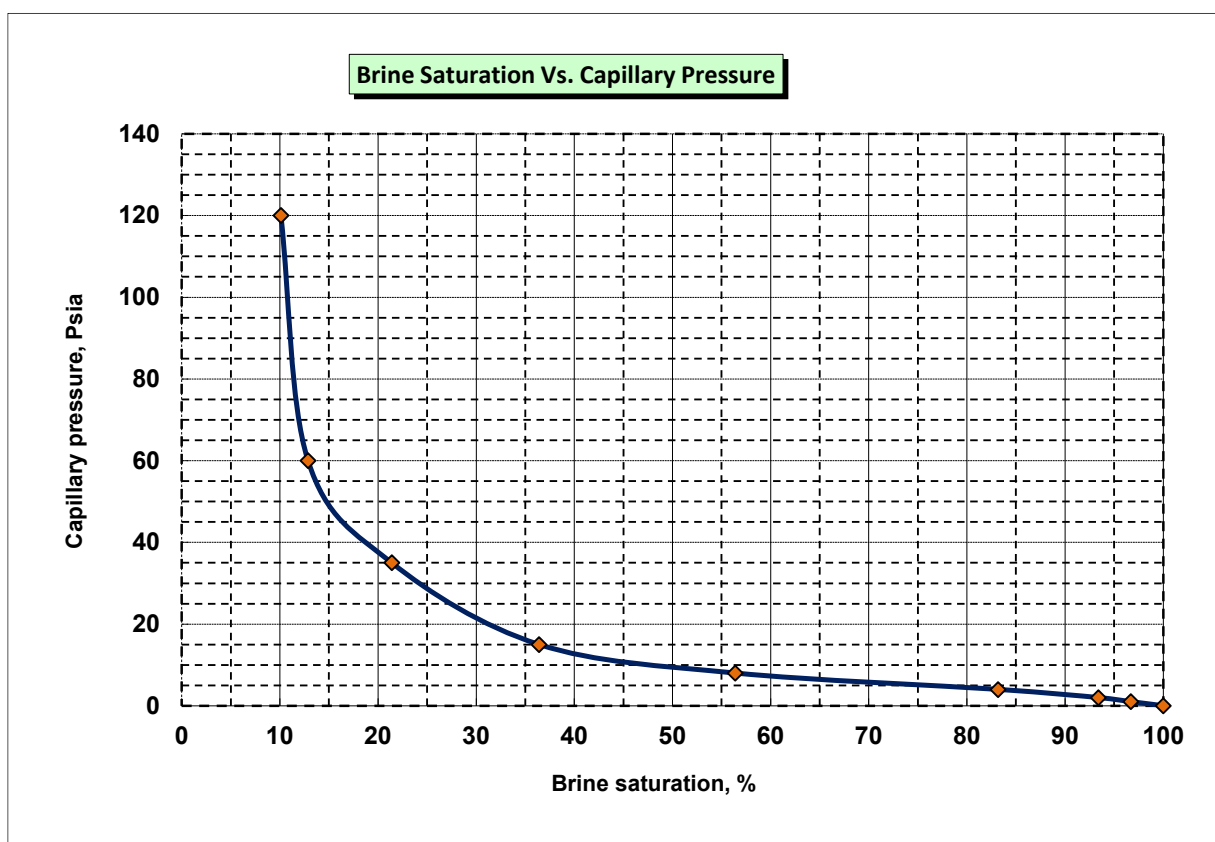


Figure (2.7.7)

Air-Water Drainage Cycle

Company	: Mellitah	Core Porosity, %	: 22.6
Well No.	: CE03-NC41	Core Air Permeability, (mD)	: 51.70
Sample No.	: 233	Irreducible Water Saturation, %	: 10.02
Salinity, ppm	: 70,000	Temperature in Chamber, C°	: 25
Brine Density(g/cc)	: 1.0495	Sample Depth (ft.in)	: 8591.04

Rotor Speed RPM	Ambient Conditions Capillary Pressure, psia	Brine Saturation fraction
0.0	0	1.0000
421	1	0.9670
596	2	0.9140
842	4	0.7340
1191	8	0.4670
1631	15	0.2960
2492	35	0.1833
3262	60	0.1289
4614	120	0.1002

Table (2.7.8)

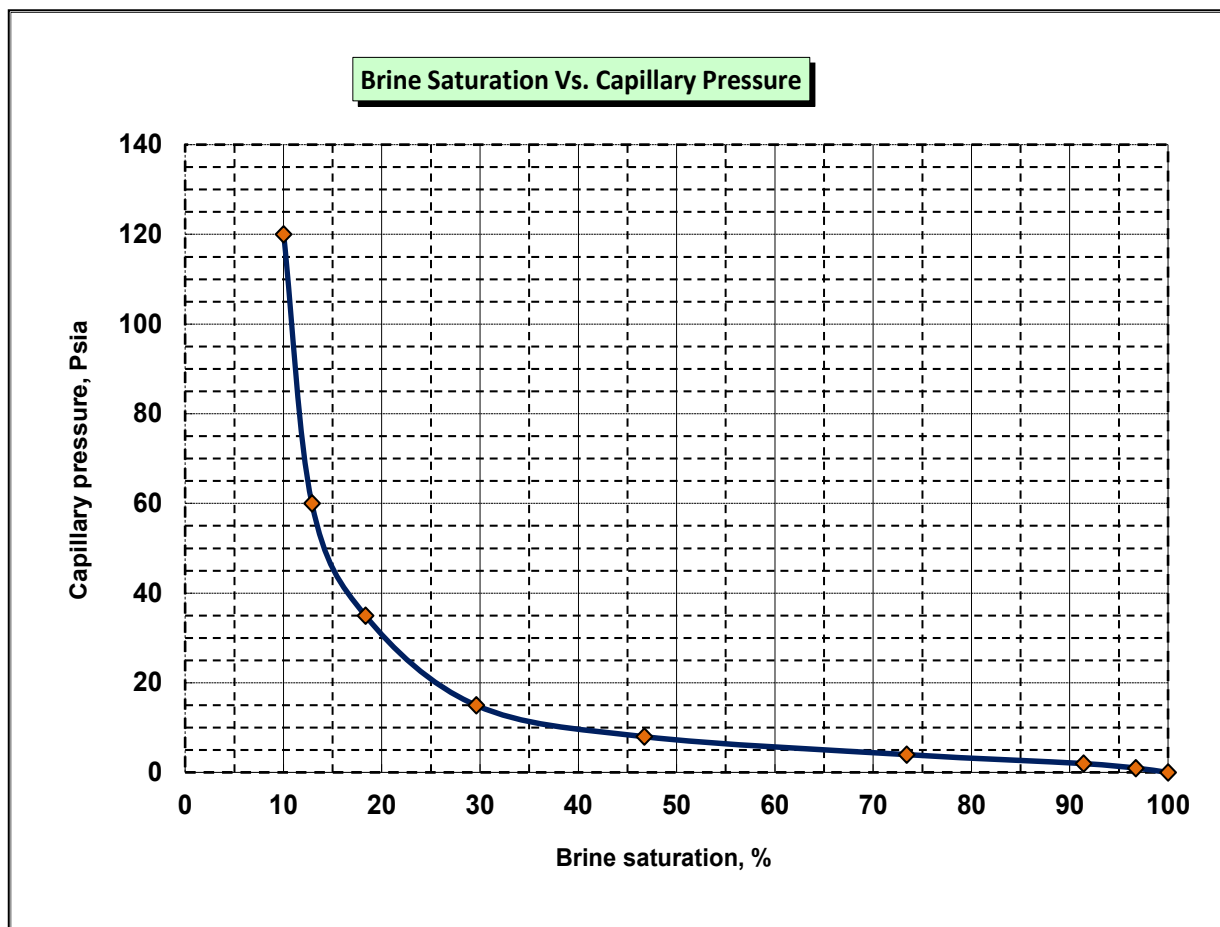


Figure (2.7.8)

Air-Water Drainage Cycle

Company	: Mellitah	Core Porosity, %	: 24.2
Well No.	: CE03-NC41	Core Air Permeability, (mD)	: 225
Sample No.	: 275	Irreducible Water Saturation, %	: 14.00
Salinity, ppm	: 70,000	Temperature in Chamber, C°	: 25
Brine Density(g/cc)	: 1.0495	Sample Depth (ft.in)	: 8636.06

Rotor Speed RPM	Ambient Conditions Capillary Pressure, psia	Brine Saturation fraction
0.0	0	1.0000
421	1	0.9570
596	2	0.9170
842	4	0.7310
1191	8	0.4650
1631	15	0.2490
2492	35	0.1662
3262	60	0.1422
4614	120	0.1400

Table (2.7.9)

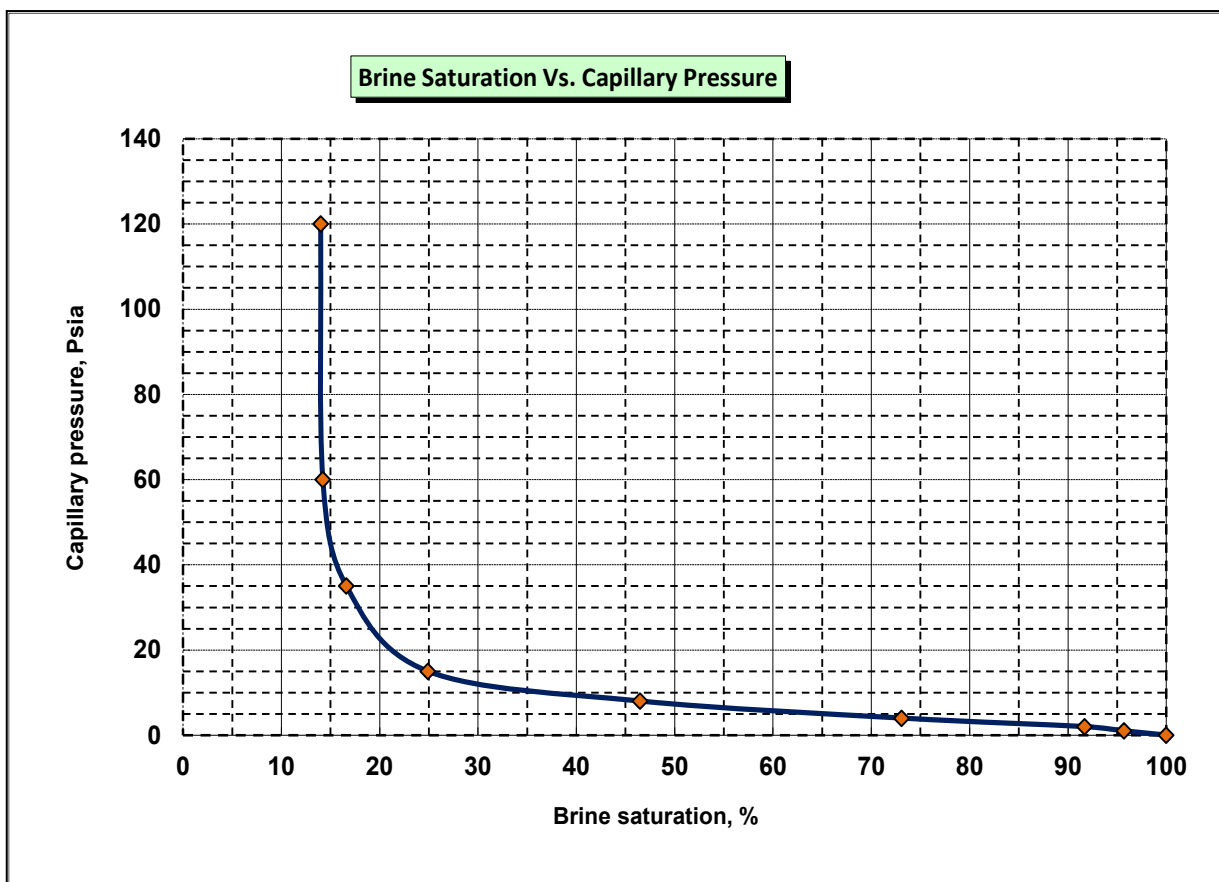


Figure (2.7.9)

Air-Water Drainage Cycle

Company	: Mellitah	Core Porosity, %	: 24.9
Well No.	: CE03-NC41	Core Air Permeability, (mD)	: 38.1
Sample No.	: 278	Irreducible Water Saturation, %	: 19.63
Salinity, ppm	: 70,000	Temperature in Chamber, C°	: 25
Brine Density(g/cc)	: 1.0495	Sample Depth (ft.in)	: 8639.01

Rotor Speed RPM	Ambient Conditions Capillary Pressure, psia	Brine Saturation fraction
0.0	0	1.0000
421	1	0.9542
596	2	0.9120
842	4	0.7640
1191	8	0.5914
1631	15	0.4436
2492	35	0.2890
3262	60	0.2171
4614	120	0.1963

Table (2.7.10)

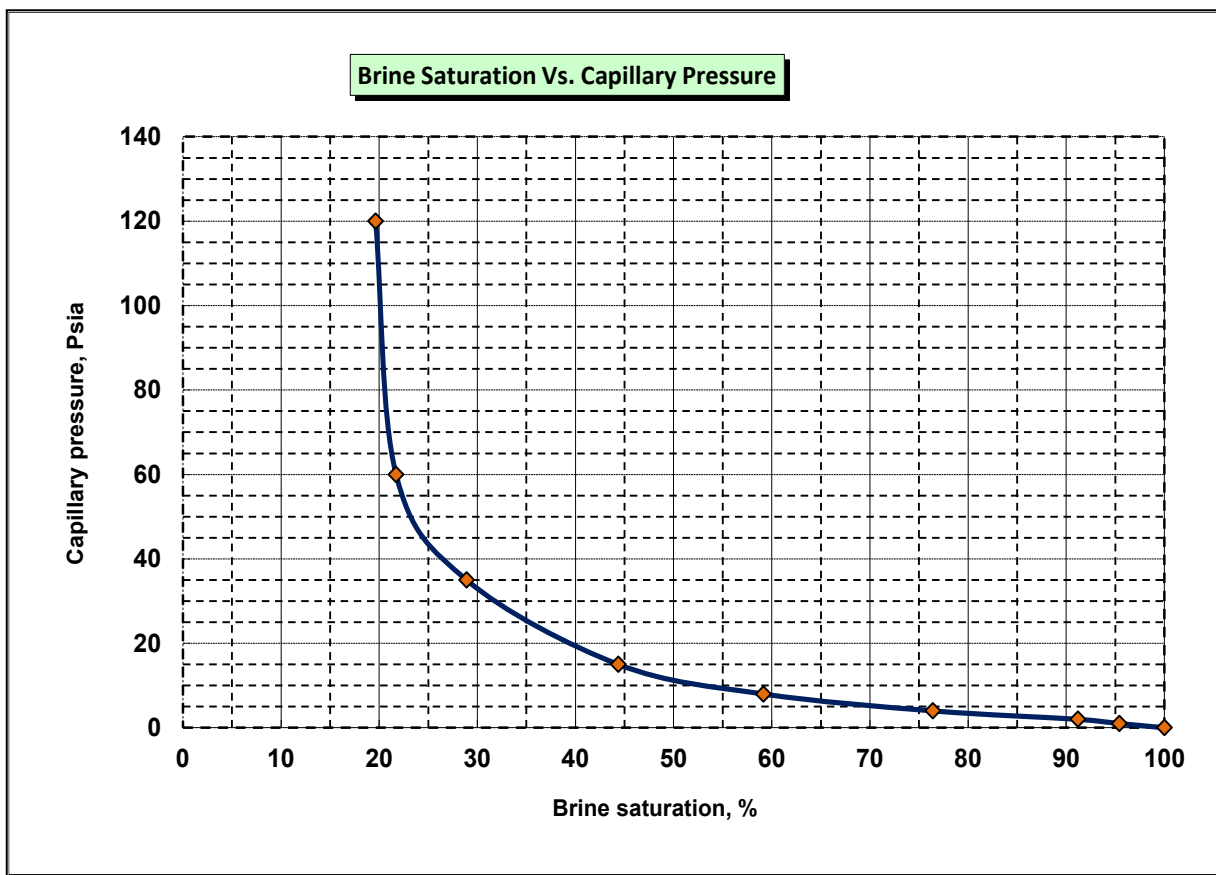


Figure (2.7.10)

Section Eight

Mercury Injection Capillary Pressure and Pore Size Distribution

MERCURY INJECTION CAPILLARY PRESSURE AND PORE SIZE DISTRIBUTION:

Ten core samples were selected from the well **CE03-NC41** to perform mercury injection capillary pressure measurements Using Micromeritics AutoPore IV 9510 Mercury Porosimeter with windows software. These core samples are representing El-Garia formation.

Prior to testing, the routine petrophysical properties of porosity, grain density and permeability to air were re-measured.

Each sample was individually weighted and a sample information file that describes the sample and gives the analysis conditions and other parameters for the analysis was created. The instrument performs semi- automatically injection of mercury at programmed pressure steps from less than one psi to 60,000 psia.

To begin the test, each sample was individually loaded into a calibrated glass penetrometer. The penetrometer consists of a sample chamber and precision-bore glass capillary whose volume was selected to be just larger than the measured routine pore volume. The outside of the glass capillary is plated with metal which acts as one plate of a capacitor. The mercury within the capillary acts as the other capacitance plate and then installs the loaded penetrometer into the low-pressure port. The first phase of the low-pressure analysis is the evacuation of gases from the penetrometer for 20-30 minutes, and then the penetrometer is backfilled automatically with Mercury.

For pressure up to 30 psia, air pressure is used, when the low-pressure analysis is complete, the penetrometer is removed from the low-pressure port and installed in a high-pressure port in which hydraulic pressure oil is used, pore volume data are calculated by determining the volume of mercury remaining in the penetrometer. As pressure increases, mercury moves into the sample's pores, vacating the stem (Drainage or Intrusion), the greater the pressure, the smaller the pore diameter into which the mercury can be forced, then the pressure is decreased gradually in steps to withdraw the mercury from the pores in pressure steps from 60,000 psia to 14 psia (Imbibition or Extrusion).

Mercury volumes are expressed in milligram per gram of dry weight of rock sample, the output data is presented in both tabular and graphical format. Graphical presentation of the data includes:

Cumulative Mercury volume versus air–Mercury capillary pressure to determine the threshold pressure. Air–Mercury capillary pressure versus wetting phase saturation for the two cycles. An accumulative volume injection curve and the incremental volume curve which is particularly useful for interpreting interconnecting pore dimensions are plotted against pore radii. Pore size is computed using La Place equation for cylindrical pores (References 5-7).

Conformance/Closure Corrections

The MICP test measures and records Hg intrusion data for each pressure step. The first pressure step is the pressure increase from the “mercury fill pressure” to the first pressure point (e.g., 1.6 - 1.8 psia). As each pressure step is sequentially performed, mercury progressively fills (i.e., “conforms to” or “closes around”) any sample surface irregularities such as micro-fractures and is, hence, recorded as Hg intrusion. At some higher-pressure step, however, sufficient pressure is achieved to cause mercury (i.e., the non-wetting phase) to intrude the largest pore throats controlling the sample’s PV. The pressure at which mercury commences to occupy the actual pore system of the sample being tested is called the “Threshold Pressure” or “closure pressure”. All intrusion data recorded up to this initial entry pressure is subtracted from the MICP raw data output as the “closure correction”.

Selection of the closure pressure for a given MICP test is determined by generating a X-Y plot of “percent sample pore volume (PV) occupied by mercury” vs. “Hg intrusion pressure”. This semi-log plot allows identification of an “inflection point” indicating a “sudden” increase in the amount of mercury intrusion. This inflection point, termed the “Threshold/closure pressure”, identifies the pressure at which mercury begins to occupy sample pore volume bounded by the largest pore throats; Pittman (1992) (Reference 8).

Baseline Error Correction Methods:

During MICP test, the penetrometer parts, the mercury, and the sample are all affected differently by the pressures exerted on them during analysis. Each material compresses to a different rate. Also, the increasing pressure within the high-pressure port can cause the temperature of one or more of these materials to rise. As a result, analysis data may show intrusion where none actually exists. For example, if the sample compresses sharply at a given pressure (compared to the compression of the mercury at that pressure), mercury moves from the penetrometer stem into the sample bulb to fill the space vacated by the shrinking sample. This reduction in the amount of mercury in the stem is interpreted by the software as intrusion.

MICP instrumentation software allows for incorporation of a “Blank Correction File” into the sample test “raw data”, which adjusts the generated MICP raw data output for equipment related compressibility effects but not for sample grain compressibility.

These baseline errors can be reduced or eliminated through one of two correction methods:

- 1- A “blank correction by formula” file contained in the MICP instrumentation software.
- 2- An “operator generated” blank correction file.

The first method (Blank Correction by Formula) was used for the entire samples selected from well CE03-NC41.

Formula Correction

Through large number of blank runs using a variety of pressure tables and penetrometers filled with mercury, Micromeritics has developed a formula for correction of each penetrometer type. The Auto Pore software provides this formula, so it can be easily incorporated into the raw generated raw data during the measurement’s procedures.

Note:

- *The core samples selected for mercury injection measurements were re-cut to approximately 1”X1” or less to fit into the sample cup of the Autopore instrumentation and in some cases pieces can be used.*
- *The first pressure value in all tables below is the Threshold Pressure.*
- *The routine core analysis was re-measured before conducting the MICP measurements.*

The generated data is presented in tables **2.8.1** through **2.8.10** and in figures **2.8.1** through **2.8.41**.

MERCURY INJECTION CAPILLARY PRESSURE AND PORE SIZE DISTRIBUTION

Client	: Mellitah Oil and Gas		
Well	: CE03-NC41		
Sample #	: 80		
Core Sample Data and Experimental Values			
Hg Pore volume, (cc)	: 1.14	Threshold Pressure, (psi)	: 7.504
Porosity, (%)	: 21.0	Sample Weight, (gm)	: 11.473
Air Perm., (mD)	: 51.7	Formation	: El-Garia
Sample Depth, (ft.in)	: 8437.06		

Pressure (psia)	Cycle	Cum. Hg Sat. (Pore Vol., %)	Wetting Phase (Sw) (Pore Vol., %)	Incr. Hg Sat. (Pore Vol., %)	Ri (microns)
7.5038	D	0.0000	100.000	0.000	14.214
8.5295	D	0.4753	99.525	0.475	12.505
10.5260	D	1.2221	98.78	0.7468	10.133
13.0304	D	2.1830	97.817	0.9610	8.1856
15.9844	D	3.995	96.005	1.8122	6.6729
19.9717	D	8.539	91.461	4.5436	5.3407
24.9547	D	15.563	84.437	7.0243	4.2742
30.0379	D	21.253	78.747	5.6895	3.5509
36.4495	D	26.202	73.798	4.9496	2.9263
46.8023	D	34.076	65.924	7.8733	2.2790
56.6461	D	41.965	58.035	7.8889	1.8830
72.0422	D	51.034	48.966	9.0690	1.4805
86.6847	D	57.276	42.724	6.2420	1.2305
112.970	D	65.796	34.204	8.5203	0.9442
137.609	D	71.896	28.104	6.0996	0.7751
171.899	D	77.757	22.243	5.8612	0.6205
217.127	D	82.579	17.421	4.8218	0.4912
269.676	D	85.642	14.358	3.0637	0.3955
326.128	D	87.791	12.209	2.1484	0.3271
417.526	D	89.493	10.507	1.7026	0.2555
516.136	D	90.531	9.4690	1.0377	0.2067
636.861	D	91.255	8.7451	0.7236	0.1675
795.100	D	91.888	8.1121	0.6338	0.1341
988.023	D	92.536	7.4642	0.6479	0.1080
1194.607	D	92.822	7.1782	0.2852	0.0893
1498.356	D	93.297	6.7031	0.4753	0.0712
1893.792	D	93.788	6.2125	0.4907	0.0563
2347.424	D	94.242	5.7578	0.4546	0.0454
2895.822	D	94.586	5.4138	0.3441	0.0368
3595.224	D	94.752	5.2483	0.1655	0.0297
4482.139	D	94.923	5.0771	0.1712	0.0238
5589.385	D	95.106	4.8941	0.1831	0.0191
6881.931	D	95.315	4.6846	0.2094	0.0155
8584.391	D	95.315	4.6846	0.0000	0.0124

Table (2.8.1)

Pressure (psia)	Cycle	Cum. Hg Sat. (Pore Vol., %)	Wetting Phase (Sw) (Pore Vol., %)	Incr. Hg Sat. (Pore Vol., %)	Ri (microns)
10583.379	D	95.315	4.6846	0.0000	0.0101
13186.539	D	95.315	4.6846	0.0000	0.0081
14782.390	D	95.315	4.6846	0.0000	0.0072
16382.332	D	95.315	4.6846	0.0000	0.0065
19981.617	D	95.315	4.6846	0.0000	0.0053
24992.207	D	95.315	4.6846	0.0000	0.0043
29991.439	D	95.315	4.6846	0.0000	0.0036
34988.617	D	95.315	4.6846	0.0000	0.0030
39986.484	D	95.315	4.6846	0.0000	0.0027
44987.816	D	95.315	4.6846	0.0000	0.0024
49985.336	D	95.315	4.6846	0.0000	0.0021
54987.020	D	95.315	4.6846	0.0000	0.0019
59985.754	D	95.315	4.6846	0.0000	0.0018
46095.152	I	95.315	4.6846	----	----
35492.145	I	95.315	4.6846	----	----
27307.291	I	95.315	4.6846	----	----
21010.561	I	95.315	4.6846	----	----
16010.941	I	95.315	4.6846	----	----
12412.865	I	95.315	4.6846	----	----
9614.012	I	95.315	4.6846	----	----
7317.126	I	95.315	4.6846	----	----
5707.553	I	95.315	4.6846	----	----
4309.691	I	95.315	4.6846	----	----
3304.693	I	95.315	4.6846	----	----
2604.193	I	95.315	4.6846	----	----
2005.643	I	95.315	4.6846	----	----
1501.503	I	95.315	4.6846	----	----
1202.919	I	95.315	4.6846	----	----
900.174	I	95.315	4.6846	----	----
700.143	I	95.315	4.6846	----	----
502.718	I	95.315	4.6846	----	----
401.029	I	95.315	4.6846	----	----
303.849	I	95.080	4.9202	----	----
242.402	I	94.000	6.0000	----	----
191.801	I	92.952	7.0479	----	----
148.036	I	89.889	10.111	----	----
112.597	I	84.666	15.334	----	----
88.3086	I	73.446	26.553	----	----
67.8241	I	57.478	42.522	----	----
52.7586	I	52.855	47.145	----	----
47.7959	I	51.066	48.934	----	----

Table (2.8.1-Continue)

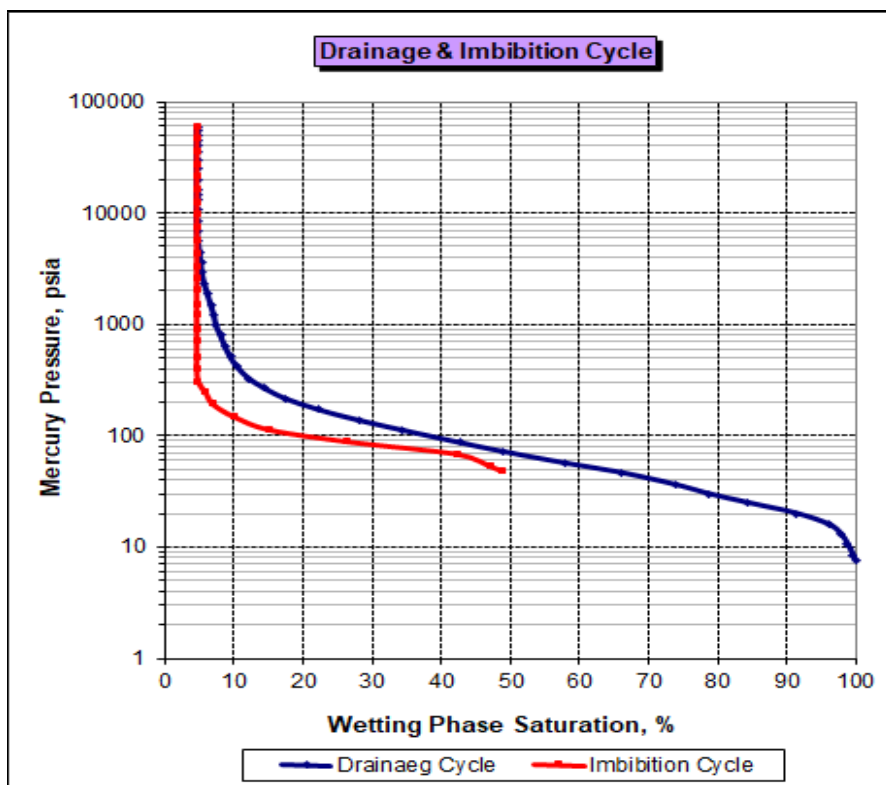


Figure (2.8.1)-Sample # 80

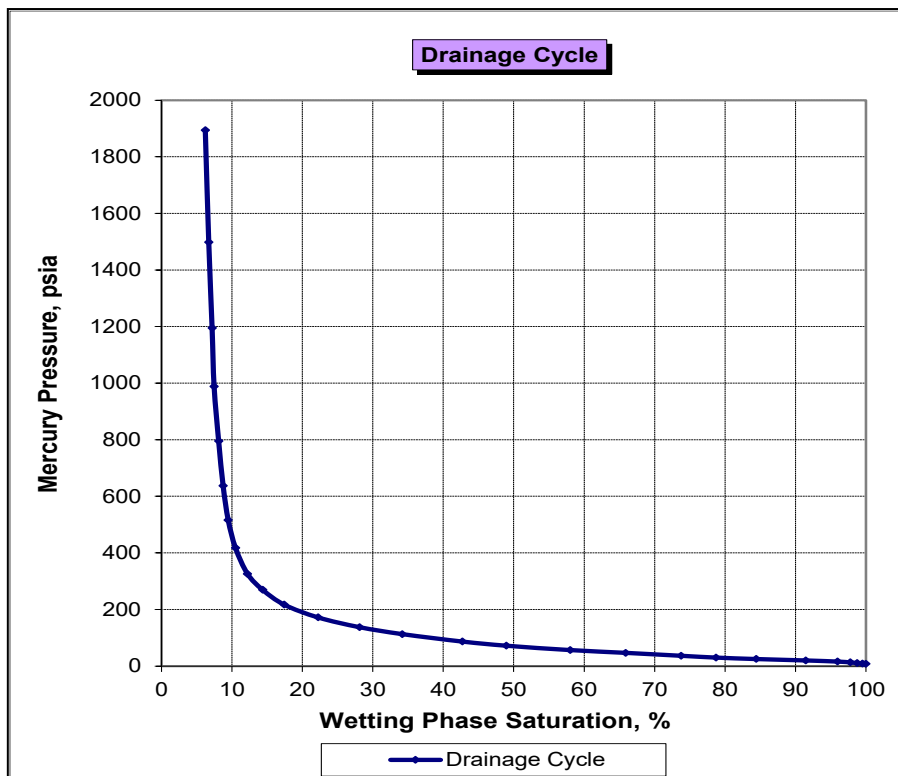


Figure (2.8.2)-Sample # 80

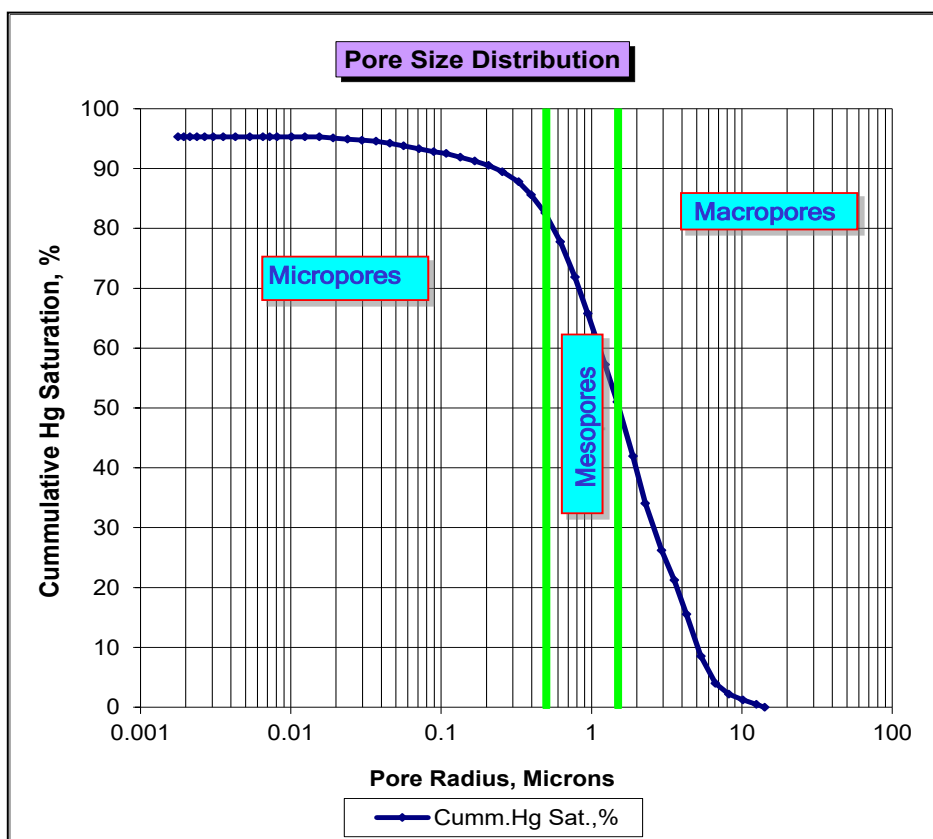


Figure (2.8.3)-Sample # 80

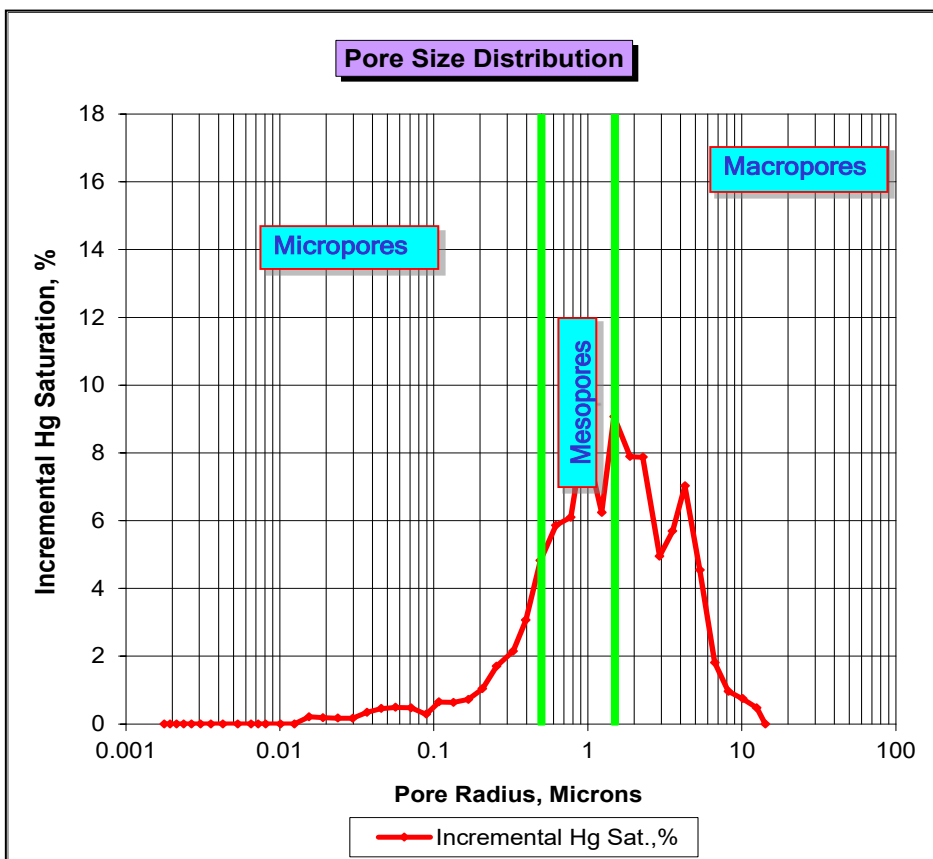


Figure (2.8.4)-Sample # 80

MERCURY INJECTION CAPILLARY PRESSURE AND PORE SIZE DISTRIBUTION

Client	: Mellitah Oil and Gas		
Well	: CE03-NC41		
Sample #	: 93		
Core Sample Data and Experimental Values			
Hg Pore volume, (cc)	: 1.11	Threshold Pressure, (psi)	: 15.959
Porosity, (%)	: 23.30	Sample Weight, (gm)	: 9.826
Air Perm., (mD)	: 18.9	Formation	: El-Garia
Sample Depth, (ft.in)	: 8450.06		

Pressure (psia)	Cycle	Cum. Hg Sat. (Pore Vol., %)	Wetting Phase (Sw) (Pore Vol., %)	Incr. Hg Sat. (Pore Vol., %)	Ri (microns)
15.95925	D	0.000	100.000	0.000	6.683
19.99844	D	0.8562	99.144	0.856	5.334
25.05211	D	2.957	97.043	2.101	4.258
34.01844	D	6.097	93.903	3.140	3.135
36.58226	D	6.812	93.188	0.715	2.916
46.11662	D	11.506	88.494	4.694	2.313
57.24352	D	19.576	80.424	8.070	1.863
71.48265	D	30.682	69.318	11.106	1.492
86.47430	D	40.001	59.999	9.318	1.233
112.33435	D	52.985	47.015	12.984	0.950
136.39018	D	62.275	37.725	9.290	0.782
171.03624	D	71.745	28.255	9.470	0.624
217.58224	D	79.431	20.569	7.686	0.490
266.96268	D	84.242	15.758	4.810	0.400
327.59070	D	87.546	12.454	3.304	0.326
419.69562	D	90.254	9.746	2.708	0.254
515.67517	D	91.849	8.151	1.594	0.207
635.10815	D	92.904	7.096	1.056	0.168
798.58466	D	93.728	6.272	0.824	0.134
985.69910	D	94.295	5.705	0.567	0.108
1201.64648	D	94.718	5.282	0.424	0.089
1496.98901	D	95.288	4.712	0.569	0.0713
1894.97021	D	95.819	4.181	0.531	0.0563
2343.55591	D	96.228	3.7721	0.409	0.0455
2892.94629	D	96.489	3.5110	0.261	0.0369
3591.77271	D	96.790	3.2095	0.301	0.0297
4488.95020	D	96.958	3.0416	0.168	0.0238
5584.33008	D	97.156	2.8442	0.197	0.0191
6883.70410	D	97.201	2.7985	0.046	0.0155
8585.46582	D	97.230	2.7699	0.029	0.0124
10584.51465	D	97.230	2.7699	0.000	0.0101
13182.19141	D	97.230	2.7699	0.000	0.0081
14784.96680	D	97.230	2.7699	0.000	0.0072
16379.51465	D	97.230	2.7699	0.000	0.0065

Table (2.8.2)

Pressure (psia)	Cycle	Cum. Hg Sat. (Pore Vol., %)	Wetting Phase (Sw) (Pore Vol., %)	Incr. Hg Sat. (Pore Vol., %)	Ri (microns)
19981.324	D	97.230	2.7699	0.000	0.0053
24991.359	D	97.230	2.7699	0.000	0.0043
29987.222	D	97.230	2.7699	0.000	0.0036
34990.722	D	97.230	2.7699	0.000	0.0030
39988.566	D	97.230	2.7699	0.000	0.0027
44989.726	D	97.230	2.7699	0.000	0.0024
49986.199	D	97.230	2.7699	0.000	0.0021
54986.289	D	97.230	2.7699	0.000	0.0019
59986.285	D	97.230	2.7699	0.000	0.0018
46104.148	I	97.230	2.7699	----	----
35503.953	I	97.230	2.7699	----	----
27306.521	I	97.230	2.7699	----	----
21010.173	I	97.230	2.7699	----	----
16006.919	I	97.230	2.7699	----	----
12406.951	I	97.230	2.7699	----	----
9612.8008	I	97.230	2.7699	----	----
7319.8457	I	97.230	2.7699	----	----
5709.1333	I	97.230	2.7699	----	----
4303.7632	I	97.230	2.7699	----	----
3306.7063	I	97.230	2.7699	----	----
2609.7014	I	97.230	2.7699	----	----
2009.4174	I	97.230	2.7699	----	----
1504.4032	I	97.230	2.7699	----	----
1202.9652	I	97.230	2.7699	----	----
901.4125	I	97.230	2.7699	----	----
704.9088	I	97.230	2.7699	----	----
504.3750	I	97.230	2.7699	----	----
399.0969	I	97.230	2.7699	----	----
302.6652	I	97.230	2.7699	----	----
242.3969	I	96.524	3.4760	----	----
191.7123	I	94.840	5.1600	----	----
147.4775	I	90.915	9.0851	----	----
113.0779	I	80.671	19.329	----	----
88.1011	I	56.167	43.833	----	----

Table (2.8.2-Continue)

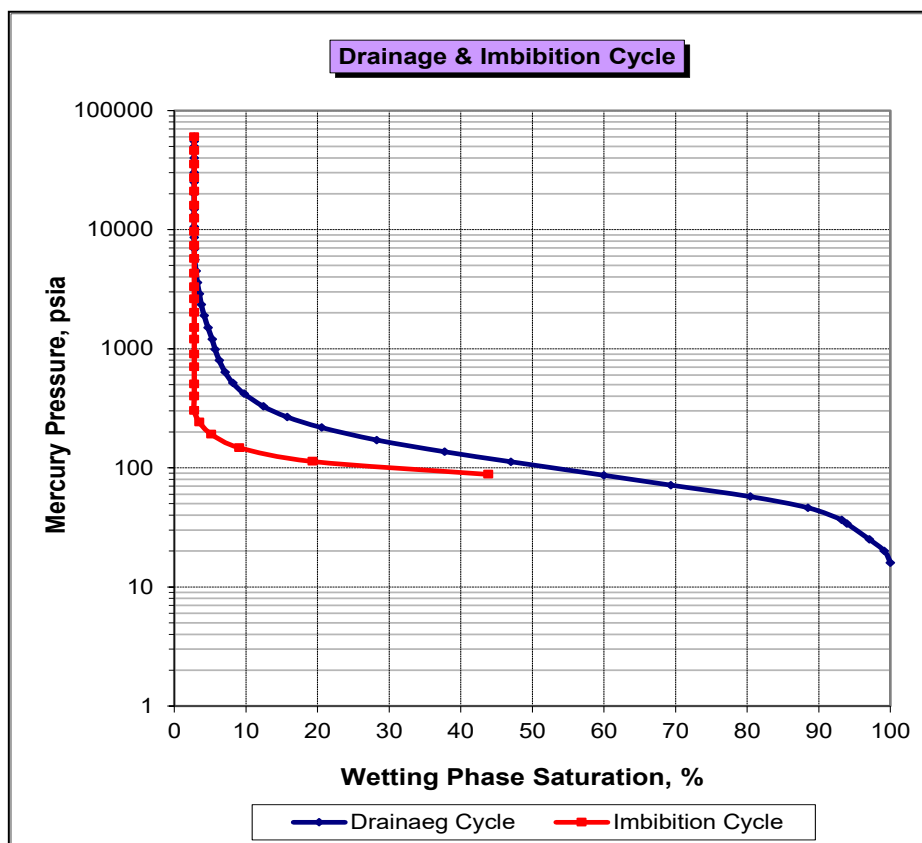


Figure (2.8.5)-Sample # 93

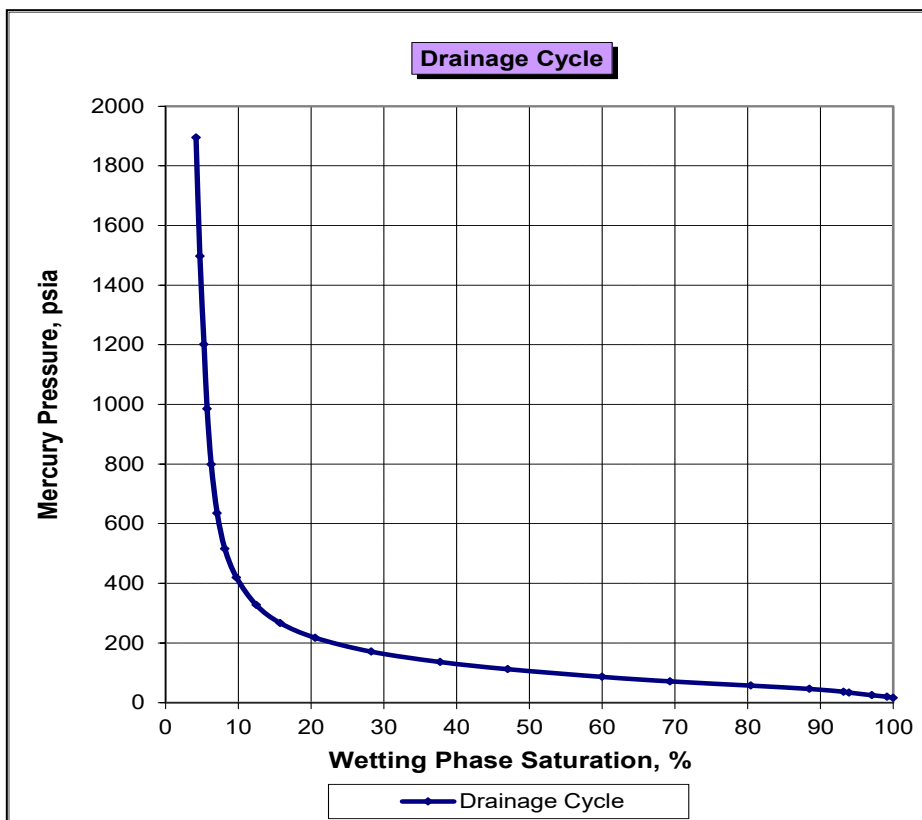


Figure (2.8.6)-Sample # 93

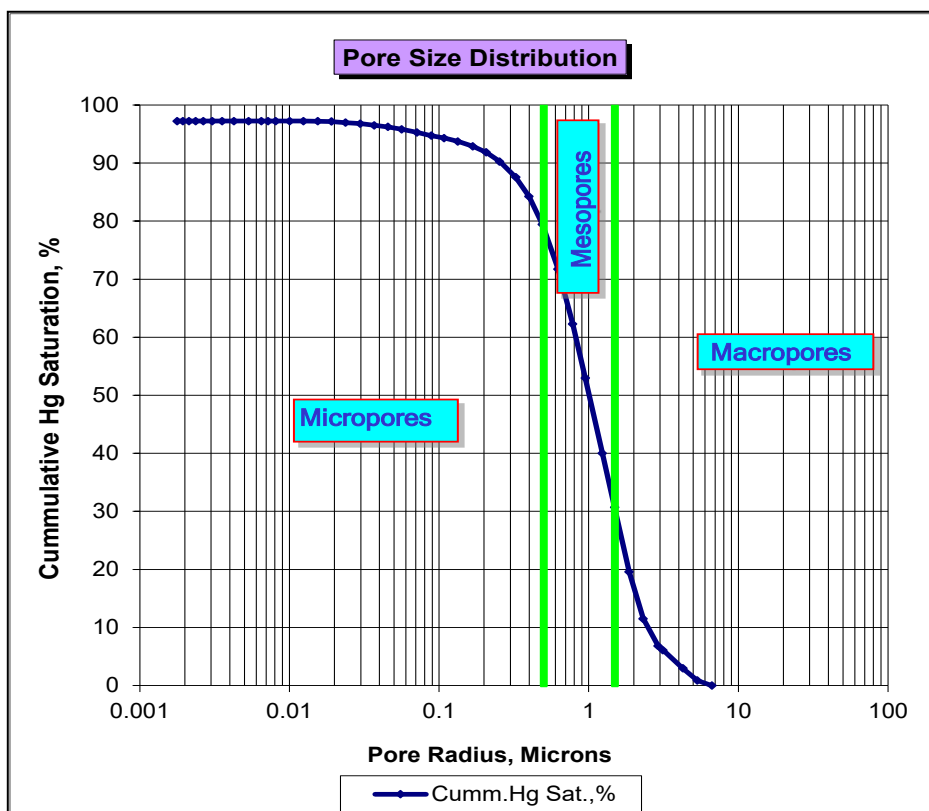


Figure (2.8.7)-Sample # 93

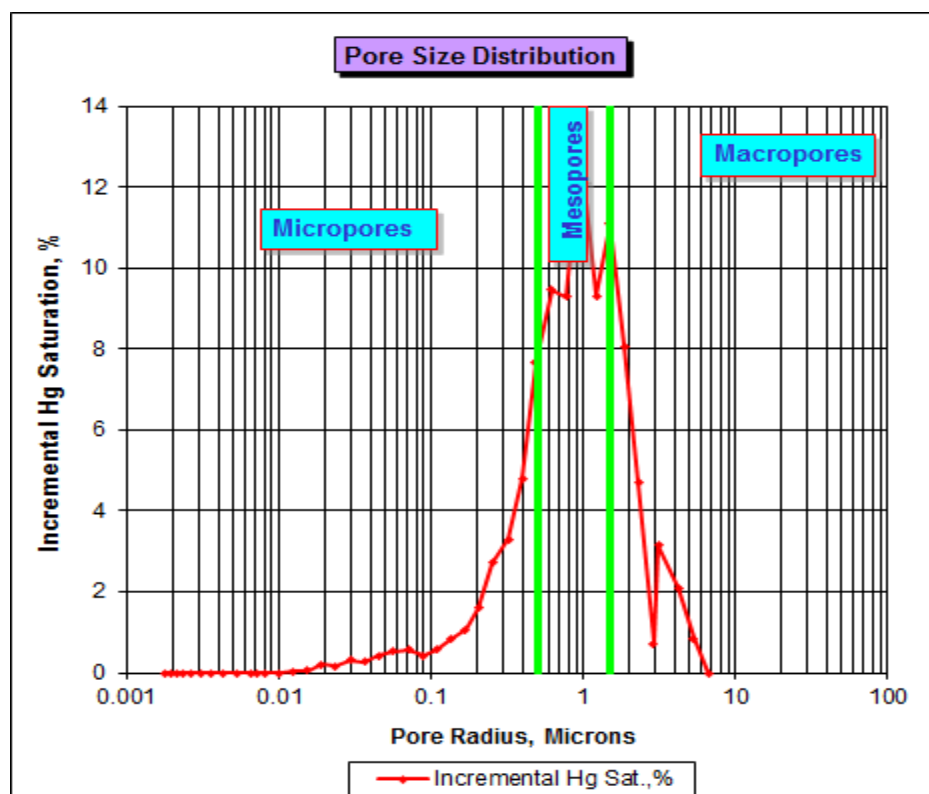


Figure (2.8.8)-Sample # 93

MERCURY INJECTION CAPILLARY PRESSURE AND PORE SIZE DISTRIBUTION

Client	: Mellitah Oil and Gas		
Well	: CE03-NC41		
Sample #	: 108		
Core Sample Data and Experimental Values			
Hg Pore volume, (cc)	: 1.28	Threshold Pressure, (psi)	: 8.535
Porosity, (%)	: 25.60	Sample Weight, (gm)	: 9.012
Air Perm., (mD)	: 62.80	Formation	: El-Garia
Sample Depth, (ft.in)	: 8465.05		

Pressure (psia)	Cycle	Cum. Hg Sat. (Pore Vol., %)	Wetting Phase (Sw) (Pore Vol., %)	Incr. Hg Sat. (Pore Vol., %)	Ri (microns)
8.5345	D	0.000	100.00	0.000	12.50
10.5258	D	0.602	99.398	0.602	10.13
13.0189	D	2.069	97.931	1.467	8.193
15.9894	D	5.332	94.668	3.264	6.671
19.9611	D	12.733	87.266	7.401	5.344
24.9558	D	20.808	79.192	8.075	4.274
30.0383	D	26.750	73.250	5.942	3.551
36.9245	D	34.662	65.338	7.912	2.889
46.8519	D	40.331	59.669	5.669	2.277
56.5136	D	45.068	54.932	4.736	1.887
71.4455	D	53.115	46.885	8.047	1.493
85.9908	D	58.816	41.184	5.701	1.240
111.482	D	66.064	33.936	7.248	0.957
136.704	D	71.371	28.629	5.307	0.780
171.806	D	76.800	23.200	5.429	0.621
217.897	D	81.893	18.107	5.093	0.490
267.590	D	85.399	14.601	3.506	0.399
329.738	D	88.062	11.938	2.663	0.323
417.910	D	90.685	9.3150	2.623	0.255
516.562	D	92.313	7.6871	1.628	0.206
635.864	D	93.333	6.6670	1.020	0.168
795.343	D	94.043	5.9570	0.710	0.134
987.661	D	94.611	5.3890	0.568	0.108
1196.879	D	95.224	4.7761	0.613	0.089
1498.295	D	95.509	4.4910	0.286	0.0712
1895.764	D	95.838	4.1620	0.329	0.0563
2348.268	D	96.224	3.7760	0.386	0.0454
2895.853	D	96.382	3.6177	0.158	0.0368
3594.728	D	96.567	3.4329	0.185	0.0297
4490.769	D	96.647	3.3529	0.080	0.0238
5582.732	D	96.846	3.1540	0.199	0.0191
6882.943	D	96.982	3.0178	0.136	0.0155
8587.321	D	97.041	2.9588	0.059	0.0124
10586.25	D	97.041	2.9588	0.000	0.0101

Table (2.8.3)

Pressure (psia)	Cycle	Cum. Hg Sat. (Pore Vol., %)	Wetting Phase (Sw) (Pore Vol., %)	Incr. Hg Sat. (Pore Vol., %)	Ri (microns)
13184.125	D	97.041	2.9588	0.000	0.0081
14782.297	D	97.041	2.9588	0.000	0.0072
16380.237	D	97.041	2.9588	0.000	0.0065
19978.389	D	97.041	2.9588	0.000	0.0053
24993.125	D	97.041	2.9588	0.000	0.0043
29991.975	D	97.041	2.9588	0.000	0.0036
34991.547	D	97.041	2.9588	0.000	0.0030
39989.547	D	97.041	2.9588	0.000	0.0027
44991.699	D	97.041	2.9588	0.000	0.0024
49984.867	D	97.041	2.9588	0.000	0.0021
54986.285	D	97.041	2.9588	0.000	0.0019
59987.879	D	97.041	2.9588	0.000	0.0018
46109.414	I	97.041	2.9588	----	----
35508.047	I	97.041	2.9588	----	----
27301.488	I	97.041	2.9588	----	----
21008.859	I	97.041	2.9588	----	----
16004.999	I	97.041	2.9588	----	----
12411.576	I	97.041	2.9588	----	----
9615.738	I	97.041	2.9588	----	----
7312.555	I	97.041	2.9588	----	----
5710.395	I	97.041	2.9588	----	----
4305.228	I	97.041	2.9588	----	----
3302.788	I	97.041	2.9588	----	----
2599.854	I	97.041	2.9588	----	----
2001.821	I	97.041	2.9588	----	----
1500.439	I	97.041	2.9588	----	----
1203.805	I	97.041	2.9588	----	----
903.303	I	97.041	2.9588	----	----
700.291	I	97.041	2.9588	----	----
503.405	I	97.041	2.9588	----	----
399.619	I	97.041	2.9588	----	----
301.467	I	97.041	2.9588	----	----
243.324	I	96.627	3.3730	----	----
192.908	I	95.448	4.5523	----	----
146.202	I	93.096	6.904	----	----
113.078	I	87.804	12.196	----	----
88.3631	I	77.358	22.642	----	----

Table (2.8.3-Continue)

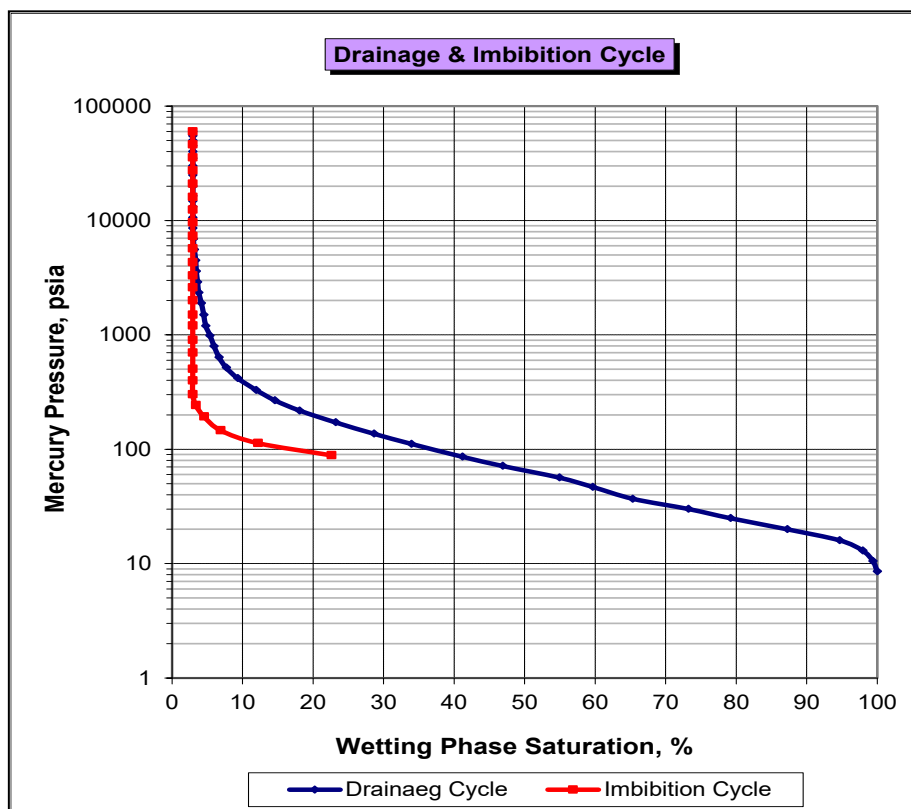


Figure (2.8.9)-Sample # 108

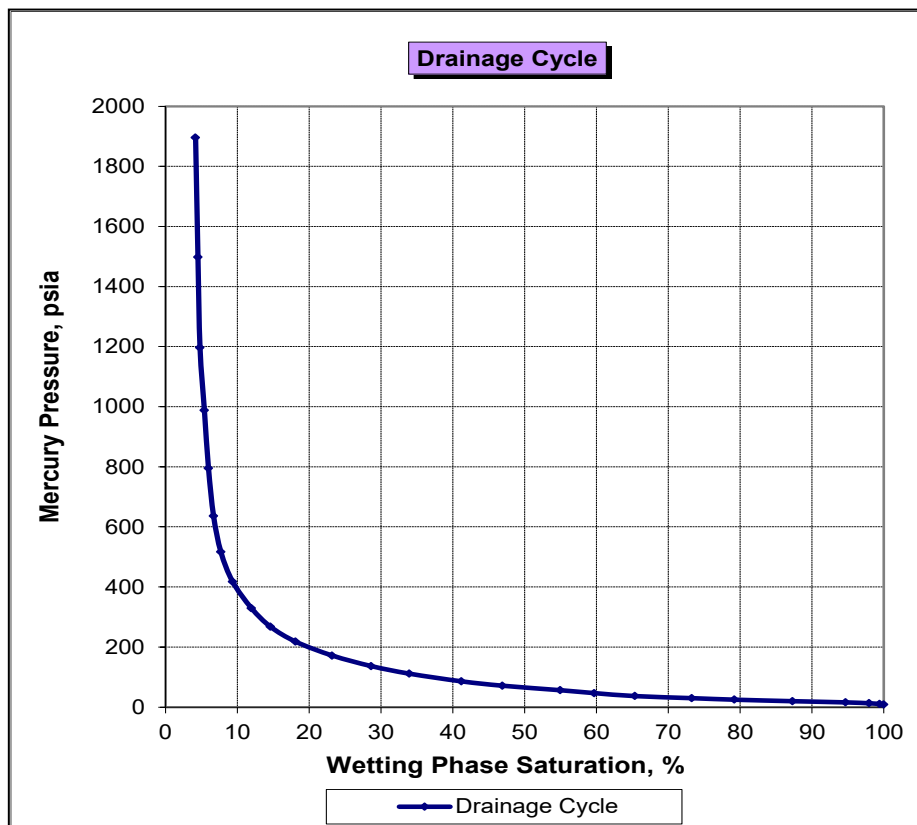


Figure (2.8.10)-Sample # 108

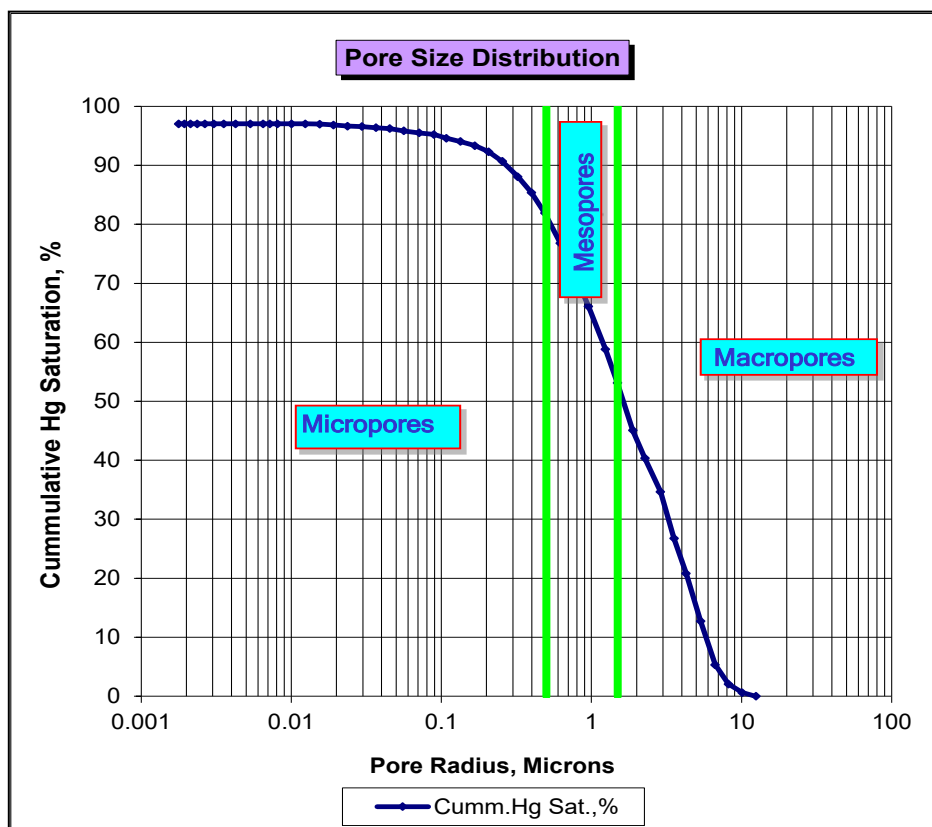


Figure (2.8.11)-Sample # 108

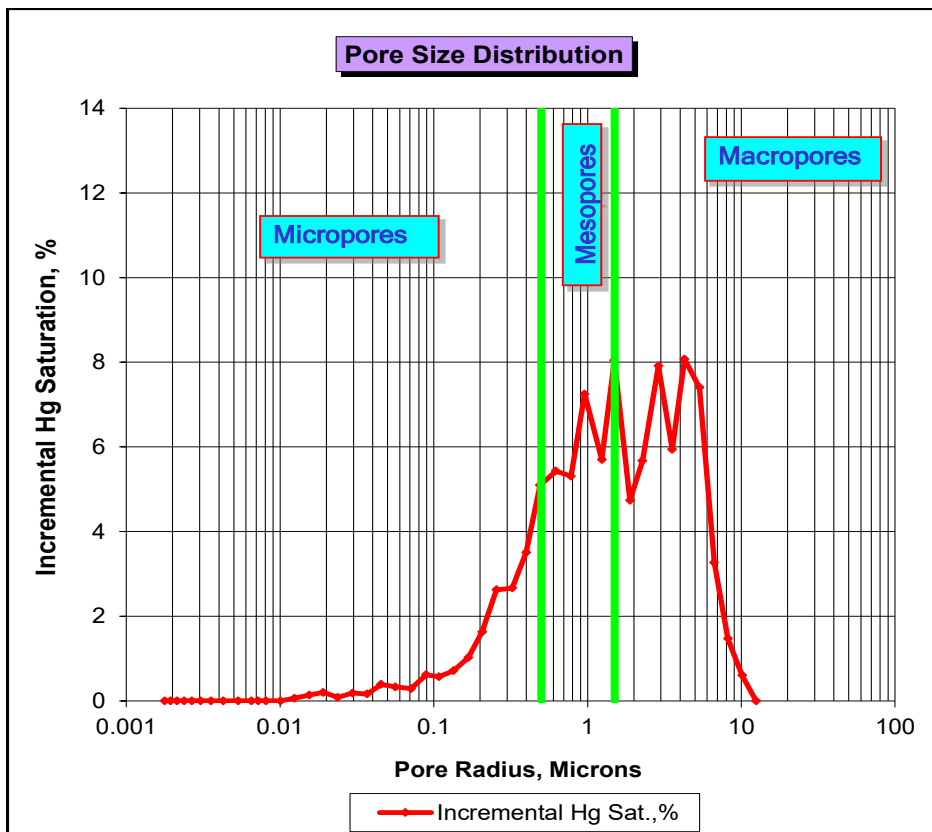


Figure (2.8.12)-Sample # 108

MERCURY INJECTION CAPILLARY PRESSURE AND PORE SIZE DISTRIBUTION

Client	: Mellitah Oil and Gas		
Well	: CE03-NC41		
Sample #	: 114		
Core Sample Data and Experimental Values			
Hg Pore volume, (cc)	: 0.95	Threshold Pressure, (psi)	: 17.977
Porosity, (%)	: 22.4	Sample Weight, (gm)	: 9.042
Air Perm., (mD)	: 32.40	Formation	: El-Garia
Sample Depth, (ft.in)	: 8471.07		

Pressure (psia)	Cycle	Cum. Hg Sat. (Pore Vol., %)	Wetting Phase (Sw) (Pore Vol., %)	Incr. Hg Sat. (Pore Vol., %)	Ri (microns)
17.9767	D	0.000	100.000	0.000	5.933
19.9861	D	0.7933	99.207	0.793	5.337
25.0685	D	3.615	96.385	2.822	4.255
30.0449	D	9.418	90.582	5.803	3.550
33.9626	D	11.590	88.410	2.172	3.141
46.0488	D	20.512	79.488	8.921	2.316
56.9883	D	32.406	67.594	11.895	1.872
70.7408	D	43.684	56.316	11.278	1.508
86.7083	D	52.533	47.467	8.848	1.230
111.5030	D	62.626	37.374	10.093	0.957
138.5049	D	69.900	30.100	7.274	0.770
172.9191	D	76.155	23.845	6.255	0.617
218.2876	D	81.482	18.518	5.327	0.489
265.9782	D	84.801	15.199	3.319	0.401
329.5045	D	87.090	12.910	2.289	0.324
416.2428	D	89.141	10.859	2.051	0.256
515.8668	D	90.955	9.0450	1.815	0.207
635.4341	D	91.610	8.3901	0.654	0.168
796.8458	D	92.761	7.2391	1.152	0.134
986.6160	D	93.721	6.2790	0.959	0.108
1196.0680	D	94.145	5.8550	0.424	0.089
1496.8257	D	94.885	5.1150	0.740	0.0713
1897.8556	D	95.690	4.3102	0.805	0.0562
2344.9695	D	96.064	3.9363	0.373	0.0455
2893.2378	D	96.108	3.8920	0.044	0.0369
3593.5242	D	96.108	3.8920	0.000	0.0297
4488.0469	D	96.247	3.7529	0.139	0.0238
5584.3013	D	96.487	3.5128	0.240	0.0191
6885.0166	D	96.487	3.5128	0.000	0.0155
8583.0020	D	96.487	3.5128	0.000	0.0124
10583.4209	D	96.611	3.3886	0.124	0.0101
13182.4854	D	96.611	3.3886	0.000	0.0081
14784.6875	D	96.611	3.3886	0.000	0.0072
16381.3867	D	96.611	3.3886	0.000	0.0065

Table (2.8.4)

Pressure (psia)	Cycle	Cum. Hg Sat. (Pore Vol., %)	Wetting Phase (Sw) (Pore Vol., %)	Incr. Hg Sat. (Pore Vol., %)	Ri (microns)
19977.82	D	96.611	3.3886	0.000	0.0053
24990.69	D	96.611	3.3886	0.000	0.0043
29987.29	D	96.611	3.3886	0.000	0.0036
34990.86	D	96.611	3.3886	0.000	0.0030
39988.24	D	96.611	3.3886	0.000	0.0027
44989.94	D	96.611	3.3886	0.000	0.0024
49988.38	D	96.611	3.3886	0.000	0.0021
54990.39	D	96.611	3.3886	0.000	0.0019
59985.05	D	96.611	3.3886	0.000	0.0018
46095.96	I	96.611	3.3886	----	----
35505.46	I	96.611	3.3886	----	----
27309.44	I	96.611	3.3886	----	----
21007.41	I	96.611	3.3886	----	----
16007.35	I	96.611	3.3886	----	----
12412.80	I	96.611	3.3886	----	----
9609.69	I	96.611	3.3886	----	----
7319.24	I	96.611	3.3886	----	----
5716.31	I	96.611	3.3886	----	----
4305.92	I	96.611	3.3886	----	----
3301.79	I	96.611	3.3886	----	----
2603.96	I	96.611	3.3886	----	----
2005.67	I	96.611	3.3886	----	----
1501.44	I	96.611	3.3886	----	----
1201.21	I	96.611	3.3886	----	----
901.65	I	96.611	3.3886	----	----
696.93	I	96.611	3.3886	----	----
503.10	I	96.611	3.3886	----	----
396.51	I	96.611	3.3886	----	----
303.34	I	96.470	3.5301	----	----
241.04	I	95.805	4.1946	----	----
192.35	I	95.220	4.7800	----	----
147.04	I	94.059	5.9410	----	----
114.11	I	92.470	7.5300	----	----
87.08	I	90.391	9.6090	----	----
67.83	I	87.353	12.647	----	----
53.09	I	83.435	16.565	----	----
32.80	I	72.314	27.686	----	----

Table (2.8.4-Continue)

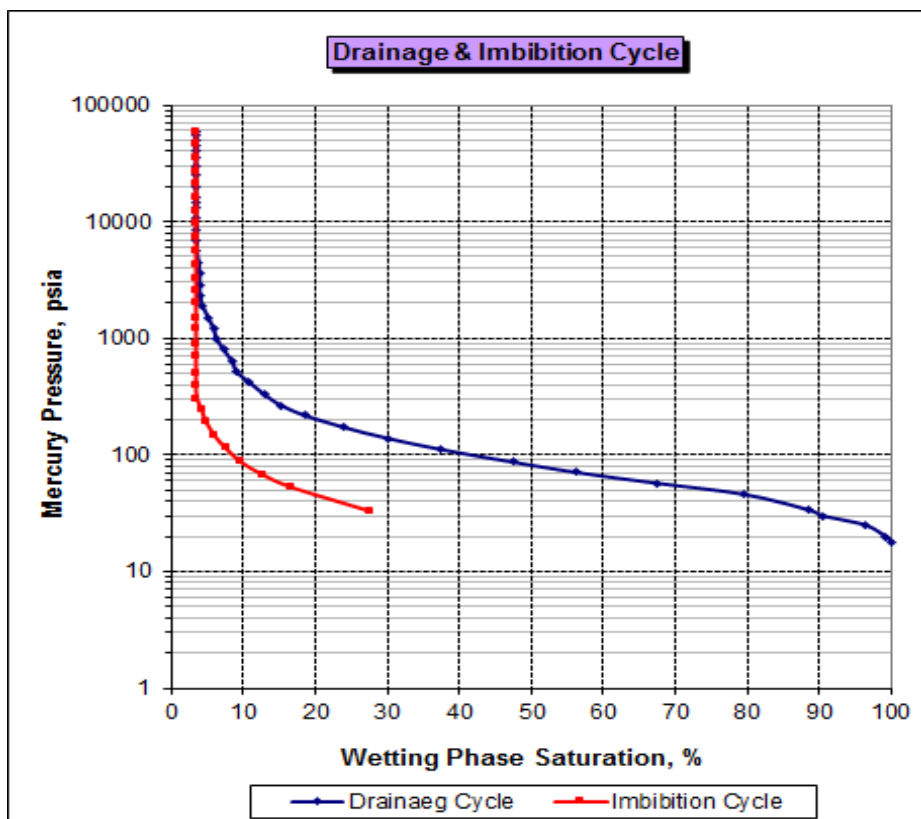


Figure (2.8.13)-Sample # 114

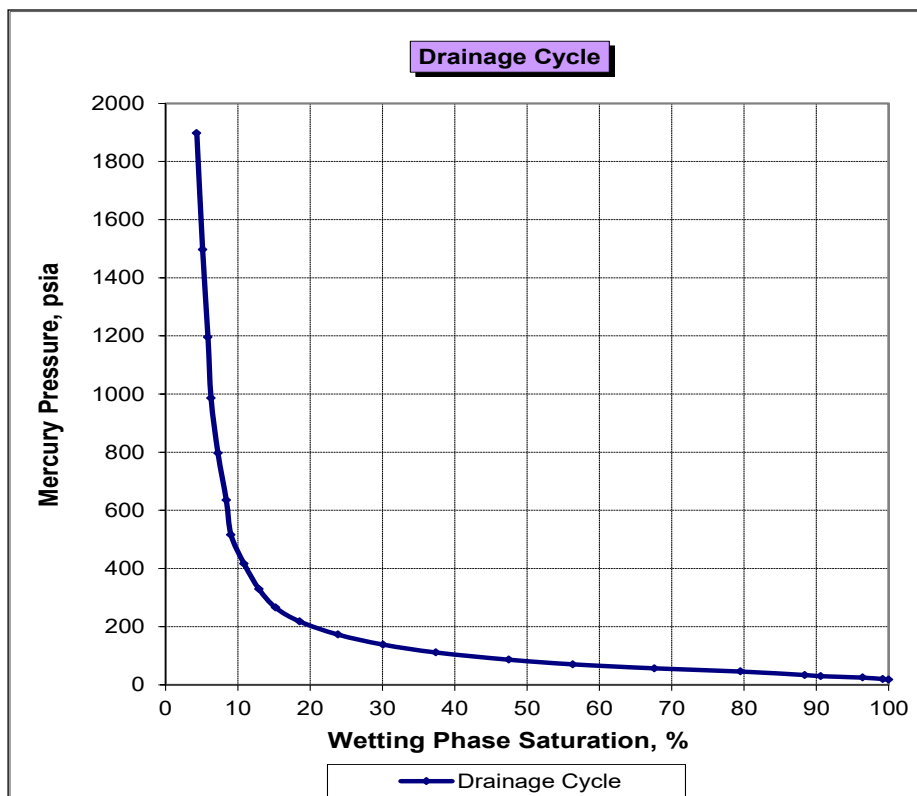


Figure (2.8.14)-Sample # 114

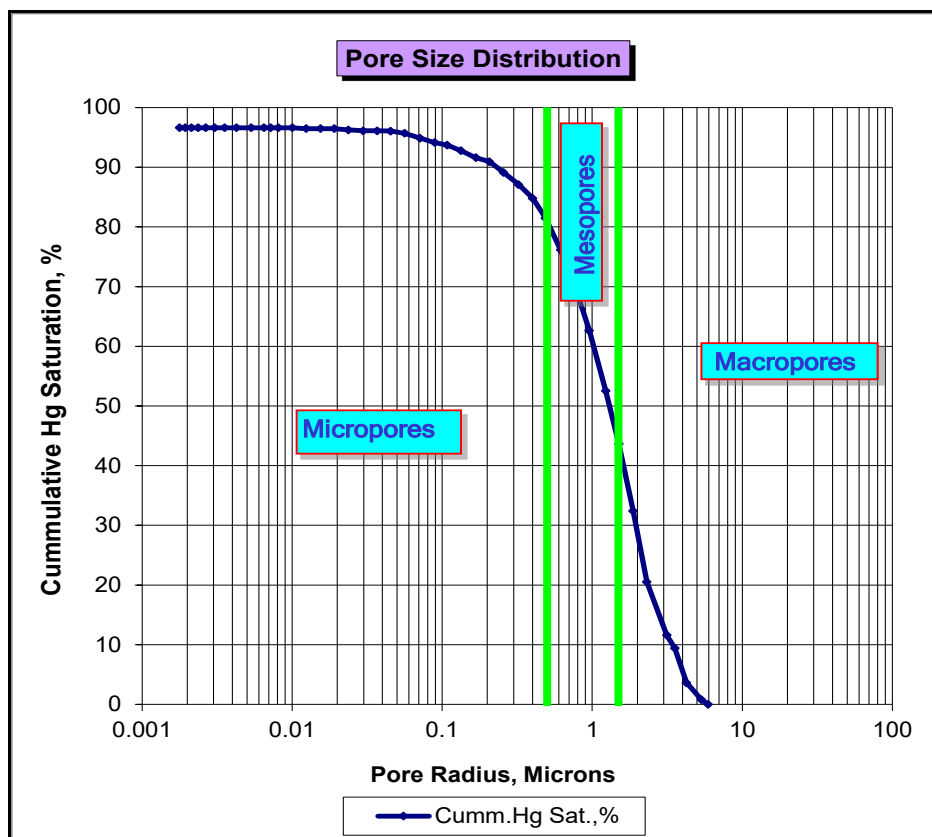


Figure (2.8.15)-Sample # 114

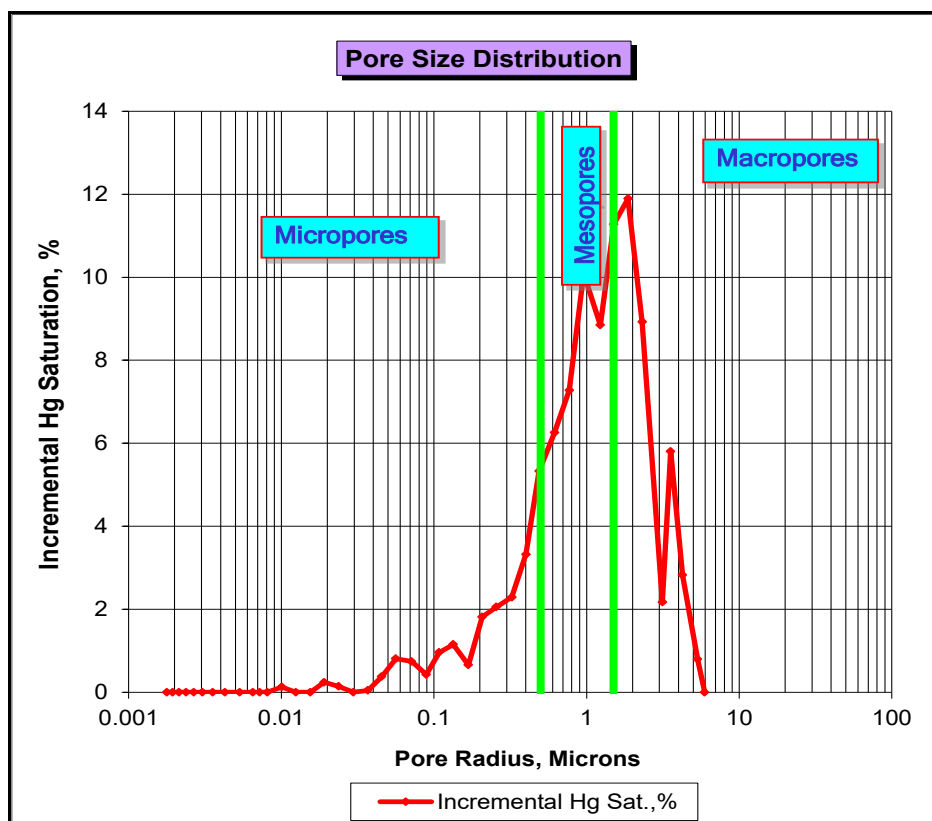


Figure (2.8.16)-Sample # 114

MERCURY INJECTION CAPILLARY PRESSURE AND PORE SIZE DISTRIBUTION

Client	: Mellitah Oil and Gas		
Well	: CE03-NC41		
Sample #	: 126		
Core Sample Data and Experimental Values			
Hg Pore volume, (cc)	: 1.48	Threshold Pressure, (psi)	: 15.981
Porosity, (%)	: 24.40	Sample Weight, (gm)	: 11.918
Air Perm., (mD)	: 39.40	Formation	: El-Garia
Sample Depth, (ft.in)	: 8483.10		

Pressure (psia)	Cycle	Cum. Hg Sat. (Pore Vol., %)	Wetting Phase (Sw) (Pore Vol., %)	Incr. Hg Sat. (Pore Vol., %)	Ri (microns)
15.9810	D	0.000	100.000	0.000	6.674
19.9797	D	0.7988	99.201	0.799	5.339
24.9626	D	2.877	97.123	2.079	4.273
30.0359	D	10.641	89.359	7.763	3.551
36.1052	D	14.780	85.220	4.140	2.954
46.8575	D	23.815	76.185	9.034	2.276
56.8833	D	32.041	67.959	8.227	1.875
71.8626	D	43.973	56.027	11.932	1.484
86.7351	D	52.365	47.635	8.392	1.230
112.1869	D	62.831	37.169	10.466	0.951
137.3855	D	69.779	30.221	6.948	0.776
173.6000	D	76.009	23.991	6.229	0.614
220.0336	D	80.993	19.007	4.984	0.485
267.5966	D	84.145	15.855	3.152	0.399
328.6202	D	86.671	13.329	2.526	0.325
420.7848	D	88.803	11.197	2.132	0.253
517.2198	D	90.158	9.8420	1.355	0.206
636.6895	D	91.269	8.7311	1.111	0.168
797.7077	D	92.027	7.9732	0.759	0.134
988.1240	D	92.758	7.2420	0.731	0.108
1194.411	D	93.472	6.5280	0.714	0.089
1500.535	D	94.094	5.9061	0.621	0.0711
1898.249	D	94.635	5.3652	0.541	0.0562
2342.754	D	95.171	4.8288	0.537	0.0455
2895.814	D	95.456	4.5445	0.284	0.0368
3592.975	D	95.769	4.2310	0.313	0.0297
4481.359	D	95.874	4.1257	0.105	0.0238
5582.126	D	96.362	3.6383	0.487	0.0191
6884.483	D	96.399	3.6014	0.037	0.0155
8584.064	D	96.399	3.6014	0.000	0.0124
10585.75	D	96.399	3.6014	0.000	0.0101
13183.02	D	96.399	3.6014	0.000	0.0081
14784.62	D	96.399	3.6014	0.000	0.0072
16381.54	D	96.399	3.6014	0.000	0.0065

Table (2.8.5)

Pressure (psia)	Cycle	Cum. Hg Sat. (Pore Vol., %)	Wetting Phase (Sw) (Pore Vol., %)	Incr. Hg Sat. (Pore Vol., %)	Ri (microns)
19981.32	D	96.399	3.6014	0.000	0.0053
24992.49	D	96.399	3.6014	0.000	0.0043
29988.52	D	96.399	3.6014	0.000	0.0036
34992.34	D	96.399	3.6014	0.000	0.0030
39989.38	D	96.399	3.6014	0.000	0.0027
44992.11	D	96.399	3.6014	0.000	0.0024
49987.25	D	96.399	3.6014	0.000	0.0021
54985.71	D	96.399	3.6014	0.000	0.0019
59984.75	D	96.399	3.6014	0.000	0.0018
46100.68	I	96.399	3.6014	----	----
35499.60	I	96.399	3.6014	----	----
27309.38	I	96.399	3.6014	----	----
21008.70	I	96.399	3.6014	----	----
16007.28	I	96.399	3.6014	----	----
12404.41	I	96.399	3.6014	----	----
9601.11	I	96.399	3.6014	----	----
7312.54	I	96.399	3.6014	----	----
5705.91	I	96.399	3.6014	----	----
4197.69	I	96.399	3.6014	----	----
3309.44	I	96.399	3.6014	----	----
2597.42	I	96.399	3.6014	----	----
2004.85	I	96.399	3.6014	----	----
1502.65	I	96.399	3.6014	----	----
1203.30	I	96.399	3.6014	----	----
904.571	I	96.399	3.6014	----	----
704.082	I	96.399	3.6014	----	----
498.616	I	96.399	3.6014	----	----
404.653	I	95.594	4.4062	----	----
301.181	I	95.468	4.5324	----	----
241.248	I	94.965	5.0347	----	----
191.285	I	94.227	5.7731	----	----
147.502	I	93.442	6.558	----	----
113.684	I	92.202	7.798	----	----
87.3646	I	90.507	9.493	----	----
67.3335	I	88.743	11.257	----	----
54.5585	I	86.286	13.714	----	----
32.9306	I	79.975	20.025	----	----

Table (2.8.5-Continue)

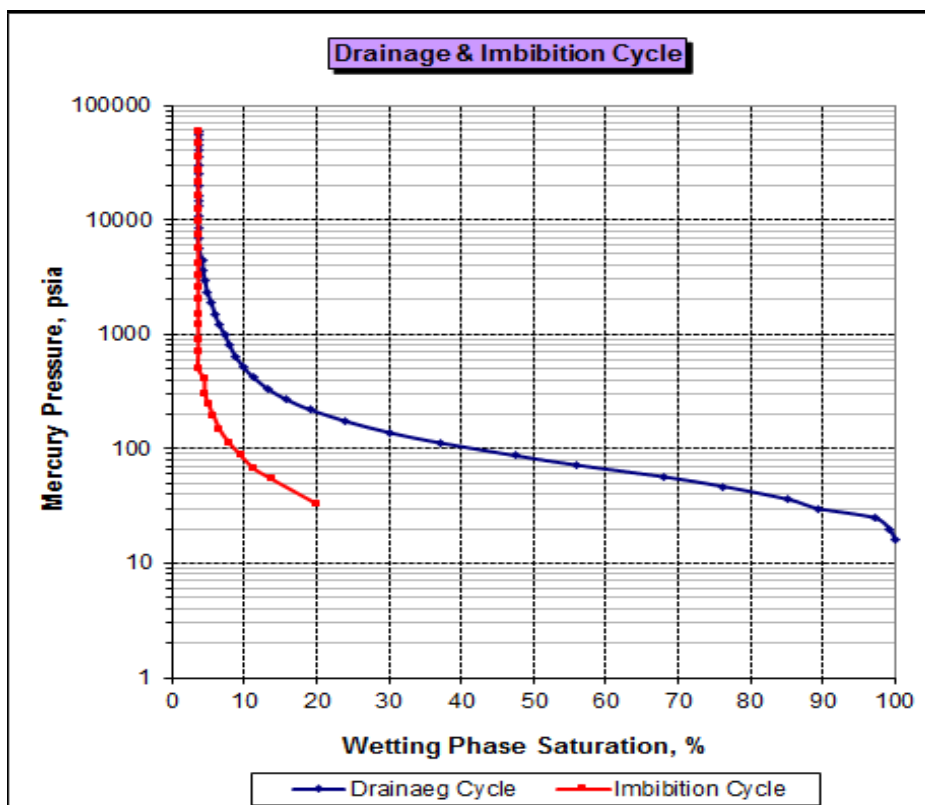


Figure (2.8.17)-Sample # 126

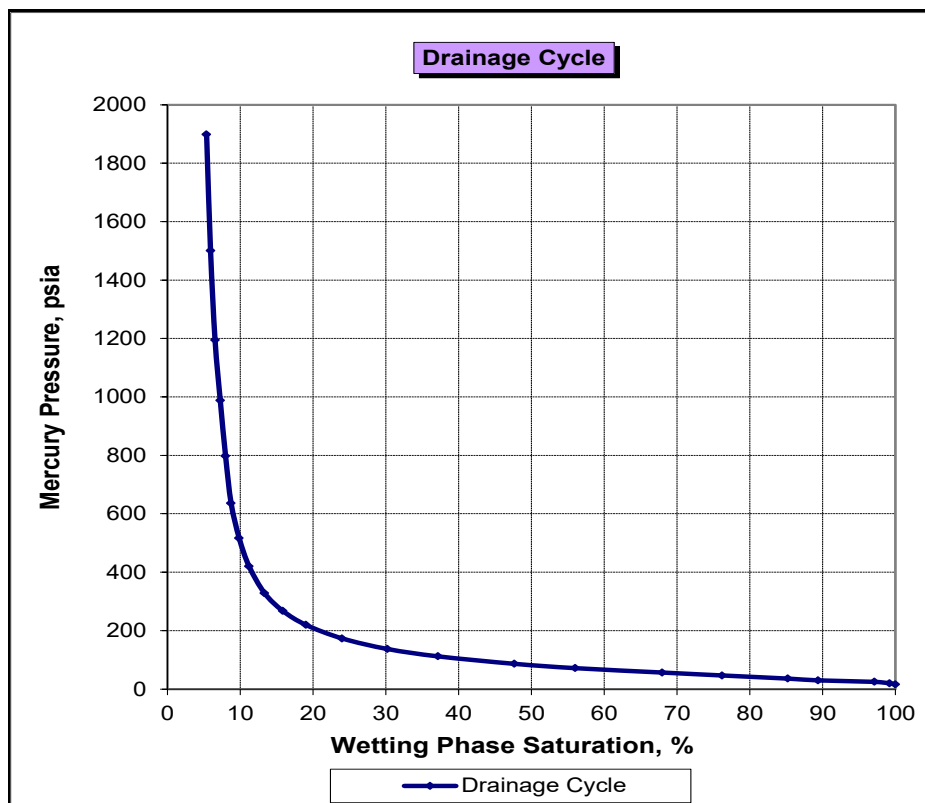


Figure (2.8.18)-Sample # 126

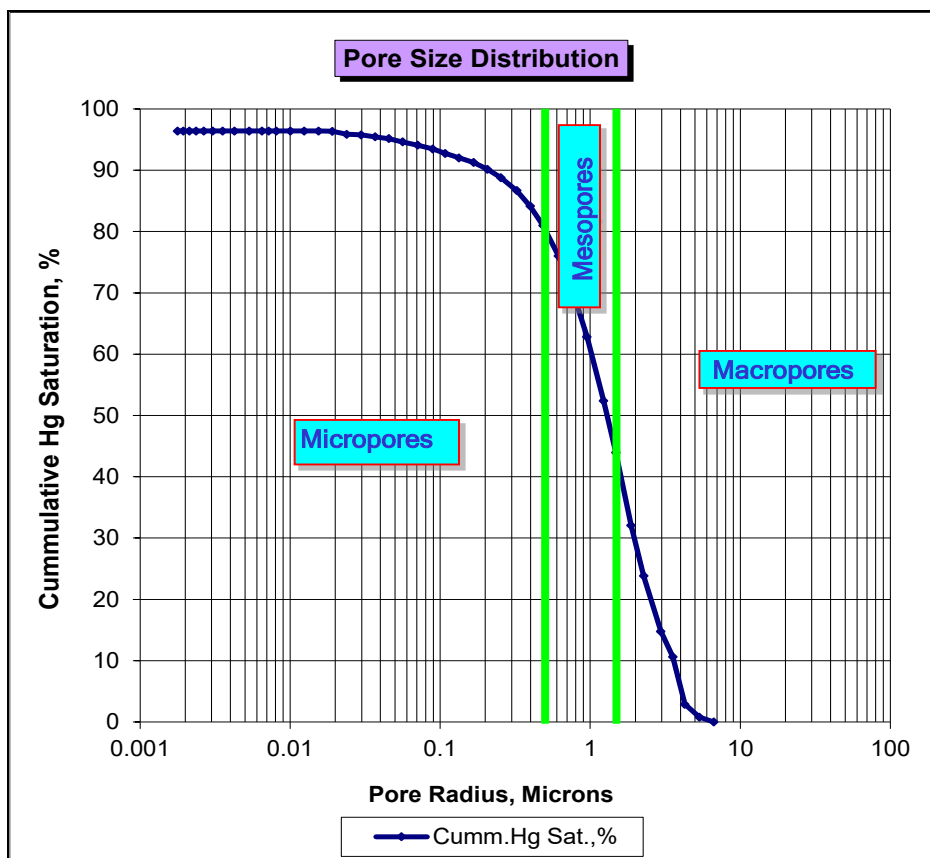


Figure (2.8.19)-Sample # 126

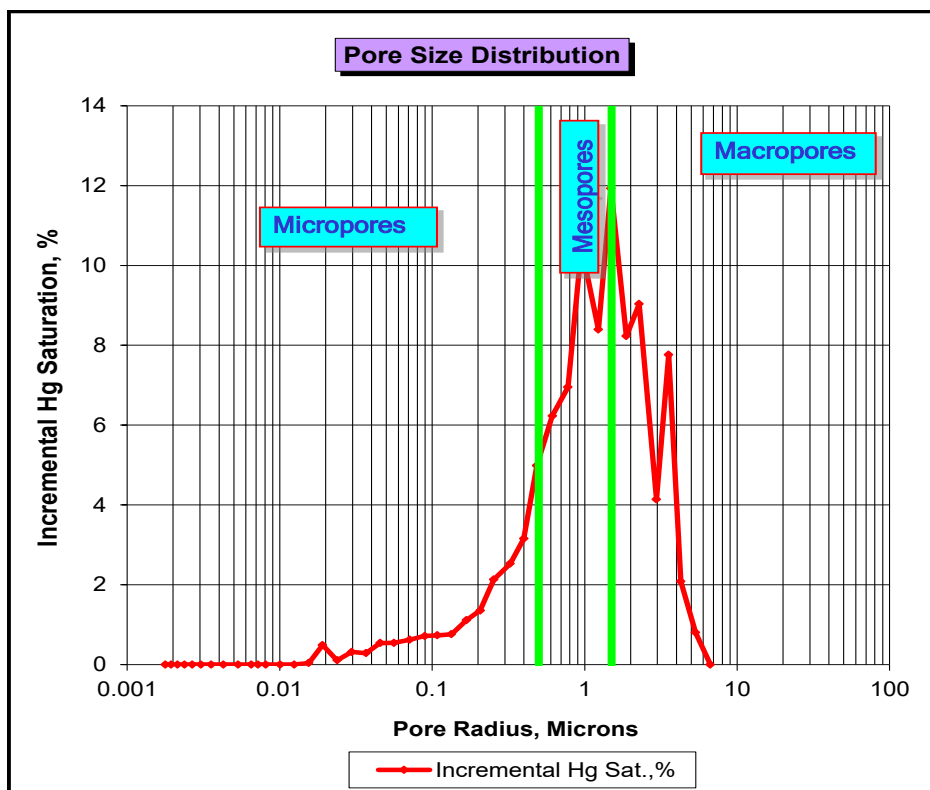


Figure (2.8.20)-Sample # 126

MERCURY INJECTION CAPILLARY PRESSURE AND PORE SIZE DISTRIBUTION

Client	: Mellitah Oil and Gas		
Well	: CE03-NC41		
Sample #	: 164		
Core Sample Data and Experimental Values			
Hg Pore volume, (cc)	: 1.42	Threshold Pressure, (psi)	: 15.976
Porosity, (%)	: 16.3	Sample Weight, (gm)	: 14.182
Air Perm., (mD)	: 4.89	Formation	: El-Garia
Sample Depth, (ft.in)	: 8521.04		

Pressure (psia)	Cycle	Cum. Hg Sat. (Pore Vol., %)	Wetting Phase (Sw) (Pore Vol., %)	Incr. Hg Sat. (Pore Vol., %)	Ri (microns)
15.9761	D	0.000	100.000	0.000	6.676
19.9770	D	0.4995	99.501	0.499	5.339
25.0578	D	1.340	98.660	0.841	4.257
30.0400	D	2.729	97.271	1.389	3.551
36.6866	D	3.721	96.279	0.992	2.907
46.3323	D	6.629	93.371	2.908	2.302
57.4291	D	15.818	84.182	9.189	1.857
72.1511	D	29.932	70.068	14.114	1.478
86.9926	D	40.895	59.105	10.963	1.226
112.1421	D	52.342	47.658	11.447	0.951
137.2489	D	60.345	39.655	8.004	0.777
172.9905	D	67.830	32.170	7.484	0.617
217.1809	D	74.230	25.770	6.400	0.491
268.0787	D	78.862	21.138	4.632	0.398
329.4727	D	82.508	17.492	3.646	0.324
417.8414	D	86.145	13.855	3.637	0.255
517.9464	D	88.671	11.329	2.525	0.206
635.7712	D	90.500	9.5001	1.829	0.168
797.7725	D	91.968	8.0320	1.468	0.134
985.8135	D	93.087	6.9131	1.119	0.108
1198.326	D	93.892	6.1081	0.806	0.089
1498.822	D	94.474	5.5260	0.582	0.0712
1898.728	D	95.228	4.7721	0.753	0.0562
2344.254	D	95.831	4.1694	0.603	0.0455
2894.932	D	96.373	3.6273	0.542	0.0368
3592.686	D	96.551	3.4494	0.178	0.0297
4500.724	D	96.669	3.3314	0.118	0.0237
5581.762	D	96.820	3.1795	0.152	0.0191
6883.903	D	96.820	3.1795	0.000	0.0155
8585.608	D	96.820	3.1795	0.000	0.0124
10582.82	D	96.820	3.1795	0.000	0.0101
13184.05	D	96.820	3.1795	0.000	0.0081
14783.38	D	96.820	3.1795	0.000	0.0072
16378.16	D	96.820	3.1795	0.000	0.0065

Table (2.8.6)

Pressure (psia)	Cycle	Cum. Hg Sat. (Pore Vol., %)	Wetting Phase (Sw) (Pore Vol., %)	Incr. Hg Sat. (Pore Vol., %)	Ri (microns)
19981.88	D	96.820	3.1795	0.000	0.0053
24991.48	D	96.820	3.1795	0.000	0.0043
29988.70	D	96.820	3.1795	0.000	0.0036
34990.07	D	96.820	3.1795	0.000	0.0030
39990.07	D	96.820	3.1795	0.000	0.0027
44985.92	D	96.820	3.1795	0.000	0.0024
49989.11	D	96.820	3.1795	0.000	0.0021
54986.33	D	96.820	3.1795	0.000	0.0019
59985.91	D	96.820	3.1795	0.000	0.0018
46090.95	I	96.820	3.1795	----	----
35502.80	I	96.820	3.1795	----	----
27307.89	I	96.820	3.1795	----	----
21008.73	I	96.820	3.1795	----	----
16009.61	I	96.820	3.1795	----	----
12409.84	I	96.820	3.1795	----	----
9605.197	I	96.820	3.1795	----	----
7303.010	I	96.820	3.1795	----	----
5707.100	I	96.820	3.1795	----	----
4105.680	I	96.820	3.1795	----	----
3297.938	I	96.820	3.1795	----	----
2598.737	I	96.820	3.1795	----	----
2006.210	I	96.820	3.1795	----	----
1497.030	I	96.820	3.1795	----	----
1200.341	I	96.688	3.3116	----	----
901.694	I	95.569	4.4310	----	----
701.825	I	95.305	4.6948	----	----
500.927	I	94.641	5.3594	----	----
401.778	I	94.117	5.8826	----	----
300.376	I	93.145	6.8549	----	----
244.109	I	92.244	7.7556	----	----
192.695	I	90.810	9.1901	----	----
147.547	I	88.193	11.807	----	----
112.544	I	83.636	16.364	----	----
87.6260	I	76.139	23.861	----	----
68.2421	I	63.506	36.494	----	----
53.0652	I	49.619	50.381	----	----

Table (2.8.6-Continue)

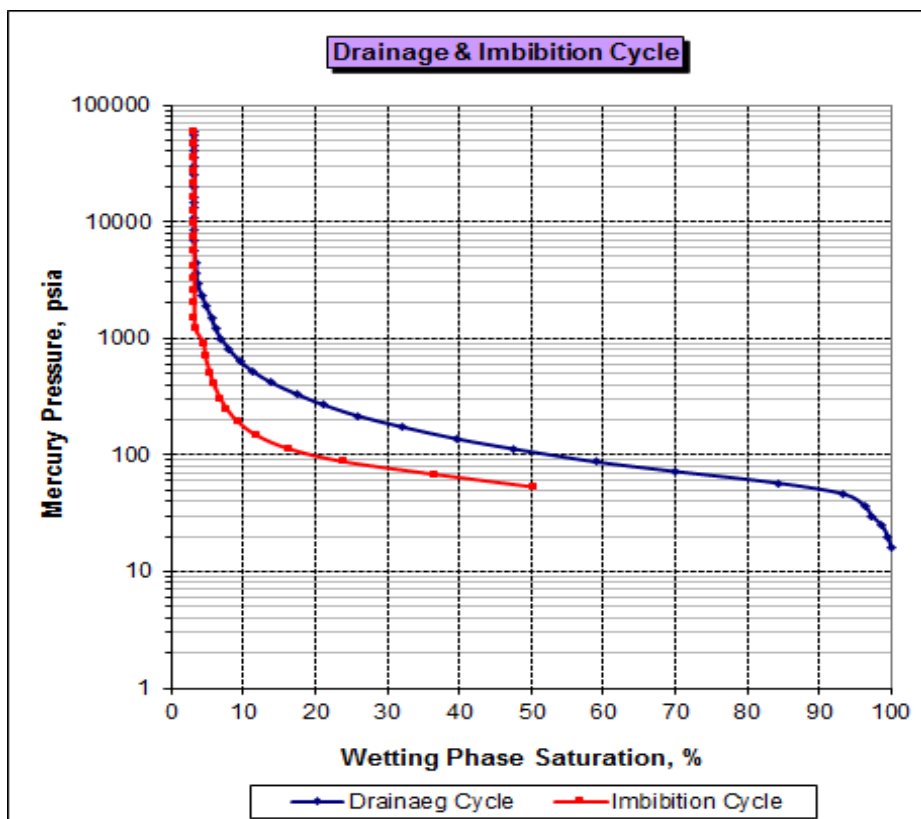


Figure (2.8.21)-Sample # 164

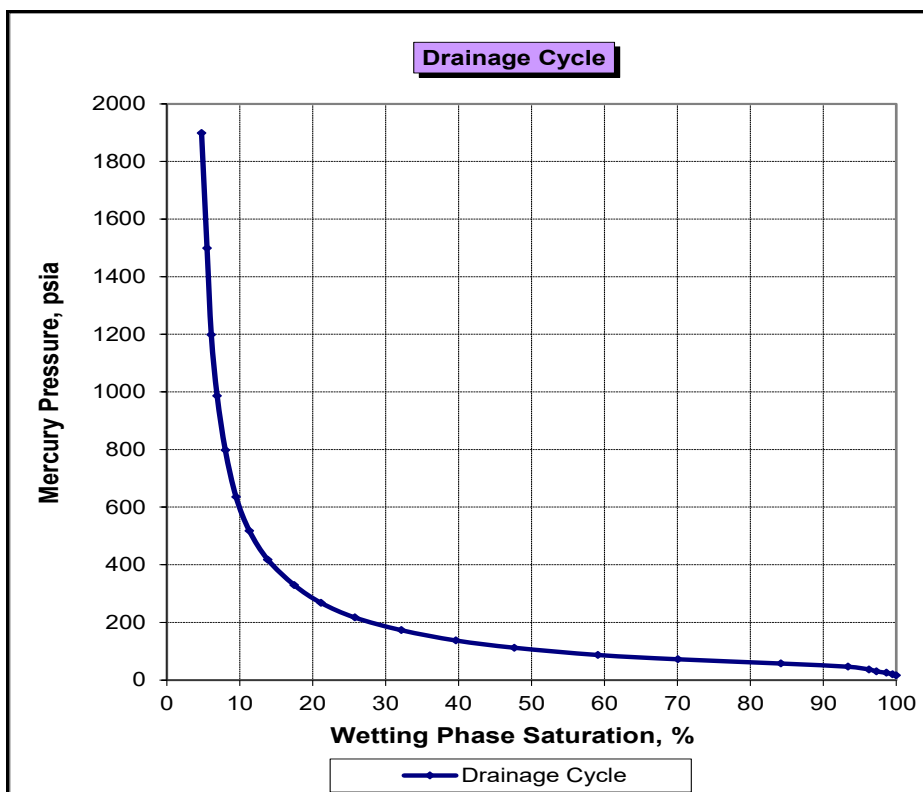


Figure (2.8.22)-Sample # 164

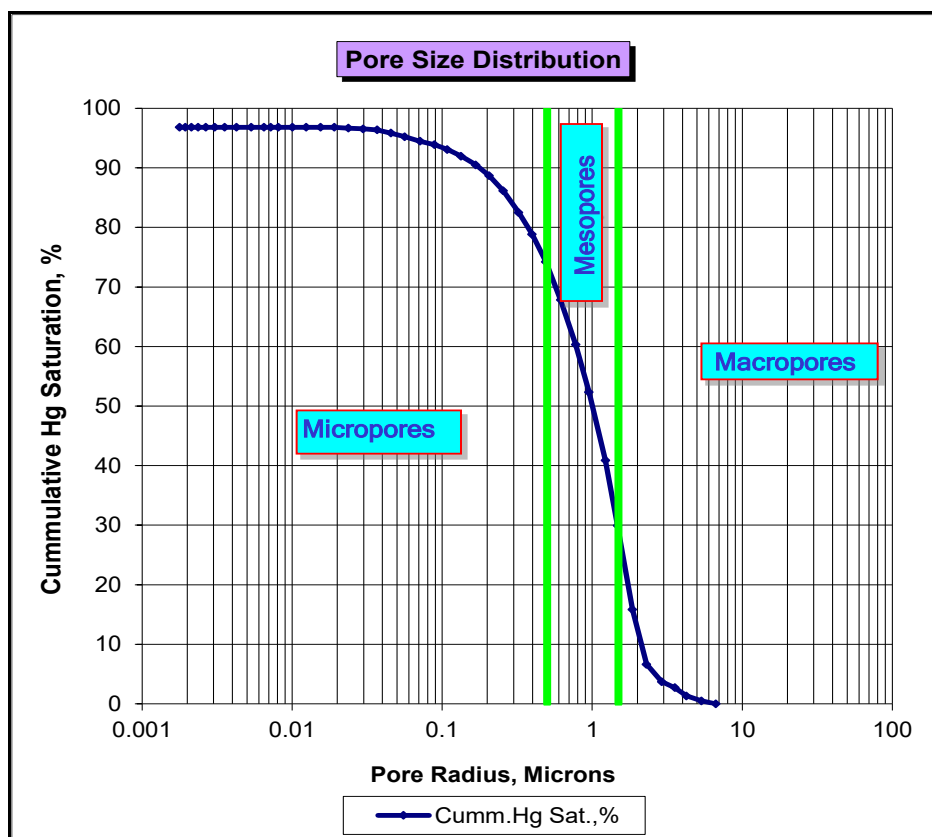


Figure (2.8.23)-Sample # 164

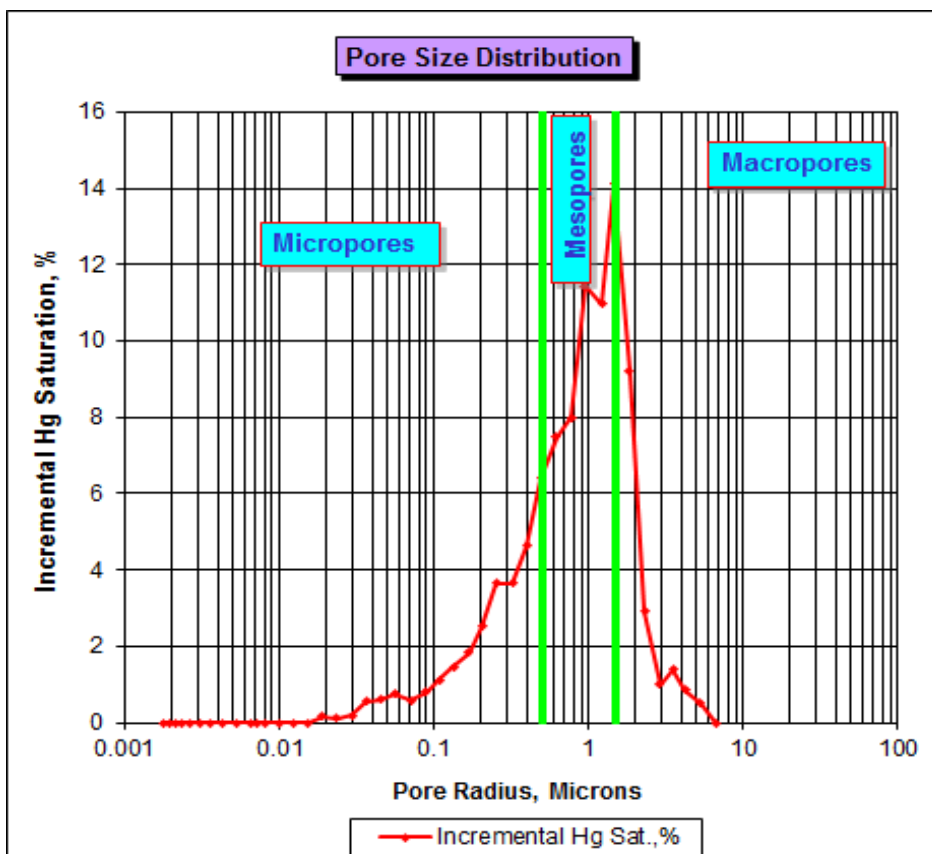


Figure (2.8.24)-Sample # 164

MERCURY INJECTION CAPILLARY PRESSURE AND PORE SIZE DISTRIBUTION

Client	: Mellitah Oil and Gas		
Well	: CE03-NC41		
Sample #	: 227		
Core Sample Data and Experimental Values			
Hg Pore volume, (cc)	: 1.14	Threshold Pressure, (psi)	: 13.015
Porosity, (%)	: 25.9	Sample Weight, (gm)	: 10.270
Air Perm., (mD)	: 126.0	Formation	: El-Garia
Sample Depth, (ft.in)	: 8585.02		

Pressure (psia)	Cycle	Cum. Hg Sat. (Pore Vol., %)	Wetting Phase (Sw) (Pore Vol., %)	Incr. Hg Sat. (Pore Vol., %)	Ri (microns)
13.0150	D	0.000	100.000	0.000	8.195
15.9629	D	3.699	96.301	3.699	6.682
19.9698	D	12.1734	87.827	8.475	5.341
28.0451	D	26.639	73.361	14.466	3.803
34.0173	D	33.979	66.021	7.340	3.136
40.6352	D	37.468	62.532	3.489	2.625
44.5689	D	40.058	59.942	2.590	2.393
53.1414	D	43.653	56.347	3.595	2.007
71.7656	D	50.160	49.840	6.508	1.486
86.1792	D	53.704	46.296	3.544	1.238
111.5807	D	58.183	41.817	4.479	0.956
136.4349	D	61.853	38.147	3.670	0.782
173.0419	D	65.306	34.694	3.453	0.616
215.6248	D	69.091	30.909	3.785	0.495
270.7241	D	72.495	27.505	3.404	0.394
330.5424	D	75.600	24.400	3.105	0.323
421.3279	D	79.009	20.991	3.409	0.253
515.6489	D	81.834	18.166	2.825	0.207
635.6830	D	84.075	15.925	2.242	0.168
797.8752	D	86.032	13.968	1.957	0.134
986.9291	D	87.856	12.144	1.824	0.108
1196.999	D	88.872	11.128	1.015	0.089
1498.226	D	90.155	9.845	1.283	0.0712
1898.706	D	91.039	8.961	0.884	0.0562
2343.941	D	91.694	8.3060	0.655	0.0455
2893.605	D	92.241	7.7594	0.547	0.0369
3595.121	D	92.870	7.1303	0.629	0.0297
4492.029	D	93.178	6.8220	0.308	0.0237
5581.426	D	93.373	6.6272	0.195	0.0191
6883.865	D	93.502	6.4982	0.129	0.0155
8582.557	D	93.653	6.3467	0.152	0.0124
10583.37	D	93.708	6.2919	0.055	0.0101
13184.11	D	93.708	6.2919	0.000	0.0081
14782.29	D	93.708	6.2919	0.000	0.0072

Table (2.8.7)

Pressure (psia)	Cycle	Cum. Hg Sat. (Pore Vol., %)	Wetting Phase (Sw) (Pore Vol., %)	Incr. Hg Sat. (Pore Vol., %)	Ri (microns)
16381.59	D	93.708	6.2919	0.000	0.0065
19982.64	D	93.708	6.2919	0.000	0.0053
24990.91	D	93.708	6.2919	0.000	0.0043
29991.66	D	93.708	6.2919	0.000	0.0036
34991.18	D	93.708	6.2919	0.000	0.0030
39990.72	D	93.708	6.2919	0.000	0.0027
44990.00	D	93.708	6.2919	0.000	0.0024
49986.76	D	93.708	6.2919	0.000	0.0021
54988.12	D	93.708	6.2919	0.000	0.0019
59987.02	I	93.708	6.2919	0.000	0.0018
46103.52	I	93.708	6.2919	----	----
35499.32	I	93.708	6.2919	----	----
27308.23	I	93.708	6.2919	----	----
21001.63	I	93.708	6.2919	----	----
16007.77	I	93.708	6.2919	----	----
12413.60	I	93.708	6.2919	----	----
9605.81	I	93.708	6.2919	----	----
7317.83	I	93.708	6.2919	----	----
5694.82	I	93.708	6.2919	----	----
4082.92	I	93.708	6.2919	----	----
3308.72	I	93.708	6.2919	----	----
2605.36	I	93.708	6.2919	----	----
1995.48	I	93.708	6.2919	----	----
1503.48	I	93.708	6.2919	----	----
1198.05	I	93.708	6.2919	----	----
902.560	I	93.708	6.2919	----	----
701.781	I	93.708	6.2919	----	----
503.501	I	93.450	6.5503	----	----
404.432	I	92.687	7.3128	----	----
303.894	I	91.097	8.9026	----	----
243.125	I	88.880	11.1203	----	----
191.798	I	85.852	14.1484	----	----
147.002	I	82.157	17.843	----	----
113.371	I	76.162	23.838	----	----
88.9722	I	66.792	33.208	----	----
67.9834	I	51.800	48.200	----	----

Table (2.8.7-Continue)

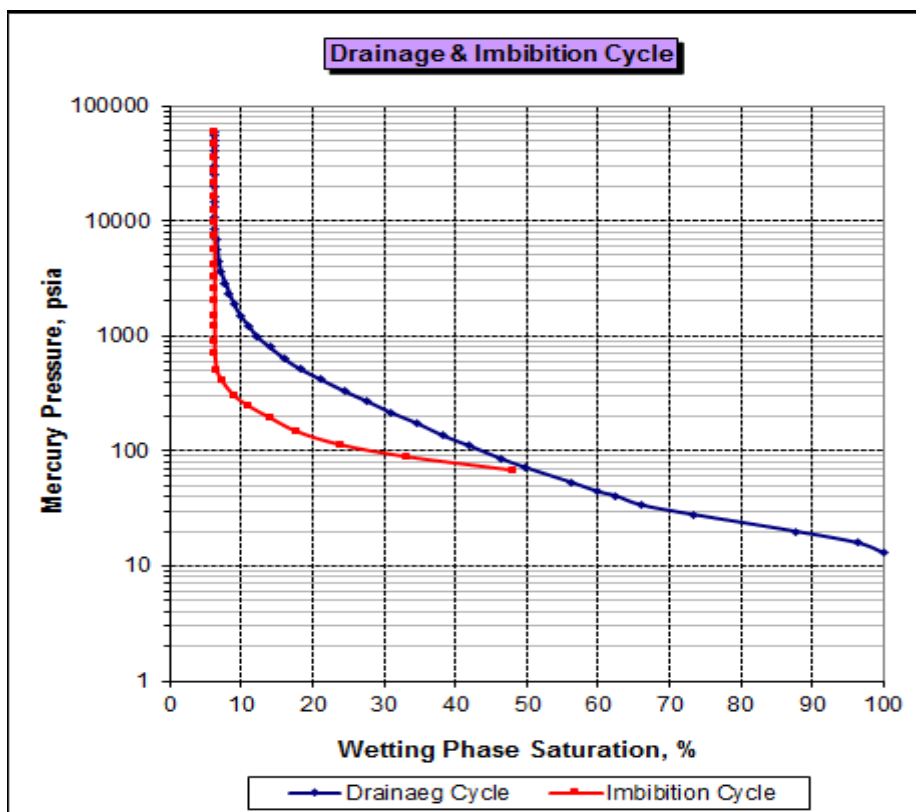


Figure (2.8.25)-Sample # 227

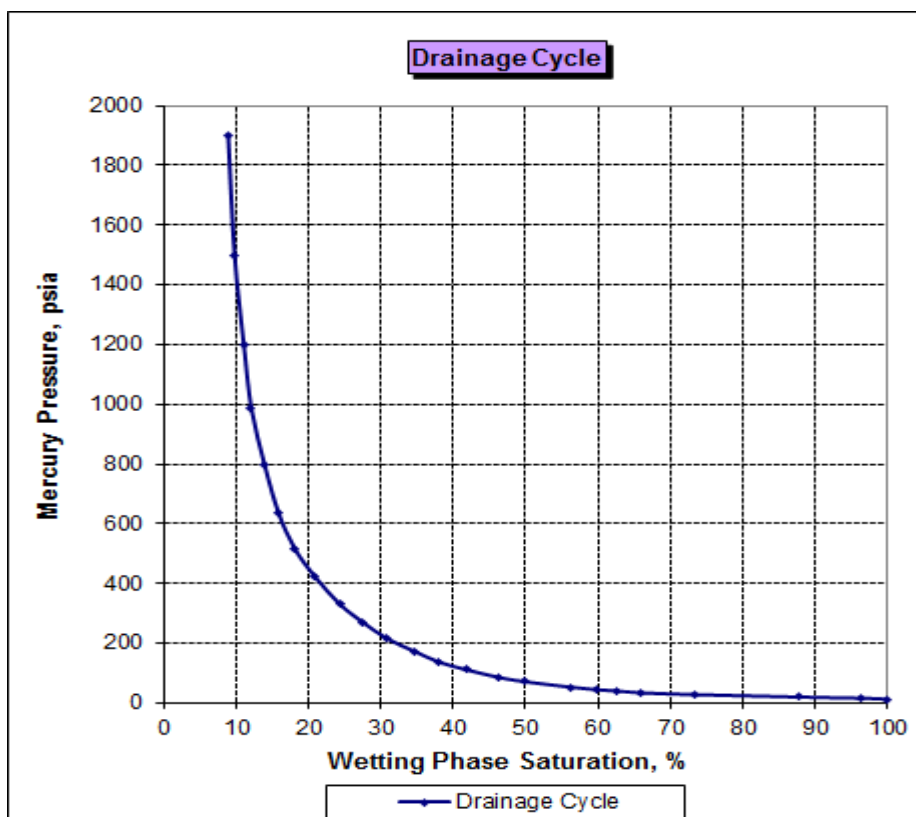


Figure (2.8.26)-Sample # 227

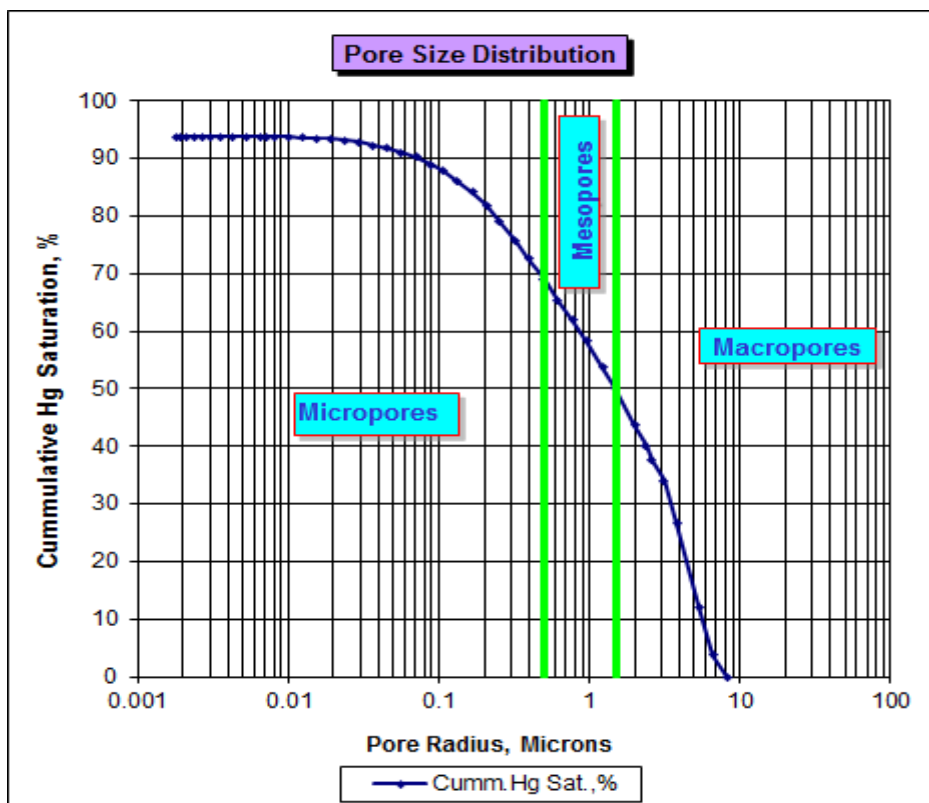


Figure (2.8.27)-Sample # 227

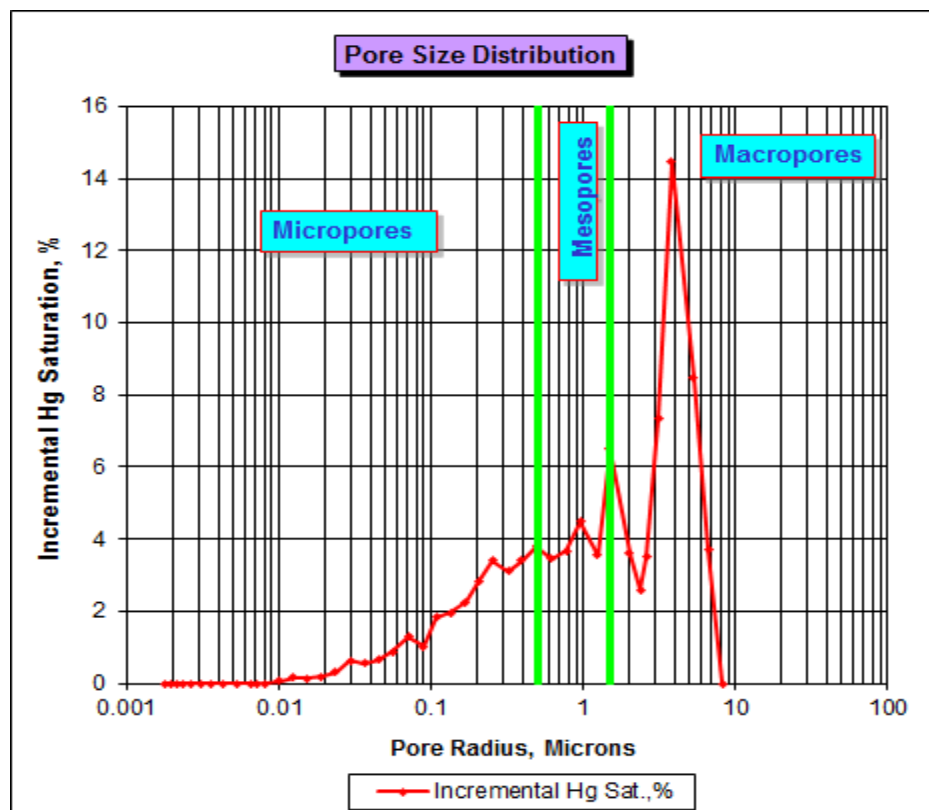


Figure (2.8.28)-Sample # 227

MERCURY INJECTION CAPILLARY PRESSURE AND PORE SIZE DISTRIBUTION

Client	: Mellitah Oil and Gas		
Well	: CE03-NC41		
Sample #	: 233		
Core Sample Data and Experimental Values			
Hg Pore volume, (cc)	: 1.20	Threshold Pressure, (psi)	: 13.013
Porosity, (%)	: 22.60	Sample Weight, (gm)	: 12.153
Air Perm., (mD)	: 51.70	Formation	: El-Garia
Sample Depth, (ft.in)	: 8591.04		

Pressure (psia)	Cycle	Cum. Hg Sat. (Pore Vol., %)	Wetting Phase (Sw) (Pore Vol., %)	Incr. Hg Sat. (Pore Vol., %)	Ri (microns)
13.0131	D	0.000	100.000	0.000	8.197
15.9668	D	0.495	99.505	0.495	6.680
19.9715	D	1.9160	98.084	1.421	5.341
25.0535	D	9.023	90.977	7.107	4.257
34.0151	D	19.469	80.531	10.44	3.136
38.8497	D	23.075	76.925	3.606	2.746
46.4241	D	26.471	73.529	3.395	2.298
52.4797	D	30.652	69.348	4.182	2.032
71.7429	D	40.831	59.169	10.17	1.487
86.9743	D	48.124	51.876	7.293	1.226
112.4813	D	55.897	44.103	7.773	0.948
136.7918	D	61.956	38.044	6.059	0.780
172.7866	D	67.611	32.389	5.656	0.617
218.4804	D	72.381	27.619	4.769	0.488
266.9107	D	76.384	23.616	4.003	0.400
330.9993	D	79.644	20.356	3.260	0.322
418.6716	D	83.371	16.629	3.727	0.255
515.7546	D	85.423	14.577	2.052	0.207
636.7894	D	87.285	12.715	1.862	0.167
796.5996	D	88.579	11.421	1.295	0.134
987.4589	D	89.680	10.320	1.100	0.108
1199.589	D	90.490	9.5100	0.810	0.089
1498.811	D	91.671	8.3291	1.181	0.0712
1897.045	D	92.456	7.5442	0.785	0.0562
2346.827	D	93.078	6.9224	0.621	0.0454
2896.018	D	93.736	6.2639	0.659	0.0368
3594.401	D	94.335	5.6652	0.599	0.0297
4486.467	D	94.532	5.4676	0.198	0.0238
5584.718	D	95.067	4.9330	0.535	0.0191
6883.784	D	95.155	4.8453	0.088	0.0155
8584.152	D	95.341	4.6586	0.187	0.0124
10583.19	D	95.571	4.4289	0.230	0.0101
13184.21	D	95.571	4.4289	0.000	0.0081
14780.30	D	95.571	4.4289	0.000	0.0072

Table (2.8.8)

Pressure (psia)	Cycle	Cum. Hg Sat. (Pore Vol., %)	Wetting Phase (Sw) (Pore Vol., %)	Incr. Hg Sat. (Pore Vol., %)	Ri (microns)
16381.42	D	95.571	4.4289	0.000	0.0065
19981.13	D	95.571	4.4289	0.000	0.0053
24991.82	D	95.571	4.4289	0.000	0.0043
29988.32	D	95.571	4.4289	0.000	0.0036
34992.86	D	95.571	4.4289	0.000	0.0030
39986.20	D	95.571	4.4289	0.000	0.0027
44989.69	D	95.571	4.4289	0.000	0.0024
49986.75	D	95.571	4.4289	0.000	0.0021
54985.06	D	95.571	4.4289	0.000	0.0019
59987.21	D	95.571	4.4289	0.000	0.0018
46100.95	I	95.571	4.4289	----	----
35501.60	I	95.571	4.4289	----	----
27306.65	I	95.571	4.4289	----	----
21007.24	I	95.571	4.4289	----	----
16006.78	I	95.571	4.4289	----	----
12413.85	I	95.571	4.4289	----	----
9610.734	I	95.571	4.4289	----	----
7310.027	I	95.571	4.4289	----	----
5707.744	I	95.571	4.4289	----	----
4308.364	I	95.571	4.4289	----	----
3299.262	I	95.571	4.4289	----	----
2594.121	I	95.571	4.4289	----	----
2004.064	I	95.571	4.4289	----	----
1502.348	I	95.571	4.4289	----	----
1204.905	I	95.571	4.4289	----	----
904.5725	I	95.571	4.4289	----	----
701.4331	I	95.571	4.4289	----	----
503.2436	I	95.571	4.4289	----	----
400.6436	I	95.175	4.8252	----	----
301.9332	I	94.066	5.9340	----	----
241.8082	I	92.619	7.3812	----	----
192.3423	I	90.477	9.5225	----	----
147.1289	I	87.104	12.896	----	----
113.2075	I	80.754	19.246	----	----
87.4494	I	67.111	32.889	----	----

Table (2.8.8-Continue)

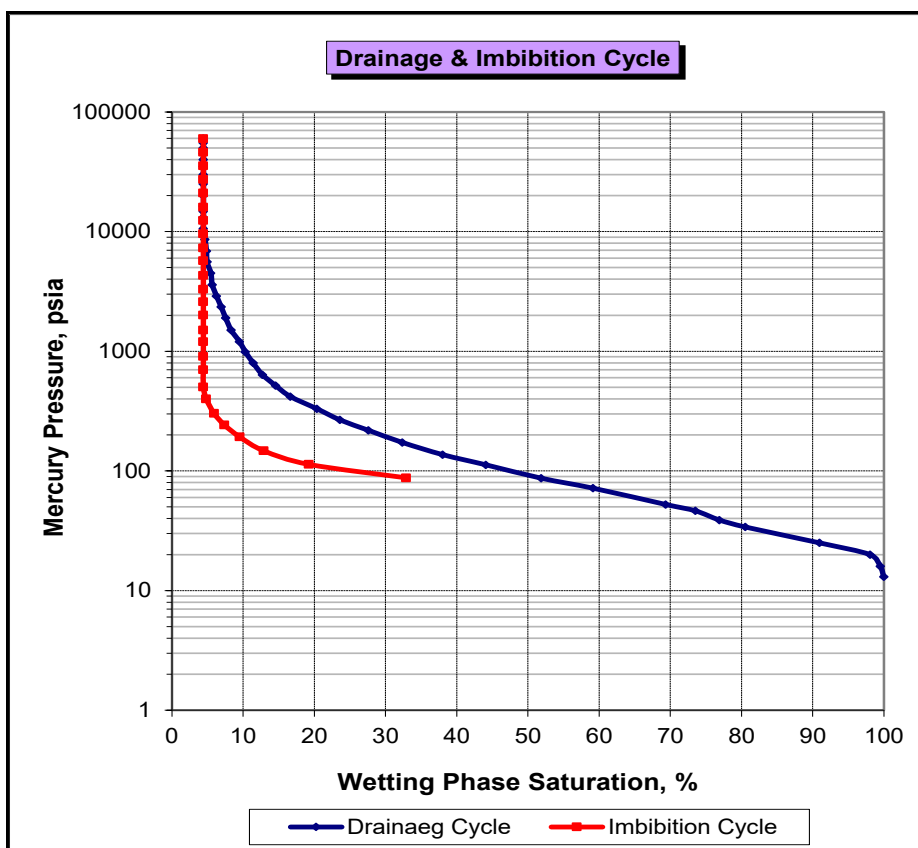


Figure (2.8.29)-Sample # 233

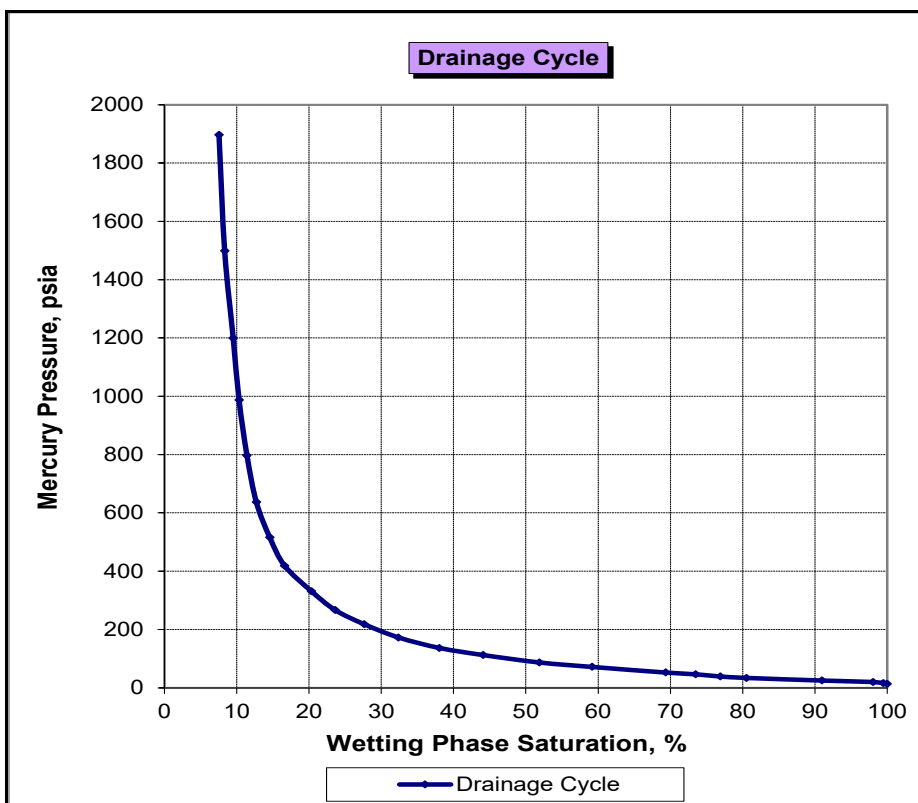


Figure (2.8.30)-Sample # 233

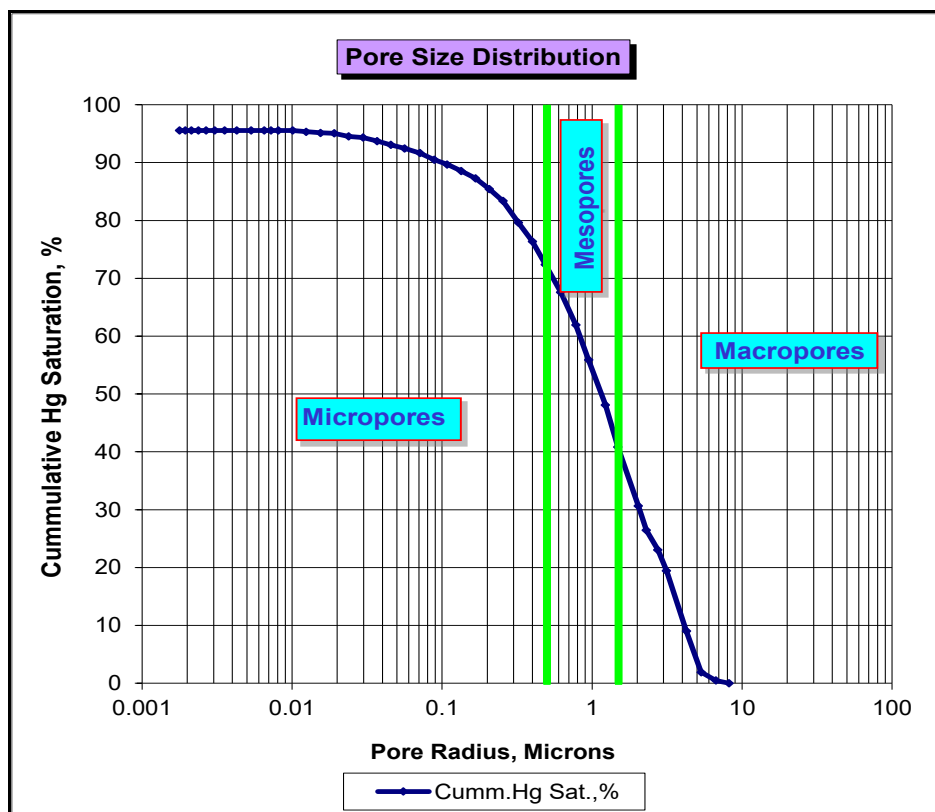


Figure (2.8.31)-Sample # 233

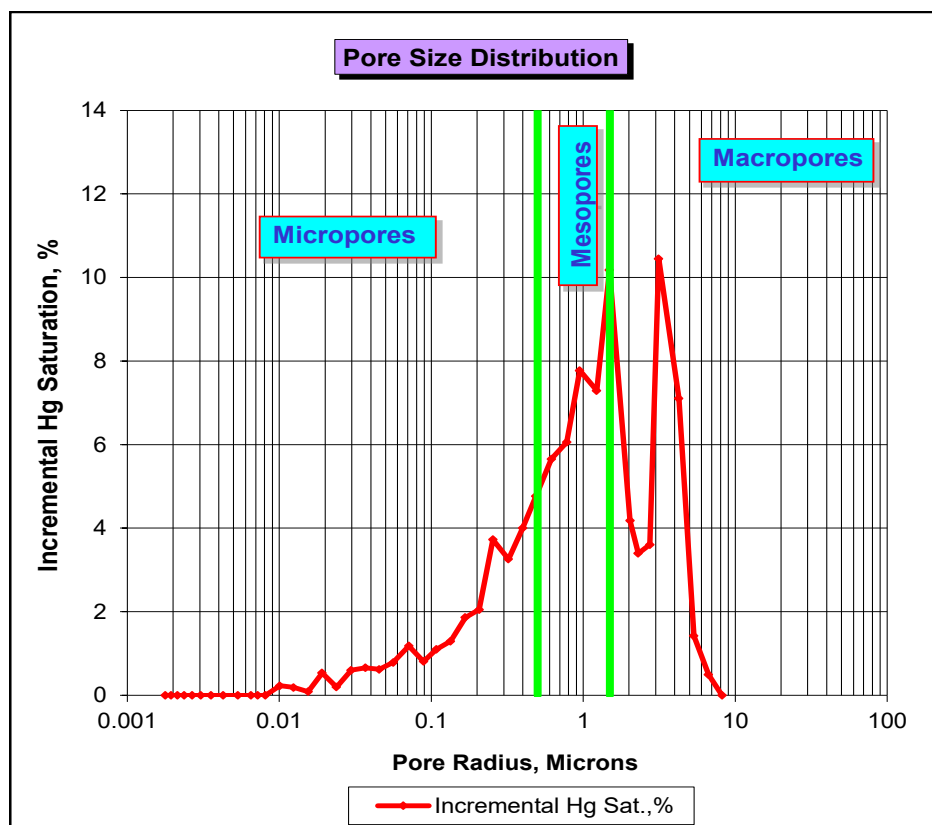


Figure (2.8.32)-Sample # 233

MERCURY INJECTION CAPILLARY PRESSURE AND PORE SIZE DISTRIBUTION

Client	: Mellitah Oil and Gas		
Well	: CE03-NC41		
Sample #	: 275		
Core Sample Data and Experimental Values			
Hg Pore volume, (cc)	: 1.24	Threshold Pressure, (psi)	: 5.518
Porosity, (%)	: 24.20	Sample Weight, (gm)	: 11.245
Air Perm., (mD)	: 225.0	Formation	: El-Garia
Sample Depth, (ft.in)	: 8636.06		

Pressure (psia)	Cycle	Cum. Hg Sat. (Pore Vol., %)	Wetting Phase (Sw) (Pore Vol., %)	Incr. Hg Sat. (Pore Vol., %)	Ri (microns)
5.51837	D	0.000	100.000	0.000	19.33
6.02204	D	2.381	97.619	2.381	17.71
7.49628	D	9.501	90.499	7.120	14.22
8.52307	D	13.936	86.064	4.435	12.51
10.5116	D	22.003	77.997	8.067	10.15
13.0296	D	30.276	69.724	8.273	8.186
15.9686	D	37.877	62.123	7.601	6.679
19.9830	D	44.711	55.288	6.835	5.338
25.0642	D	50.421	49.579	5.710	4.256
36.0323	D	54.427	45.573	4.006	2.960
37.2271	D	55.101	44.899	0.674	2.865
46.2165	D	56.225	43.775	1.124	2.308
53.2718	D	57.237	42.763	1.012	2.002
63.2335	D	58.689	41.311	1.452	1.687
87.0782	D	61.339	38.661	2.650	1.225
112.2003	D	63.746	36.254	2.407	0.951
140.7694	D	65.876	34.124	2.130	0.758
169.4128	D	67.647	32.353	1.771	0.630
200.8776	D	68.656	31.344	1.009	0.531
220.0233	D	69.295	30.705	0.639	0.485
326.6609	D	71.607	28.393	2.312	0.327
412.0470	D	73.226	26.774	1.618	0.259
515.9117	D	75.178	24.822	1.952	0.207
640.4840	D	77.869	22.131	2.691	0.167
796.7731	D	80.012	19.988	2.143	0.134
986.9216	D	81.985	18.015	1.974	0.108
1200.322	D	83.019	16.981	1.033	0.089
1496.785	D	84.268	15.732	1.249	0.0713
1897.337	D	85.563	14.437	1.295	0.0562
2344.444	D	86.591	13.409	1.028	0.0455
2894.879	D	87.256	12.744	0.665	0.0368
3592.099	D	87.837	12.162	0.582	0.0297
4495.186	D	88.176	11.824	0.339	0.0237
5581.814	D	88.341	11.659	0.165	0.0191

Table (2.8.9)

Pressure (psia)	Cycle	Cum. Hg Sat. (Pore Vol., %)	Wetting Phase (Sw) (Pore Vol., %)	Incr. Hg Sat. (Pore Vol., %)	Ri (microns)
6882.68	D	88.444	11.5557	0.103	0.0155
8584.57	D	88.652	11.3483	0.207	0.0124
10583.20	D	88.883	11.1165	0.232	0.0101
13182.48	D	88.889	11.1114	0.005	0.0081
14782.60	D	88.889	11.1114	0.000	0.0072
16381.45	D	88.889	11.1114	0.000	0.0065
19983.01	D	88.889	11.1114	0.000	0.0053
24993.36	D	88.889	11.1114	0.000	0.0043
29991.32	D	88.889	11.1114	0.000	0.0036
34987.29	D	88.889	11.1114	0.000	0.0030
39990.73	D	88.889	11.1114	0.000	0.0027
44985.83	D	88.889	11.1114	0.000	0.0024
49987.21	D	88.889	11.1114	0.000	0.0021
54989.34	D	88.889	11.1114	0.000	0.0019
59988.03	D	88.889	11.1114	0.000	0.0018
46091.24	I	88.889	11.1114	----	----
35507.22	I	88.889	11.1114	----	----
27306.64	I	88.889	11.1114	----	----
21005.38	I	88.889	11.1114	----	----
16008.30	I	88.889	11.1114	----	----
12410.90	I	88.889	11.1114	----	----
9613.07	I	88.889	11.1114	----	----
7304.05	I	88.889	11.1114	----	----
5703.07	I	88.889	11.1114	----	----
4306.14	I	88.889	11.1114	----	----
3302.84	I	88.889	11.1114	----	----
2609.48	I	88.889	11.1114	----	----
2000.82	I	88.889	11.1114	----	----
1503.58	I	88.889	11.1114	----	----
1196.47	I	88.889	11.1114	----	----
900.749	I	87.342	12.6576	----	----
700.617	I	86.759	13.2411	----	----
502.874	I	85.710	14.2904	----	----
401.824	I	84.958	15.0424	----	----
303.180	I	83.045	16.9550	----	----
241.785	I	81.673	18.3272	----	----
192.456	I	79.406	20.5939	----	----
147.580	I	77.081	22.9190	----	----
113.197	I	74.518	25.4821	----	----
88.1393	I	70.609	29.3910	----	----
70.402	I	63.164	36.8361	----	----

Table (2.8.9-Continue)

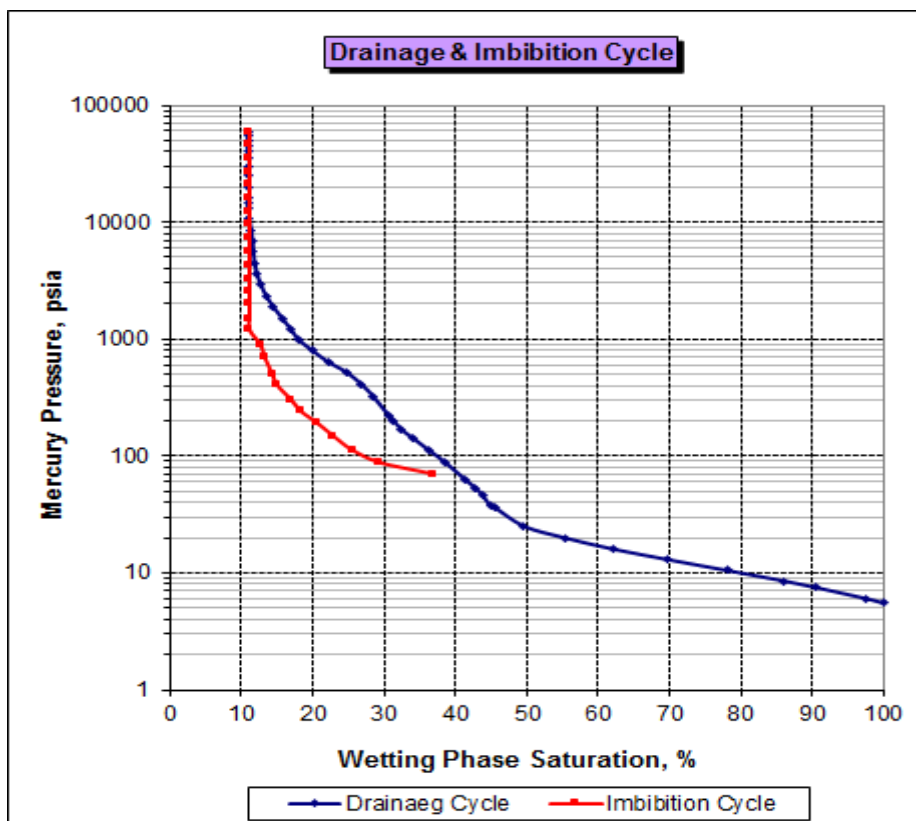


Figure (2.8.33)-Sample # 275

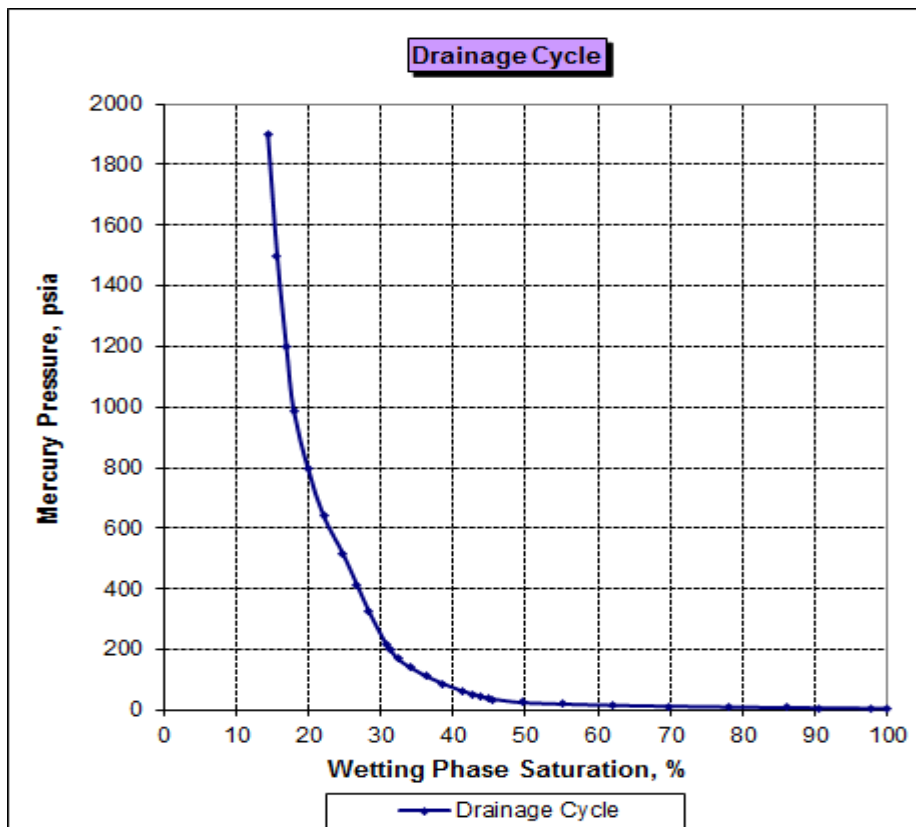


Figure (2.8.34)-Sample # 275

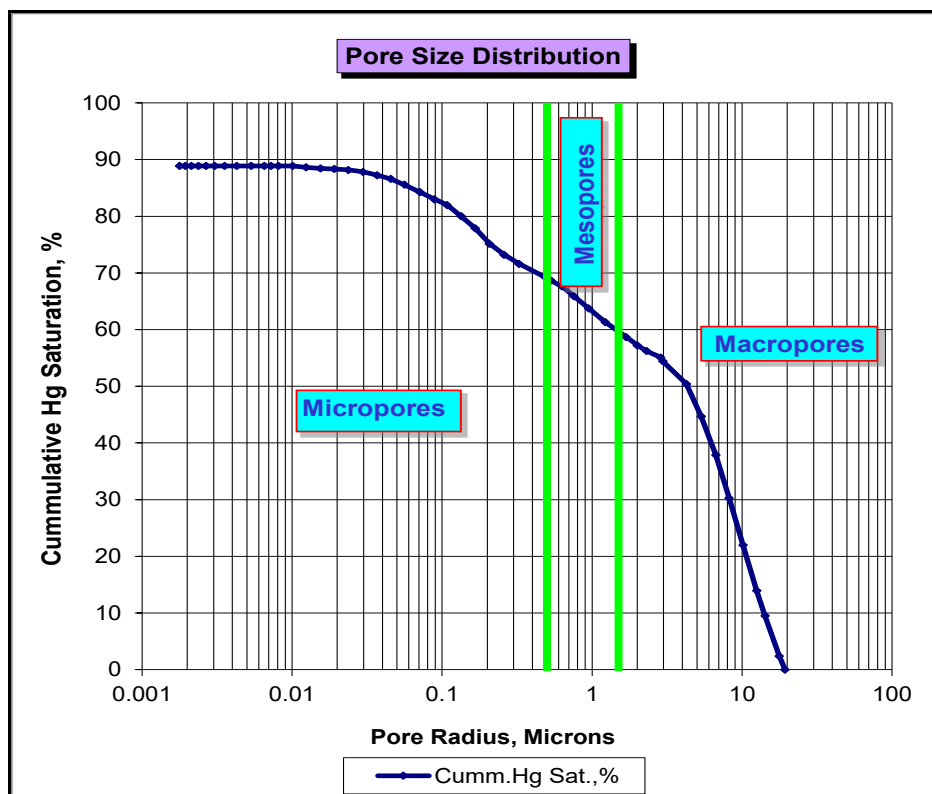


Figure (2.8.35)-Sample # 275

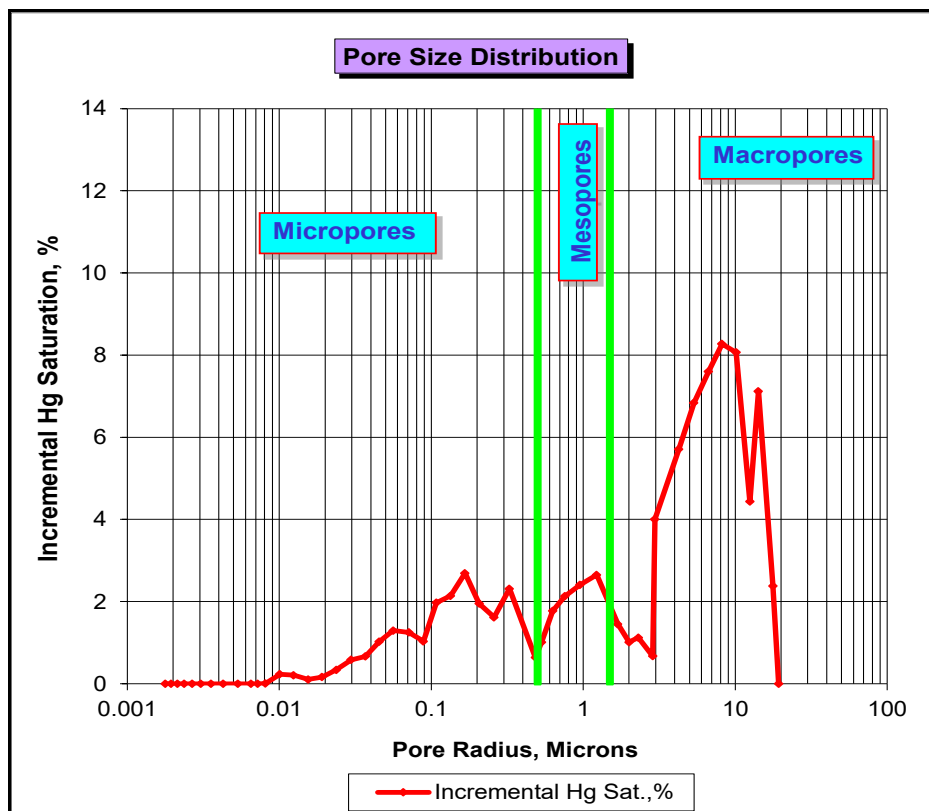


Figure (2.8.36)-Sample # 275

MERCURY INJECTION CAPILLARY PRESSURE AND PORE SIZE DISTRIBUTION

Client	: Mellitah Oil and Gas		
Well	: CE03-NC41		
Sample #	: 278		
Core Sample Data and Experimental Values			
Hg Pore volume, (cc)	: 0.83	Threshold Pressure, (psi)	: 13.030
Porosity, (%)	: 24.90	Sample Weight, (gm)	: 7.350
Air Perm., (mD)	: 38.10	Formation	: El-Garia
Sample Depth, (ft.in)	: 8639.01		

Pressure (psia)	Cycle	Cum. Hg Sat. (Pore Vol., %)	Wetting Phase (Sw) (Pore Vol., %)	Incr. Hg Sat. (Pore Vol., %)	Ri (microns)
13.0300	D	0.000	100.000	0.000	8.186
15.9749	D	0.599	99.401	0.599	6.677
19.9843	D	2.2218	97.778	1.623	5.337
24.9701	D	6.492	93.508	4.270	4.272
30.0363	D	10.829	89.171	4.337	3.551
35.7033	D	12.778	87.222	1.949	2.987
47.1645	D	19.588	80.412	6.810	2.261
57.0978	D	25.820	74.180	6.232	1.868
71.1972	D	32.644	67.356	6.824	1.498
85.6648	D	37.505	62.495	4.861	1.245
111.5331	D	43.811	56.189	6.306	0.956
137.2845	D	48.552	51.448	4.741	0.777
172.7061	D	54.586	45.414	6.035	0.618
215.5091	D	61.675	38.325	7.089	0.495
268.2946	D	67.861	32.139	6.185	0.398
328.6172	D	74.222	25.778	6.361	0.325
417.1125	D	80.791	19.209	6.569	0.256
517.5712	D	85.294	14.706	4.503	0.206
638.7850	D	88.456	11.544	3.162	0.167
795.2734	D	90.618	9.3820	2.163	0.134
985.8002	D	92.316	7.6841	1.698	0.108
1200.578	D	93.353	6.6474	1.037	0.089
1497.294	D	93.966	6.0343	0.613	0.0712
1897.677	D	94.733	5.2672	0.767	0.0562
2344.341	D	95.053	4.9472	0.320	0.0455
2894.393	D	95.617	4.3834	0.564	0.0369
3594.534	D	95.711	4.2889	0.094	0.0297
4484.935	D	95.852	4.1484	0.141	0.0238
5582.510	D	96.022	3.9779	0.171	0.0191
6884.243	D	96.357	3.6433	0.335	0.0155
8582.454	D	96.357	3.6433	0.000	0.0124
10584.691	D	96.357	3.6433	0.000	0.0101
13182.642	D	96.357	3.6433	0.000	0.0081
14782.079	D	96.357	3.6433	0.000	0.0072

Table (2.8.10)

Pressure (psia)	Cycle	Cum. Hg Sat. (Pore Vol., %)	Wetting Phase (Sw) (Pore Vol., %)	Incr. Hg Sat. (Pore Vol., %)	Ri (microns)
16381.653	D	96.357	3.6433	0.000	0.0065
19982.875	D	96.357	3.6433	0.000	0.0053
24992.344	D	96.357	3.6433	0.000	0.0043
29990.279	D	96.357	3.6433	0.000	0.0036
34990.152	D	96.357	3.6433	0.000	0.0030
39988.88	D	96.357	3.6433	0.000	0.0027
44986.35	D	96.357	3.6433	0.000	0.0024
49987.63	D	96.357	3.6433	0.000	0.0021
54987.57	D	96.357	3.6433	0.000	0.0019
59985.33	D	96.357	3.6433	0.000	0.0018
46104.79	I	96.357	3.6433	----	----
35504.24	I	96.357	3.6433	----	----
27309.68	I	96.357	3.6433	----	----
21008.98	I	96.357	3.6433	----	----
16011.21	I	96.357	3.6433	----	----
12411.64	I	96.357	3.6433	----	----
9612.402	I	96.357	3.6433	----	----
7317.248	I	96.357	3.6433	----	----
5713.443	I	96.357	3.6433	----	----
4302.443	I	96.357	3.6433	----	----
3307.563	I	96.357	3.6433	----	----
2604.880	I	96.357	3.6433	----	----
2004.766	I	96.357	3.6433	----	----
1505.568	I	96.357	3.6433	----	----
1205.372	I	96.357	3.6433	----	----
899.129	I	96.357	3.6433	----	----
703.599	I	96.357	3.6433	----	----
498.453	I	96.357	3.6433	----	----
397.5693	I	96.357	3.6433	----	----
303.0836	I	95.924	4.0763	----	----
239.6604	I	94.652	5.3480	----	----
190.7071	I	92.737	7.2632	----	----
146.7557	I	89.395	10.605	----	----
111.4977	I	83.639	16.361	----	----
87.4083	I	76.579	23.421	----	----
68.65387	I	68.564	31.436	----	----
53.71202	I	60.618	39.382	----	----
32.96341	I	48.715	51.285	----	----

Table (2.8.10-Continue)

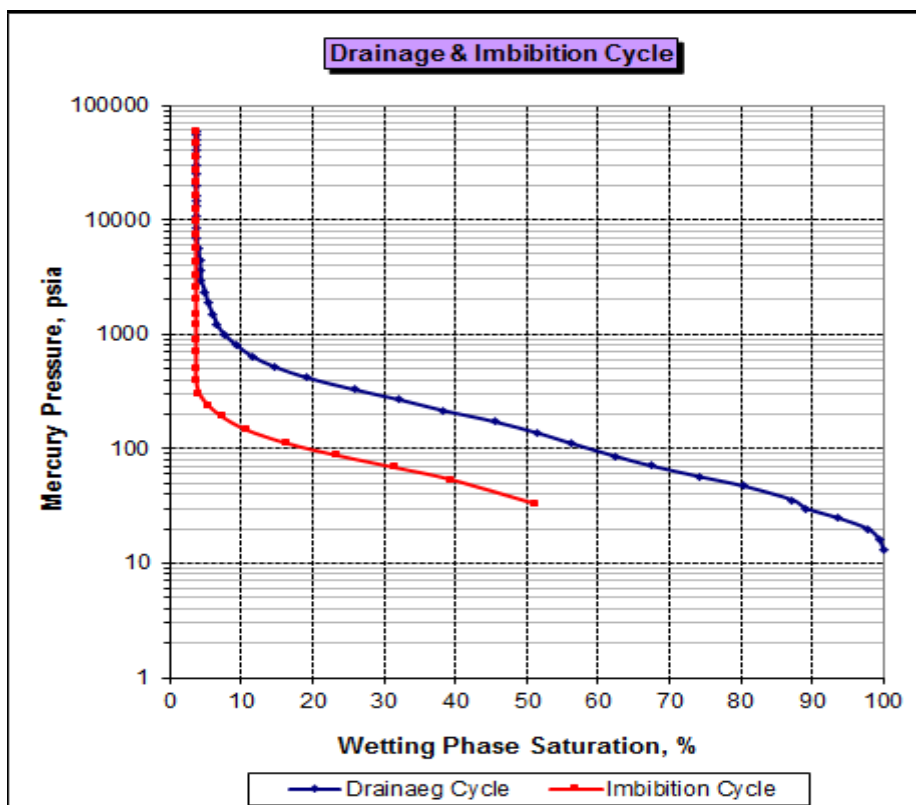


Figure (2.8.37)-Sample # 278

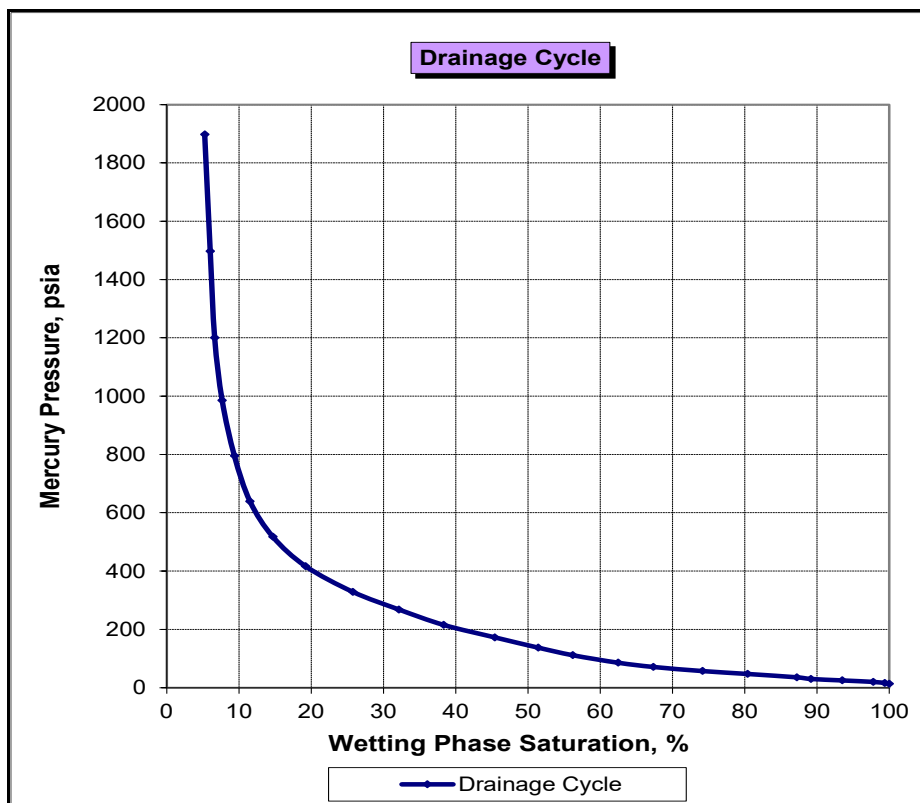


Figure (2.8.38)-Sample # 278

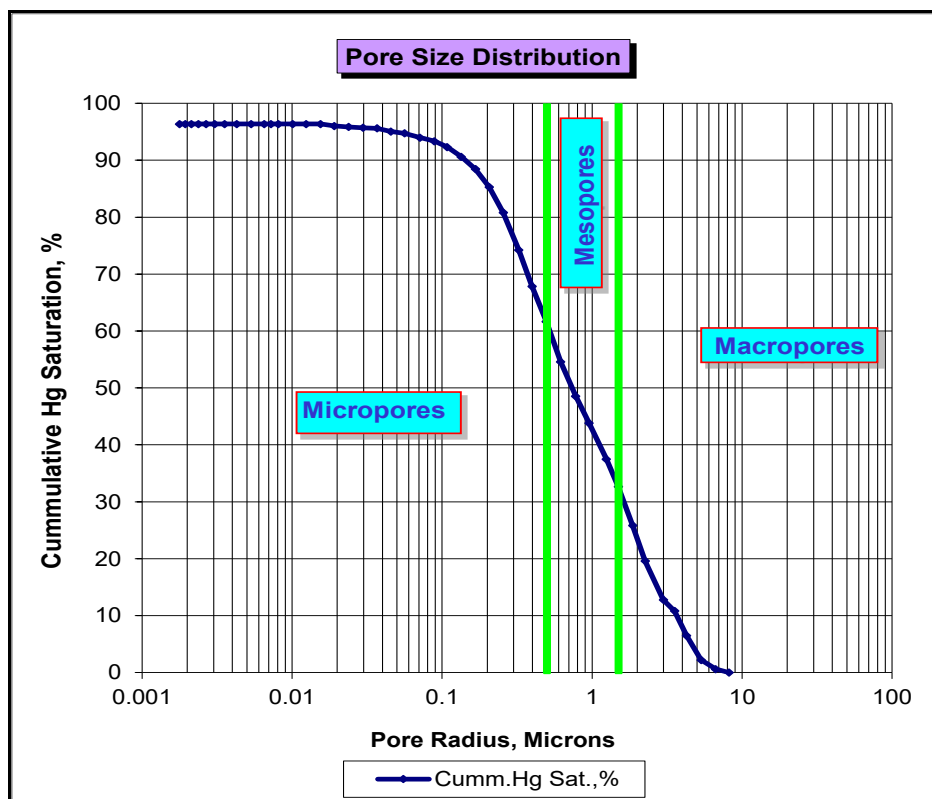


Figure (2.8.39)-Sample # 278

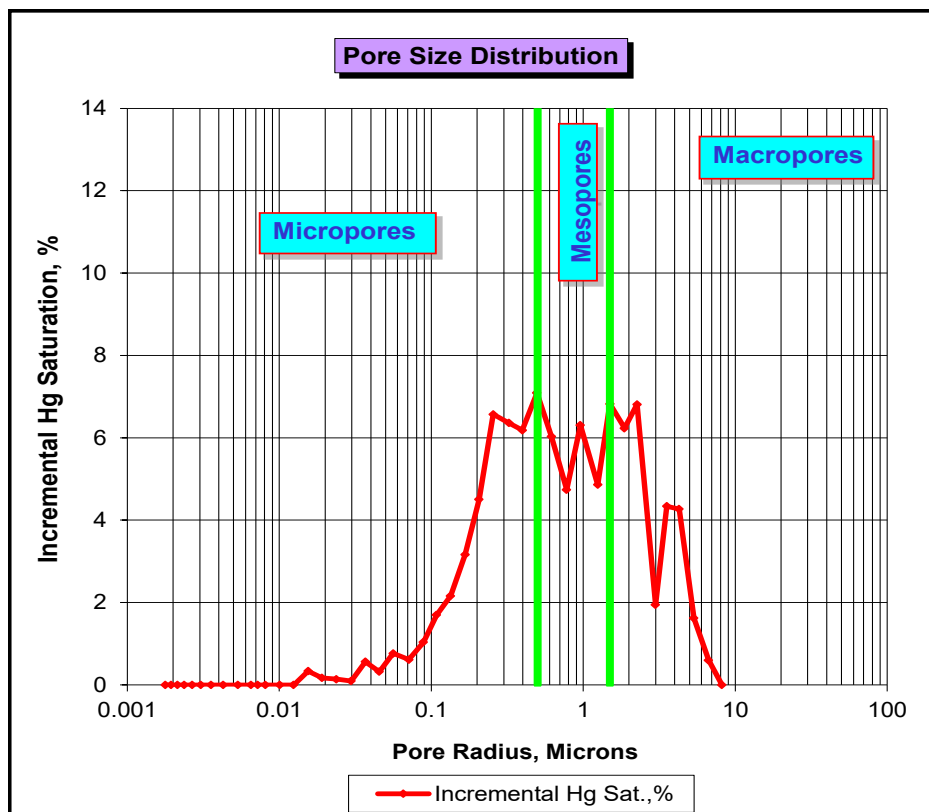


Figure (2.8.40)-Sample # 278

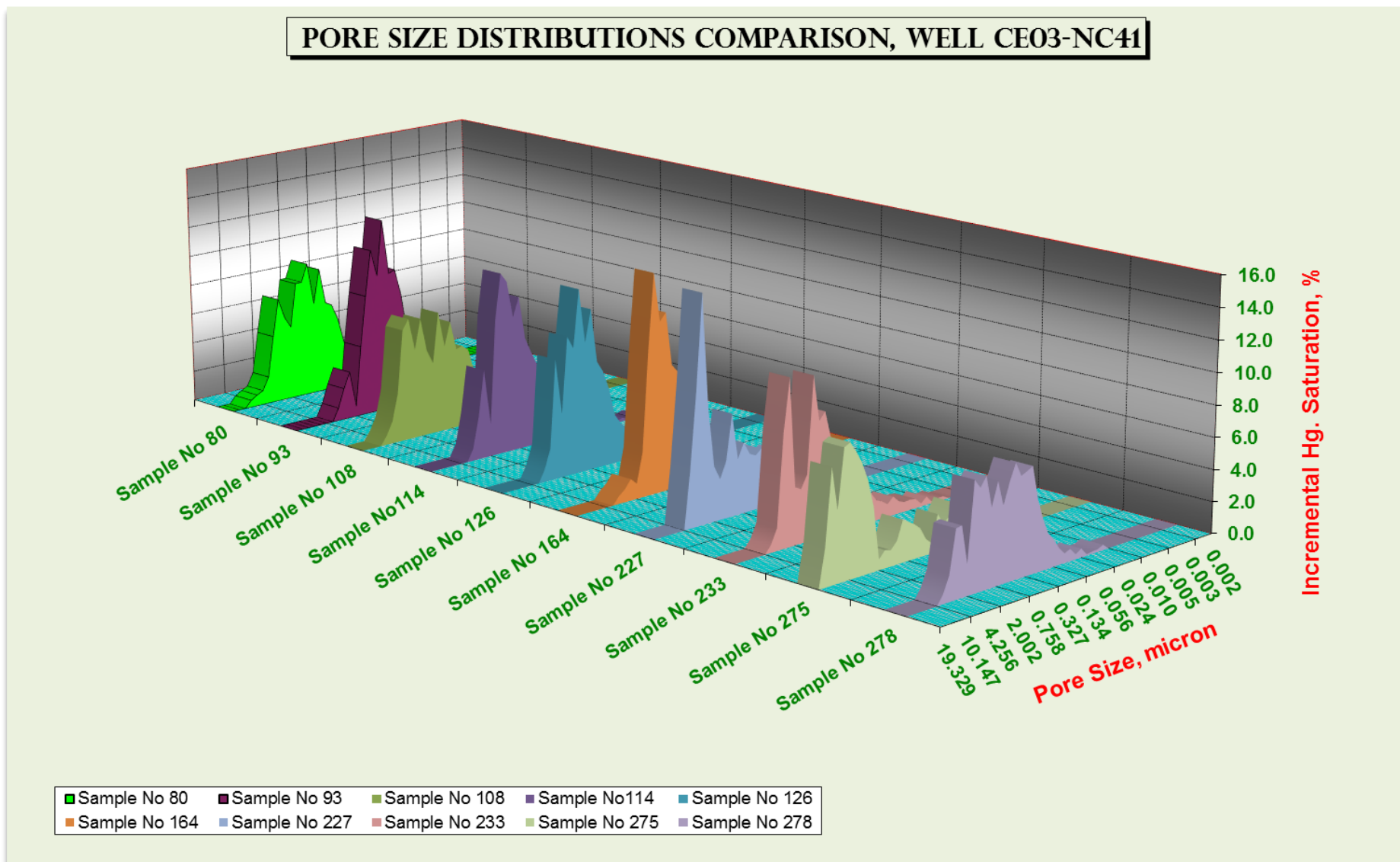


Figure (2.8.41)



مليته للنفط والغاز
Mellitah Oil & Gas

Mellitah Oil & Gas Company



الوطنية للنفط
حوث النفط

Section Nine

Unsteady State Gas-Water Relative Permeability (At Reservoir Conditions)

LABORATORY PROCEDURES

Unsteady-State Gas-Water and Gas-Oil Relative Permeability

1. Five samples were selected by Mellitah Oil Company from the well **CE03-NC41** to perform unsteady-state gas-water and gas-oil relative permeability test at reservoir conditions by using X-Ray system.
2. Samples underwent solvent extraction of hydrocarbons to ensure they were reverted to a water-wet wettability preference, then leached of salts with methanol and dried to a stable weight. Basic properties including grain density, pore volume, and permeability to air were measured at reservoir net confining stress.
3. Simulated formation brine containing (**70,000 ppm**) sodium chloride was used as the synthetic formation brine based on information provided by Mellitah. The brine was mixed and filtered to 0.45 microns and degassed. Fluid parameters including viscosity and density were measured at ambient and elevated temperature.
4. The samples were evacuated of air and pressure saturated with synthetic formation brine.
5. Teflon tape was used to rap and stabilize each sample.
6. Each sample was loaded into an X-ray core holder, and the appropriate net confining stress (**3858 psi**) was applied. The sample was scanned prior to connecting injection lines. Injection lines were attached, and a pore pressure was established by injecting tagged water through the system and through the sample against a backpressure until differential pressure stabilized.
7. Tagged water, filtered to 0.45 microns, and tagged with 28.2% by weight Sodium Iodide, was injected through the sample at a constant rate. Differential pressure was monitored until equilibrium was achieved. Effective permeability to water was determined at two injection rates while X-ray scanning the sample. The measurements were conducted at ambient and reservoir temperature (**278 °F**).
8. Humidified nitrogen was injected at a constant pressure, while scanning the sample. The volumes out were recorded as a function of elapsed time throughout the test. Once the volume out and x-ray scans stabilized, effective permeability to gas at initial water saturation was measured at two injection rates.
9. Wettability restoration was done for the five samples by (Ageing) in dead crude oil under reservoir pressure and temperature for 21 days.

10. Tagged crude oil (for samples 126&275) and tagged laboratory oil (for samples 80, 93&108) filtered to 0.45 microns, and tagged with 12.5% by weight 1-Iododecane, was injected through the sample and system against backpressure. The oil was injected at a suitable constant rate until an equilibrium differential pressure was observed. After equilibrium was achieved, effective permeability to oil at initial water saturation was determined at two injection rates while scanning the sample.
11. Humidified nitrogen was injected at a constant pressure, while scanning the sample. The volumes out were recorded as a function of elapsed time throughout the test. Once the volume out and x-ray scans stabilized, effective permeability to gas at residual oil saturation was measured at two injection rates.
12. Pore pressure and overburden pressure were slowly released. The sample was unloaded, weighed, and submitted to Dean-Stark and soxhlet extraction with toluene as a check on residual fluid volumes, leached of salts with methanol, and dried to a constant weight in a convection oven.
13. The sample was then loaded into the X-ray core holder and gas, tagged oil, and tagged water base scans at 100% saturations were measured, with the sample saturated with, then cleaned of, each fluid as needed.
14. Each sample was extracted of any final fluid, leached of salts, and dried in a convection oven until the weight stabilized.
15. Unsteady-State gas-water and gas-oil relative permeability relationships at reservoir conditions were calculated from collected data, sample and fluid parameters by using **Sendra** software (Reference 9) which is a two-phase 1D black-oil simulation model. It is used for analyzing single SCAL experiments as well as several SCAL experiments simultaneously.

The analysis can be performed by either simulation or estimation. Simulation is forward simulations of a multi-phase flow experiment, while Estimation utilizes an optimization routine to history match the experimental data obtained in the laboratory. Sendra uses different types of correlations to history match the experimental data, the type of correlation to be used may vary with which type of experiments that is analyzed as well as which type of experimental data that is used throughout the estimation. The available correlations are as follows: Corey, LET, Chierici, Burdine and Sigmund – McCaffery.

- The results of gas-water relative permeability are presented in tables 2.9.1 through 2.9.5 and in figures 2.9.1 through 2.9.20 and the results of gas-oil relative permeability are presented in tables 2.10.1 through 2.10.5 and in figures 2.10.1 through 2.10.20.

GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.0754
Well	: CE03-NC41	Permeability to water., mD	: 11.4
Sample #	: 80	Perm. to gas at irreducible water sat., mD	: 7.4
Porosity, %	: 21.0	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 51.7	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8437.06
Field	: Bahr Essalam		

Gas Saturation (fraction)	Relative Permeability to Water, (fraction)	Relative Permeability to Gas, (fraction)	Relative Permeability Ratio
0.000000	1.000000	0.000000	0.000000
0.018869	9.00E-01	0.000000	0.000000
0.037737	8.09E-01	0.000005	0.000006
0.056606	7.25E-01	0.000021	0.00003
0.075474	6.48E-01	0.000061	0.00009
0.094343	5.78E-01	0.000140	0.00024
0.113211	5.14E-01	0.000274	0.00053
0.132080	4.56E-01	0.000485	0.00106
0.150949	4.03E-01	0.000795	0.00197
0.169817	3.55E-01	0.001229	0.00346
0.188686	3.12E-01	0.001815	0.00581
0.207554	2.73E-01	0.002583	0.00944
0.226423	2.39E-01	0.003564	0.01493
0.245291	2.08E-01	0.004793	0.02309
0.264160	1.80E-01	0.006305	0.03506
0.283029	1.55E-01	0.008138	0.05248
0.301897	1.33E-01	0.010333	0.07759
0.320766	0.113832	0.012932	0.11360
0.339634	0.096815	0.015977	0.16503
0.358503	0.081907	0.019516	0.23827
0.377371	0.068902	0.023594	0.34240
0.396240	0.057611	0.028262	0.49060
0.415109	0.047858	0.033570	0.70150
0.433977	0.039479	0.039572	1.00240
0.452846	0.032322	0.046321	1.43310

Table (2.9.1)

GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.0754
Well	: CE03-NC41	Permeability to water., mD	: 11.4
Sample #	: 80	Perm. to gas at irreducible water sat., mD	: 7.4
Porosity, %	: 21.0	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 51.7	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8437.06
Field	: Bahr Essalam		

Gas Saturation (fraction)	Relative Permeability to Water, (fraction)	Relative Permeability to Gas, (fraction)	Relative Permeability Ratio
0.471714	0.026247	0.053873	2.053
0.490583	0.021126	0.062286	2.948
0.509451	0.016841	0.071621	4.253
0.528320	0.013284	0.081936	6.17
0.547188	0.010358	0.093296	9.01
0.566057	0.007973	0.105765	13.26
0.584926	0.006052	0.119407	19.7
0.603794	0.004522	0.134291	29.7
0.622663	0.003319	0.150485	45.3
0.641531	0.002388	0.168060	70
0.660400	0.001680	0.187087	111
0.679268	0.001151	0.207640	1.80E+02
0.698137	0.000765	0.229794	3.00E+02
0.717006	0.000491	0.253624	5.17E+02
0.735874	0.000302	0.279210	9.25E+02
0.754743	0.000176	0.306629	1.74E+03
0.773611	0.000097	0.335963	3.47E+03
0.792480	0.000049	0.367294	7.50E+03
0.811348	0.000022	0.400705	1.80E+04
0.830217	0.000009	0.436281	4.95E+04
0.849086	0.000003	0.474109	1.68E+05
0.867954	0.000001	0.514275	7.90E+05
0.886823	0.000000	0.556870	6.77E+06
0.905691	0.000000	0.601983	2.51E+08
0.924560	0.000000	0.649707	*****

Table (2.9.1)-Continue

GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.0754
Well	: CE03-NC41	Permeability to water., mD	: 11.4
Sample #	: 80	Perm. to gas at irreducible water sat., mD	: 7.4
Porosity, %	: 21.0	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 51.7	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8437.06
Field	: Bahr Essalam		

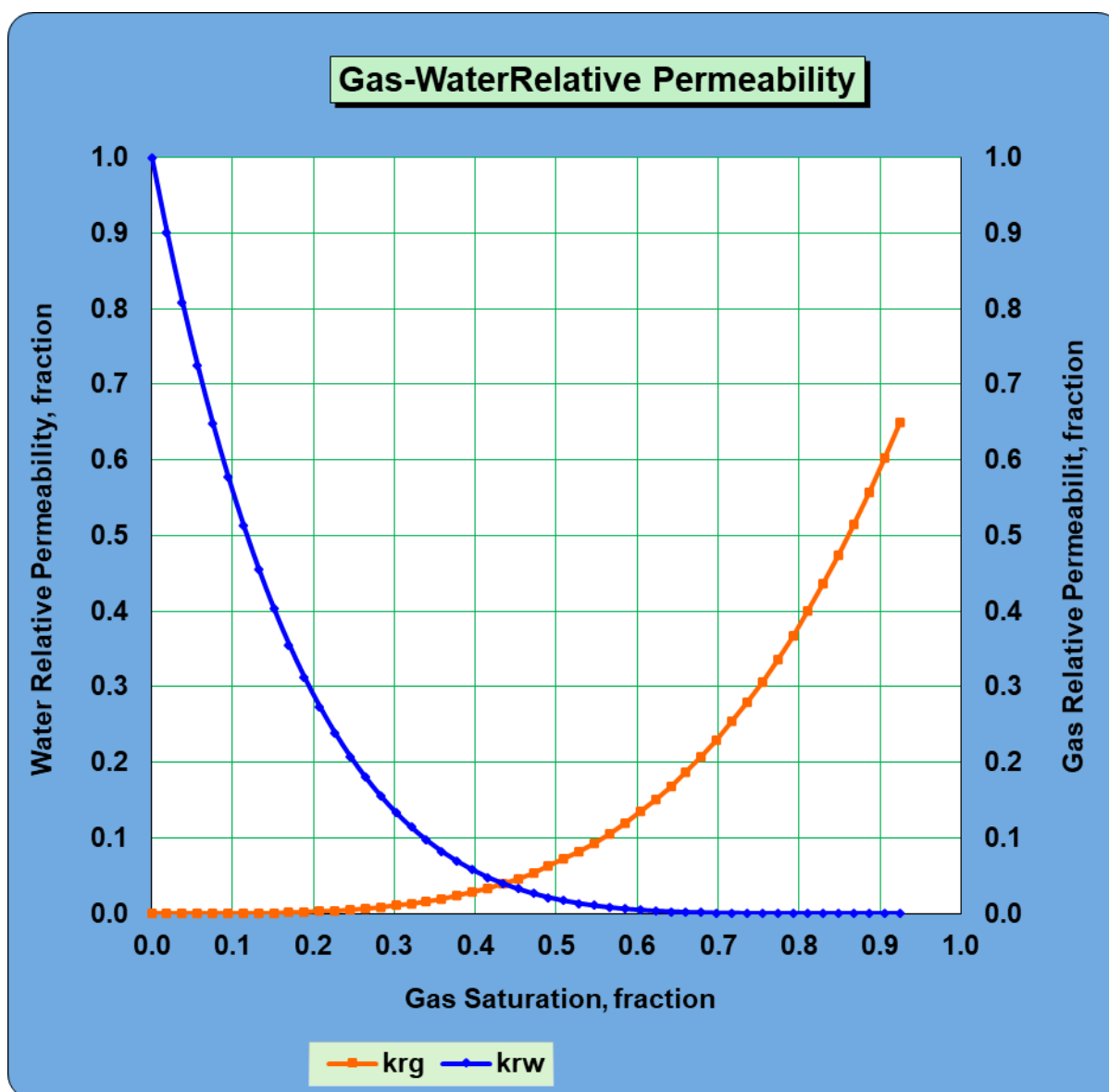


Figure (2.9.1)

GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.0754
Well	: CE03-NC41	Permeability to water., mD	: 11.4
Sample #	: 80	Perm. to gas at irreducible water sat., mD	: 7.4
Porosity, %	: 21.0	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 51.7	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8437.06
Field	: Bahr Essalam		

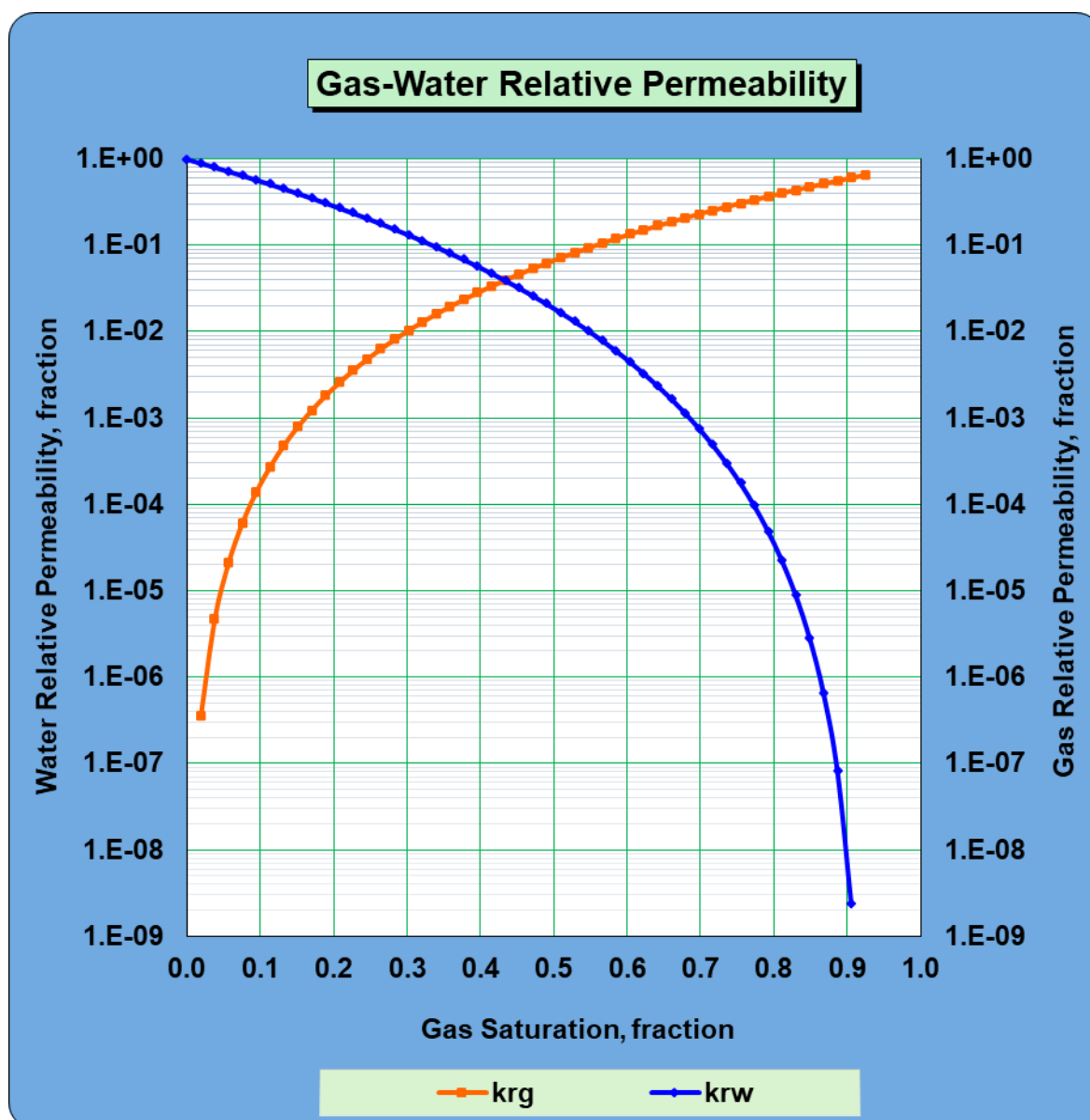


Figure (2.9.2)

GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.0754
Well	: CE03-NC41	Permeability to water., mD	: 11.4
Sample #	: 80	Perm. to gas at irreducible water sat., mD	: 7.4
Porosity, %	: 21.0	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 51.7	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8437.06
Field	: Bahr Essalam		

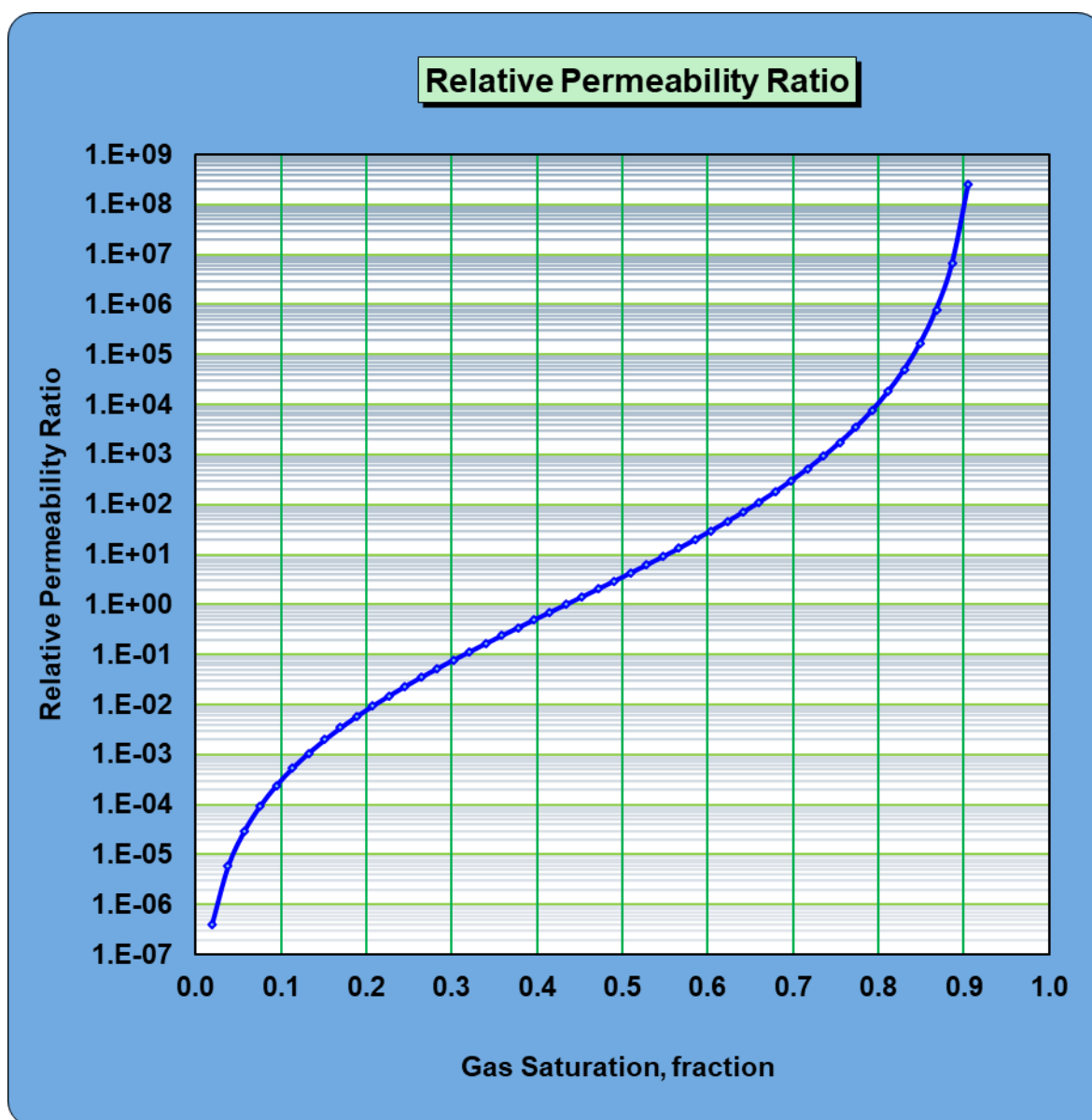


Figure (2.9.3)

GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.0754
Well	: CE03-NC41	Permeability to water., mD	: 11.4
Sample #	: 80	Perm. to gas at irreducible water sat., mD	: 7.4
Porosity, %	: 21.0	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 51.7	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8437.06
Field	: Bahr Essalam		

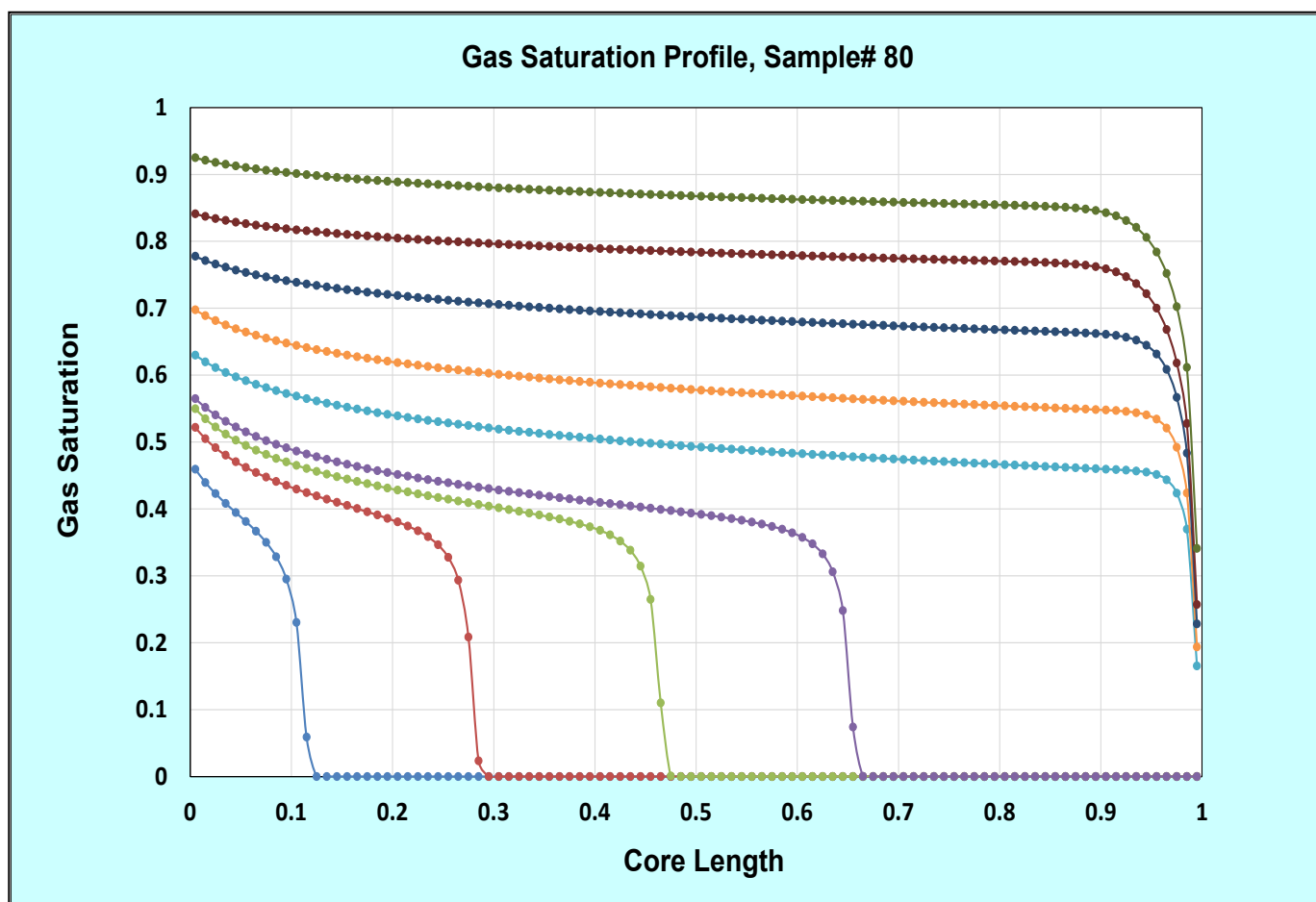


Figure (2.9.4)

GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1800
Well	: CE03-NC41	Permeability to water., mD	: 6.5
Sample #	: 93	Perm. to gas at irreducible water sat., mD	: 5.2
Porosity, %	: 23.3	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 18.9	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8450.06
Field	: Bahr Essalam		

Gas Saturation (fraction)	Relative Permeability to Water, (fraction)	Relative Permeability to Gas, (fraction)	Relative Permeability Ratio
0.000000	1.000000	0.000000	0.000000
0.016735	8.69E-01	0.000006	0.000006
0.033469	7.52E-01	0.000046	0.000061
0.050204	6.50E-01	0.000159	0.000245
0.066939	5.59E-01	0.000383	0.000684
0.083673	4.80E-01	0.000756	0.001576
0.100408	4.10E-01	0.001319	0.003216
0.117143	3.49E-01	0.002111	0.006045
0.133878	2.96E-01	0.003173	0.010709
0.150612	2.50E-01	0.004545	0.018155
0.167347	2.11E-01	0.006269	0.029760
0.184082	1.76E-01	0.008385	0.047523
0.200816	1.47E-01	0.010934	0.074342
0.217551	1.22E-01	0.013959	0.114417
0.234286	0.100663	0.017500	0.173851
0.251020	0.082598	0.021601	0.261519
0.267755	0.067375	0.026302	0.390385
0.284490	0.054615	0.031647	0.579452
0.301224	0.043977	0.037676	0.856733
0.317959	0.035160	0.044434	1.26375
0.334694	0.027899	0.051961	1.86251
0.351429	0.021958	0.060302	2.74626
0.368163	0.017132	0.069499	4.05658
0.384898	0.013243	0.079594	6.0104
0.401633	0.010133	0.090631	8.9440

Table (2.9.2)

GAS-WATER RELATIVE PERMEABILITY

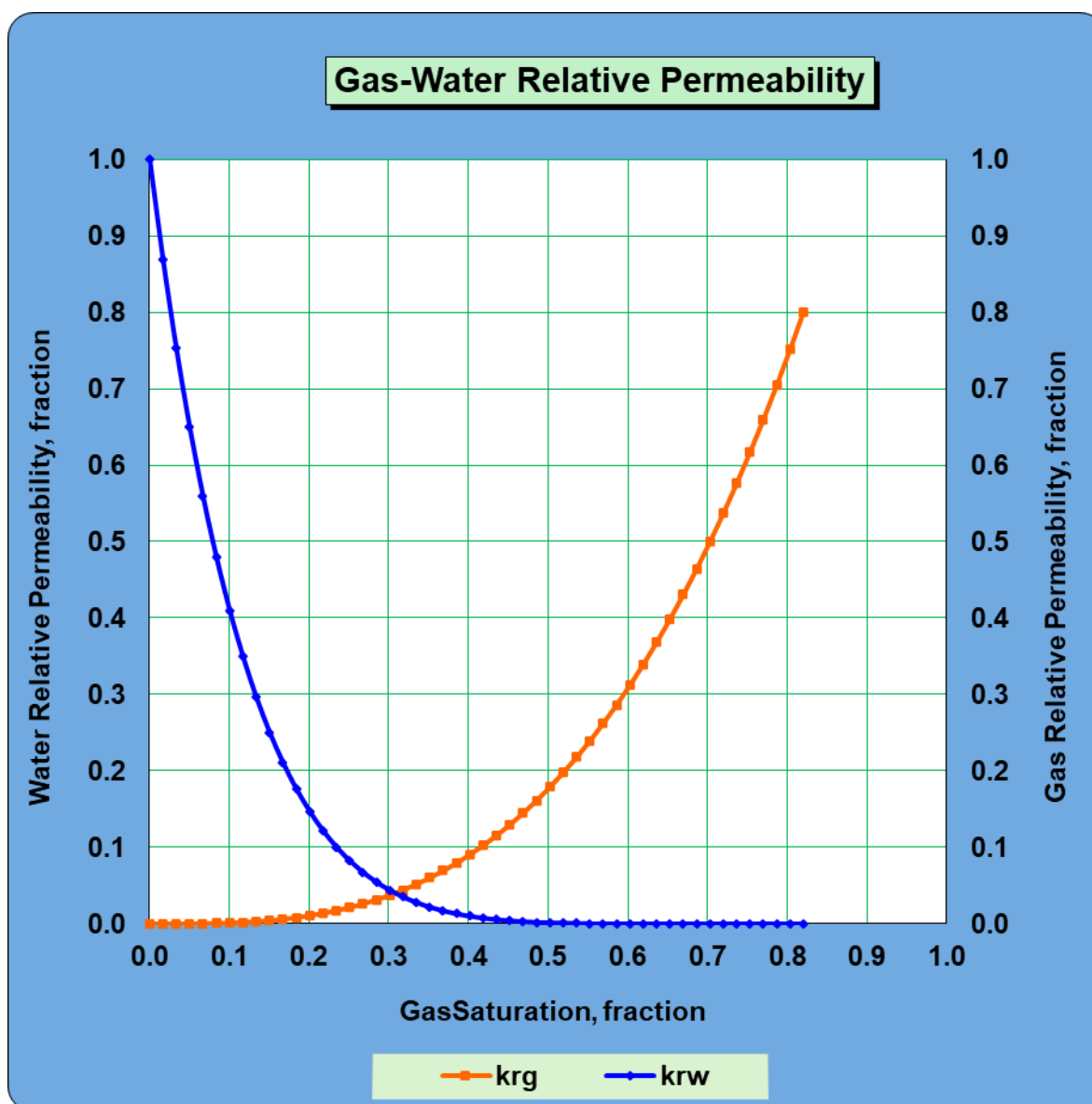
Client	: Mellitah	Irreducible water saturation, fraction	: 0.1800
Well	: CE03-NC41	Permeability to water., mD	: 6.5
Sample #	: 93	Perm. to gas at irreducible water sat., mD	: 5.2
Porosity, %	: 23.3	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 18.9	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8450.06
Field	: Bahr Essalam		

Gas Saturation (fraction)	Relative Permeability to Water, (fraction)	Relative Permeability to Gas, (fraction)	Relative Permeability Ratio
0.418367	0.007670	0.102652	13.3844
0.435102	0.005736	0.115702	20.1696
0.451837	0.004236	0.129823	30.651
0.468571	0.003084	0.145058	47.043
0.485306	0.002210	0.161451	73.043
0.502041	0.001558	0.179046	114.950
0.518776	0.001077	0.197885	183.73
0.535510	0.000729	0.218014	298.98
0.552245	0.000482	0.239475	496.68
0.568980	0.000310	0.262312	8.45E+02
0.585714	0.000194	0.286569	1.48E+03
0.602449	0.000117	0.312290	2.67E+03
0.619184	0.000068	0.339520	5.01E+03
0.635918	0.000037	0.368301	9.85E+03
0.652653	0.000020	0.398679	2.04E+04
0.669388	0.000010	0.430697	4.53E+04
0.686122	0.000004	0.464400	1.09E+05
0.702857	0.000002	0.499832	2.92E+05
0.719592	0.000001	0.537038	8.98E+05
0.736327	0.000000	0.576061	3.34E+06
0.753061	0.000000	0.616946	1.64E+07
0.769796	0.000000	0.659738	1.25E+08
0.786531	0.000000	0.704481	2.12E+09
0.803265	0.000000	0.751220	2.57E+11
0.820000	0.000000	0.800000	*****

Table (2.9.2)-Continue

GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1800
Well	: CE03-NC41	Permeability to water., mD	: 6.5
Sample #	: 93	Perm. to gas at irreducible water sat., mD	: 5.2
Porosity, %	: 23.3	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 18.9	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8450.06
Field	: Bahr Essalam		



GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1800
Well	: CE03-NC41	Permeability to water., mD	: 6.5
Sample #	: 93	Perm. to gas at irreducible water sat., mD	: 5.2
Porosity, %	: 23.3	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 18.9	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8450.06
Field	: Bahr Essalam		

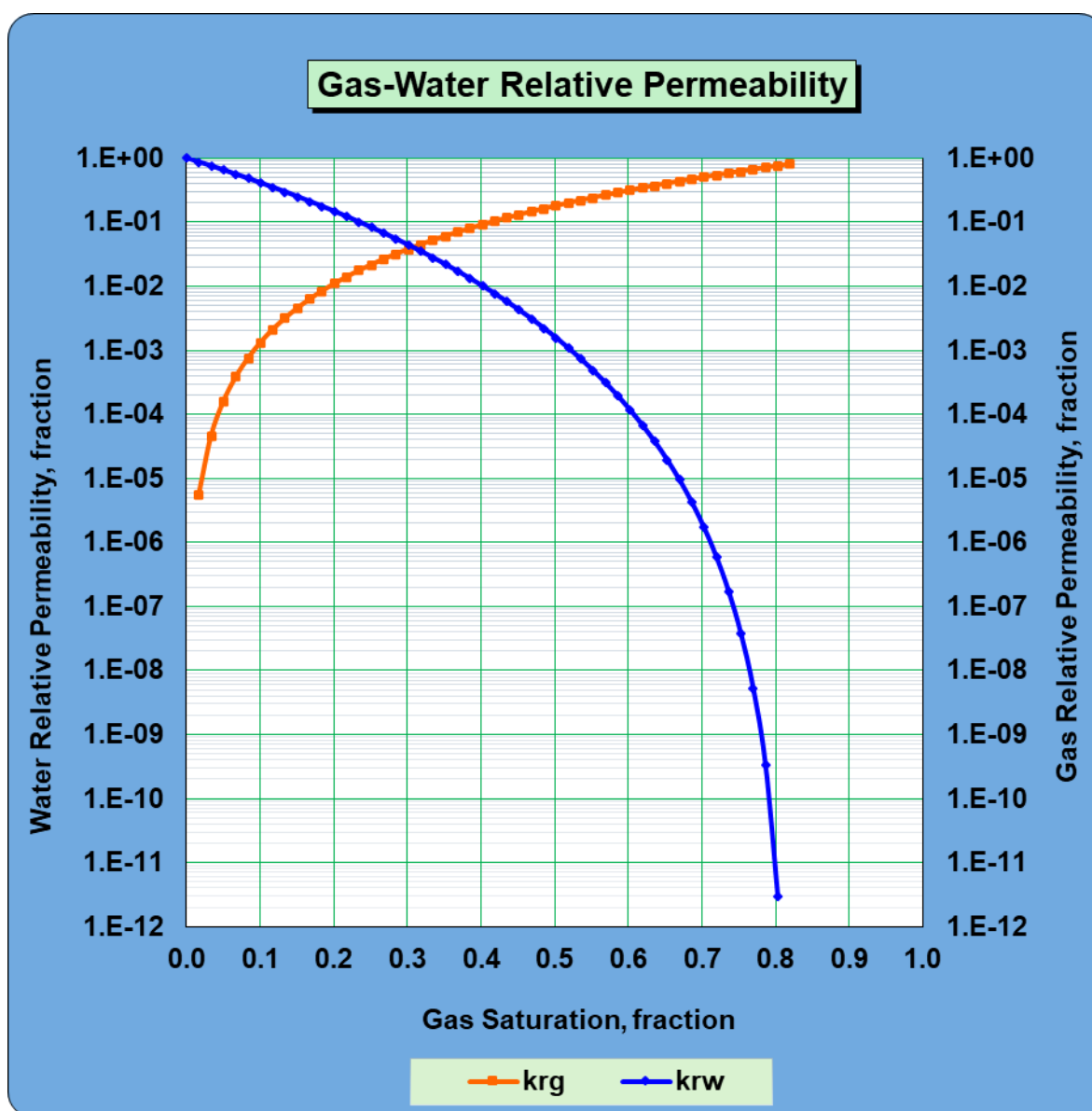


Figure (2.9.6)

GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1800
Well	: CE03-NC41	Permeability to water., mD	: 6.5
Sample #	: 93	Perm. to gas at irreducible water sat., mD	: 5.2
Porosity, %	: 23.3	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 18.9	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8450.06
Field	: Bahr Essalam		

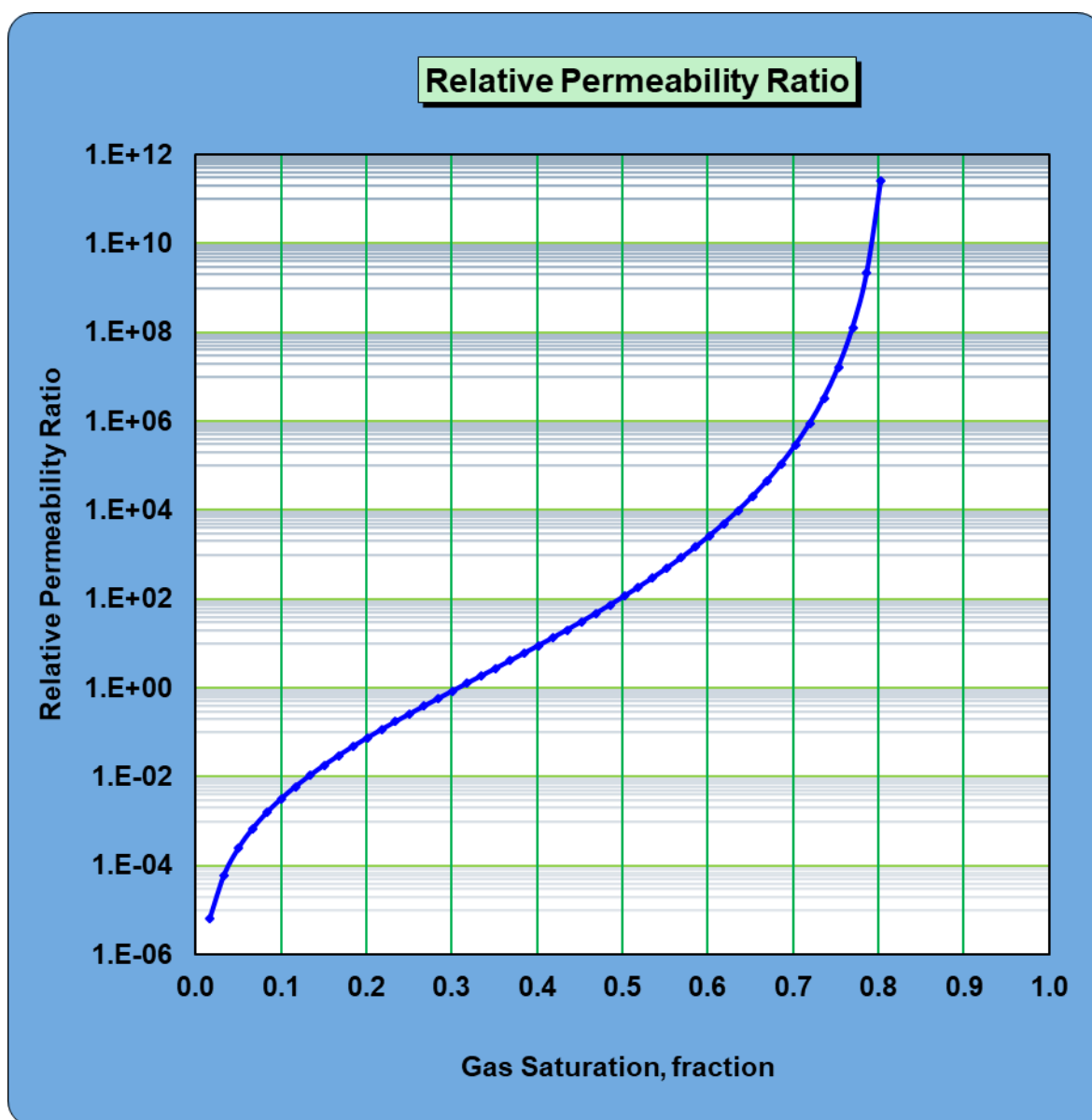


Figure (2.9.7)

GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1800
Well	: CE03-NC41	Permeability to water., mD	: 6.5
Sample #	: 93	Perm. to gas at irreducible water sat., mD	: 5.2
Porosity, %	: 23.3	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 18.9	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8450.06
Field	: Bahr Essalam		

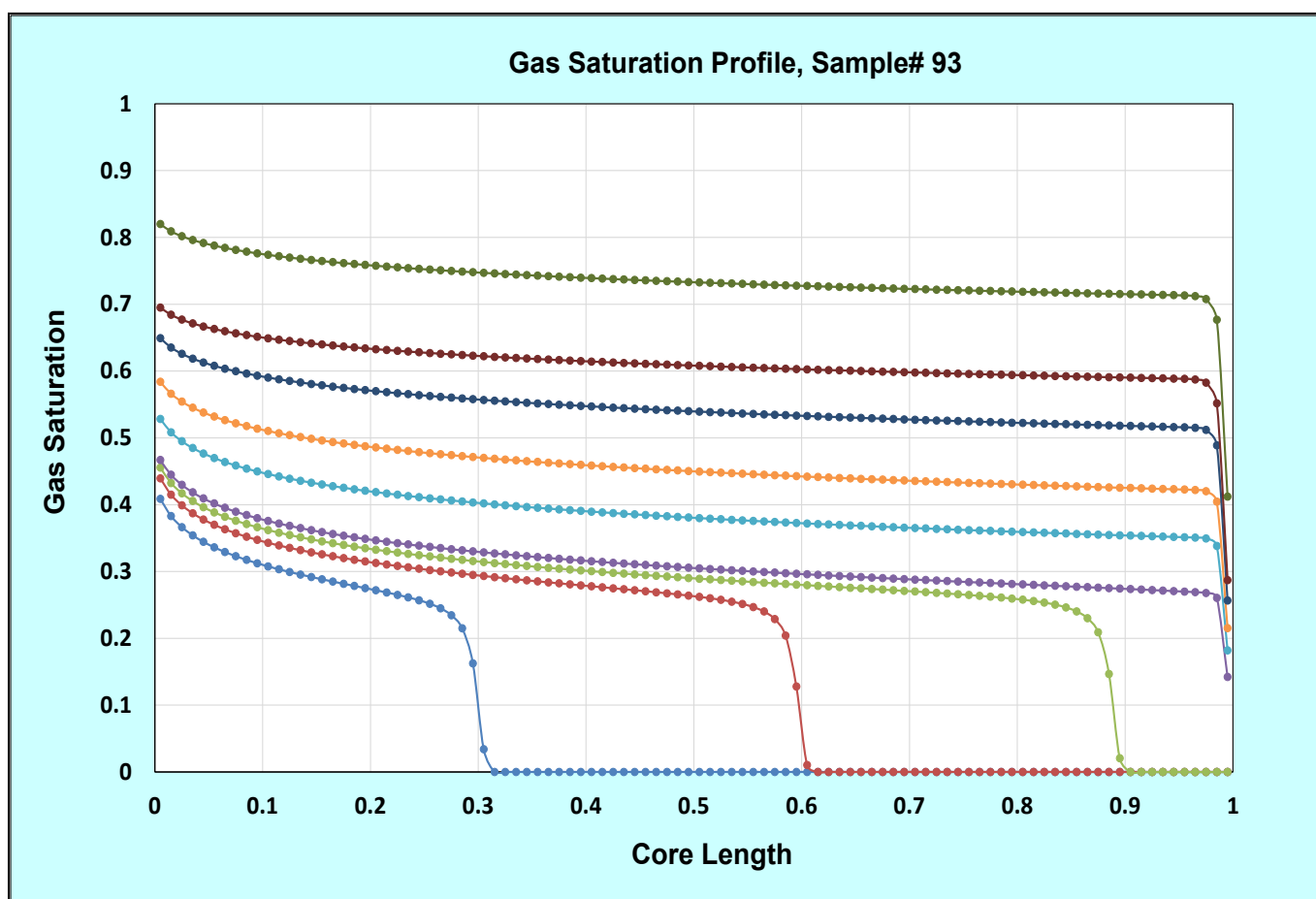


Figure (2.9.8)

GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1523
Well	: CE03-NC41	Permeability to water., mD	: 20.06
Sample #	: 108	Perm. to gas at irreducible water sat., mD	: 13.0
Porosity, %	: 25.6	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 62.8	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8465.05
Field	: Bahr Essalam		

Gas Saturation (fraction)	Relative Permeability to Water, (fraction)	Relative Permeability to Gas, (fraction)	Relative Permeability Ratio
0.000000	1.000000	0.000000	0.000000
0.017300	9.03E-01	0.000000	0.000000
0.034600	8.14E-01	0.000003	0.000004
0.051900	7.32E-01	0.000016	0.000021
0.069199	6.57E-01	0.000047	0.000071
0.086499	5.88E-01	0.000109	0.000185
0.103799	5.25E-01	0.000218	0.000416
0.121099	4.67E-01	0.000392	0.000840
0.138399	0.414668	0.000652	0.001573
0.155699	0.367063	0.001021	0.002782
0.172999	0.323922	0.001525	0.004709
0.190299	0.284924	0.002193	0.007696
0.207598	0.249765	0.003054	0.012228
0.224898	0.218156	0.004143	0.018989
0.242198	0.189823	0.005493	0.028939
0.259498	0.164504	0.007144	0.043427
0.276798	0.141955	0.009134	0.064346
0.294098	0.121942	0.011506	0.094360
0.311398	0.104246	0.014305	0.137219
0.328698	0.088661	0.017575	0.198226
0.345997	0.074993	0.021366	0.28491
0.363297	0.063061	0.025729	0.40800
0.380597	0.052694	0.030715	0.58290
0.397897	0.043734	0.036381	0.83188
0.415197	0.036033	0.042782	1.18731

Table (2.9.3)

GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1523
Well	: CE03-NC41	Permeability to water., mD	: 20.06
Sample #	: 108	Perm. to gas at irreducible water sat., mD	: 13.0
Porosity, %	: 25.6	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 62.8	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8465.05
Field	: Bahr Essalam		

Gas Saturation (fraction)	Relative Permeability to Water, (fraction)	Relative Permeability to Gas, (fraction)	Relative Permeability Ratio
0.432497	0.029454	0.049978	1.69681
0.449797	0.023871	0.058029	2.43096
0.467097	0.019166	0.066998	3.4957
0.484396	0.015232	0.076950	5.0519
0.501696	0.011970	0.087951	7.3474
0.518996	0.009292	0.100072	10.7700
0.536296	0.007114	0.113381	15.9370
0.553596	0.005365	0.127953	23.851
0.570896	0.003977	0.143860	36.176
0.588196	0.002891	0.161181	55.746
0.605495	0.002056	0.179993	87.524
0.622795	0.001426	0.200376	140.50
0.640095	0.000961	0.222413	231.55
0.657395	0.000625	0.246188	393.89
0.674695	0.000390	0.271786	696.23
0.691995	0.000232	0.299295	1290.0
0.709295	0.000130	0.328804	2535.4
0.726595	0.000067	0.360406	5374
0.743894	0.000031	0.394192	12584
0.761194	0.000013	0.430259	3.38E+04
0.778494	0.000004	0.468702	1.11E+05
0.795794	0.000001	0.509620	4.99E+05
0.813094	0.000000	0.553115	4.01E+06
0.830394	0.000000	0.599287	1.33E+08
0.847694	0.000000	0.648241	*****

Table (2.9.3)-Continue

GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1523
Well	: CE03-NC41	Permeability to water., mD	: 20.06
Sample #	: 108	Perm. to gas at irreducible water sat., mD	: 13.0
Porosity, %	: 25.6	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 62.8	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8465.05
Field	: Bahr Essalam		

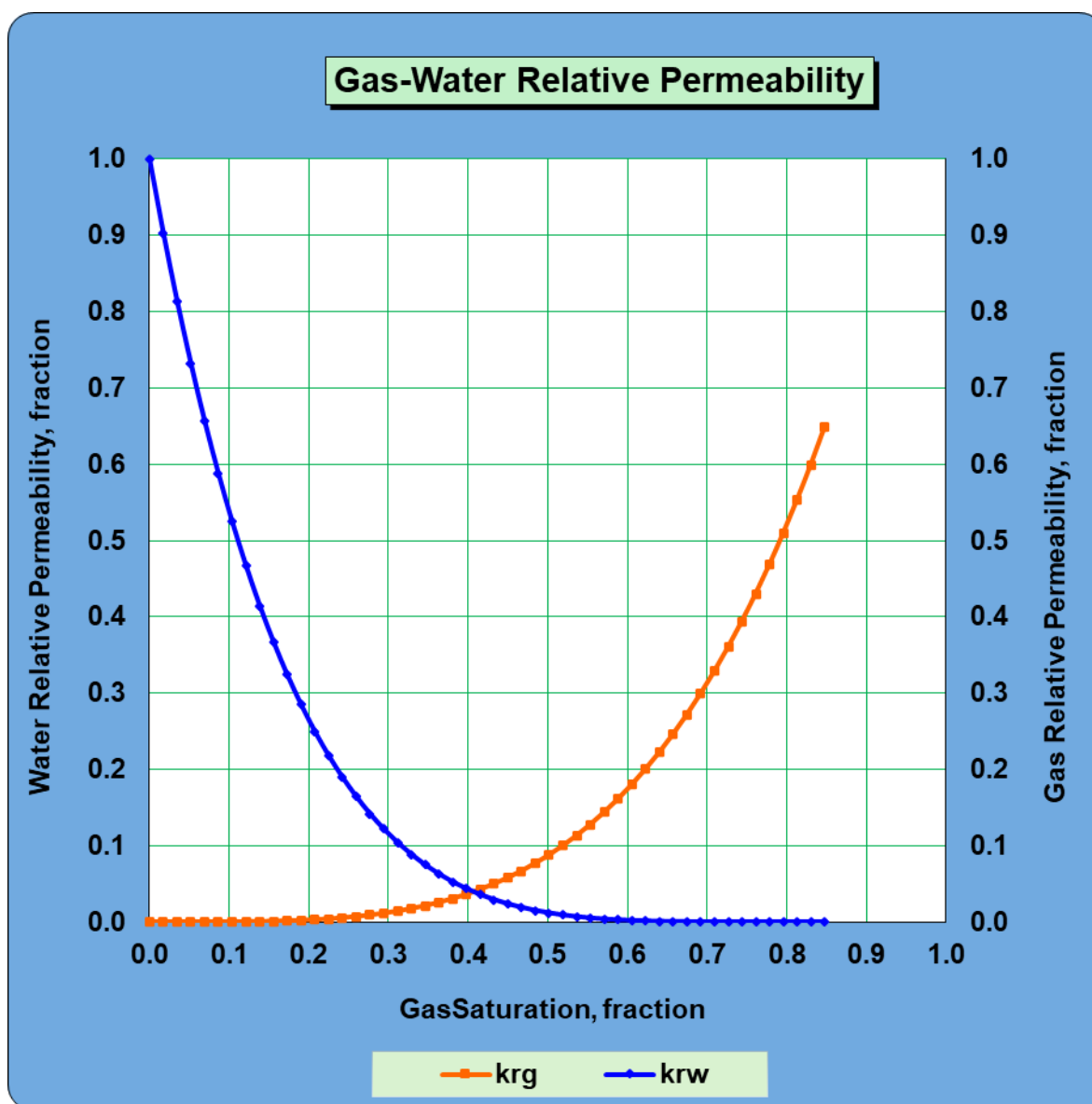
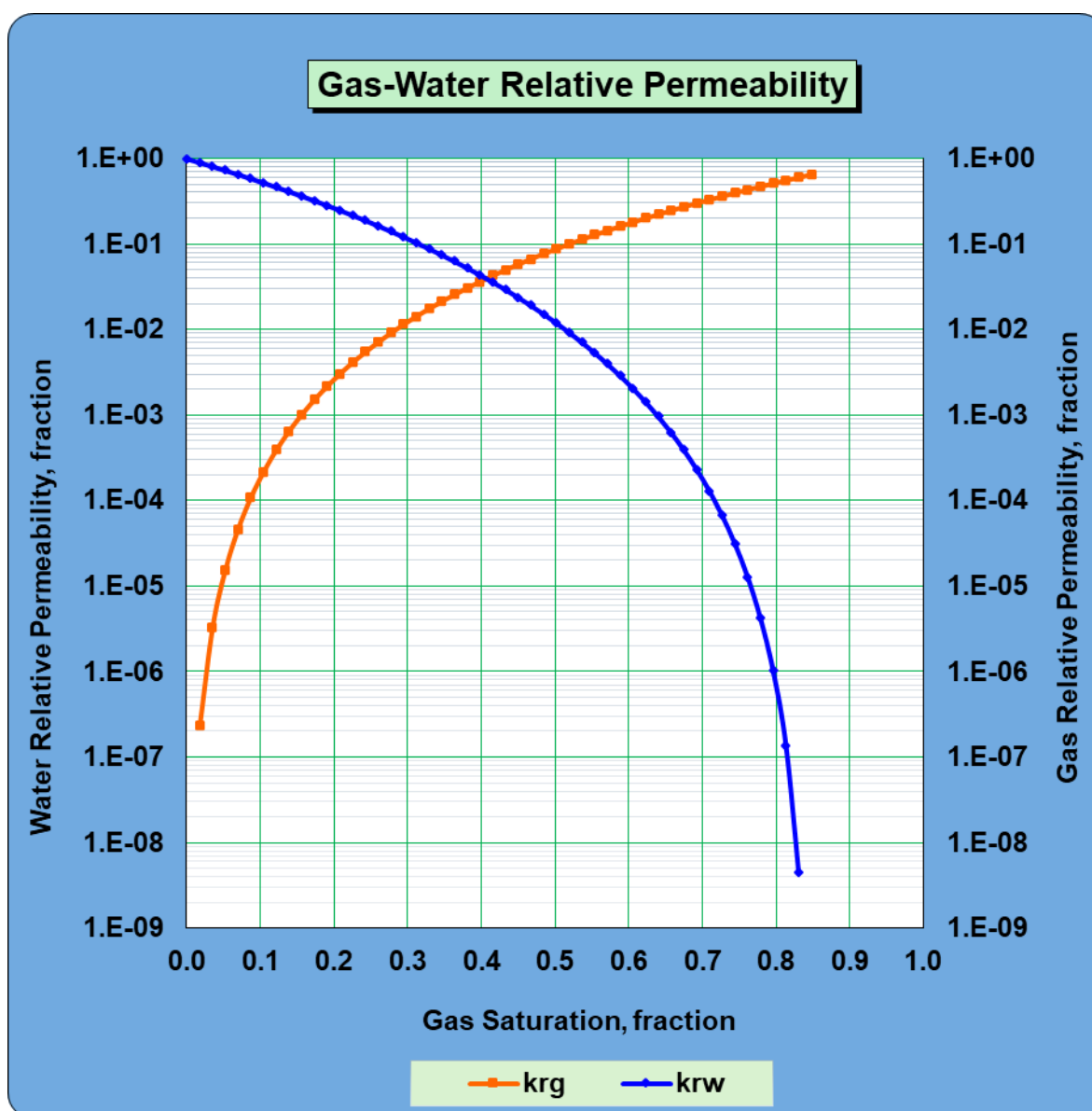


Figure (2.9.9)

GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1523
Well	: CE03-NC41	Permeability to water., mD	: 20.06
Sample #	: 108	Perm. to gas at irreducible water sat., mD	: 13.0
Porosity, %	: 25.6	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 62.8	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8465.05
Field	: Bahr Essalam		



GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1523
Well	: CE03-NC41	Permeability to water., mD	: 20.06
Sample #	: 108	Perm. to gas at irreducible water sat., mD	: 13.0
Porosity, %	: 25.6	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 62.8	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8465.05
Field	: Bahr Essalam		

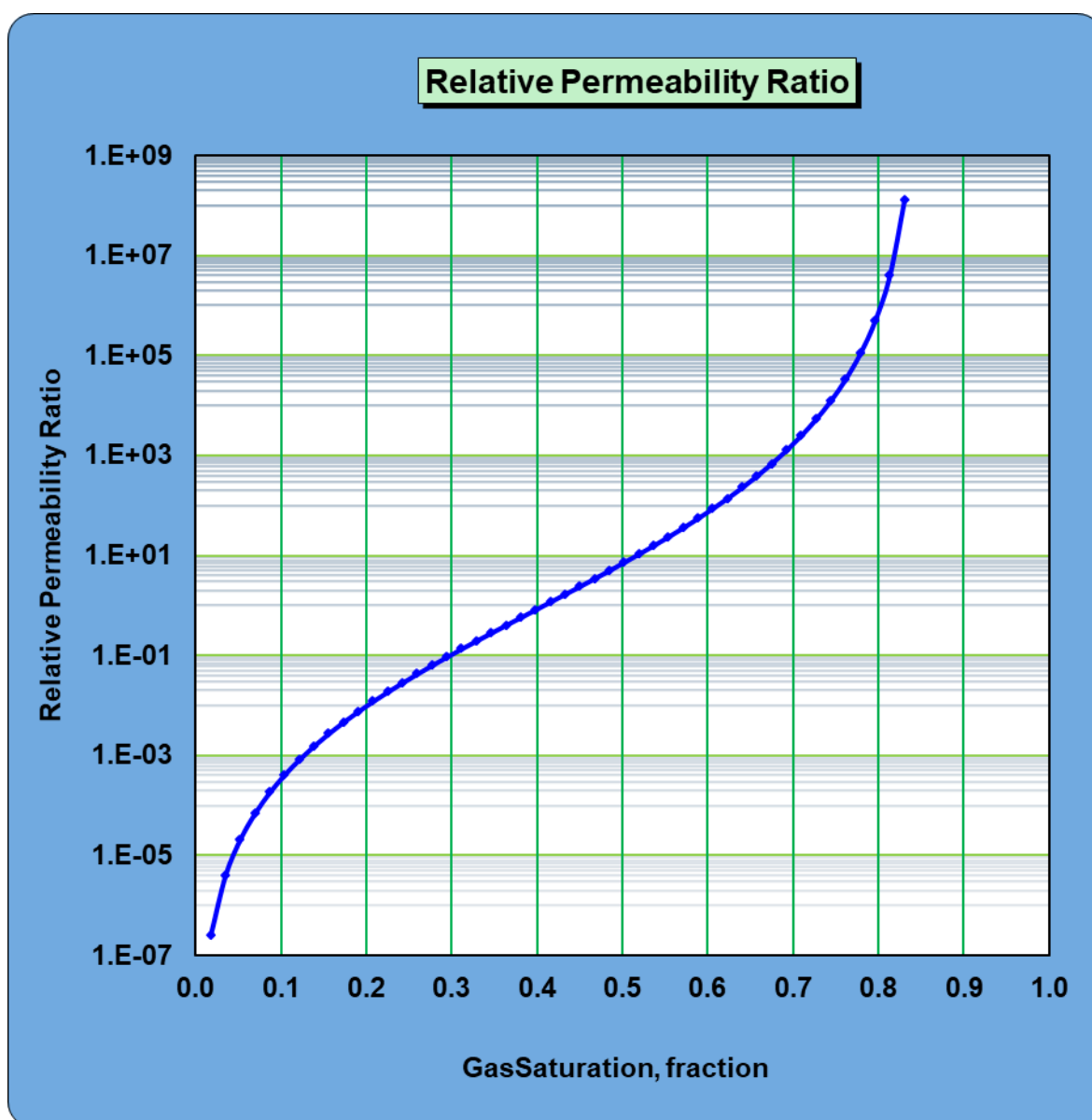


Figure (2.9.11)

GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1523
Well	: CE03-NC41	Permeability to water., mD	: 20.06
Sample #	: 108	Perm. to gas at irreducible water sat., mD	: 13.0
Porosity, %	: 25.6	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 62.8	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8465.05
Field	: Bahr Essalam		

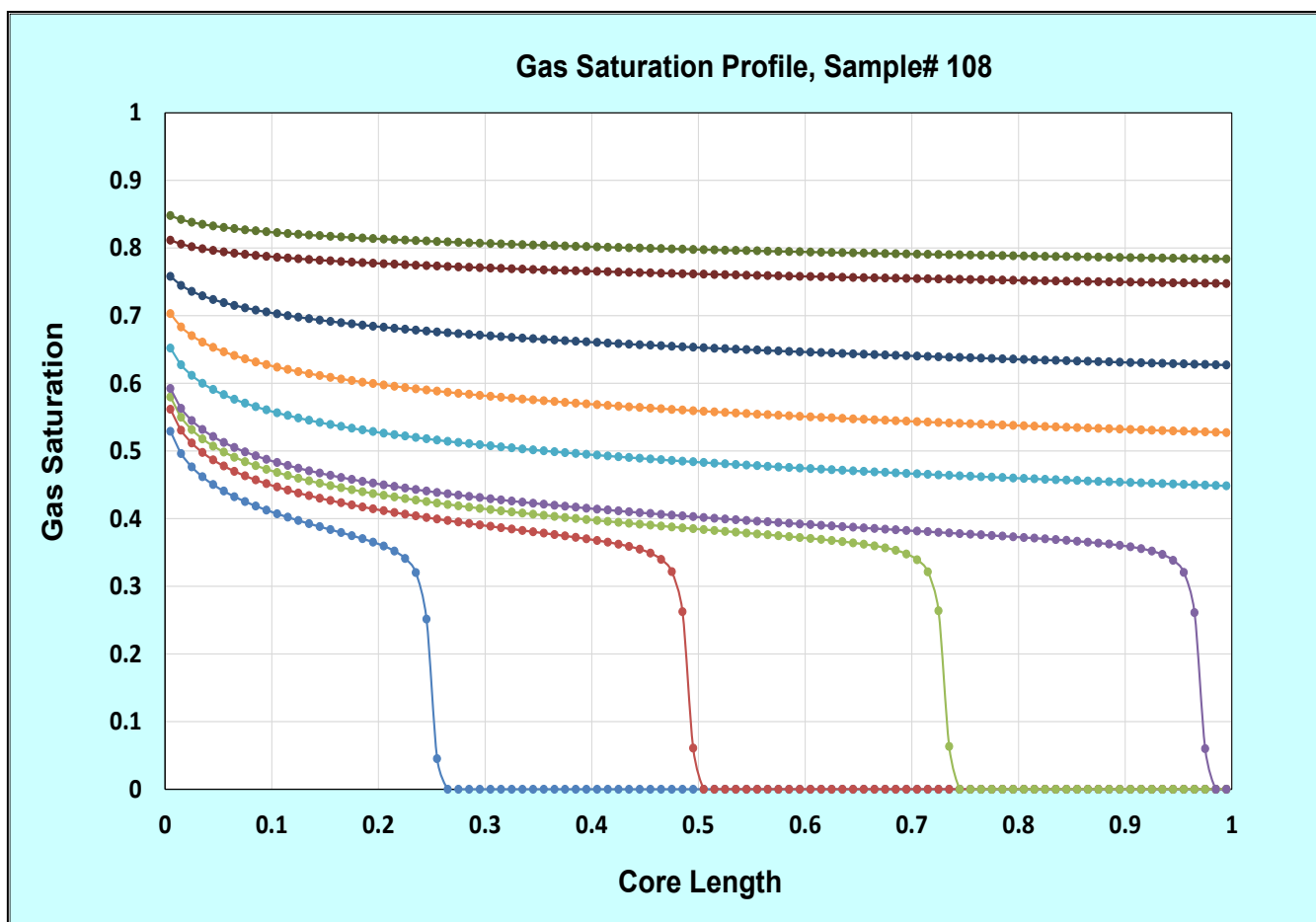


Figure (2.9.12)

GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1200
Well	: CE03-NC41	Permeability to water., mD	: 4.99
Sample #	: 126	Perm. to gas at irreducible water sat., mD	: 3.42
Porosity, %	: 24.4	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 39.4	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8483.10
Field	: Bahr Essalam		

Gas Saturation (fraction)	Relative Permeability to Water, (fraction)	Relative Permeability to Gas, (fraction)	Relative Permeability Ratio
0.000000	1.000000	0.000000	0.000000
0.017959	9.08E-01	0.000000	0.000000
0.035918	8.22E-01	0.000004	0.000004
0.053878	7.43E-01	0.000017	0.000023
0.071837	6.70E-01	0.000050	0.000075
0.089796	6.03E-01	0.000117	0.000194
0.107755	5.41E-01	0.000234	0.000433
0.125714	4.85E-01	0.000421	0.000869
0.143673	4.33E-01	0.000699	0.001616
0.161633	3.85E-01	0.001094	0.002839
0.179592	3.42E-01	0.001632	0.004773
0.197551	0.302734	0.002345	0.007746
0.215510	0.267071	0.003264	0.012221
0.233469	0.234801	0.004424	0.018842
0.251429	0.205683	0.005863	0.028506
0.269388	0.179485	0.007621	0.042459
0.287347	0.155989	0.009739	0.062432
0.305306	0.134984	0.012262	0.090839
0.323265	0.116273	0.015236	0.131040
0.341224	0.099666	0.018712	0.187742
0.359184	0.084986	0.022739	0.26755
0.377143	0.072065	0.027370	0.37980
0.395102	0.060742	0.032663	0.53773
0.413061	0.050869	0.038674	0.76026
0.431020	0.042306	0.045462	1.07461

Table (2.9.4)

GAS-WATER RELATIVE PERMEABILITY

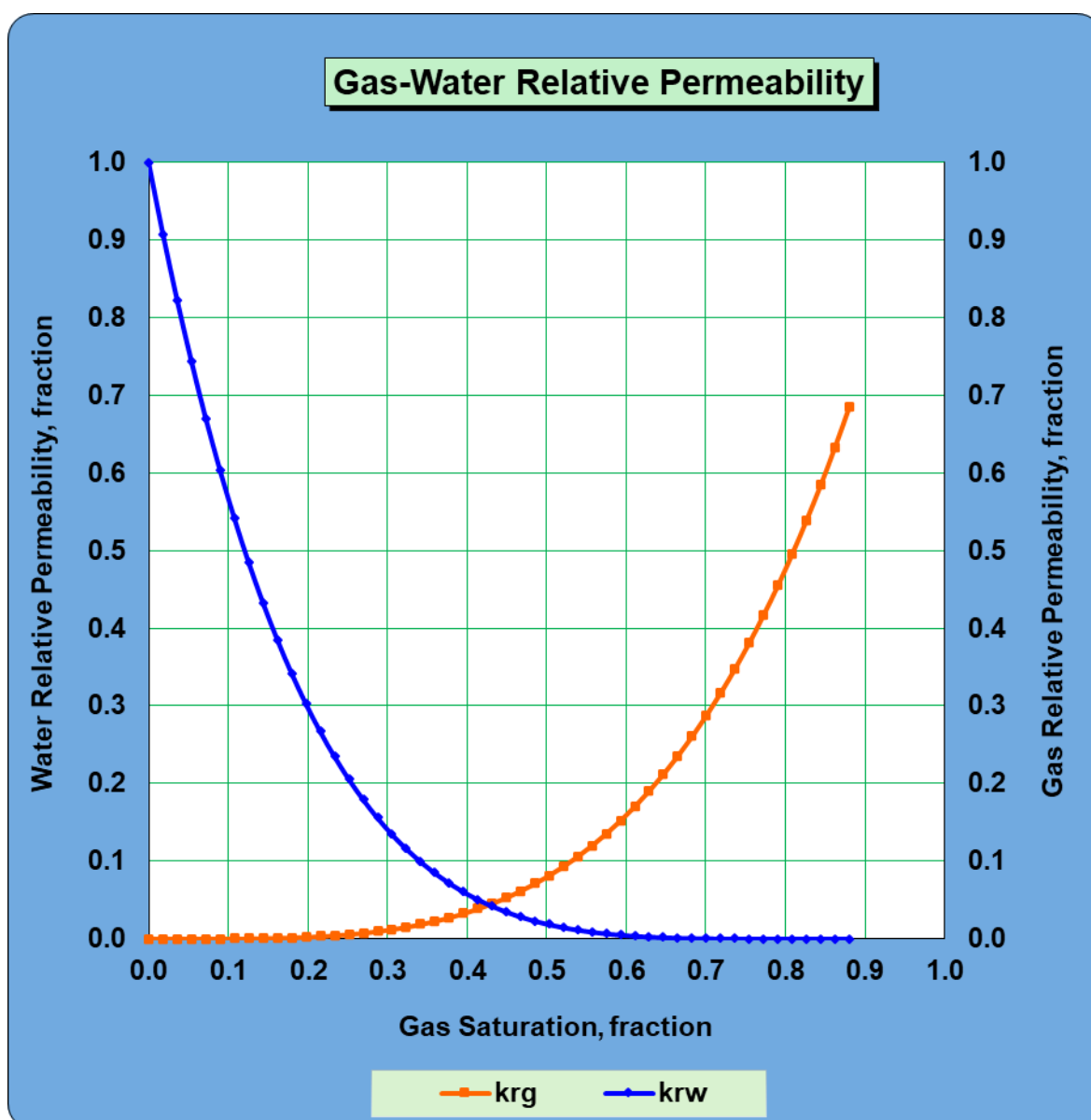
Client	: Mellitah	Irreducible water saturation, fraction	: 0.1200
Well	: CE03-NC41	Permeability to water., mD	: 4.99
Sample #	: 126	Perm. to gas at irreducible water sat., mD	: 3.42
Porosity, %	: 24.4	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 39.4	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8483.10
Field	: Bahr Essalam		

Gas Saturation (fraction)	Relative Permeability to Water, (fraction)	Relative Permeability to Gas, (fraction)	Relative Permeability Ratio
0.448980	0.034920	0.053091	1.5204
0.466939	0.028589	0.061623	2.1555
0.484898	0.023199	0.071126	3.0659
0.502857	0.018643	0.081667	4.3807
0.520816	0.014822	0.093317	6.2957
0.538776	0.011647	0.106147	9.114
0.556735	0.009034	0.120232	13.309
0.574694	0.006905	0.135649	19.644
0.592653	0.005193	0.152476	29.360
0.610612	0.003834	0.170792	44.54
0.628571	0.002773	0.190680	68.77
0.646531	0.001957	0.212225	108.44
0.664490	0.001343	0.235512	175.3
0.682449	0.000893	0.260630	292.0
0.700408	0.000570	0.287669	504
0.718367	0.000348	0.316720	911
0.736327	0.000200	0.347877	1741
0.754286	0.000107	0.381237	3.57E+03
0.772245	0.000052	0.416896	8.07E+03
0.790204	0.000022	0.454954	2.07E+04
0.808163	0.000008	0.495513	6.45E+04
0.826122	0.000002	0.538675	2.71E+05
0.844082	0.000000	0.584547	1.98E+06
0.862041	0.000000	0.633234	5.57E+07
0.880000	0.000000	0.684845	

Table (2.9.4)-Continue

GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1200
Well	: CE03-NC41	Permeability to water., mD	: 4.99
Sample #	: 126	Perm. to gas at irreducible water sat., mD	: 3.42
Porosity, %	: 24.4	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 39.4	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8483.10
Field	: Bahr Essalam		



GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1200
Well	: CE03-NC41	Permeability to water., mD	: 4.99
Sample #	: 126	Perm. to gas at irreducible water sat., mD	: 3.42
Porosity, %	: 24.4	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 39.4	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8483.10
Field	: Bahr Essalam		

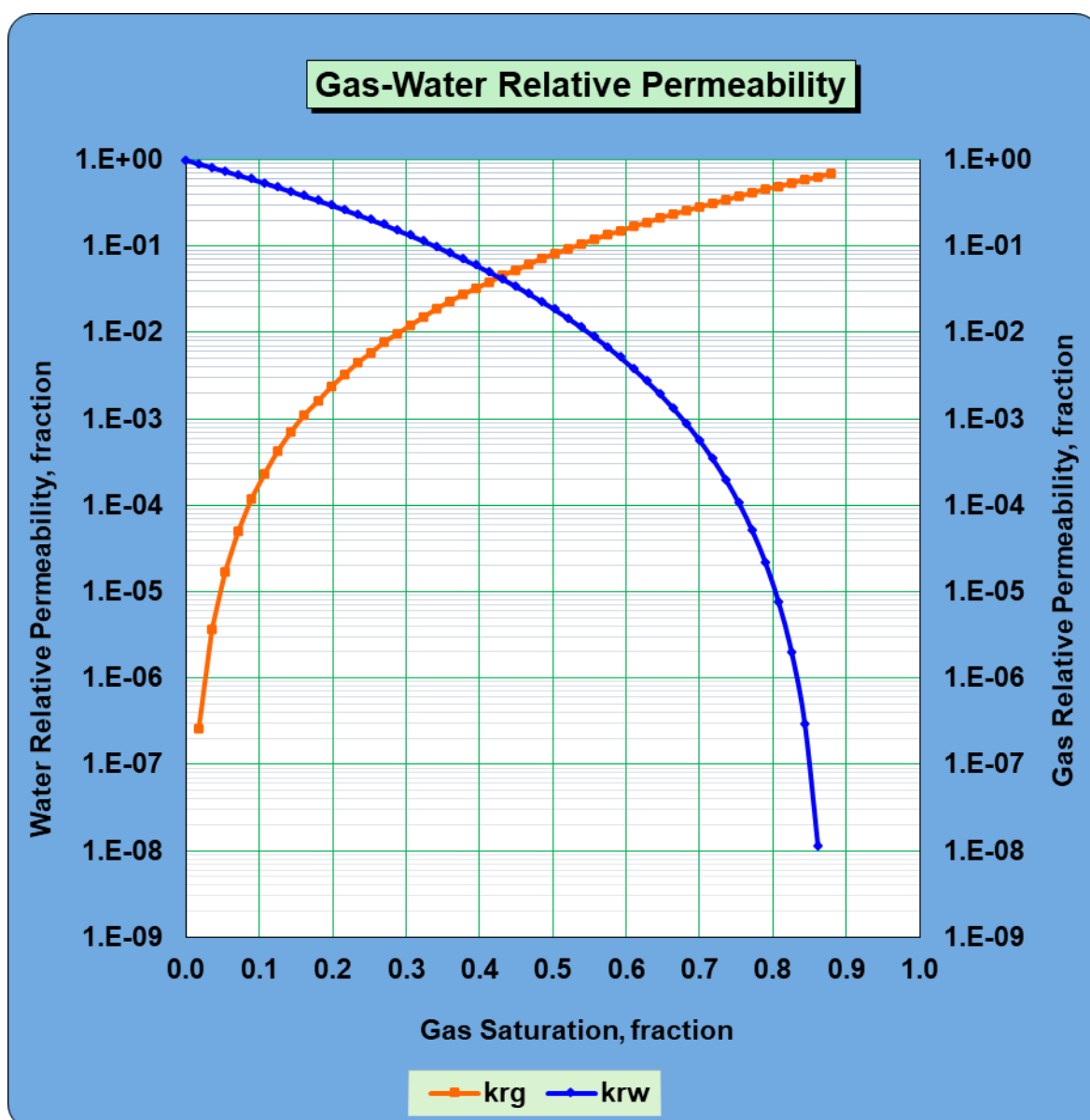


Figure (2.9.14)

GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1200
Well	: CE03-NC41	Permeability to water., mD	: 4.99
Sample #	: 126	Perm. to gas at irreducible water sat., mD	: 3.42
Porosity, %	: 24.4	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 39.4	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8483.10
Field	: Bahr Essalam		

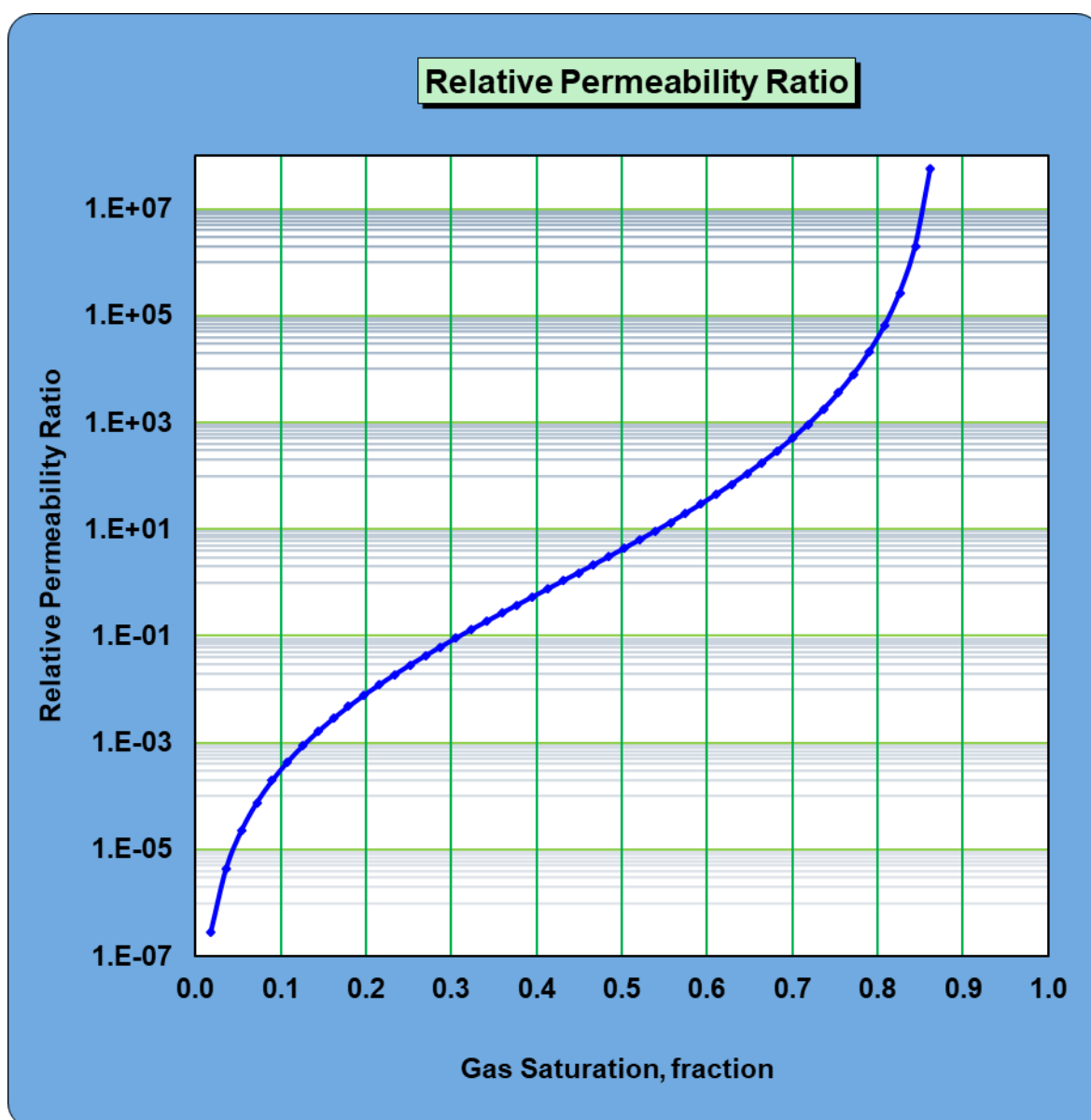


Figure (2.9.15)

GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1200
Well	: CE03-NC41	Permeability to water., mD	: 4.99
Sample #	: 126	Perm. to gas at irreducible water sat., mD	: 3.42
Porosity, %	: 24.4	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 39.4	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8483.10
Field	: Bahr Essalam		

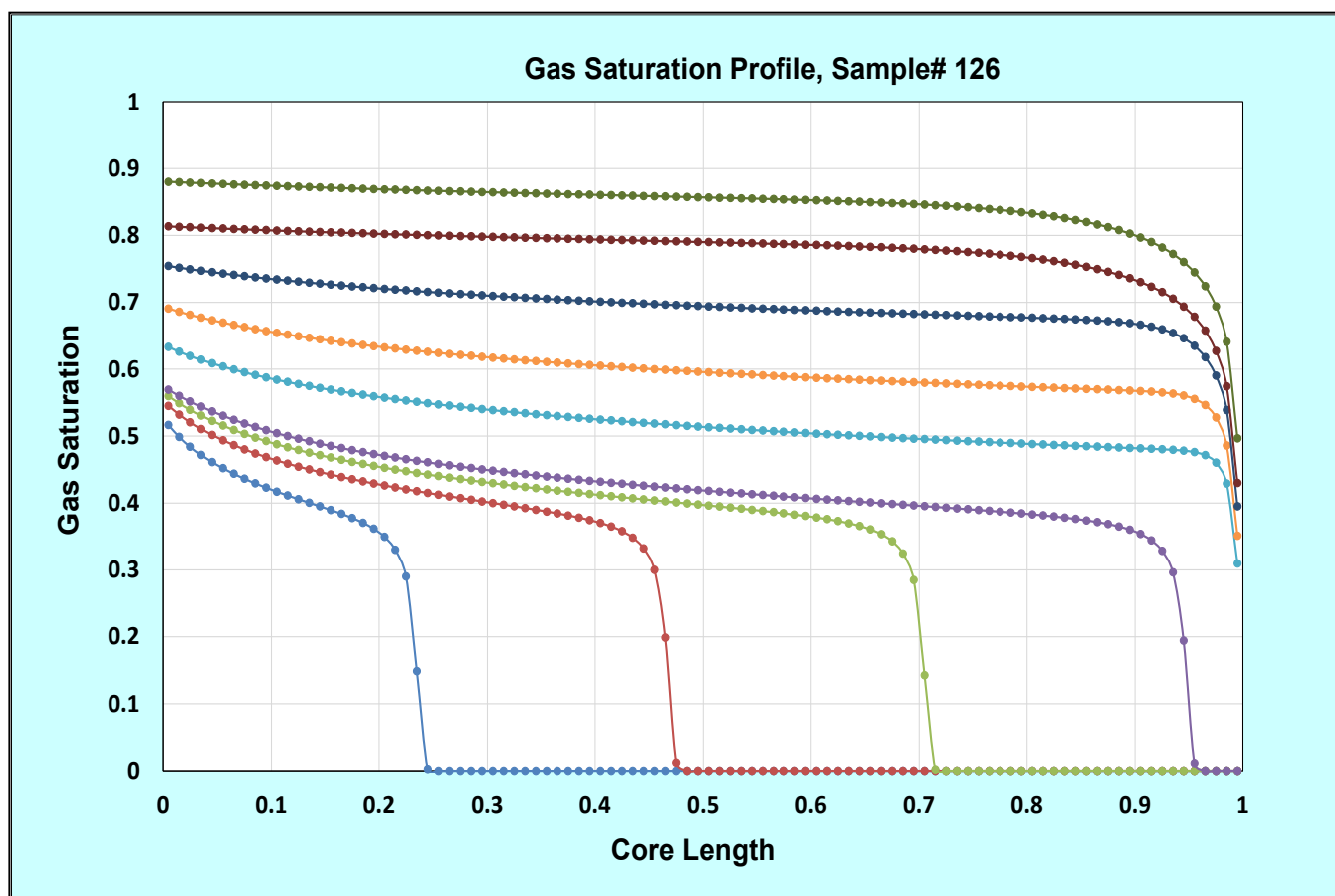


Figure (2.9.16)

GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1800
Well	: CE03-NC41	Permeability to water., mD	: 27.0
Sample #	: 275	Perm. to gas at irreducible water sat., mD	: 17.55
Porosity, %	: 24.2	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 225	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8636.06
Field	: Bahr Essalam		

Gas Saturation (fraction)	Relative Permeability to Water, (fraction)	Relative Permeability to Gas, (fraction)	Relative Permeability Ratio
0.000000	1.000000	0.000000	0.000000
0.016735	8.93E-01	0.000004	0.000004
0.033469	7.95E-01	0.000032	0.000041
0.050204	7.06E-01	0.000113	0.000161
0.066939	6.26E-01	0.000277	0.000442
0.083673	5.53E-01	0.000552	0.000998
0.100408	4.88E-01	0.000971	0.001992
0.117143	4.28E-01	0.001566	0.003655
0.133878	3.75E-01	0.002368	0.006312
0.150612	3.28E-01	0.003411	0.010414
0.167347	2.85E-01	0.004727	0.016590
0.184082	0.247020	0.006351	0.025711
0.200816	0.213320	0.008316	0.038984
0.217551	0.183479	0.010657	0.058081
0.234286	0.157143	0.013407	0.085317
0.251020	0.133985	0.016602	0.123909
0.267755	0.113697	0.020277	0.178339
0.284490	0.095995	0.024466	0.254871
0.301224	0.080614	0.029206	0.362294
0.317959	0.067311	0.034532	0.513016
0.334694	0.055861	0.040479	0.72464
0.351429	0.046057	0.047085	1.02233
0.368163	0.037707	0.054384	1.44228
0.384898	0.030639	0.062414	2.03708
0.401633	0.024694	0.071211	2.88373

Table (2.9.5)

GAS-WATER RELATIVE PERMEABILITY

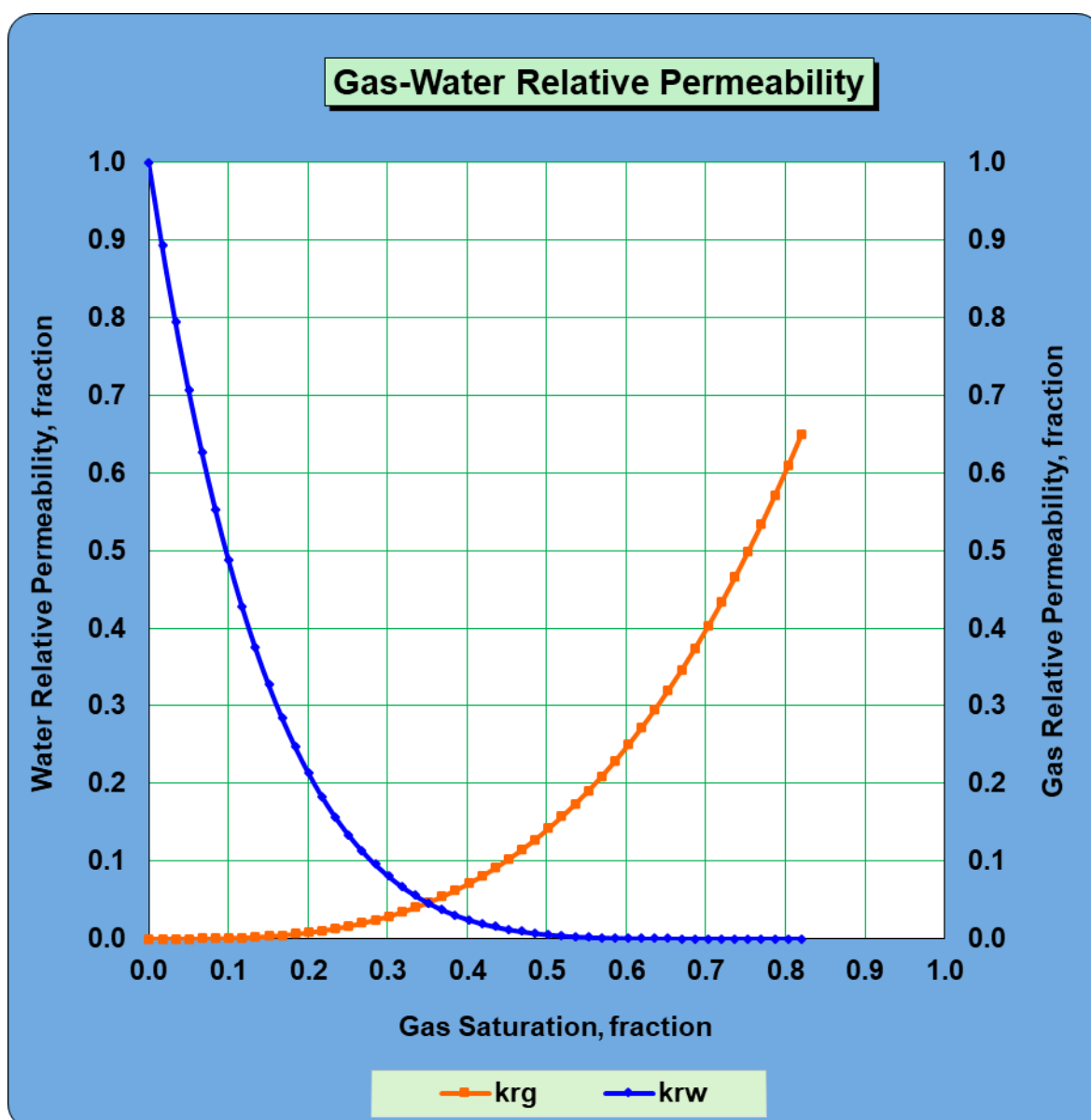
Client	: Mellitah	Irreducible water saturation, fraction	: 0.1800
Well	: CE03-NC41	Permeability to water., mD	: 27.0
Sample #	: 275	Perm. to gas at irreducible water sat., mD	: 17.55
Porosity, %	: 24.2	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 225	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8636.06
Field	: Bahr Essalam		

Gas Saturation (fraction)	Relative Permeability to Water, (fraction)	Relative Permeability to Gas, (fraction)	Relative Permeability Ratio
0.418367	0.019728	0.080811	4.0963
0.435102	0.015611	0.091252	5.8454
0.451837	0.012225	0.102570	8.3903
0.468571	0.009465	0.114803	12.1290
0.485306	0.007237	0.127988	17.6841
0.502041	0.005458	0.142162	26.045
0.518776	0.004054	0.157363	38.813
0.535510	0.002961	0.173628	58.645
0.552245	0.002121	0.190996	90.041
0.568980	0.001487	0.209503	140.85
0.585714	0.001018	0.229188	225.20
0.602449	0.000677	0.250090	369.39
0.619184	0.000436	0.272246	624.5
0.635918	0.000270	0.295695	1094.6
0.652653	0.000160	0.320474	2004
0.669388	0.000090	0.346624	3869
0.686122	0.000047	0.374181	7983
0.702857	0.000022	0.403185	1.79E+04
0.719592	0.000010	0.433676	4.50E+04
0.736327	0.000004	0.465690	1.32E+05
0.753061	0.000001	0.499269	4.82E+05
0.769796	0.000000	0.534449	2.51E+06
0.786531	0.000000	0.571272	2.50E+07
0.803265	0.000000	0.609776	1.21E+09
0.820000	0.000000	0.650000	*****

Table (2.9.5)-Continue

GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1800
Well	: CE03-NC41	Permeability to water., mD	: 27.0
Sample #	: 275	Perm. to gas at irreducible water sat., mD	: 17.55
Porosity, %	: 24.2	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 225	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8636.06
Field	: Bahr Essalam		



GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1800
Well	: CE03-NC41	Permeability to water., mD	: 27.0
Sample #	: 275	Perm. to gas at irreducible water sat., mD	: 17.55
Porosity, %	: 24.2	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 225	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8636.06
Field	: Bahr Essalam		

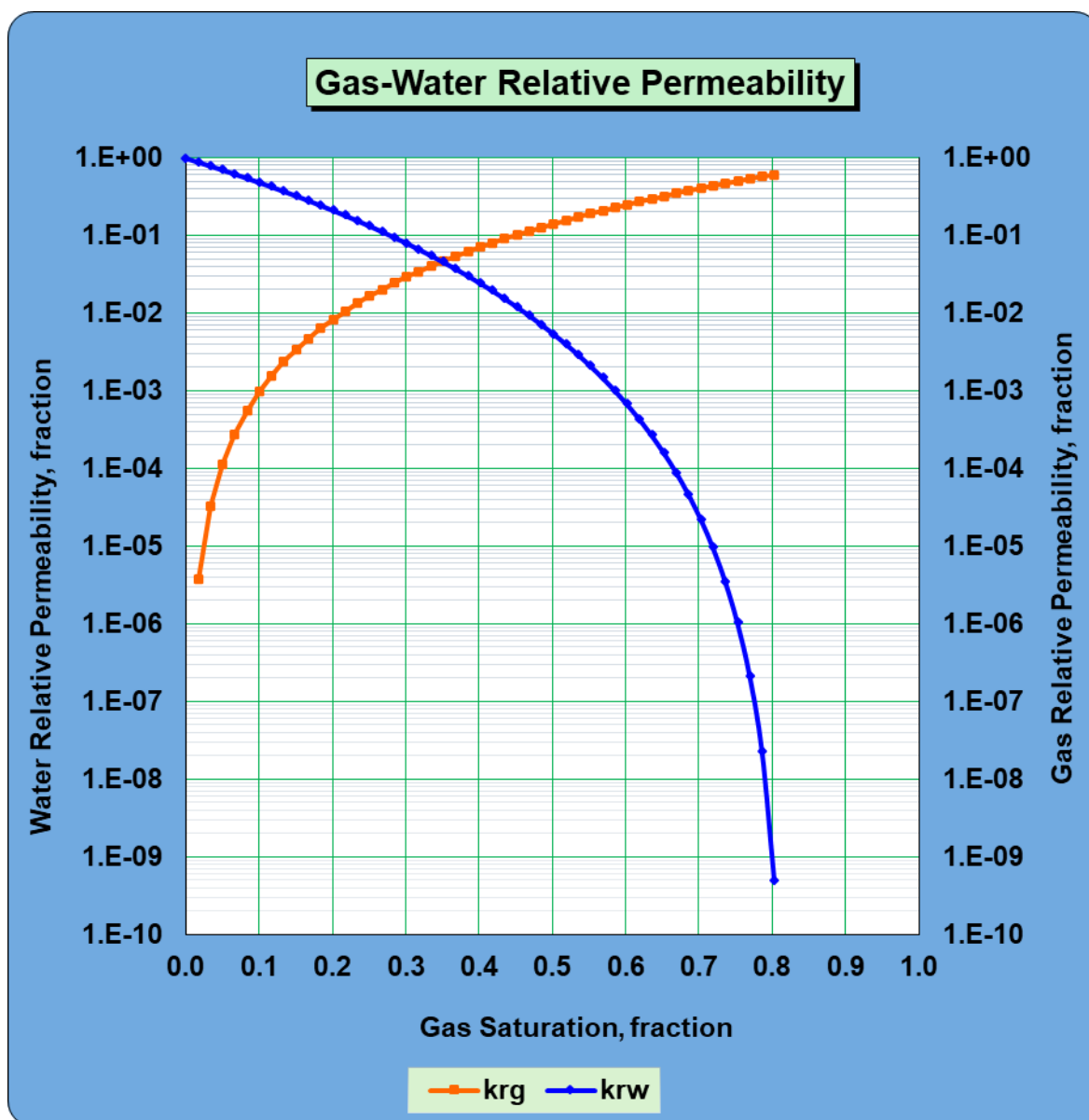


Figure (2.9.18)

GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1800
Well	: CE03-NC41	Permeability to water., mD	: 27.0
Sample #	: 275	Perm. to gas at irreducible water sat., mD	: 17.55
Porosity, %	: 24.2	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 225	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8636.06
Field	: Bahr Essalam		

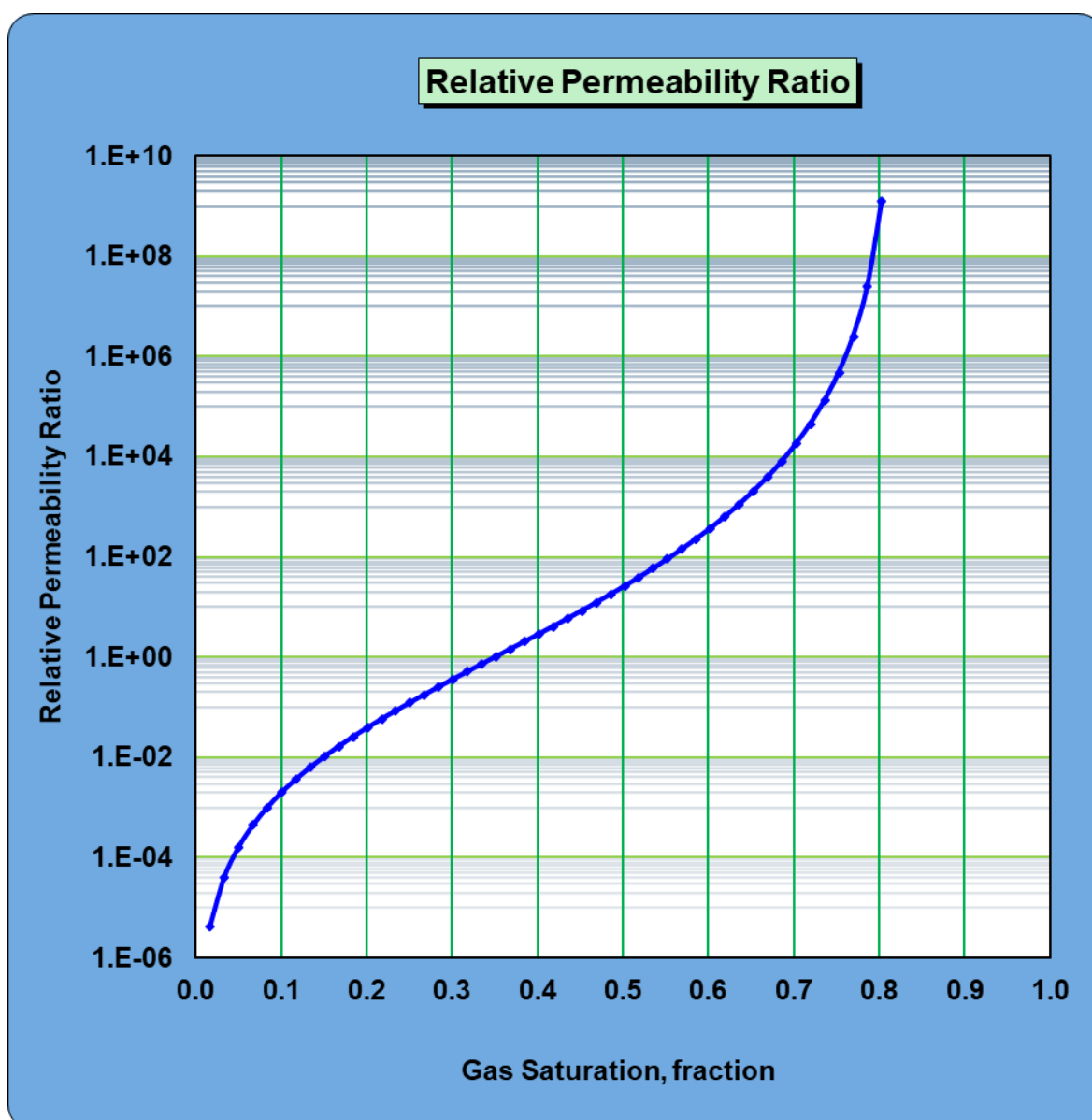


Figure (2.9.19)

GAS-WATER RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1800
Well	: CE03-NC41	Permeability to water., mD	: 27.0
Sample #	: 275	Perm. to gas at irreducible water sat., mD	: 17.55
Porosity, %	: 24.2	Net Confining Pressure, psi	: 3858
Permeability to air, mD	: 225	Temperature, °F	: 278
Formation	: Elgaria	Sample Depth, ft.in	: 8636.06
Field	: Bahr Essalam		

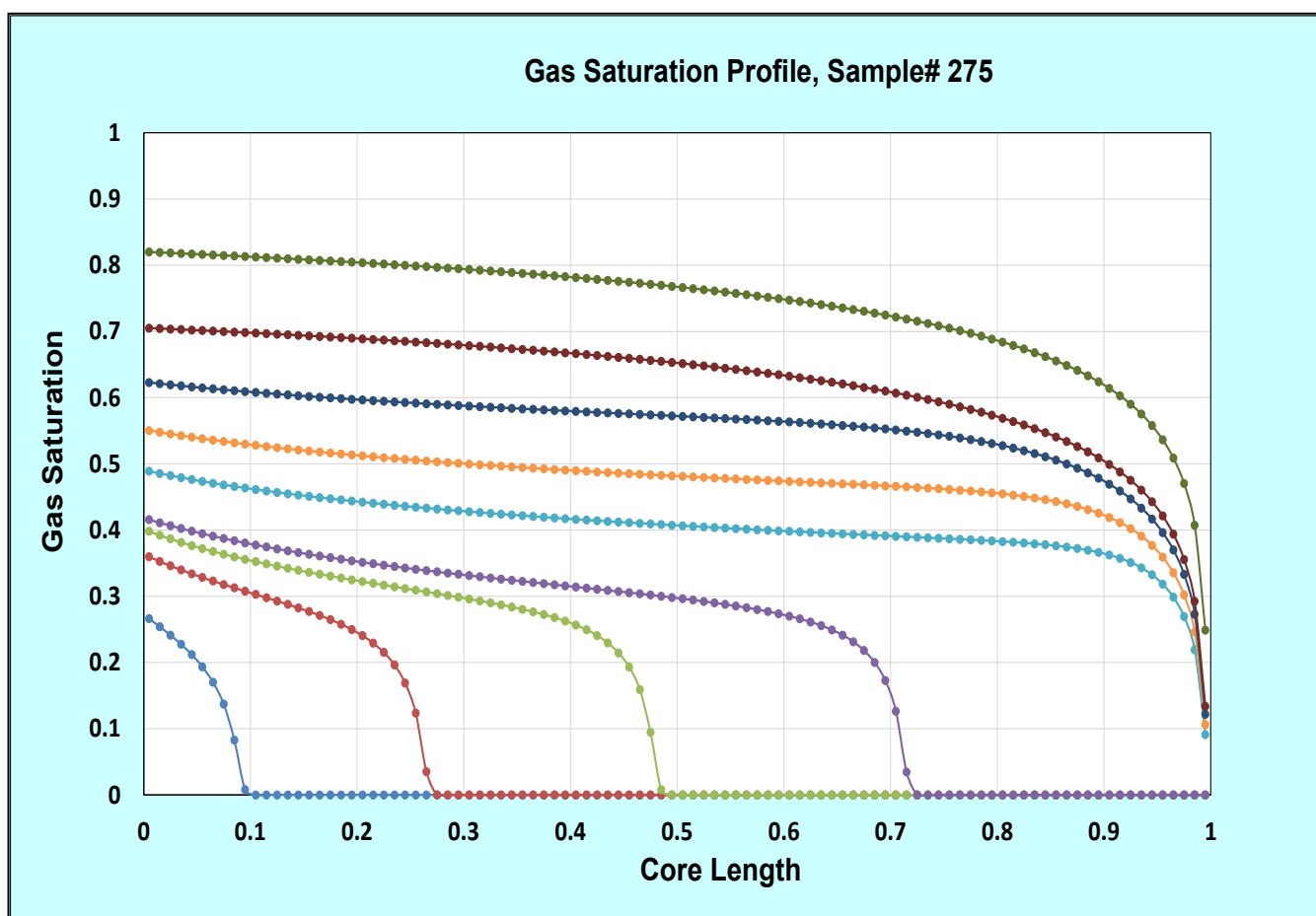


Figure (2.9.20)

Section Ten

Unsteady State Gas-Oil Relative Permeability (At Reservoir Conditions)

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.0754
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 9.32
Sample #	: 80	Perm. to gas at residual oil saturation, mD	: 6.685
Porosity, %	: 21.0	Residual oil saturation (Sorg), fraction	: 0.2271
Permeability to air, mD	: 51.7	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8437.06

Gas Saturation (fraction)	Relative Permeability to Oil, (fraction)	Relative Permeability to Gas, (fraction)	Relative Permeability Ratio
0.000000	1.000000	0.000000	0.000000
0.014234	0.908140	0.000000	0.000000
0.028468	0.823047	0.000003	0.000003
0.042702	0.744350	0.000013	0.000018
0.056936	0.671693	0.000041	0.000061
0.071170	0.604730	0.000098	0.000162
0.085404	0.543130	0.000199	0.000366
0.099638	0.486573	0.000363	0.000746
0.113872	0.434752	0.000611	0.001405
0.128106	0.387372	0.000967	0.002497
0.142340	0.344148	0.001459	0.004238
0.156574	0.304808	0.002115	0.006940
0.170808	0.269094	0.002970	0.011037
0.185041	0.236754	0.004058	0.017141
0.199275	0.207550	0.005418	0.026105
0.213509	0.181256	0.007091	0.039121
0.227743	0.157654	0.009120	0.057850
0.241977	0.136538	0.011553	0.084614
0.256211	0.117712	0.014438	0.122655
0.270445	0.100988	0.017827	0.176528
0.284679	0.086192	0.021775	0.252635
0.298913	0.073156	0.026339	0.360039
0.313147	0.061722	0.031579	0.511625
0.327381	0.051742	0.037556	0.725834
0.341615	0.043077	0.044337	1.029252
0.355849	0.035596	0.051989	1.460536

Table (2.10.1)

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.0754
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 9.32
Sample #	: 80	Perm. to gas at residual oil saturation, mD	: 6.685
Porosity, %	: 21.0	Residual oil saturation (Sorg), fraction	: 0.2271
Permeability to air, mD	: 51.7	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8437.06

Gas Saturation (fraction)	Relative Permeability to Oil, (fraction)	Relative Permeability to Gas, (fraction)	Relative Permeability Ratio
0.370083	0.029176	0.060582	2.076437
0.384317	0.023703	0.070188	2.961119
0.398551	0.019072	0.080884	4.240983
0.412785	0.015184	0.092746	6.108354
0.427019	0.011947	0.105856	8.860170
0.441253	0.009280	0.120297	12.963234
0.455487	0.007105	0.136154	19.16431
0.469721	0.005352	0.153514	28.68466
0.483955	0.003958	0.172469	43.57071
0.498189	0.002867	0.193112	67.34648
0.512423	0.002028	0.215538	106.27582
0.526656	0.001395	0.239845	171.90479
0.540890	0.000929	0.266134	286.4505
0.555124	0.000595	0.294507	494.8558
0.569358	0.000364	0.325071	893.7004
0.583592	0.000210	0.357932	1706.3069
0.597826	0.000112	0.393202	3498.456
0.612060	0.000055	0.430994	7881.062
0.626294	0.000023	0.471422	20209.112
0.640528	0.000008	0.514604	62587.22
0.654762	0.000002	0.560660	261557.98
0.668996	0.000000	0.609713	1891879.4
0.683230	0.000000	0.661888	52397305.3
0.697464	0.000000	0.717313	*****

Table (2.10.1)-Continue

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.0754
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 9.32
Sample #	: 80	Perm. to gas at residual oil saturation, mD	: 6.685
Porosity, %	: 21.0	Residual oil saturation (Sorg), fraction	: 0.2271
Permeability to air, mD	: 51.7	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8437.06

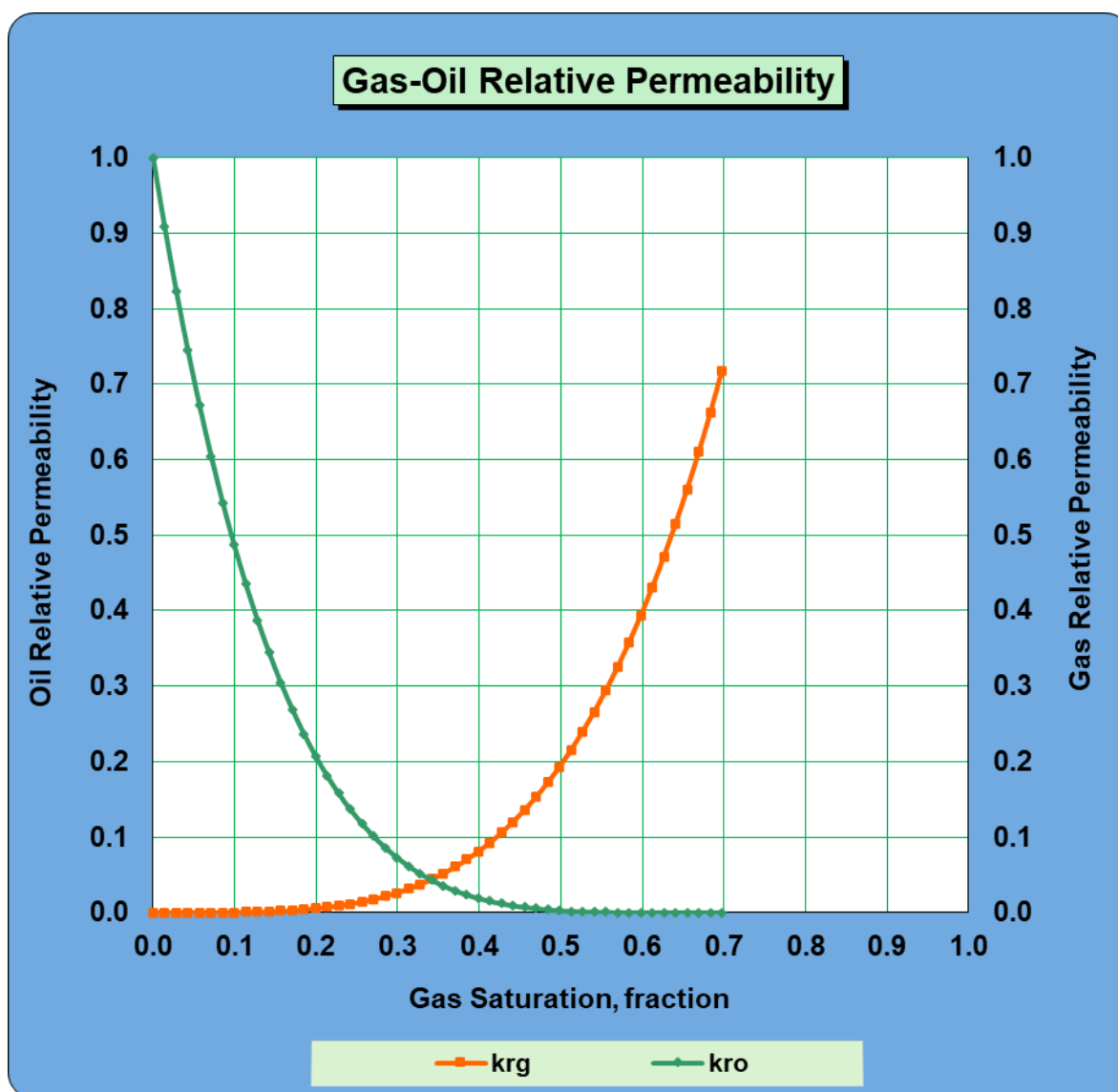


Figure (210.1)

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.0754
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 9.32
Sample #	: 80	Perm. to gas at residual oil saturation, mD	: 6.685
Porosity, %	: 21.0	Residual oil saturation (Sorg), fraction	: 0.2271
Permeability to air, mD	: 51.7	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8437.06

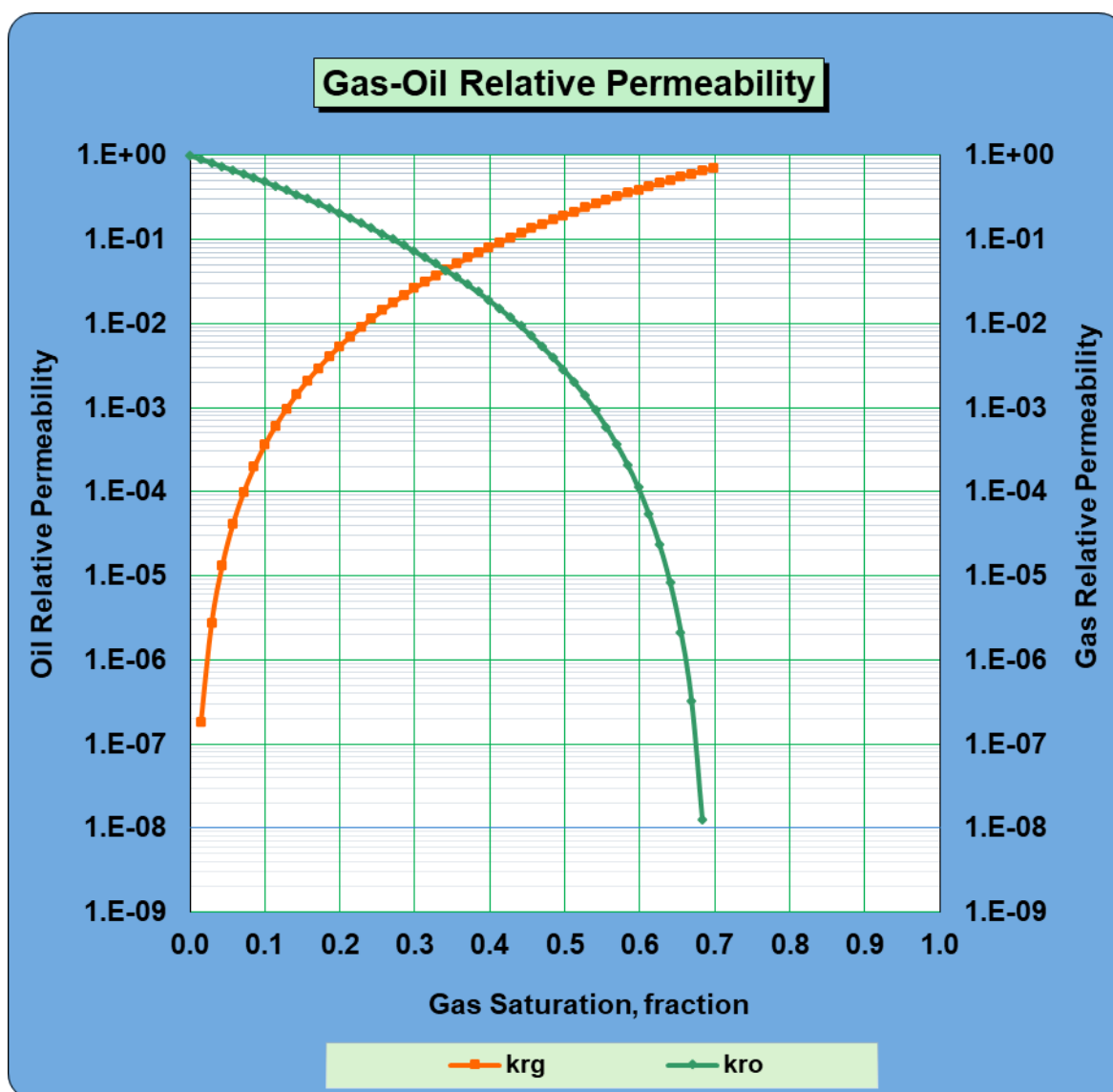


Figure (2.10.2)

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.0754
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 9.32
Sample #	: 80	Perm. to gas at residual oil saturation, mD	: 6.685
Porosity, %	: 21.0	Residual oil saturation (Sorg), fraction	: 0.2271
Permeability to air, mD	: 51.7	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8437.06

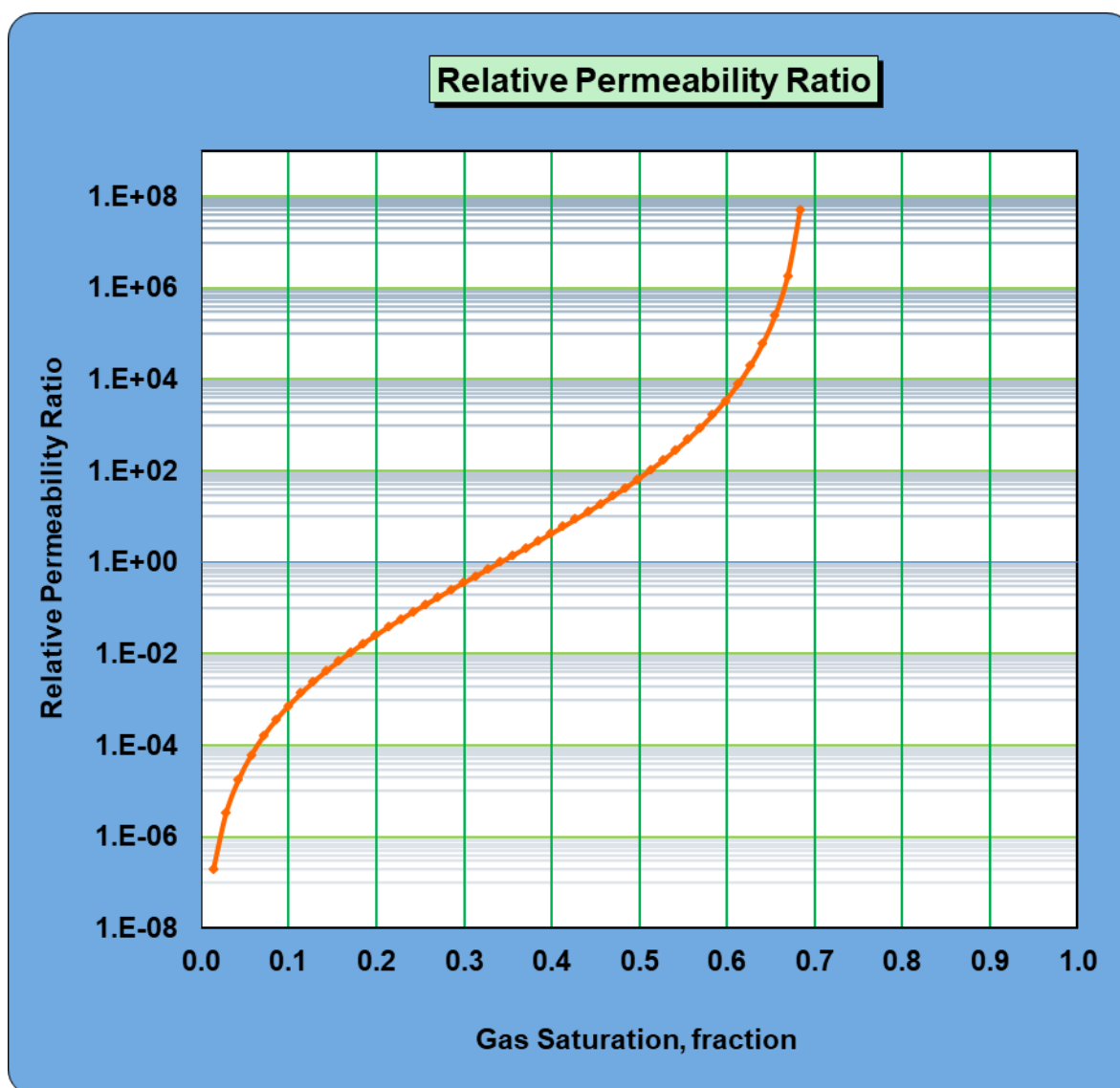


Figure (2.10.3)

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.0754
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 9.32
Sample #	: 80	Perm. to gas at residual oil saturation, mD	: 6.685
Porosity, %	: 21.0	Residual oil saturation (Sorg), fraction	: 0.2271
Permeability to air, mD	: 51.7	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8437.06

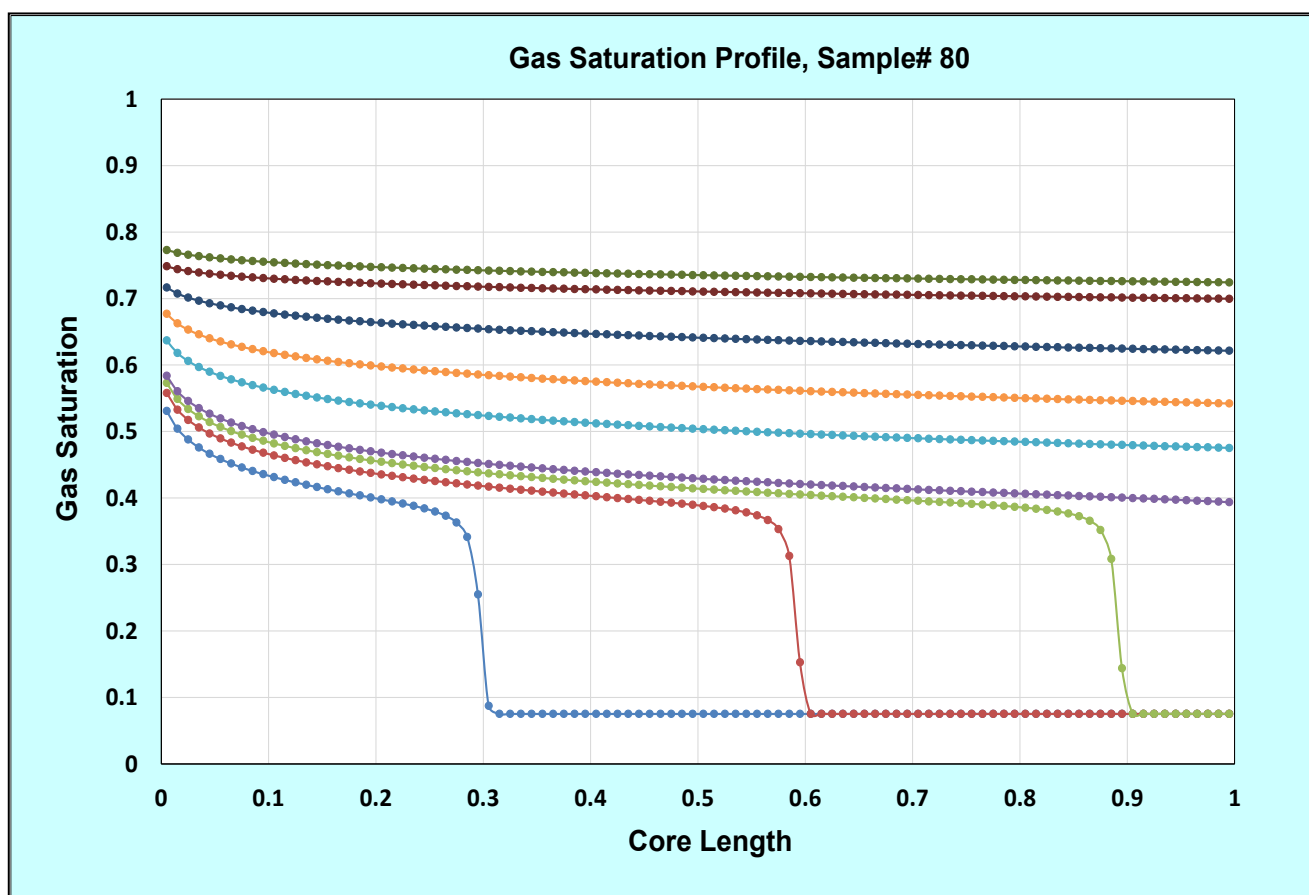


Figure (2.10.4)

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1800
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 4.22
Sample #	: 93	Perm. to gas at residual oil saturation, mD	: 3.334
Porosity, %	: 23.3	Residual oil saturation (Sorg), fraction	: 0.2400
Permeability to air, mD	: 18.9	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8450.06

Gas Saturation (fraction)	Relative Permeability to Oil, (fraction)	Relative Permeability to Gas, (fraction)	Relative Permeability Ratio
0.0000	1.0000	0.000E+00	0.000E+00
0.0118	0.9480	8.050E-10	8.491E-10
0.0237	0.8977	3.216E-08	3.582E-08
0.0355	0.8491	2.780E-07	3.274E-07
0.0473	0.8021	1.285E-06	1.602E-06
0.0592	0.7567	4.210E-06	5.564E-06
0.0710	0.7130	1.111E-05	1.558E-05
0.0829	0.6708	2.522E-05	3.759E-05
0.0947	0.6302	0.0001	8.142E-05
0.1065	0.5912	0.0001	1.624E-04
0.1184	0.5537	0.0002	3.038E-04
0.1302	0.5176	0.0003	5.395E-04
0.1420	0.4831	0.0004	9.183E-04
0.1539	0.4500	0.0007	1.509E-03
0.1657	0.4183	0.0010	2.408E-03
0.1776	0.3881	0.0015	3.747E-03
0.1894	0.3592	0.0020	5.706E-03
0.2012	0.3317	0.0028	8.532E-03
0.2131	0.3055	0.0038	1.255E-02
0.2249	0.2806	0.0051	1.822E-02
0.2367	0.2570	0.0067	2.614E-02
0.2486	0.2347	0.0087	3.711E-02
0.2604	0.2136	0.0112	5.222E-02
0.2722	0.1937	0.0141	7.295E-02
0.2841	0.1750	0.0177	1.013E-01
0.2959	0.1574	0.0220	1.399E-01

Table (2.10.2)

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1800
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 4.22
Sample #	: 93	Perm. to gas at residual oil saturation, mD	: 3.334
Porosity, %	: 23.3	Residual oil saturation (Sorg), fraction	: 0.2400
Permeability to air, mD	: 18.9	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8450.06

Gas Saturation (fraction)	Relative Permeability to Oil, (fraction)	Relative Permeability to Gas, (fraction)	Relative Permeability Ratio
0.3078	0.1410	0.0271	1.924E-01
0.3196	0.1257	0.0332	2.639E-01
0.3314	0.1114	0.0402	3.612E-01
0.3433	0.0982	0.0485	4.939E-01
0.3551	0.0860	0.0581	6.756E-01
0.3669	0.0747	0.0692	9.253E-01
0.3788	0.0645	0.0819	1.270E+00
0.3906	0.0551	0.0964	1.751E+00
0.4024	0.0466	0.1130	2.425E+00
0.4143	0.0390	0.1319	3.383E+00
0.4261	0.0322	0.1532	4.762E+00
0.4380	0.0262	0.1773	6.779E+00
0.4498	0.0209	0.2043	9.787E+00
0.4616	0.0163	0.2346	1.438E+01
0.4735	0.0124	0.2684	2.162E+01
0.4853	0.0091	0.3061	3.345E+01
0.4971	0.0065	0.3479	5.374E+01
0.5090	0.0043	0.3943	9.079E+01
0.5208	0.0027	0.4456	1.645E+02
0.5327	0.0015	0.5022	3.305E+02
0.5445	0.0007	0.5645	7.826E+02
0.5563	0.0003	0.6329	2.508E+03
0.5682	0.0000	0.7079	1.689E+04
0.5800	0.0000	0.7900	*****

Table (2.10.2)-Continue

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1800
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 4.22
Sample #	: 93	Perm. to gas at residual oil saturation, mD	: 3.334
Porosity, %	: 23.3	Residual oil saturation (Sorg), fraction	: 0.2400
Permeability to air, mD	: 18.9	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8450.06

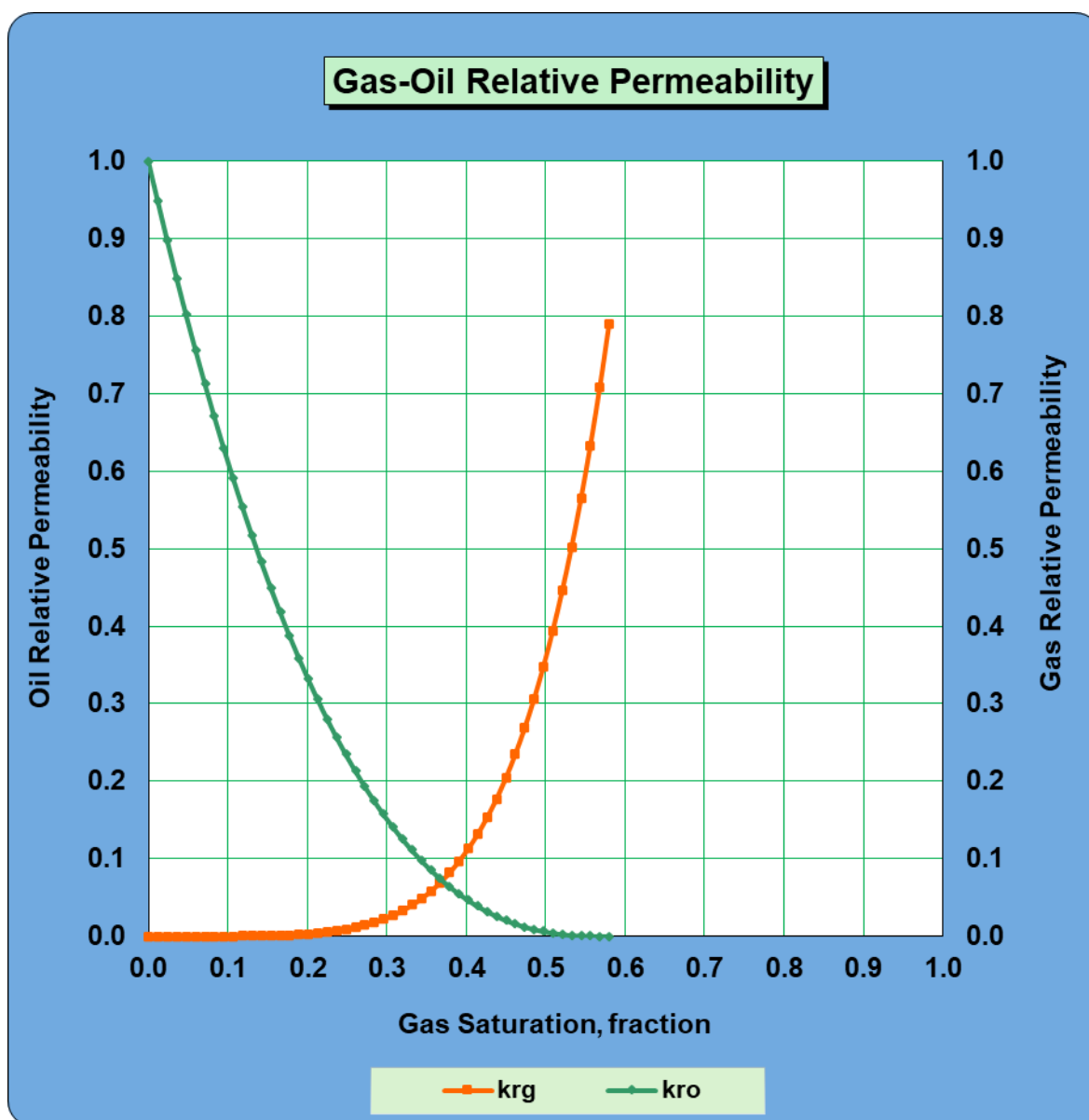


Figure (2.10.5)

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1800
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 4.22
Sample #	: 93	Perm. to gas at residual oil saturation, mD	: 3.334
Porosity, %	: 23.3	Residual oil saturation (Sorg), fraction	: 0.2400
Permeability to air, mD	: 18.9	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8450.06

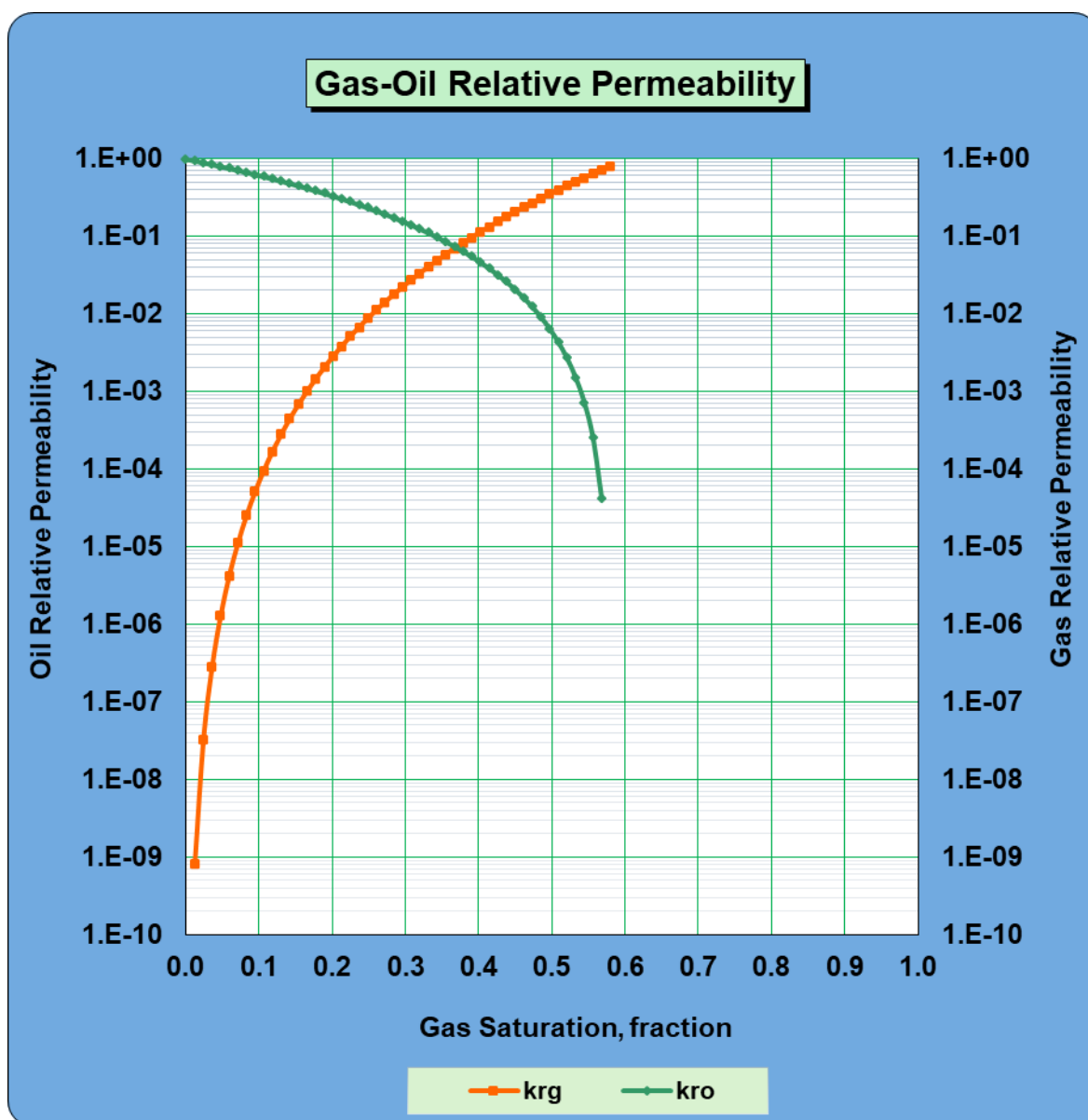


Figure (2.10.6)

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1800
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 4.22
Sample #	: 93	Perm. to gas at residual oil saturation, mD	: 3.334
Porosity, %	: 23.3	Residual oil saturation (Sorg), fraction	: 0.2400
Permeability to air, mD	: 18.9	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8450.06

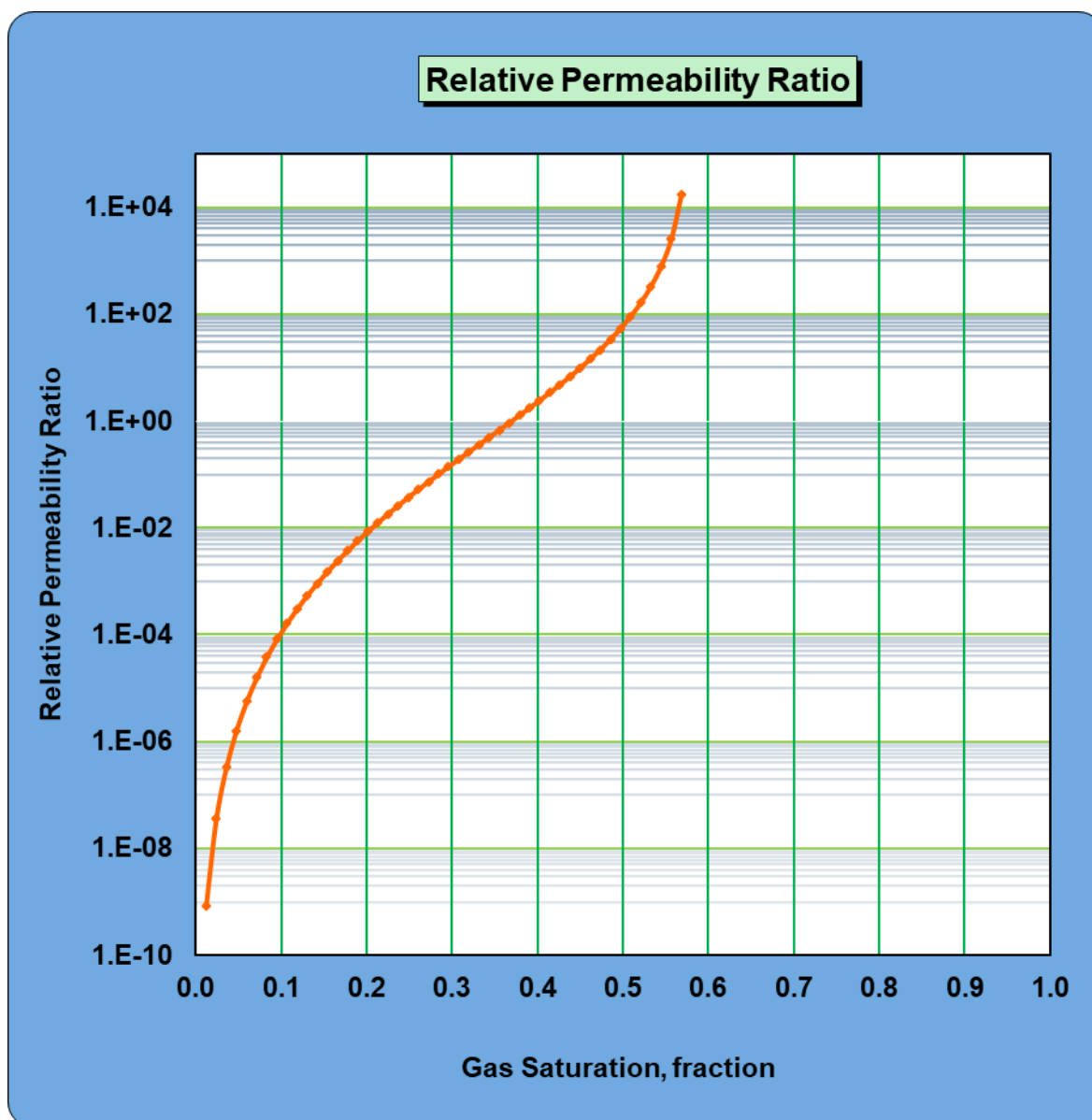


Figure (2.10.7)

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1800
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 4.22
Sample #	: 93	Perm. to gas at residual oil saturation, mD	: 3.334
Porosity, %	: 23.3	Residual oil saturation (Sorg), fraction	: 0.2400
Permeability to air, mD	: 18.9	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8450.06

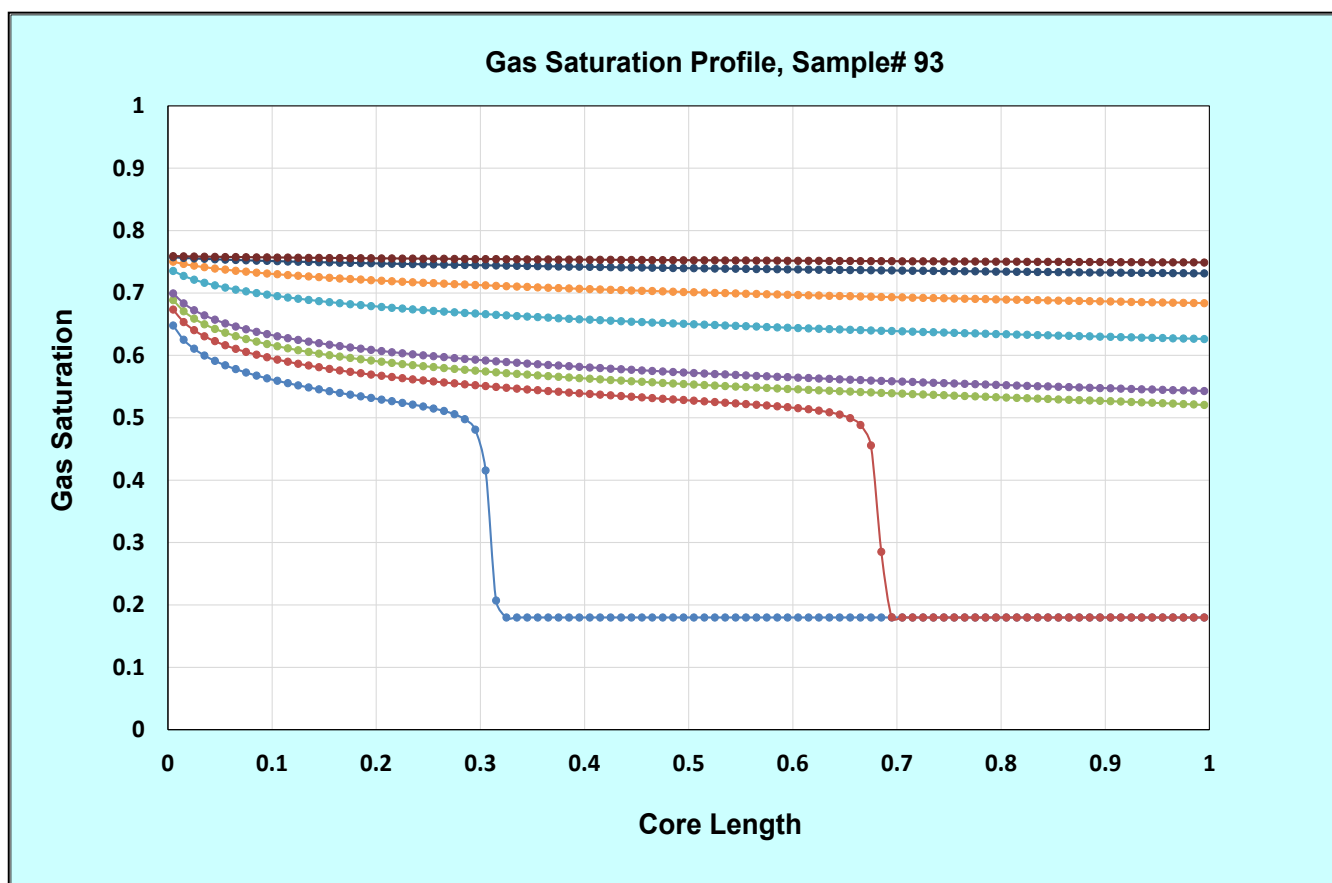


Figure (2.10.8)

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1523
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 10.97
Sample #	: 108	Perm. to gas at residual oil saturation, mD	: 8.666
Porosity, %	: 25.6	Residual oil saturation (Sorg), fraction	: 0.2497
Permeability to air, mD	: 62.8	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8465.05

Gas Saturation (fraction)	Relative Permeability to Oil, (fraction)	Relative Permeability to Gas, (fraction)	Relative Permeability Ratio
0.0000	1.0000	0.000E+00	0.000E+00
0.0122	0.9400	6.091E-09	6.480E-09
0.0244	0.8825	1.697E-07	1.923E-07
0.0366	0.8273	1.188E-06	1.436E-06
0.0488	0.7746	4.727E-06	6.103E-06
0.0610	0.7241	1.380E-05	1.905E-05
0.0732	0.6758	3.310E-05	4.898E-05
0.0854	0.6297	0.0001	1.102E-04
0.0976	0.5858	0.0001	2.248E-04
0.1098	0.5440	0.0002	4.260E-04
0.1220	0.5042	0.0004	7.622E-04
0.1342	0.4664	0.0006	1.302E-03
0.1464	0.4305	0.0009	2.142E-03
0.1587	0.3966	0.0014	3.414E-03
0.1709	0.3644	0.0019	5.303E-03
0.1831	0.3341	0.0027	8.055E-03
0.1953	0.3055	0.0037	1.201E-02
0.2075	0.2785	0.0049	1.762E-02
0.2197	0.2532	0.0065	2.550E-02
0.2319	0.2295	0.0084	3.647E-02
0.2441	0.2073	0.0107	5.164E-02
0.2563	0.1866	0.0135	7.252E-02
0.2685	0.1673	0.0169	1.011E-01
0.2807	0.1494	0.0209	1.402E-01
0.2929	0.1328	0.0257	1.934E-01

Table (2.10.3)

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1523
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 10.97
Sample #	: 108	Perm. to gas at residual oil saturation, mD	: 8.666
Porosity, %	: 25.6	Residual oil saturation (Sorg), fraction	: 0.2497
Permeability to air, mD	: 62.8	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8465.05

Gas Saturation (fraction)	Relative Permeability to Oil, (fraction)	Relative Permeability to Gas, (fraction)	Relative Permeability Ratio
0.3173	0.1034	0.0377	3.647E-01
0.3295	0.0905	0.0452	4.995E-01
0.3417	0.0787	0.0538	6.839E-01
0.3539	0.0680	0.0637	9.369E-01
0.3661	0.0583	0.0750	1.286E+00
0.3783	0.0496	0.0877	1.770E+00
0.3905	0.0418	0.1022	2.447E+00
0.4027	0.0348	0.1185	3.402E+00
0.4149	0.0287	0.1367	4.765E+00
0.4271	0.0233	0.1571	6.736E+00
0.4393	0.0187	0.1799	9.632E+00
0.4516	0.0147	0.2051	1.397E+01
0.4638	0.0113	0.2332	2.061E+01
0.4760	0.0085	0.2641	3.107E+01
0.4882	0.0062	0.2982	4.813E+01
0.5004	0.0044	0.3358	7.716E+01
0.5126	0.0029	0.3769	1.293E+02
0.5248	0.0018	0.4220	2.299E+02
0.5370	0.0011	0.4713	4.435E+02
0.5492	0.0005	0.5249	9.650E+02
0.5614	0.0002	0.5833	2.542E+03
0.5736	0.0001	0.6468	9.512E+03
0.5858	0.0000	0.7156	8.418E+04
0.5980	0.0000	0.7900	*****

Table (2.10.3)-Continue

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1523
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 10.97
Sample #	: 108	Perm. to gas at residual oil saturation, mD	: 8.666
Porosity, %	: 25.6	Residual oil saturation (Sorg), fraction	: 0.2497
Permeability to air, mD	: 62.8	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8465.05

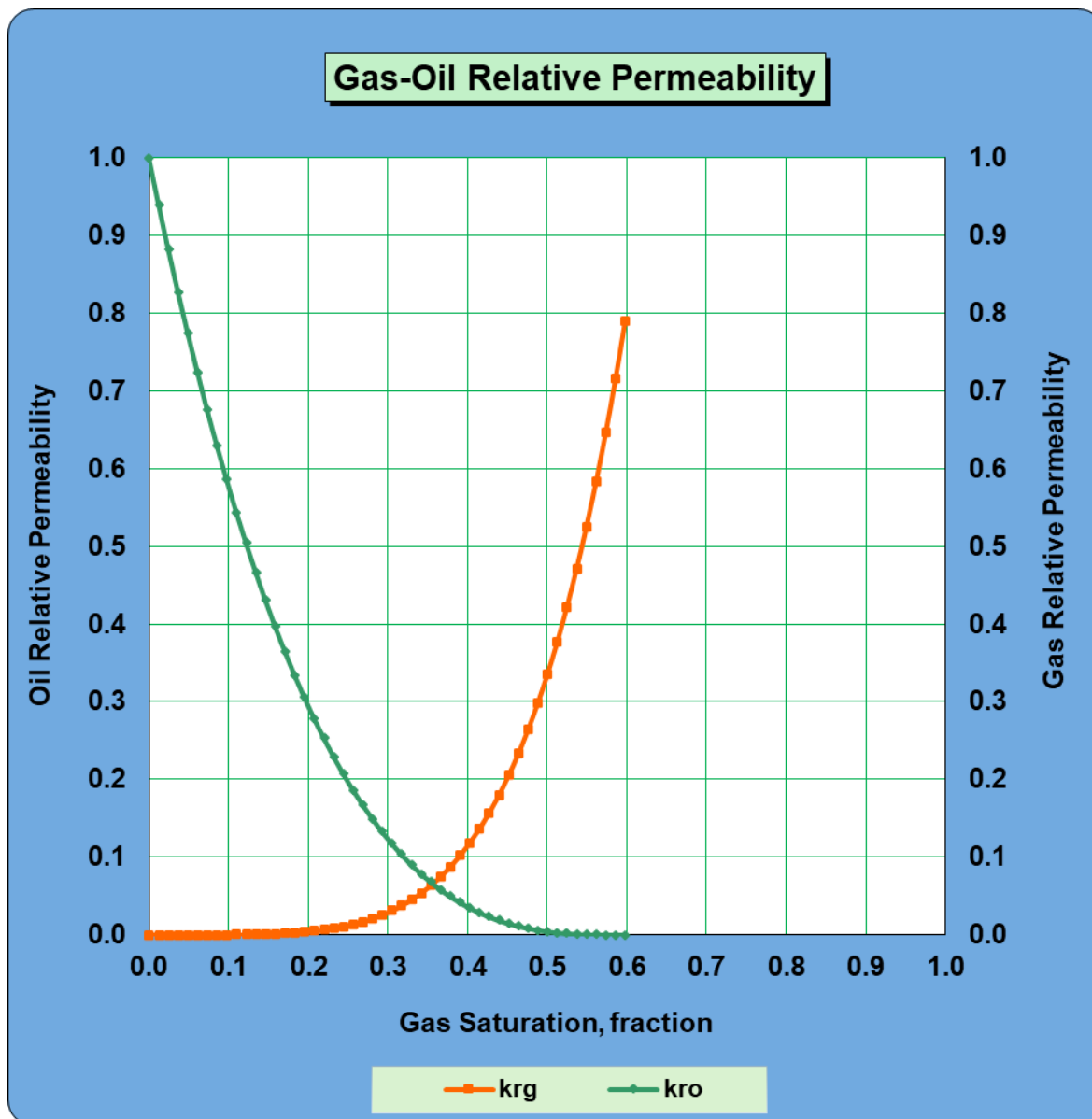


Figure (2.10.9)

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1523
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 10.97
Sample #	: 108	Perm. to gas at residual oil saturation, mD	: 8.666
Porosity, %	: 25.6	Residual oil saturation (Sorg), fraction	: 0.2497
Permeability to air, mD	: 62.8	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8465.05

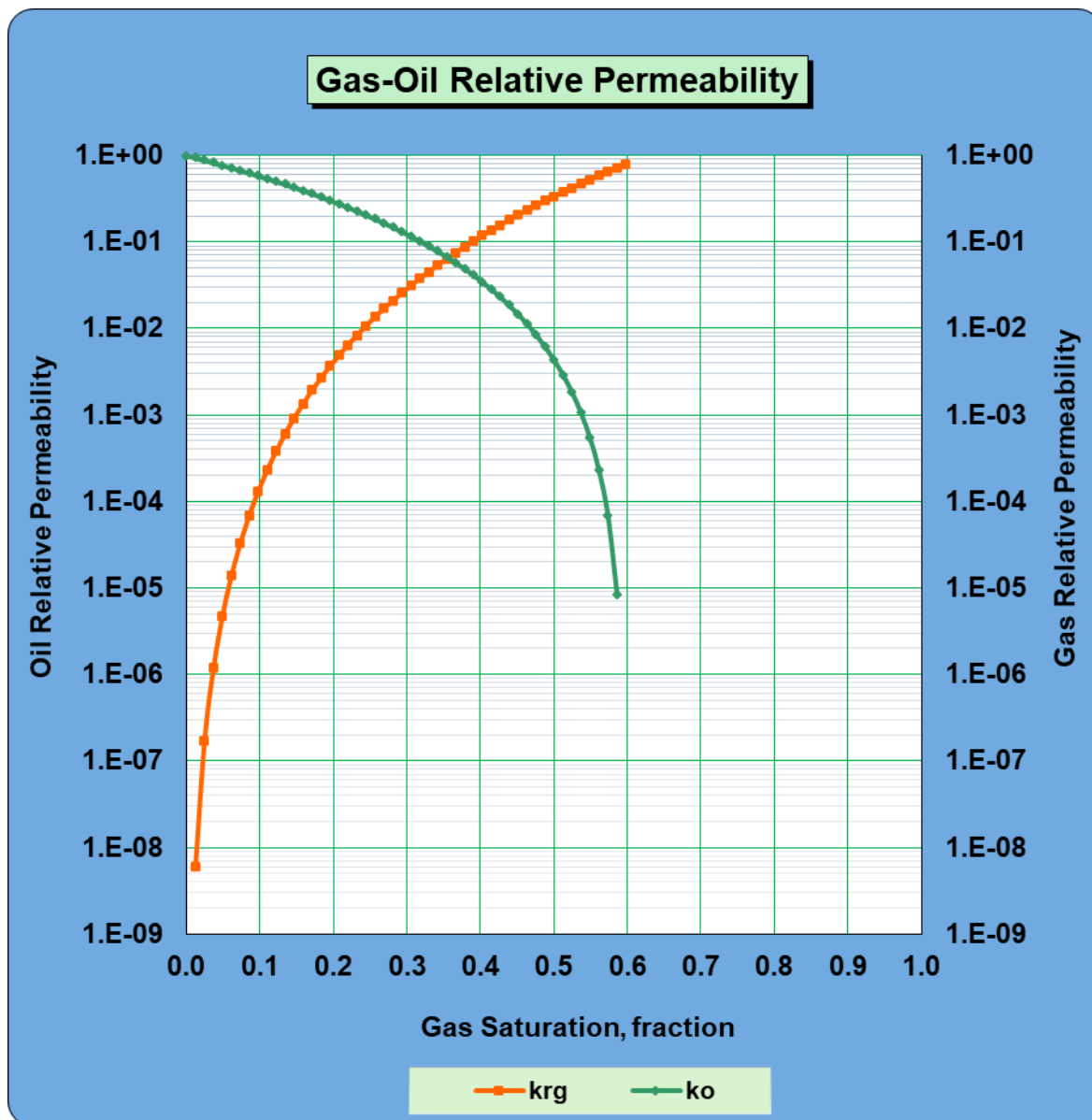


Figure (2.10.10)

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1523
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 10.97
Sample #	: 108	Perm. to gas at residual oil saturation, mD	: 8.666
Porosity, %	: 25.6	Residual oil saturation (S _{org}), fraction	: 0.2497
Permeability to air, mD	: 62.8	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8465.05

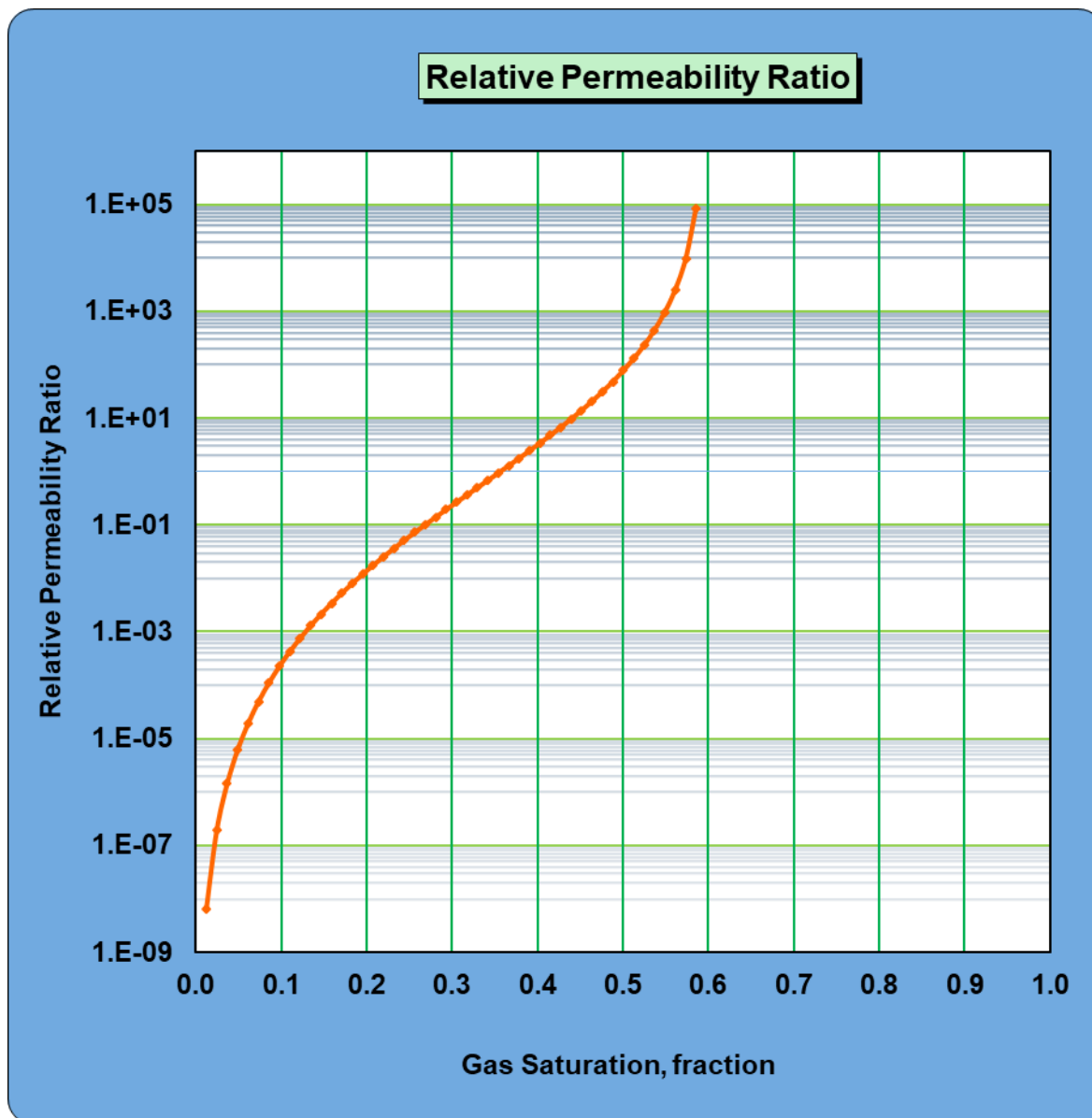


Figure (2.10.11)

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1523
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 10.97
Sample #	: 108	Perm. to gas at residual oil saturation, mD	: 8.666
Porosity, %	: 25.6	Residual oil saturation (Sorg), fraction	: 0.2497
Permeability to air, mD	: 62.8	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8465.05

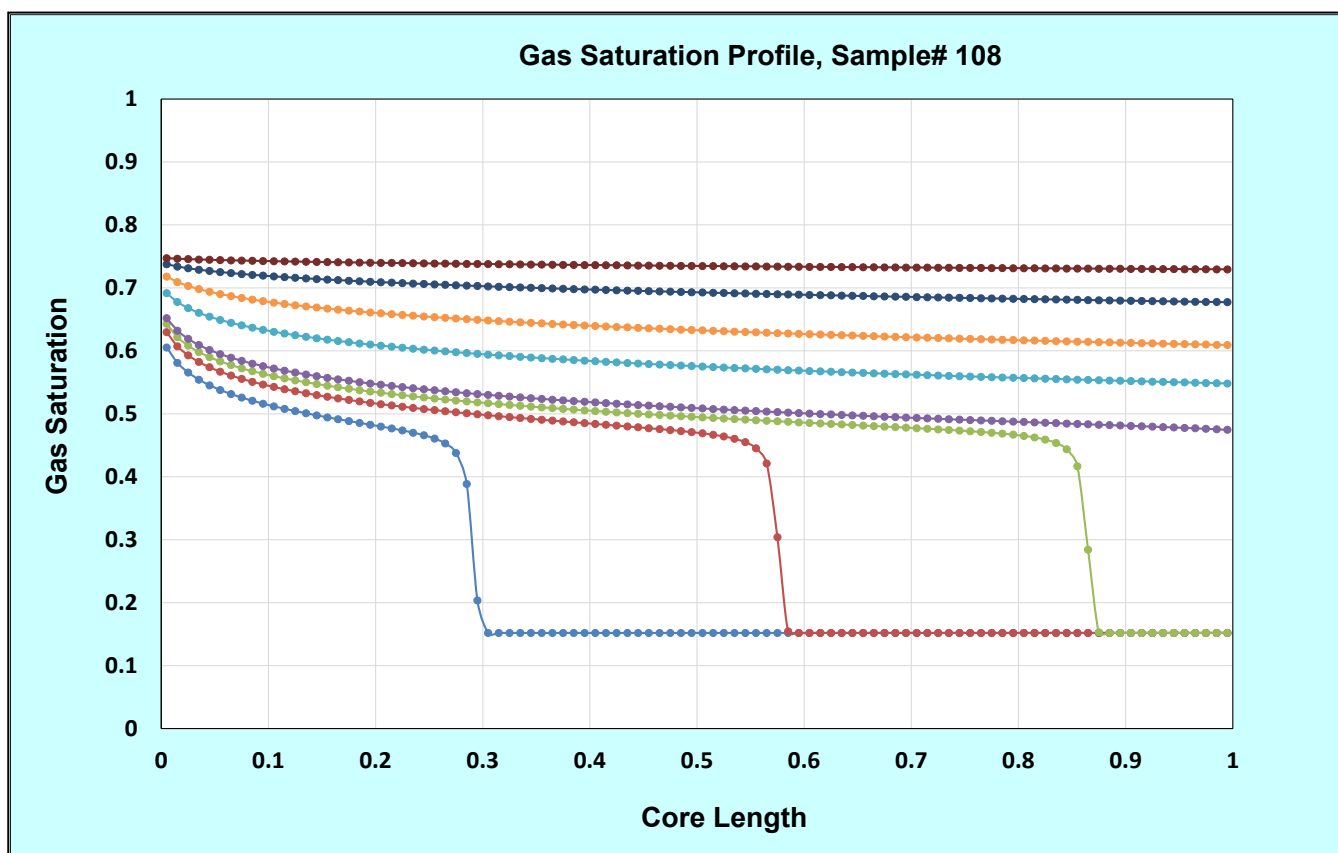


Figure (2.10.12)

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1200
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 1.84
Sample #	: 126	Perm. to gas at residual oil saturation, mD	: 1.49
Porosity, %	: 24.4	Residual oil saturation (Sorg), fraction	: 0.2200
Permeability to air, mD	: 39.4	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8483.10

Gas Saturation (fraction)	Relative Permeability to Oil, (fraction)	Relative Permeability to Gas, (fraction)	Relative Permeability Ratio
0.0000	1.0000	0.0000	0.000E+00
0.0135	0.9361	0.0000	1.602E-05
0.0269	0.8752	0.0001	1.193E-04
0.0404	0.8170	0.0003	3.978E-04
0.0539	0.7615	0.0007	9.551E-04
0.0673	0.7086	0.0014	1.917E-03
0.0808	0.6584	0.0023	3.438E-03
0.0943	0.6106	0.0035	5.707E-03
0.1078	0.5653	0.0051	8.960E-03
0.1212	0.5224	0.0070	1.348E-02
0.1347	0.4817	0.0095	1.964E-02
0.1482	0.4433	0.0124	2.787E-02
0.1616	0.4070	0.0158	3.873E-02
0.1751	0.3729	0.0197	5.290E-02
0.1886	0.3407	0.0243	7.124E-02
0.2020	0.3105	0.0294	9.482E-02
0.2155	0.2822	0.0353	1.250E-01
0.2290	0.2558	0.0418	1.634E-01
0.2424	0.2311	0.0491	2.123E-01
0.2559	0.2080	0.0571	2.743E-01
0.2694	0.1867	0.0659	3.530E-01
0.2829	0.1668	0.0755	4.528E-01
0.2963	0.1485	0.0860	5.794E-01
0.3098	0.1316	0.0974	7.404E-01
0.3233	0.1161	0.1098	9.457E-01
0.3367	0.1019	0.1231	1.208E+00

Table (2.10.4)

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1200
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 1.84
Sample #	: 126	Perm. to gas at residual oil saturation, mD	: 1.49
Porosity, %	: 24.4	Residual oil saturation (Sorg), fraction	: 0.2200
Permeability to air, mD	: 39.4	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8483.10

Gas Saturation (fraction)	Relative Permeability to Oil, (fraction)	Relative Permeability to Gas, (fraction)	Relative Permeability Ratio
0.3502	0.0889	0.1374	1.545E+00
0.3637	0.0771	0.1527	1.980E+00
0.3771	0.0664	0.1690	2.544E+00
0.3906	0.0568	0.1865	3.281E+00
0.4041	0.0482	0.2051	4.251E+00
0.4176	0.0406	0.2248	5.540E+00
0.4310	0.0338	0.2457	7.270E+00
0.4445	0.0278	0.2678	9.621E+00
0.4580	0.0226	0.2911	1.286E+01
0.4714	0.0182	0.3157	1.739E+01
0.4849	0.0143	0.3417	2.386E+01
0.4984	0.0111	0.3689	3.328E+01
0.5118	0.0084	0.3975	4.737E+01
0.5253	0.0062	0.4275	6.911E+01
0.5388	0.0044	0.4589	1.039E+02
0.5522	0.0030	0.4917	1.624E+02
0.5657	0.0020	0.5261	2.663E+02
0.5792	0.0012	5.619E-01	4.658E+02
0.5927	0.0007	5.992E-01	8.903E+02
0.6061	0.0003	6.382E-01	1.936E+03
0.6196	0.0001	6.787E-01	5.170E+03
0.6331	0.0000	7.208E-01	2.010E+04
0.6465	0.0000	7.646E-01	1.959E+05
0.6600	0.0000	8.100E-01	*****

Table (2.10.4)-Continue

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1200
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 1.84
Sample #	: 126	Perm. to gas at residual oil saturation, mD	: 1.49
Porosity, %	: 24.4	Residual oil saturation (Sorg), fraction	: 0.2200
Permeability to air, mD	: 39.4	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8483.10

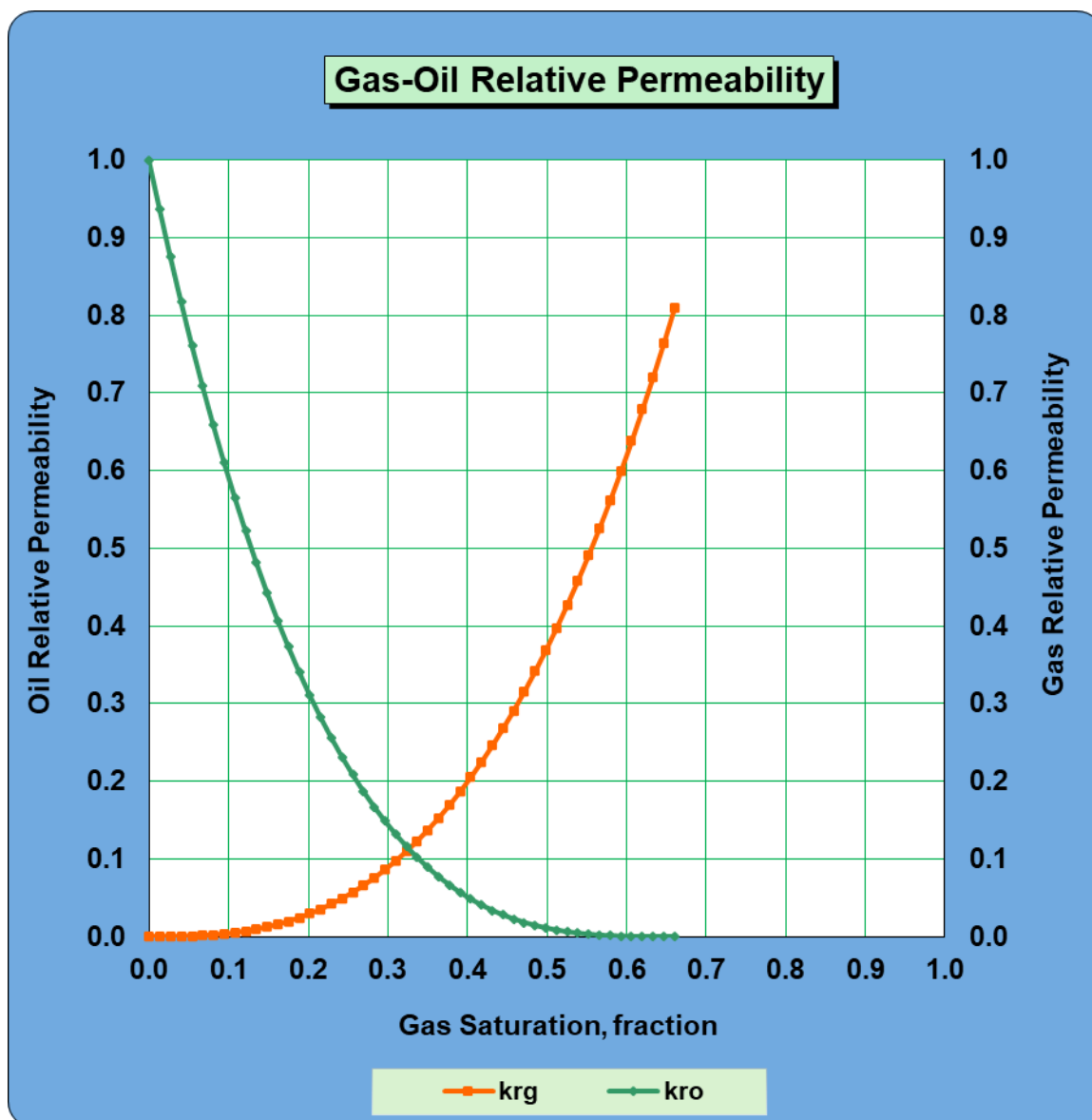


Figure (2.10.13)

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1200
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 1.84
Sample #	: 126	Perm. to gas at residual oil saturation, mD	: 1.49
Porosity, %	: 24.4	Residual oil saturation (S _{org}), fraction	: 0.2200
Permeability to air, mD	: 39.4	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8483.10

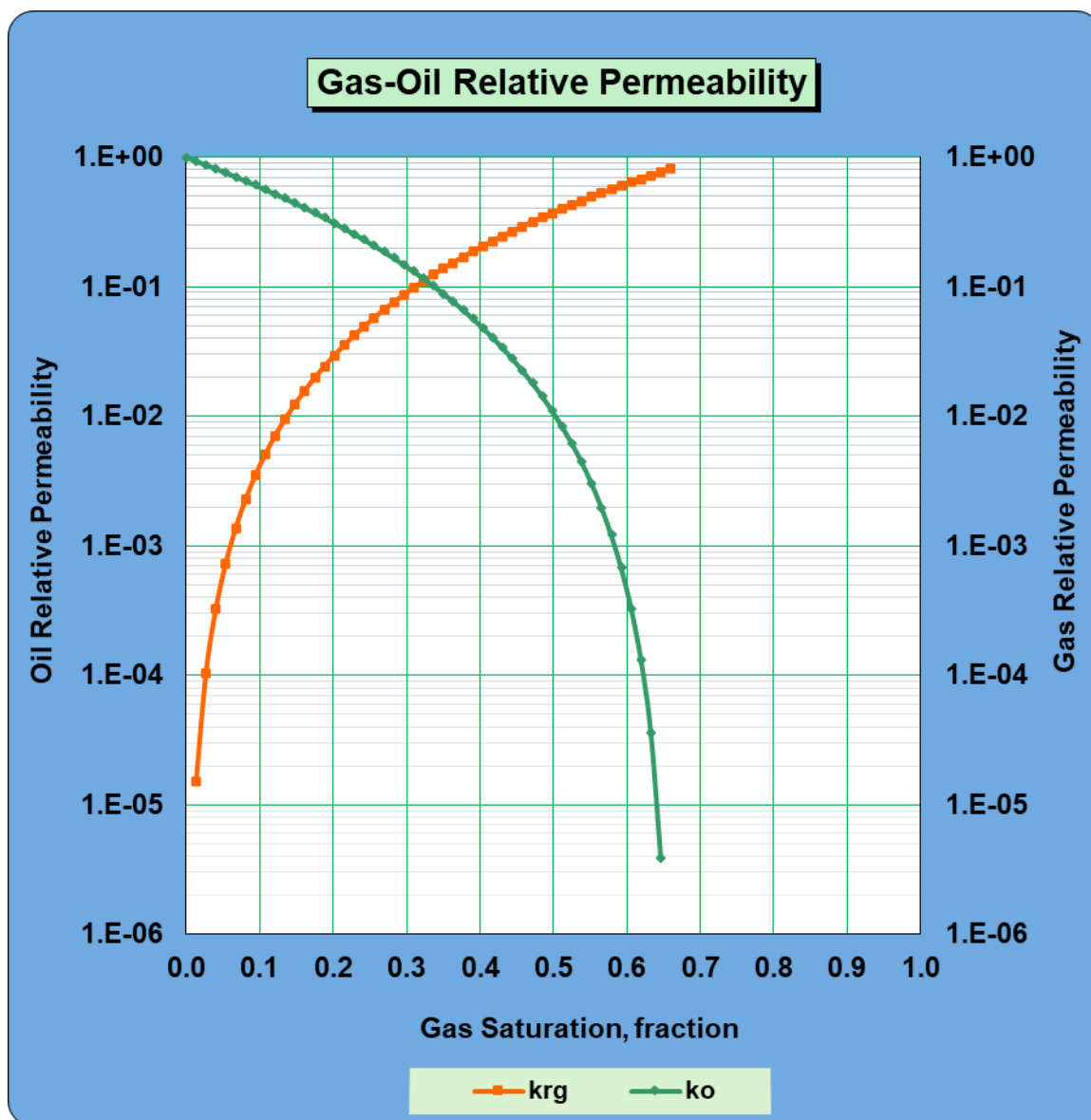


Figure (2.10.14)

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1200
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 1.84
Sample #	: 126	Perm. to gas at residual oil saturation, mD	: 1.49
Porosity, %	: 24.4	Residual oil saturation (Sorg), fraction	: 0.2200
Permeability to air, mD	: 39.4	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8483.10

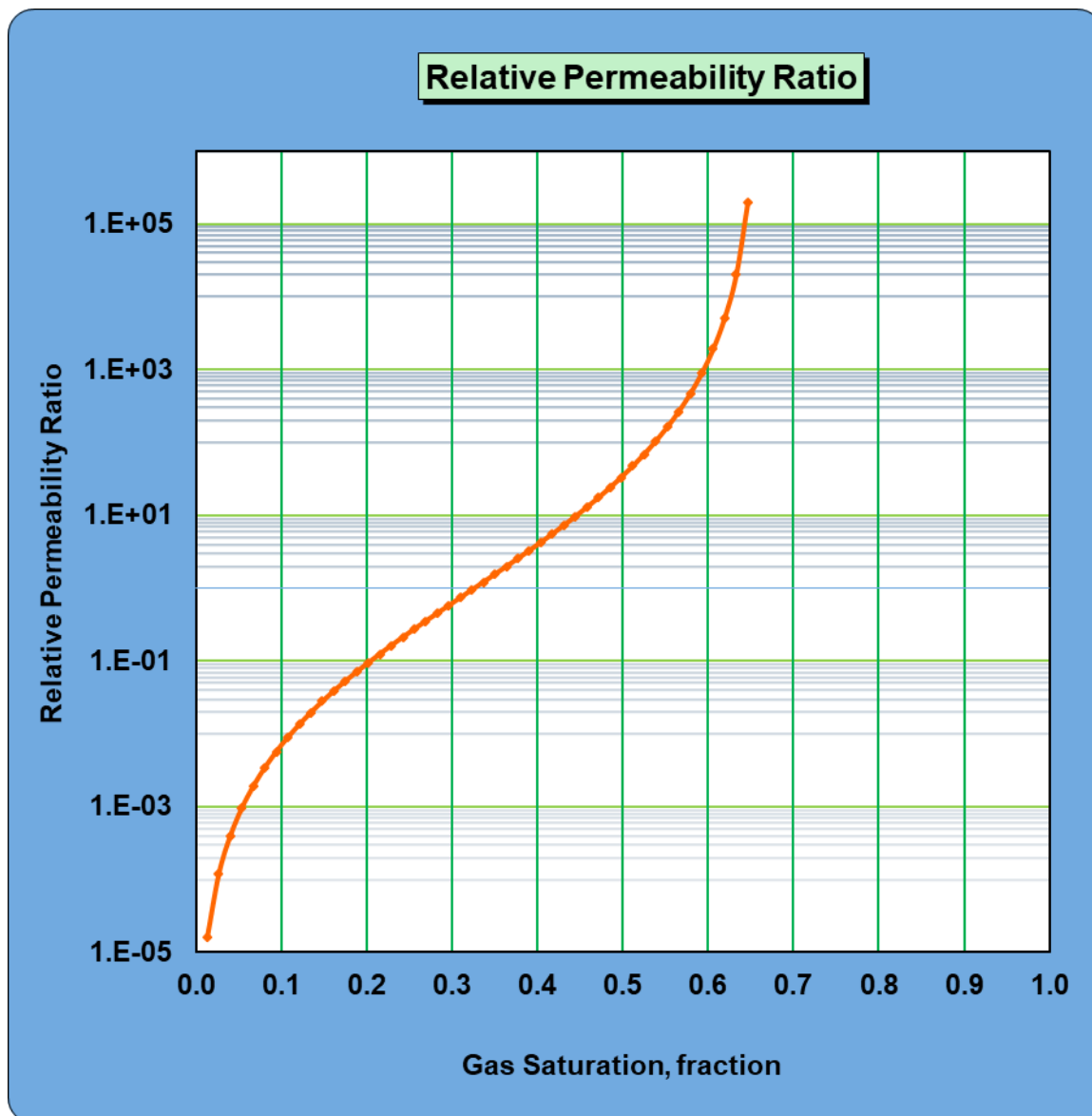


Figure (2.10.15)

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1200
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 1.84
Sample #	: 126	Perm. to gas at residual oil saturation, mD	: 1.49
Porosity, %	: 24.4	Residual oil saturation (Sorg), fraction	: 0.2200
Permeability to air, mD	: 39.4	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8483.10

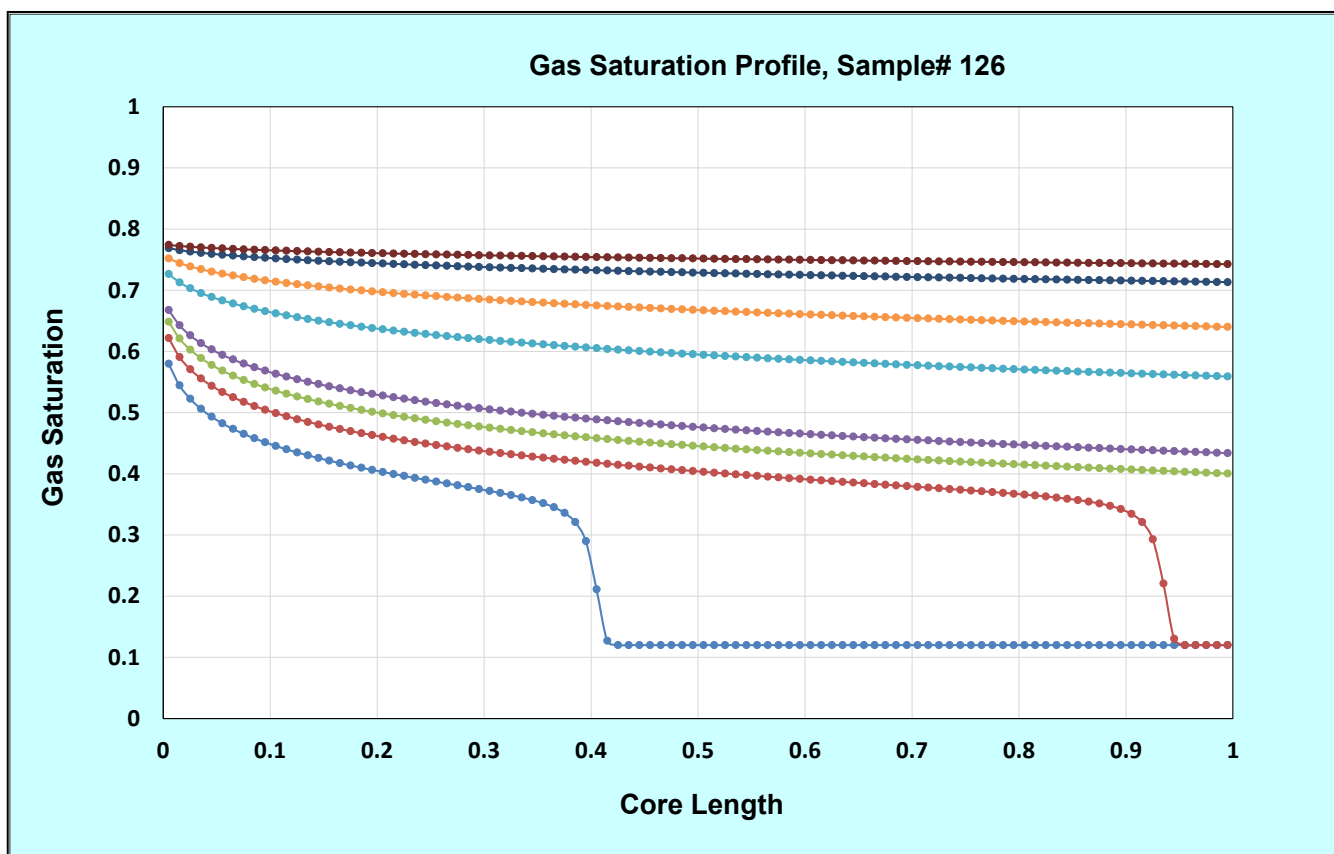


Figure (2.10.16)

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1800
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 20.91
Sample #	: 275	Perm. to gas at residual oil saturation, mD	: 13.59
Porosity, %	: 24.2	Residual oil saturation (Sorg), fraction	: 0.1650
Permeability to air, mD	: 225	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8636.06

Gas Saturation (fraction)	Relative Permeability to Oil, (fraction)	Relative Permeability to Gas, (fraction)	Relative Permeability Ratio
0.0000	1.0000	0.0000	9.372E-57
0.0134	0.9020	0.0000	8.750E-07
0.0267	0.8119	0.0000	1.100E-05
0.0401	0.7291	0.0000	5.062E-05
0.0535	0.6533	0.0001	1.547E-04
0.0668	0.5838	0.0002	3.779E-04
0.0802	0.5204	0.0004	8.024E-04
0.0936	0.4627	0.0007	1.548E-03
0.1069	0.4101	0.0011	2.787E-03
0.1203	0.3625	0.0017	4.762E-03
0.1337	0.3194	0.0025	7.814E-03
0.1470	0.2805	0.0035	1.242E-02
0.1604	0.2455	0.0047	1.925E-02
0.1738	0.2141	0.0063	2.921E-02
0.1871	0.1859	0.0081	4.358E-02
0.2005	0.1608	0.0103	6.414E-02
0.2139	0.1385	0.0129	9.334E-02
0.2272	0.1188	0.0160	1.346E-01
0.2406	0.1014	0.0195	1.927E-01
0.2540	0.0860	0.0236	2.743E-01
0.2673	0.0726	0.0282	3.889E-01
0.2807	0.0609	0.0335	5.498E-01
0.2941	0.0508	0.0394	7.760E-01
0.3074	0.0421	0.0461	1.095E+00
0.3208	0.0346	0.0535	1.546E+00
0.3342	0.0282	0.0617	2.187E+00

Table (2.10.5)

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1800
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 20.91
Sample #	: 275	Perm. to gas at residual oil saturation, mD	: 13.59
Porosity, %	: 24.2	Residual oil saturation (Sorg), fraction	: 0.1650
Permeability to air, mD	: 225	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8636.06

Gas Saturation (fraction)	Relative Permeability to Oil, (fraction)	Relative Permeability to Gas, (fraction)	Relative Permeability Ratio
0.3476	0.0228	0.0707	3.104E+00
0.3609	0.0182	0.0807	4.425E+00
0.3743	0.0145	0.0917	6.341E+00
0.3877	0.0113	0.1037	9.151E+00
0.4010	0.0088	0.1167	1.332E+01
0.4144	0.0067	0.1309	1.957E+01
0.4278	0.0050	0.1463	2.911E+01
0.4411	0.0037	0.1629	4.389E+01
0.4545	0.0027	0.1809	6.729E+01
0.4679	0.0019	0.2002	1.051E+02
0.4812	0.0013	0.2209	1.681E+02
0.4946	0.0009	0.2432	2.761E+02
0.5080	0.0006	0.2670	4.683E+02
0.5213	0.0004	0.2924	8.259E+02
0.5347	0.0002	0.3195	1.528E+03
0.5481	0.0001	0.3483	3.003E+03
0.5614	0.0001	0.3790	6.369E+03
0.5748	0.0000	4.115E-01	1.495E+04
0.5882	0.0000	4.460E-01	4.031E+04
0.6015	0.0000	4.825E-01	1.331E+05
0.6149	0.0000	5.211E-01	6.057E+05
0.6283	0.0000	5.618E-01	4.959E+06
0.6416	0.0000	6.047E-01	1.708E+08
0.6550	0.0000	6.500E-01	*****

Table (2.10.5)-Continue

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1800
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 20.91
Sample #	: 275	Perm. to gas at residual oil saturation, mD	: 13.59
Porosity, %	: 24.2	Residual oil saturation (Sorg), fraction	: 0.1650
Permeability to air, mD	: 225	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8636.06

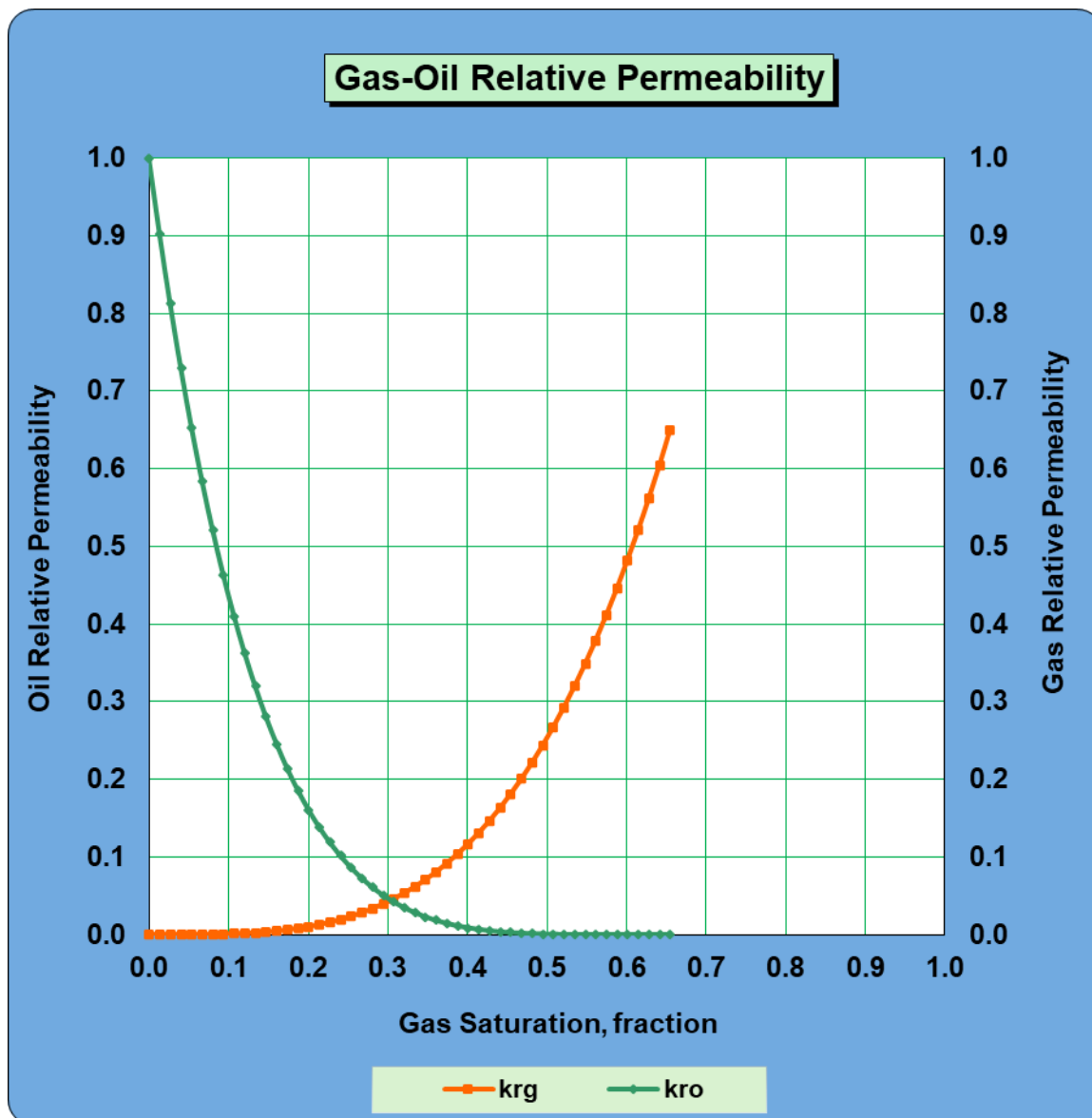


Figure (2.10.17)

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1800
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 20.91
Sample #	: 275	Perm. to gas at residual oil saturation, mD	: 13.59
Porosity, %	: 24.2	Residual oil saturation (Sorg), fraction	: 0.1650
Permeability to air, mD	: 225	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8636.06

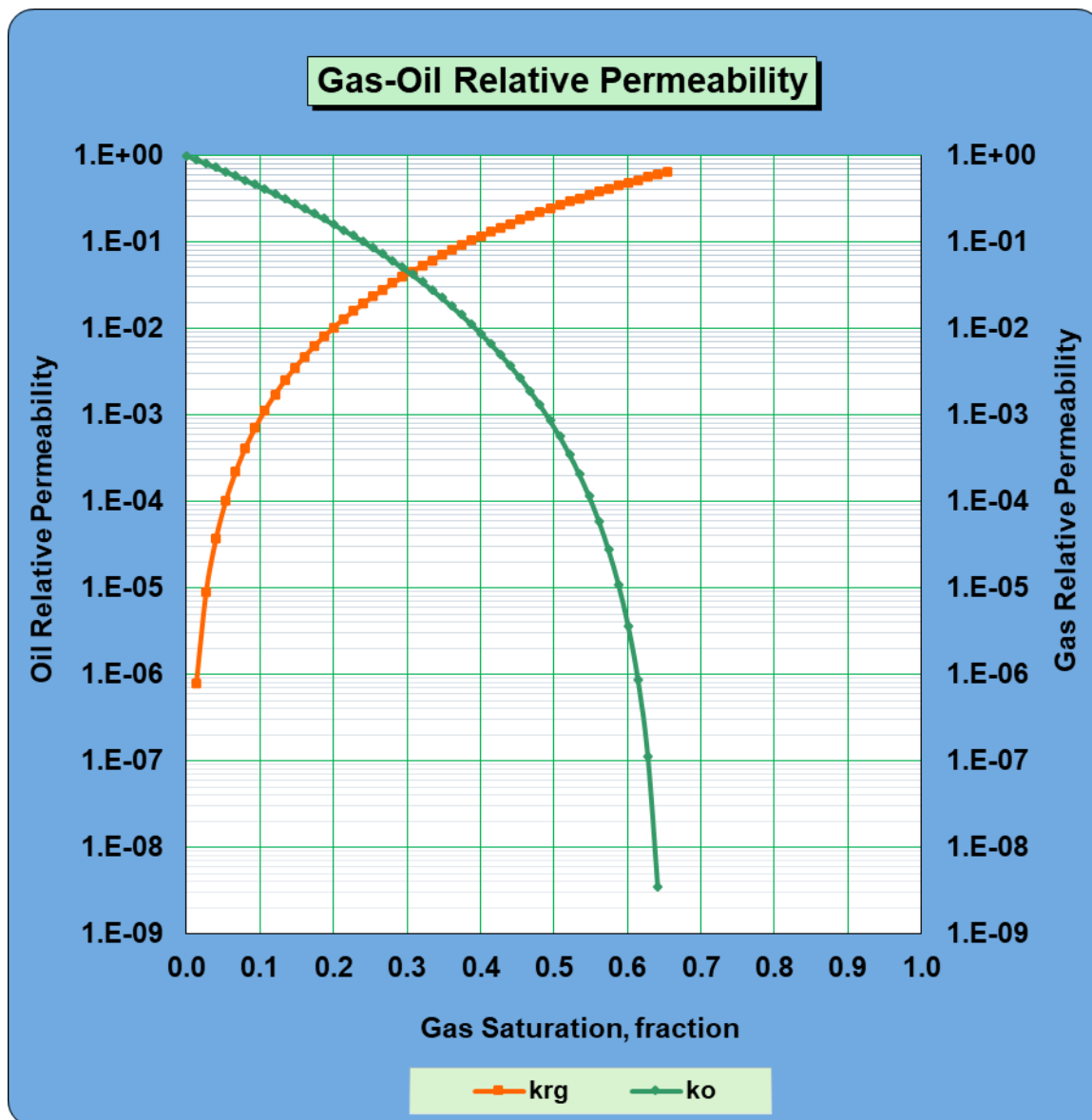


Figure (2.10.18)

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1800
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 20.91
Sample #	: 275	Perm. to gas at residual oil saturation, mD	: 13.59
Porosity, %	: 24.2	Residual oil saturation (Sorg), fraction	: 0.1650
Permeability to air, mD	: 225	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8636.06

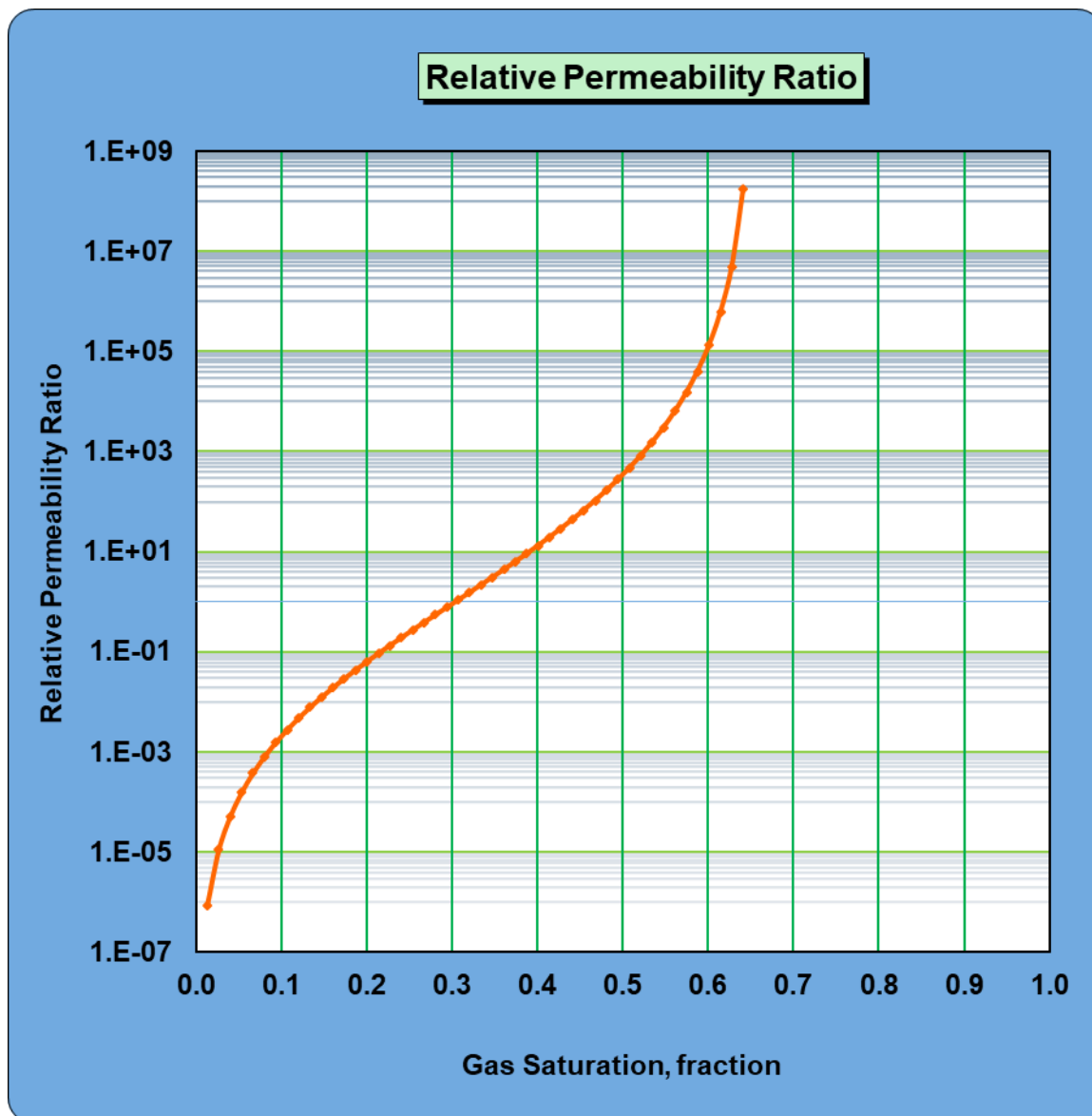


Figure (2.10.19)

GAS-OIL RELATIVE PERMEABILITY

Client	: Mellitah	Irreducible water saturation, fraction	: 0.1800
Well	: CE03-NC41	Perm. to oil at irreducible water sat., mD	: 20.91
Sample #	: 275	Perm. to gas at residual oil saturation, mD	: 13.59
Porosity, %	: 24.2	Residual oil saturation (Sorg), fraction	: 0.1650
Permeability to air, mD	: 225	Net Confining Pressure, psi	: 3858
Formation	: Elgaria	Temperature, °F	: 278
Field	: Bahr Essalam	Sample Depth, ft.in	: 8636.06

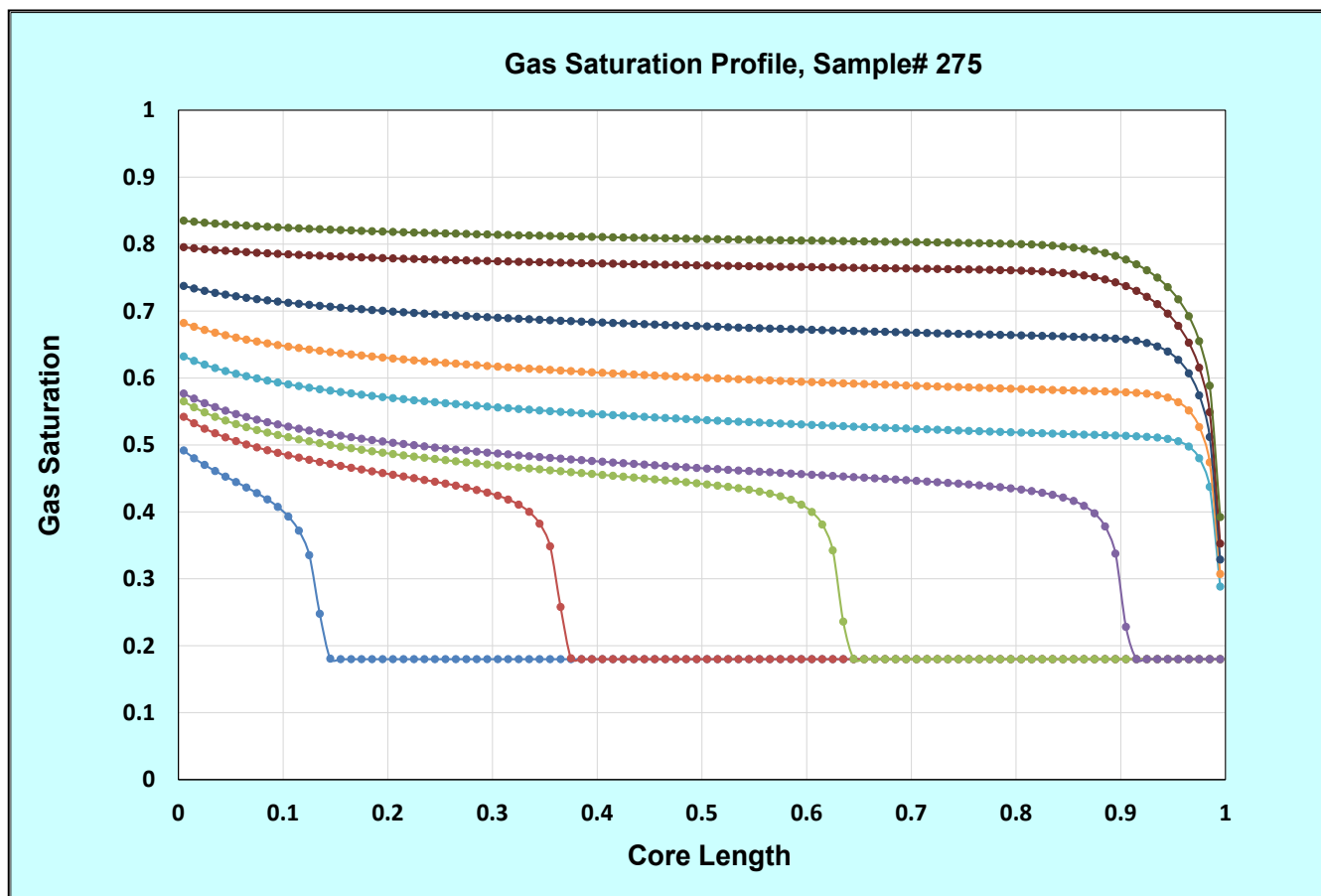


Figure (2.10.20)

Conclusions

Conclusions

PART ONE: CONVENTIONAL CORE ANALYSIS (CCA)

In conclusion, after doing all the required conventional core analysis, the results obtained show the following:

1- Porosity and Permeability (CCA)

In this study the controls on porosity and permeability of Elgaria Formation for well CE03-NC41, are considered with respect to their texture and cementation, their petrophysical classification and the effect of subdivision of the petrophysical rock types. The porosity and permeability of the study which are determined from the laboratory are highly variable across the whole volume of the reservoir being poor to fair in some intervals and good in other intervals this may be due to heterogeneity of the reservoir. Horizontal permeability range between (0.225-2382 mD), most the samples are vuggy, and fractured while the vertical permeability classified as poor to good according to the values obtained (0.194-3711 mD) but most of them are vuggy and low. Finally, the porosity values (4.15%-30.82%) indicate poor to good classification.

- **Additionally**, many plug samples are fractured, vuggy, broken, and low permeability.

PART TWO: SPECIAL CORE ANALYSIS (SCAL)

1. Effect of overburden pressure on porosity and permeability

- Ten plug samples, representing El-Garia formation, were selected from the well # CE03-NC41, to investigate the effect of overburden pressure on effective porosity and permeability. Porosity ranges between 21.0% and 24.9% and permeabilities between 4.89 mD and 225 mD. The porosity, and permeability were measured at overburden pressure at different pressures as in section 1. The generated data were presented in Tables (2.1.1-2.1.10), and in Figures (2.1.1-2.1.10). The data show a decrease in porosity with increasing overburden pressure. The data for core plugs exhibit a reduction in absolute permeability with increasing overburden pressure. The reductions in permeability values at 26.6 Mpa (3858 psi) for samples # 80, and 278 as

an example, which selected from well **CE03-NC41**, are 11.86%, and 8.39%, respectively. These reductions in absolute permeabilities with increasing overburden pressure can be attributed to the decrease in pore size and compressibility of the rock. It is evident from most figures related to the subjected well that most of reduction in porosity and takes place in the pressure range of 0- 26.6 Mpa (0-3858 psi). The lesser decrease in porosity at higher stress values is attributed to the increase in rigidity of the rock matrix due to compaction of grains at higher stress. A possible explanation is a limiting value of elastic deformation of the matrix much slower type of deformation occurs.

2. Formation resistivity factor at ambient conditions and at overburden Pressure

- In performing laboratory measurements, changes have been observed in the resistivity of fluid filled reservoir rocks because of changing overburden pressure conditions. These changes may result from changing the internal pore structure and an increase in tortuosity and decrease in the effective cross-sectional area that is available for the flow of electric current, Wyble (Reference 10). Most previous studies showed that the formation resistivity factor and cementation exponent increase with overburden pressure. In this work, both formation resistivity factor (FF) and cementation exponent (m) were found to increase with confining pressure for Elgaria formation. In fact, this relation is very clear in this study in section (3) and section (4) of this report. The value of cementation exponent at overburden pressure increased from 1.56 to 1.76 as a result of increasing the pressure up to 3858 psi. The overburden pressure will reduce the bulk volume, at low overburden pressures, fissures start to close with small compression in mineral grains. As the overburden pressure increases, the rock undergoes bulk compression resulting from pore and grain deformation. As the pressure is depleted in a reservoir, the effective overburden increases causing a reduction in pore volume. The results indicate the rock is compacted due to overburden pressure, the matrix is under stress and porosity decreases, and therefore the cementation exponent will change. This change may result from changing the internal pore structure and a decrease in the effective cross-sectional area that is available to flow the electric current. The reason for the observed variation in cementation factor has been attributed to several factors such as:

- Degree of cementation.
- Shape, sorting and packing of particles system
- Type of pore system (intergranular, fractured, vuggy and microporous system)
- Tortuosity
- Constrictions existing in the porous system
- Conductive solids and clay minerals
- Compaction due to overburden pressure

3. Resistivity Index at ambient conditions and at overburden Pressure

- Mungan and Moore (Reference 11) pointed out that an Archie's saturation exponent as high as 9.0 can be calculated when the conductive liquid is non-wetting. For strongly water-wet rocks, a unique relationship exists between Archie's saturation exponent and resistivity as described by Archie's law. At lower saturation, Archie's saturation exponent becomes larger and more saturation dependent. They attributed the change of saturation exponent (n) to higher values at lower water saturation to the fact that water is becoming discontinuous and not contributing to the flow of current.
- In this work, the average value saturation exponent (n) for the samples of Elgaria formation was increased from 1.71 (at ambient conditions) to 2.06 when the pressure increased up to 3858 psi, for more details see figures (2.5.11 and 2.6.11) in sections five and six in this report.

4. Capillary Pressure Measurements

- Capillary pressure characteristics of reservoir rocks are dependent on grain size, grain shape, packing, sorting and environment of deposition and diagenesis. These geological parameters affect the pore throat radius, often referred to as the pore size distribution within the rock. In the present study, the centrifuge capillary pressure (section 7) shows the drainage cycle for air -water system, we can note that the low permeability samples gave higher irreducible water saturation than high permeability samples in most cases. As the overburden applied on the samples, the amount of water comes out from the sample, therefore the water saturation will decrease and saturation exponent (n) increase as well, this fact supports our result in item (3) as mentioned above.

- Drainage and imbibition curves in mercury injection capillary curves were obtained (see section 8). Pressure is introduced into the apparatus, which allows the mercury to enter the rock sample. The pressure is introduced at a series of increasing pressure increments. The magnitude of pressure and the incremental volume of mercury injected into the sample are monitored and recorded after equilibrium. As the pressure increases, mercury moves into the sample's pores, vacating the stem (Drainage or Intrusion). The greater the pressure, the smaller the pore diameter into which the mercury can be forced. An imbibition curve can be produced by incrementally decreasing the pressure gradually in steps to withdraw the mercury from the pores in pressure steps from 60,000 psi to 14 psi (Imbibition or Extrusion).

Pore throat radius can be determined from mercury injection tests and may be used to categorize the rock by pore type. macro, meso, and micro. Micro-porosity in reservoir rocks has been described as the pore system whose average pore diameter is less than 0.0625mm, Pittman (Reference 8). From the results of mercury injection capillary pressure measurements (up to 60,000 psi), indicate the selected samples have pore size distributions with three different types of pores (macropores, mesopores and micropores). High pressure mercury porosimeter is useful in determining the relative proportion of micro porosity in reservoir rocks.

5. Relative Permeability

- Relative permeability data are used to predict the most effective hydrocarbon displacement mechanisms in a reservoir and how to get most of the oil or gas out of the reservoir as quickly and cheaply as possible. In this study, the test of gas-water and gas-oil relative permeability at reservoir conditions was conducted for five samples. The test results show that the effective water permeability in the gas-water flooding was higher than the effective permeability to oil at initial water saturation in the gas-oil flooding. Also, the effective permeability to gas at initial water saturation (S_{wi}) was higher than the effective permeability to gas at residual oil saturation (S_{org}). The percentage of residual oil saturation ranged from 16.50 to 24.97%, with sample number (275), which had the highest permeability (225 mD), showing the lowest value of residual oil saturation (16.5%).

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